

OBG Question-Engraved Concept Book

Obstetrics & Gynaecology

NEET-PG · INI-CET · FMGE · UPSC CMS

Built from **1,063 PYQs** · 12 Chapters · 65,000+ words
NEET-PG 2026 Edition

“A brilliant senior teaching you the secrets of the exam.”

OBG Question-Engraved Concept Book

A Complete High-Yield NEET-PG / INI-CET Preparation Book

Obstetrics & Gynaecology — All 1,063 PYQs Integrated

"A brilliant senior is teaching me the secrets of the exam."

How to Use This Book

This is not a textbook. This is a **concept book built from questions**.

Every major concept in OBG has been reverse-engineered from the 1,063 PYQ MCQs (NEET-PG 2015–2025) in this database. Each chapter:

1. **Starts with examiner psychology** — what does this topic's question pattern look like?
2. **Builds the concept** — deep but simple, with analogies and clinical stories
3. **Embeds MCQs into teaching** — concept → question → trap → elimination → takeaway
4. **Removes bias** — trains you against psychological traps
5. **Ends with rapid revision** — 20-second summaries, tables, volatile facts

Study Strategy

- **First pass:** Read sequentially, attempt each MCQ before reading the explanation
- **Second pass:** Jump to 🔍 Examiner Notes and ⚡ Revision sections only
- **Day before exam:** Read only the ⚡ Ultra High-Yield Revision layers

Negative Marking Reminder (+4/–1)

- If you can eliminate even 1 option → expected value positive → **always attempt**
- This book trains your elimination instinct systematically

Book Structure

Part	Chapters	Topics
Part I — Obstetrics	1–4	Normal pregnancy, Antenatal care, HDP, Labour
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Total: 65,739 words | 1,063 questions integrated

PART I — OBSTETRICS

Chapter 1: Normal Pregnancy — Anatomy, Embryology, Physiology & Placentation

Big Picture First

Normal pregnancy is the single most tested foundation in NEET-PG OBG. Examiners love it because it rewards students who truly understand the *logic* of embryology rather than rote memorisation — and it cruelly punishes those who have only half-learned the facts. A question on "when does the morula enter the uterine cavity?" sounds elementary, but if you confuse it with blastocyst formation or implantation timing, you lose four marks. That's two and a half wrong answers wiped out.

The three areas examiners return to most obsessively are: (1) the embryological timeline — zygote → morula → blastocyst → implantation, with exact day numbers; (2) the physiological changes of pregnancy, especially the "which one DECREASES?" traps; and (3) the anatomy of female pelvic structures, especially Wolffian and Müllerian duct remnants, uterine blood supply, and levator ani. These areas yield disproportionately high returns per hour of study.

The most misunderstood area, without question, is the distinction between ectopic pregnancy site for *fertilisation* versus ectopic pregnancy site for *longest survival* — many students get these backwards. Similarly, the distinction between epoophoron (cranial Wolffian remnant) and paroophoron (caudal remnant), and their respective clinical sequelae, trips up a large fraction of candidates. Once you understand the anatomical logic, these stop being memory items and become reasoning items.

Question Pattern Map

Recurring PYQ trends: - Embryological timeline (day of entry into uterus, day of implantation, cell stage at entry) - "Which parameter DECREASES in pregnancy?" — almost always a trap on respiratory or cardiovascular physiology - hCG peak timing — examined nearly every alternate year - Ectopic pregnancy: site of fertilisation vs. site of longest survival (two different questions, two different answers) - Wolffian duct anatomy: Gartner's duct location, epoophoron vs. paroophoron - Human placenta classification (all-of-the-above answer) - Amniotic fluid contribution (maternal vs. fetal, before vs. after 20 weeks)

Common examiner traps: - FRC vs. vital capacity (only FRC decreases; vital capacity is maintained) - Ampulla = fertilisation; Interstitium = longest survival (not the same) - "Superfetation" vs. "superfecundation" (one word changes the entire concept) - Epoophoron vs. paroophoron — both Wolffian remnants but different clinical manifestations - hCG peaks at 8–10 weeks, NOT at 11–13 weeks (the NT scan window is a distractor)

How questions are typically framed: - "Which of the following DECREASES in pregnancy?" → pick the odd one out from a list of increasing values - "Longest ectopic survival is in..." → pick interstitium, not ampulla - "Conceptus enters uterus at which cell stage?" → 16-cell morula, not 8-cell or blastocyst - "True statement regarding Organ of Rosenmüller?" → multiple true-sounding options, needs knowing the ONE specific correct detail

Core Concept Deep Dive

1. Female Reproductive Embryology: The Wolffian-Müllerian Story

Think of embryological development like two parallel railway tracks — the Wolffian (mesonephric) track and the Müllerian (paramesonephric) track. In females, testosterone is absent, so the Wolffian track regresses while the Müllerian track develops into the fallopian tubes, uterus, cervix, and upper vagina. But the Wolffian track doesn't disappear completely — it leaves *remnants*, and those remnants are the source of persistent NEET-PG questions.

Wolffian duct remnants in females: - **Gartner's duct:** the most caudal remnant; runs in the anterolateral wall of the vagina and within the *leaves of the broad ligament*. It can form Gartner's duct cysts in the lateral vaginal wall. - **Epoophoron (Organ of Rosenmüller):** the cranial end of the Wolffian (mesonephric) body; lies in the mesovarium between the ovary and fallopian tube; has vertical tubules lined by cuboidal cells. - **Paroophoron:** the caudal remnant of the Wolffian body; can form *paraovarian cysts*.

 **Examiner Note:** The question "Wolffian duct remnants in females are found in?" specifically points to Gartner's duct — answer is the broad ligament. Do NOT confuse this with the ovarian fossae or pouches.

MCQ Integration — Embryological Anatomy

Q1 — Wolffian duct remnants

Wolffian (mesonephric) duct remnants in females are found in:

- A. Pouch of Douglas
- B. Leaves of broad ligament
- C. Uterovesical pouch
- D. Iliac fossa

► Answer & Explanation:

****B. Leaves of broad ligament**** — Gartner's duct (mesonephric duct remnant) lies in the broad ligament. May form Gartner's duct cysts in the lateral vaginal wall.

Why students get trapped: Option A (Pouch of Douglas) is the posterior cul-de-sac and is a common "gynae structure" distractor. Option D (iliac fossa) sounds anatomically plausible for a retroperitoneal structure. The key is knowing that the Wolffian duct runs in a lateral position, embedded in the broad ligament on its way from kidney to bladder.

Elimination strategy: Eliminate C (uterovesical pouch = anterior cul-de-sac, unrelated to embryological remnants) and D (iliac fossa = not a gynae structure). Between A and B, remember: Gartner's duct is in the *lateral vaginal wall and broad ligament* — the pouch of Douglas is posterior, midline.

Key takeaway: Gartner's duct = Wolffian remnant = broad ligament + lateral vaginal wall. Paroophoron forms paraovarian cysts. Epoophoron is in the mesovarium.

Q3 — Organ of Rosenmüller (Epoophoron)

Which of the following is TRUE regarding the Organ of Rosenmuller (epoophoron)?

- A. It forms paraovarian cysts
- B. It consists of horizontal tubules in the mesovarium
- C. Tubules are lined by columnar cells
- D. It represents the cranial end of the Wolffian (mesonephric) body

► **Answer & Explanation:**

****D. It represents the cranial end of the Wolffian body**** — Epoophoron (Organ of Rosenmuller) = cranial remnant of the mesonephric (Wolffian) body; lies in the mesovarium with vertical tubules lined by cuboidal cells. Paroophoron (the caudal remnant) can form paraovarian cysts.

Why students get trapped: This question has four plausible-sounding options. Option A trips up students who know epoophoron is a Wolffian remnant but forget it is the *cranial* part — paraovarian cysts come from paroophoron (the caudal remnant). Option B and C are anatomically wrong details (tubules are vertical, not horizontal; lining is cuboidal, not columnar) — classic examiner technique of putting partially correct statements.

Key takeaway: Epoophoron = cranial, vertical tubules, cuboidal lining, mesovarium. Paroophoron = caudal, forms paraovarian cysts.

Remnant	Origin	Location	Clinical Significance
Gartner's duct	Wolffian duct	Broad ligament / lateral vaginal wall	Gartner's duct cysts
Epoophoron	Cranial Wolffian body	Mesovarium	Rarely symptomatic
Paroophoron	Caudal Wolffian body	Mesosalphinx	Paraovarian cysts

2. The Embryological Timeline: A Story of Days

This is pure exam gold. Learn it as a narrative, not a list.

Day 0 (Fertilisation): The ovulated secondary oocyte (arrested in metaphase II) meets a spermatozoon in the *ampulla of the fallopian tube*. Fertilisation triggers completion of meiosis II and extrusion of the second polar body. The fertilised ovum is now a zygote with 46 chromosomes.

Day 1–3 (Cleavage): The zygote undergoes rapid mitotic divisions (cleavage) as it travels along the fallopian tube toward the uterus. The cells get smaller with each division (no growth of the whole embryo yet). By day 3, there are 8–12 cells still enclosed in the zona pellucida.

Day 3–4 (Morula): At the 16-cell stage, the embryo is called a *morula* (from Latin "mulberry"). This is when it enters the uterine cavity. It is still enclosed in the zona pellucida.

Day 5 (Blastocyst): Fluid accumulates and a cavity forms (blastocoel). Now called blastocyst with two distinct populations: inner cell mass (ICM/embryoblast → embryo) and trophoblast (→ placenta). The zona pellucida begins to thin and eventually *hatches* from it.

Day 6–7 (Implantation begins): The free blastocyst attaches to the posterior wall of the uterine cavity, most commonly the posterior fundus. In a regular 28-day cycle (ovulation day 14), implantation occurs on cycle days 20–22.

Day 8–12 (Deep implantation): The blastocyst burrows into the endometrium (secretory phase, thickened ~15–20 mm). Trophoblast differentiates into cytotrophoblast and syncytiotrophoblast. hCG is first detectable in maternal blood by day 8–10, rising exponentially.

 **Examiner Note:** The question says "zygote with zona pellucida reaches uterine cavity" — this specifies the morula stage (day 4), not the blastocyst (day 5, when zona hatches). This distinction is the trap.

MCQ Integration — Embryological Timeline

Q284 — Zygote reaches uterine cavity

Zygote with zona pellucida reaches the uterine cavity by:

- A. 2 days
- B. 4 days
- C. 5 days
- D. 6 days

► Answer & Explanation:

****B. 4 days**** — Morula (16-cell) enters uterus on day 4; implantation begins ~day 6–7 after hatching from zona.

Q285 — Conceptus enters uterus at cell stage

Conceptus enters the uterine cavity at which cell stage?

- A. 4 cells
- B. 8 cells
- C. 16 cells (morula)
- D. 32 cells

► Answer & Explanation:

****C. 16 cells (morula)**** — At day 3–4, 16-cell morula enters the uterine cavity; cavitates into blastocyst on day 5.

Q286 — Day of implantation

Implantation occurs on which menstrual cycle day?

- A. 5–7
- B. 14–18
- C. 20–22
- D. 26–28

► Answer & Explanation:

****C. 20–22**** — Ovulation day 14 + ~6–7 days for implantation → day 20–22 of a 28-day cycle.

Why students get confused: Option B (14–18) is a trap because day 14 = ovulation. Students think "implantation happens soon after ovulation" and pick 14–18. But the embryo needs 6–7 days after fertilisation to reach the uterus and implant — that means day 20–22.

Key takeaway chain: Fertilisation day 14 → Morula enters uterus day 18 → Blastocyst day 19 → Implantation day 20–22. Always add 6–7 days to ovulation for implantation.

Q287 — First polar body formation

The first polar body is formed after:

- A. Mitosis
- B. First meiosis
- C. Second meiosis
- D. Fertilisation

► **Answer & Explanation:**

****B. First meiosis**** — 1st polar body extruded at completion of meiosis I (just before ovulation). 2nd polar body forms after fertilisation (completion of meiosis II).

The complete oocyte meiosis story: The primary oocyte begins meiosis I in fetal life and is *arrested in prophase I* until puberty. At ovulation, LH surge triggers completion of meiosis I → extrusion of 1st polar body → secondary oocyte arrested in metaphase II. Fertilisation triggers completion of meiosis II → extrusion of 2nd polar body.

Event	Stage
1st polar body extruded	After meiosis I (just before ovulation)
2nd polar body extruded	After meiosis II (triggered by fertilisation)
Oocyte arrested from birth to LH surge	Prophase I
Secondary oocyte at ovulation arrested in	Metaphase II

3. hCG: The Pregnancy Hormone Timeline

Human chorionic gonadotropin (hCG) is produced by the syncytiotrophoblast, first detectable ~8–10 days after fertilisation. It doubles every 48 hours in early normal pregnancy (useful for diagnosing ectopic if it doesn't double). It peaks at gestational weeks 8–10 (~100,000 mIU/mL), then declines and plateaus.

Why does hCG peak matter clinically? - Morning sickness is worst at 8–10 weeks — coincides with hCG peak (hCG stimulates TSH-like receptors → transient hyperthyroidism in severe cases) - Molar pregnancies: hCG is disproportionately elevated - Ectopic pregnancy: hCG rises slower than expected (sub-doubling) - After 10 weeks, declining hCG coincides with improvement in nausea

Q288 — Peak serum hCG

Peak serum hCG is attained at which gestational age?

- A. 7–9 weeks
- B. 11–13 weeks
- C. 20 weeks
- D. 25 weeks

► **Answer & Explanation:**

****A. 7–9 weeks**** — hCG peaks at 8–10 weeks (~100,000 mIU/mL), then declines and plateaus.

Why option B is the trap: 11–13 weeks is the nuchal translucency scan window — a very memorable milestone in the first trimester. Students conflate "first trimester peak" with the NT scan timing. But hCG peaks earlier, at 7–9 weeks, and by 11–13 weeks it is already declining.

4. Physiological Changes of Pregnancy: The "What Decreases?" Battlefield

Pregnancy causes sweeping physiological changes across every system. The examiner's favourite tactic is to ask "which of these DECREASES?" because most parameters increase. Know the exceptions cold.

Cardiovascular changes: - Cardiac output: ↑ 40–50% (increases from early first trimester) - Heart rate: ↑ ~15–20 bpm - Stroke volume: ↑ - Blood pressure: ↓ in first and second trimester (peripheral vasodilation by progesterone), rises back to normal in third trimester - SVR (systemic vascular resistance): ↓ significantly - ECG: left axis deviation (uterus pushes heart upward and to left); NOT right axis deviation - LVEDD: ↑ (increased preload from expanded blood volume)

Respiratory changes: - Tidal volume: ↑ 30–40% - Respiratory rate: unchanged - Minute ventilation: ↑ 40% - Inspiratory capacity: ↑ - Vital capacity: unchanged (or slightly increased) - **Functional residual capacity (FRC): ↓ ~20%** ← The only lung volume that falls - Residual volume: ↓ - Total lung capacity: ↓ slightly - PaO₂: ↑ (due to increased ventilation) - PaCO₂: ↓ (hyperventilation → compensated respiratory alkalosis)

Q273 — Decreased parameter in pregnancy

Which respiratory parameter DECREASES in pregnancy?

- A. Respiratory rate
- B. Vital capacity
- C. Functional residual capacity (FRC)
- D. Inspiratory capacity

► Answer & Explanation:

****C. Functional residual capacity**** — FRC decreases ~20% (diaphragm elevation by gravid uterus); minute ventilation, tidal volume, inspiratory capacity all INCREASE. Respiratory rate is unchanged. PCO₂ drops (compensated respiratory alkalosis).

Q856 — CVS changes in pregnancy

During pregnancy, true statement about CVS is:

- A. Cardiac output decreases
- B. Right axis deviation
- C. Increase in left ventricular end diastolic diameter
- D. All of the above

► Answer & Explanation:

****C. Increase in left ventricular end diastolic diameter**** — Pregnancy causes increased plasma volume and preload, leading to increased LVEDD (physiological cardiac dilatation); cardiac output increases and axis shifts left, not right.

Why B (right axis deviation) is tempting: Students remember "the heart is pushed by the uterus" and think the heart rotates to the right. But the uterus pushes the heart *up and to the left* → left axis deviation on ECG.

Clinical importance of FRC decrease: Lower FRC means less oxygen reserve during apnoea (e.g., induction of anaesthesia). Pregnant patients desaturate much faster than non-pregnant patients during airway management — a critical anaesthesia principle.

5. Uterine Blood Supply: "Water Under the Bridge"

The uterine artery arises from the anterior division of the internal iliac artery. It travels medially in the base of the broad ligament and, at the level of the internal os, crosses *over* the ureter — the famously remembered "water under the bridge" (ureter = water, uterine artery = bridge). This crossing is surgically critical: during hysterectomy, the ureter can be inadvertently ligated when the uterine artery is clamped.

Q289 — Uterine artery origin

The uterine artery is a branch of:

- A. Left common iliac
- B. Internal iliac (anterior division)
- C. Internal pudendal
- D. Ovarian artery

► Answer & Explanation:

****B. Internal iliac (anterior division)**** — Anterior division of internal iliac → uterine artery; crosses ureter ("water under the bridge").

Key learning: The internal pudendal artery (option C) is also from the anterior division of the internal iliac, but it supplies the perineum, not the uterus. The ovarian artery (option D) arises directly from the aorta and anastomoses with the uterine artery at the uterine cornu.

6. Ectopic Pregnancy: The Two Questions Every Examiner Loves

There are two completely different questions about ectopic pregnancy and the fallopian tube, and mixing them up costs marks:

Question type 1: "Most common site of fertilisation in the fallopian tube?" Answer: **Ampulla** — This is where normal fertilisation happens, and also the most common site of ectopic implantation (~70% of ectopics).

Question type 2: "In which part does ectopic pregnancy survive longest?" Answer: **Interstitial (interstitial/cornual region)** — The thick myometrium surrounding this segment supports growth up to 14–16 weeks before catastrophic rupture. This is why interstitial ectopics have the highest mortality.

Q2 — Fertilisation site

The most common site of fertilisation in the fallopian tube is:

- A. Ampulla
- B. Isthmus

- C. Interstitial segment
- D. Infundibulum

► **Answer & Explanation:**

****A. Ampulla**** — Widest part of the fallopian tube; fertilisation occurs here within ~24 hours of ovulation. Ectopic pregnancies most commonly occur in the ampulla too.

Q290 — Longest ectopic survival

In which part of the fallopian tube does ectopic pregnancy survive longest?

- A. Isthmus
- B. Ampulla
- C. Cornua
- D. Interstitium

► **Answer & Explanation:**

****D. Interstitium (interstitial/cornual)**** — Thick myometrium supports growth up to 16 weeks before catastrophic rupture; highest mortality among ectopics.

Ectopic site survival hierarchy: Interstitium (longest, ~16 wks) > Ampulla (~6–8 wks, most common) > Isthmus (~6 wks, earliest rupture due to narrow lumen)

7. The Human Placenta: A Multi-Descriptor Organ

The human placenta is described using three classification systems simultaneously, and the examiner loves the "all of the above" answer here.

Classification axis	Human placenta type	Explanation
Shape	Discoid	Single disc shape (vs. zonary in dogs, diffuse in horses)
Maternal-fetal interface	Haemochorial	Maternal blood directly bathes trophoblast (most intimate; 3 layers lost vs. 6 in epitheliochorial)
Decidual shedding	Deciduate	Maternal decidua is shed with placenta at delivery (vs. adeciduate in horses)

Q291 — Human placenta type

Human placenta is:

- A. Discoid
- B. Haemochorial
- C. Deciduate
- D. All of the above

► **Answer & Explanation:**

****D. All of the above**** — Human placenta is discoid in shape, hemochorial (maternal blood contacts trophoblast directly), and deciduate (maternal decidua shed at delivery).

8. Amniotic Fluid: The Changing Story

Before 20 weeks, amniotic fluid is mainly a *transudate of maternal plasma* through the fetal skin (which is not yet keratinised — the skin is thin and permeable before keratinisation at ~22–25 weeks).

After 20 weeks, the primary source becomes **fetal urine**. The fetus produces urine at ~400–1200 mL/day. Fetal swallowing removes fluid (~400–500 mL/day) and fetal lung liquid contributes (~350 mL/day).

Clinical correlation: - **Oligohydramnios** after 20 weeks = renal agenesis (Potter sequence), posterior urethral valves, IUGR, PROM - **Polyhydramnios** = fetal swallowing problem (oesophageal atresia, duodenal atresia, anencephaly) or maternal diabetes (osmotic diuresis)

Q293 — Amniotic fluid contribution after 20 weeks

After 20 weeks of gestation, amniotic fluid is mainly contributed by:

- A. Maternal plasma ultrafiltrate
- B. Fetal urine
- C. Fetal lung fluid
- D. Fetal skin

► Answer & Explanation:

****B. Fetal urine**** — Beyond 20 weeks, fetal urine production dominates; renal agenesis → severe oligohydramnios (Potter sequence).

9. Twinning: Superfetation vs. Superfecundation

Superfecundation: Fertilisation of two ova released in the same cycle by two different acts of coitus (possibly by different fathers — "heteropaternal superfecundation"). Both twins are the same gestational age. Does NOT rule out twins of the same sex.

Superfetation: Fertilisation of a second ovum during an *already established pregnancy*. Theoretically requires ovulation during pregnancy. The two fetuses would be of *different gestational ages*. Twins of the same age and sex rule OUT superfetation (same age = not superfetation).

Q292 — Same age/sex twins rule out

Twin pregnancy of same age and sex rules out:

- A. Superfetation
- B. Maternal twins
- C. Superfecundation
- D. None of the above

► Answer & Explanation:

****A. Superfetation**** — Superfetation = fertilisation of a second ovum during an ongoing pregnancy (different gestational ages). Same age/sex rules it out.

10. Pelvic Floor Anatomy: Levator Ani

The pelvic floor (pelvic diaphragm) is formed by the levator ani muscles plus the coccygeus (ischiococcygeus). Levator ani = pubococcygeus + iliococcygeus. Pubococcygeus includes the puborectalis (the sling muscle around the anorectal junction). Damage during childbirth → pelvic organ prolapse, stress incontinence, faecal incontinence.

Q405 — Levator ani anatomy

Which muscles form the levator ani (the main component of the pelvic diaphragm)?

- A. Obturator internus, piriformis, coccygeus
- B. Pubococcygeus, iliococcygeus, ischiococcygeus (coccygeus)
- C. Puborectalis, bulbospongiosus, superficial transverse perinei
- D. External sphincter, deep transverse perinei, ischiocavernosus

► Answer & Explanation:

****B. Pubococcygeus, iliococcygeus, ischiococcygeus (coccygeus)**** — The levator ani is composed of pubococcygeus (including puborectalis) and iliococcygeus; together with coccygeus (ischiococcygeus) they form the pelvic diaphragm. Damage during childbirth predisposes to pelvic organ prolapse and stress incontinence.

Distractor analysis: Option A mixes pelvic wall muscles (obturator internus, piriformis) with pelvic floor muscles. Option C mixes puborectalis (which IS part of levator ani) with perineal muscles (bulbospongiosus, transverse perinei). Option D is entirely the urogenital triangle.

11. Clinical Sequelae of Pelvic Floor Damage: Stress Incontinence and Heavy Menstrual Bleeding

Damage to the pelvic floor (levator ani, urethral supports) during vaginal delivery predisposes to *stress urinary incontinence* — leakage with increases in intra-abdominal pressure (cough, sneeze, laugh). Understanding the surgical options for stress incontinence is a natural extension of pelvic floor anatomy.

Stress incontinence treatment ladder: 1. Conservative: pelvic floor exercises (Kegel), physiotherapy (first-line for mild-moderate) 2. Surgical if conservative fails: **mid-urethral tape (TVT or TOT)** — gold standard, > 85% cure rate - TVT (tension-free vaginal tape): retropubic - TOT (transobturator tape): transobturator approach 3. Burch colposuspension: laparoscopic/open; similar efficacy but more invasive 4. Anterior colporrhaphy: lower long-term cure rate 5. Periurethral injections: short-term; lowest cure rate

Q406 — Stress incontinence: surgical treatment

A 50-year-old woman with stress urinary incontinence has failed pelvic floor physiotherapy for 6 months. Urodynamics confirms urethral hypermobility. Which surgery has the HIGHEST long-term cure rate?

- A. Anterior colporrhaphy (anterior repair)
- B. Burch colposuspension
- C. Mid-urethral tape (TVT or TOT)
- D. Periurethral collagen injection

► Answer & Explanation:

****C. Mid-urethral tape (TVT or TOT)**** — Mid-urethral slings have become the gold-standard surgical treatment for stress incontinence with > 85% cure rates and are preferred over Burch colposuspension (similar efficacy but more invasive) and anterior repair (lower cure rate).

Why B (Burch) is tempting: Burch colposuspension was the gold standard before TVT became available (post-1995). Students who have read older textbooks may still think Burch is the best. The contemporary answer is mid-urethral tape. When the question specifies "highest cure rate" and "gold standard," TVT/TOT is the answer in current practice.

12. Heavy Menstrual Bleeding (AUB-E): Non-Hormonal Treatment

Heavy menstrual bleeding without structural pathology (AUB-E, endometrial dysfunction) is managed non-hormonally in women who decline hormonal treatment.

Non-hormonal options: - **Tranexamic acid** (antifibrinolytic): 1 g TDS during menstruation — reduces blood loss by 40–50%; first-line non-hormonal - NSAIDs (mefenamic acid): 20–25% reduction; second non-hormonal option - Hormonal options (LNG-IUS, OCPs) are preferred if contraception also required

Q407 — DUB: non-hormonal treatment

A 38-year-old woman with heavy regular periods and normal pelvic ultrasound and endometrial biopsy (AUB-E) does not wish to use hormonal treatment. What is the MOST APPROPRIATE non-hormonal first-line treatment?

- A. Norethisterone 5 mg three times daily cyclically
- B. Tranexamic acid 1 g three times daily during menstruation
- C. Danazol 200 mg daily
- D. GnRH agonist monthly injection

► Answer & Explanation:

****B. Tranexamic acid 1 g three times daily during menstruation**** — Tranexamic acid (antifibrinolytic) reduces menstrual blood loss by 40–50% and is the first-line *non-hormonal* pharmacological treatment for HMB. NSAIDs (mefenamic acid) are an alternative non-hormonal option.

Why A is wrong: Norethisterone IS a hormonal treatment (progestogen). The patient specifically declined hormonal treatment — option A violates that. Options C (danazol) and D (GnRH agonist) are also hormonal and have significant side-effect profiles; they are not first-line.

Bias Removal Section

Common psychological mistakes:

1. **The "most common" bias:** When asked where ectopic pregnancy lasts longest, students default to "ampulla" because it is the most common site overall. But "most common site" and "longest survival" are asking different things — resist the pull of the familiar.

2. **The "first trimester = hCG peak" bias:** Students know hCG peaks in the first trimester and also know NT is measured at 11–13 weeks. The mind conflates these two first-trimester events. Always remember: hCG peaks at 7–9 weeks (earlier in first trimester), not at the NT scan window.
3. **The absolute-word trap for FRC:** "Which respiratory parameter does NOT change?" Respiratory rate genuinely does not change — a surprising fact. Students guess it must change, leading them to mark it as the answer when the question asks for the one that decreases.
4. **CVS: Right vs. left axis:** The word "pushed" makes students think rightward. But the enlarged uterus pushes the diaphragm up and *rotates the heart to the left* — left axis deviation on ECG.
5. **Partial-correct traps:** In the epoophoron question, options B (horizontal tubules) and C (columnar lining) each have ONE wrong word — the tubules are vertical and the lining is cuboidal. Examiners use this to trap students who know "epoophoron is in the mesovarium" but haven't studied the histological detail.

Variations & Twists

Twisted versions of "ectopic survival": - "An ectopic pregnancy presenting with catastrophic haemorrhage at 14 weeks was most likely in which location?" → Interstitium - "After an IVF transfer, an ectopic pregnancy is found in the cornual region — which ectopic has the highest risk of rupture mortality?" → Interstitial (same answer)

Twisted versions of "FRC in pregnancy": - "A pregnant patient requires emergency intubation. Why does she desaturate faster than a non-pregnant patient?" → Because FRC is reduced → less oxygen reserve during apnoea - "Which of these lung volumes does pregnancy decrease?" → FRC and residual volume

Twisted versions of "morula timing": - "A blastocyst implants on day 20. Fertilisation occurred on day ___?" → Day 14 (ovulation) - "In an IVF cycle, embryo transfer is done at the 5-day blastocyst stage. This corresponds to which stage of natural development?" → After zona hatching, ready for implantation

Memory Engineering

For the embryological timeline:

"Fertilisation → Morula enters uterus → Blastocyst → Implantation" = F-M-B-I = "Friendly Mother Brings Infant" Days: 0 – 4 – 5 – 6/7 (or cycle days 14 – 18 – 19 – 20–22)

For polar bodies:

"1st polar body = 1st meiosis (before ovulation); 2nd polar body = 2nd meiosis (after fertilisation)" Visual: Imagine the oocyte "splitting" first at ovulation, then again when sperm arrives.

For Wolffian remnants:

"Gartner's duct = Gone to the broad ligament (lateral, midvaginal wall)" "Epoophoron = cranial End of Wolffian body" "Paroophoron = caudal remnant, makes Paraovarian cysts"

For "what decreases in pregnancy":

*"Only **FRC Falls**" (among respiratory volumes) "SVR and DBP **Drop**" (among cardiovascular)"*

For hCG peak:

*"hCG peaks at **8–10** weeks — this is when morning sickness is worst. After NT scan (11–13 weeks), hCG is already coming down."*

For ectopic survival:

*"The thick **Interstitium** = **Impressive** survival (up to 16 weeks), **Incredibly** dangerous rupture"*

For "water under the bridge":

Visualise the uterine artery as a bridge over a river (ureter = river of urine). The bridge crosses above the water.

Ultra High-Yield Revision Layer

"What the Examiner REALLY Wants"

The examiner wants to know if you can distinguish: (a) fertilisation site from longest ectopic survival site; (b) day of morula entry (4) from blastocyst formation (5) from implantation (20–22); (c) FRC as the one thing that decreases among respiratory volumes; (d) Gartner's duct (broad ligament) from other Wolffian remnants; (e) left axis deviation in pregnancy, not right.

20-Second Revision Bullets

- Fertilisation: ampulla
- Morula enters uterus: day 4 (16-cell stage)
- Blastocyst: day 5
- Implantation: days 20–22 of 28-day cycle; endometrium ~15–20 mm
- 1st polar body: after meiosis I (just before ovulation)
- 2nd polar body: after meiosis II (after fertilisation)
- hCG peaks: 7–9 weeks
- FRC: only respiratory volume that decreases (~20%)
- Respiratory rate: unchanged in pregnancy
- Cardiac output: increases 40–50%
- ECG in pregnancy: left axis deviation
- LVEDD: increases (physiological cardiac dilatation)
- Uterine artery: anterior division of internal iliac; crosses over ureter
- Ectopic longest survival: interstitium (up to 16 weeks)

- Amniotic fluid after 20 weeks: fetal urine
- Human placenta: discoid + haemochorial + deciduate (all three)
- Superfetation = different gestational ages; same age twins rule it OUT
- Gartner's duct: in leaves of broad ligament
- Epoophoron: cranial Wolffian remnant, mesovarium, vertical tubules, cuboidal lining
- Paroophoron: caudal Wolffian remnant → paraovarian cysts
- Levator ani: pubococcygeus + iliococcygeus + coccygeus

Top Traps List

1. Ampulla = fertilisation; Interstitium = longest survival (not the same)
2. hCG peaks 7–9 weeks (NOT at 11–13 weeks NT window)
3. FRC decreases; vital capacity does NOT decrease
4. ECG in pregnancy: LEFT axis (not right)
5. Morula (16-cell) enters uterus day 4; blastocyst forms day 5
6. Epoophoron = cranial; paroophoron = caudal (for paraovarian cysts)
7. Superfetation = different ages (same age rules it out)

High-Yield Comparison Table: Fallopian Tube Segments & Ectopic

Segment	Width	Ectopic frequency	Survival before rupture
Interstitium	Narrowest	Rare (~2%)	Longest (~16 wks) — highest mortality
Isthmus	Narrow	~12%	Earliest (~6 wks)
Ampulla	Widest	~70%	~6–8 weeks
Infundibulum	Funnel	Rare	Variable

Most Volatile Facts

- Endometrial thickness at implantation: sources quote 10–20 mm (secretory phase); exam answer is typically 15–20 mm
- Amniotic fluid volume peaks at ~34 weeks (~800–1000 mL), then declines
- hCG doubling time in early normal pregnancy: every 48 hours (useful for ectopic diagnosis when it fails to double)
- CVS in pregnancy: CO increases as early as 5 weeks (even before significant uterine enlargement)

Chapter 2: Antenatal Care — Screening, Fetal Surveillance, Doppler, NST, BPP, Bishop Score, IUGR, Preterm & Rhesus

Big Picture First

Antenatal care (ANC) is the single most clinically practical chapter in NEET-PG OBG, and it generates some of the highest question density in recent papers (2016–2024). What makes it simultaneously exciting and dangerous is that it spans three very different domains — epidemiological protocols (how many visits, when, what to screen), clinical interpretation (is this NST reactive? is this Doppler finding alarming?), and emergency decision-making (should I deliver this baby now?). Examiners can attack from any angle.

The topics that recur with near-annual frequency are: nuchal translucency timing and its association with Down syndrome, the NST reactive criteria (≥ 2 accelerations, ≥ 15 bpm, ≥ 15 sec), the biophysical profile score with the critical detail of which parameter is *lost last* in hypoxia, umbilical artery Doppler interpretation (normal $\rightarrow$ elevated S/D $\rightarrow$ absent $\rightarrow$ reversed end-diastolic flow), and betamethasone indications. The Rhesus incompatibility sequence — sensitisation, Kleihauer-Betke test, anti-D prophylaxis — appears in almost every recent paper.

The most common mistake students make in this chapter is learning isolated facts rather than *clinical sequences*. Antenatal care is a story: you screen $\rightarrow$ you detect an abnormality $\rightarrow$ you surveil $\rightarrow$ you decide when to act. If you understand the story, every individual question becomes a logical step in a sequence rather than an isolated memory item.

Question Pattern Map

Recurring PYQ trends (year-by-year analysis): - NST reactive criteria: appeared in 2016, 2018, 2021, 2023 (virtually every year) - BPP parameter lost last in hypoxia: 2016, 2019 - Nuchal translucency timing (11–13 weeks): 2019, 2022, 2023 - Quadruple test timing (15–20 weeks): 2017, 2018, 2021 - Betamethasone indications ("all EXCEPT PPH"): 2016, 2017 - CVS vs. amniocentesis timing for Down syndrome: 2016, 2022, 2023 - IUGR investigation of choice (umbilical artery Doppler): 2021, 2023, 2024 - Anti-D / Kleihauer-Betke: 2022, 2024 - Tocolytic of choice (nifedipine): 2023, 2024

Common examiner traps: - NST reactive: "2 accelerations in 20 minutes" — students forget the amplitude (≥ 15 bpm) and duration (≥ 15 sec) criteria and just write "2 in 20" - BPP: tone is lost LAST (most primitive), NST is lost FIRST (most recently acquired neurological function) - Betamethasone: all three neonatal benefits are true (lung maturity + IVH + PVL); PPH is the odd one out - Uterine artery Doppler at 14 weeks detects *early-onset* pre-eclampsia, NOT late-onset - Quadruple test is second-trimester (15–20 wks); NT/double marker is first-trimester - Kleihauer-Betke tests for fetal RBCs in maternal blood — NOT maternal antibody titre

How questions are framed: - "NST is reactive when..." → give ALL three criteria - "All EXCEPT..." → Betamethasone's role - "Best diagnostic test for Down syndrome at 11 weeks..." → CVS (not amniocentesis, not triple test) - "Investigation of choice for IUGR monitoring..." → Umbilical artery Doppler - "Ominous Doppler finding requiring immediate delivery in IUGR..." → Reversed a-wave in ductus venosus

Core Concept Deep Dive

1. ANC Schedule: The Global Standard Just Changed

The WHO updated its ANC model in 2016, and this has been a PYQ favourite since. The key change: the older 4-visit "focused ANC" model was replaced by an 8-contact model. This is not a minor administrative detail — the change was driven by evidence that 8 contacts reduces perinatal mortality compared to 4 visits.

India's own schedule (Ministry of Health) uses a different framework with visits at: first contact (as early as possible, ideally <12 weeks), 14–26 weeks, 28–34 weeks, and 36 weeks onward, but the WHO 2016 model of 8 contacts is what gets tested.

Q378 — ANC schedule (WHO 2016)

The WHO 2016 antenatal care model recommends a minimum of how many ANC contacts for an uncomplicated pregnancy?

- A. 4 contacts
- B. 6 contacts
- C. 8 contacts
- D. 10 contacts

► Answer & Explanation:

****C. 8 contacts**** — The WHO 2016 ANC model (replacing the older 4-visit focused ANC model) recommends a minimum of 8 contacts for uncomplicated singleton pregnancies, to reduce perinatal mortality.

Why option A is the trap: 4 was the prior standard ("focused ANC"). Students who haven't updated their knowledge pick 4 out of habit. The current answer is 8.

2. Nutritional Advice in Pregnancy: The Caloric Increase Rule

Energy requirements increase in pregnancy, but the amounts vary by trimester, and examiners test the specific numbers.

Standard recommendations (ICMR/WHO): - 1st trimester: No additional calories needed above baseline - 2nd trimester: +300 kcal/day additional - 3rd trimester: +300 kcal/day additional (some texts say slightly higher)

Q982 — Nutritional advice at 12 weeks

A primigravida woman at 12 weeks of gestation comes to the antenatal clinic for nutritional advice. Which of the following will you recommend?

- A. Additional 300 kcal in 2nd trimester

- B. Additional 300 kcal in 1st trimester
- C. Additional 400 kcal in 3rd trimester
- D. Additional 300 kcal throughout the pregnancy

► **Answer & Explanation:**

****A. Additional 300 kcal in 2nd trimester****

Key distinction: At 12 weeks (first trimester), the advice is forward-looking — you counsel her about what she will need *coming up* (i.e., in the second trimester: +300 kcal). Option D "throughout pregnancy" is wrong because the first trimester doesn't need extra calories, and option C states an incorrect amount for the third trimester.

3. Anaemia in Pregnancy: Knowing the Cutoffs

Anaemia in pregnancy is the most common complication globally, and WHO has a specific grading system that differs from the non-pregnant adult cutoffs.

Normal lower limit of Hb in pregnancy: - First and third trimester: 11.0 g/dL - Second trimester: 10.5 g/dL (physiological dilutional anaemia is maximal here)

WHO grades of anaemia in pregnancy: - Mild: Hb 10.0–10.9 g/dL - Moderate: Hb 7.0–9.9 g/dL - Severe: Hb < 7.0 g/dL - Very severe: Hb < 4.0 g/dL

Q379 — Anaemia in pregnancy: WHO classification

According to WHO criteria, what haemoglobin level defines moderate anaemia in pregnancy?

- A. Hb < 11.0 g/dL
- B. Hb 7.0–9.9 g/dL
- C. Hb 10.0–10.9 g/dL
- D. Hb < 7.0 g/dL

► **Answer & Explanation:**

****B. Hb 7.0–9.9 g/dL**** — WHO classification: mild = Hb 10.0–10.9 g/dL; moderate = Hb 7.0–9.9 g/dL; severe = Hb < 7.0 g/dL. Normal lower limit in pregnancy is 11.0 g/dL (first/third trimester) or 10.5 g/dL (second trimester).

Why option A is tempting: Hb < 11.0 g/dL is the *definition of anaemia in pregnancy* (the cutoff for diagnosing anaemia), but the question asks for the grade "moderate" specifically. Once you know someone has anaemia, 10.0–10.9 is mild, 7.0–9.9 is moderate.

4. First-Trimester Screening: The Nuchal Translucency Story

At 11+0 to 13+6 weeks of gestation, a specialised ultrasound measures the *nuchal translucency* (NT) — the fluid-filled space at the back of the fetal neck. This measurement, combined with maternal serum biochemistry (free β -hCG and PAPP-A), constitutes the *combined first-trimester screening test* (also called the "double marker test" when including NT + 2 serum markers).

Normal NT: < 3 mm (or < 95th centile for crown-rump length) **Significance of raised NT (> 3 mm):** - Most commonly: Down syndrome (Trisomy 21) — the most common chromosomal aneuploidy - Also: other trisomies

(18, 13), Turner syndrome, cardiac defects, skeletal dysplasias - NT is a *screening* test; raised NT → offer diagnostic testing (CVS at 11–13 weeks or amniocentesis at 15–18 weeks)

Q924 — Nuchal translucency timing

Nuchal translucency on USG is detected at:

- A. 11–13 weeks
- B. 18–20 weeks
- C. 8–10 weeks
- D. 20–22 weeks

► Answer & Explanation:

****A. 11–13 weeks**** — First-trimester screening window (11+0 to 13+6 weeks). NT >3 mm suggests Down syndrome, cardiac defects, or other aneuploidy.

Q521 — Increased NT: most common cause

Increased nuchal translucency at 13-week fetal ultrasound is MOST COMMONLY associated with:

- A. Turner syndrome
- B. Down syndrome (Trisomy 21)
- C. Hydrocephalus
- D. Klinefelter syndrome

► Answer & Explanation:

****B. Down syndrome (Trisomy 21)**** — NT > 3 mm ($\geq$ 95th centile) is most commonly associated with Trisomy 21. All chromosomal aneuploidies and many cardiac/structural defects can raise NT.

Why Turner syndrome (A) is the trap: Turner syndrome (45X) is associated with cystic hygroma — a severe version of increased NT — and students correctly recall this association. But the question asks for the *most common* cause of raised NT, and Trisomy 21 is by far more prevalent than Turner syndrome. Turner = severe NT / cystic hygroma; Trisomy 21 = most common cause of NT elevation overall.

5. Screening Timeline: First vs. Second Trimester Tests

This is one of the most high-yield table-worthy topics. The examiner loves to ask the timing of various screening tests.

Test	Timing	What it detects
NT (nuchal translucency)	11–13+6 weeks	Down syndrome (Trisomy 21), other aneuploidies
Double marker (NT + PAPP-A + free β -hCG)	11–13 weeks	Down syndrome screening (combined test)
CVS (diagnostic)	10–13 weeks	Karyotype (definitive)
Amniocentesis (diagnostic)	15–18 weeks (ideally)	Karyotype, AFP, enzyme assays
Triple marker (AFP + hCG + uE3)	15–20 weeks	Down, Edwards, NTDs
Quadruple marker (+ inhibin A)	15–20 weeks	Improved Down syndrome detection
Anomaly scan (detailed anatomy)	18–20 weeks	Structural defects
Glucose challenge test (GCT)	24–28 weeks	GDM screening

 **Examiner Note:** CVS gives you karyotype at 10–13 weeks; amniocentesis gives you karyotype at 15–18 weeks. If someone has had a previously affected pregnancy and wants the earliest diagnostic test, the answer is CVS, not amniocentesis, and NOT triple/double marker (which are screening, not diagnostic).

Q963 — First-trimester dating

First antenatal visit, LMP 2 months ago — most accurate dating parameter?

- A. BPD
- B. CRL (crown-rump length)
- C. Femur length
- D. Abdominal circumference

► Answer & Explanation:

****B. CRL**** — Most accurate in first trimester (5–13 weeks); ± 5 days accuracy. BPD/FL/AC used in 2nd and 3rd trimester. CRL measured by TVS in early weeks.

Q656 — Down syndrome: diagnosis at 11 weeks

A woman with a previous Down syndrome pregnancy (abortion) is now 11 weeks pregnant. Best diagnostic modality to exclude Down syndrome at this gestation?

- A. Amniocentesis
- B. Chorionic villus sampling (CVS)
- C. Double marker test
- D. Triple marker test

► Answer & Explanation:

****B. Chorionic villus sampling**** — CVS (10–13 weeks): earliest invasive test for karyotype. Amniocentesis is performed at 15–18 weeks. Double/triple marker are screening tests, not diagnostic.

Why this question is a trap: The patient has had a previous affected pregnancy, so she needs *diagnosis*, not screening — eliminate C and D immediately. Then between A (amniocentesis) and B (CVS): the patient is 11 weeks, amniocentesis is done at 15–18 weeks. CVS is the *only* option that gives a definitive karyotype at 11 weeks.

The quadruple test timing (second trimester):

Q890 — Quadruple test timing

Best time to do the quadruple test (quad screen) is:

- A. 8–12 weeks
- B. 11–15 weeks
- C. 15–20 weeks
- D. 18–22 weeks

► Answer & Explanation:

****C. 15–20 weeks**** — The quad screen is a second-trimester test performed between 15 and 20 weeks, ideally in conjunction with first-trimester combined screening.

Memory anchor: Quad test = second trimester = 15–20 weeks. NT = first trimester = 11–13 weeks. These are the two most commonly confused timing questions.

6. Thalassaemia Screening: The Stepwise Approach

Thalassaemia screening follows a two-step cascade: screen first, then confirm.

Step 1 (screening): $MCV \leq 80$ fL — this cheap, routinely available parameter is the initial screen. If MCV is low, move to Step 2.

Step 2 (confirmation): HPLC (high-performance liquid chromatography) — gold standard for identifying haemoglobin variants (HbA2 elevated in β -thalassaemia trait; HbH inclusions in α -thalassaemia)

Step 3 (if both parents are carriers): Offer prenatal diagnosis — CVS or amniocentesis for fetal genotyping.

Q569 — Thalassaemia in pregnancy: screening investigation

Screening investigation for thalassaemia in a pregnant woman is:

- A. Haemoglobin electrophoresis
- B. HPLC (high-performance liquid chromatography)
- C. Mean corpuscular volume (MCV) ≤ 80 fL
- D. Osmotic fragility test

► Answer & Explanation:

****C. Mean corpuscular volume (MCV) ≤ 80 fL**** — Initial screening: $MCV < 80$ fL triggers HPLC/haemoglobin electrophoresis to identify thalassaemia trait. Both parents tested; if both carriers → offer prenatal diagnosis (CVS/amniocentesis).

Key distinction: HPLC is the *diagnostic/confirmatory* test; MCV is the *screening* test. The question asks for "screening investigation" — answer is MCV, not HPLC. The osmotic fragility test is for hereditary spherocytosis, not thalassaemia.

7. Uterine Artery Doppler in ANC: Predicting Pre-eclampsia

The uterine artery Doppler at 11–14 weeks is one component of the *first-trimester combined screening for pre-eclampsia* (along with mean arterial pressure, PAPP-A, and patient history). Abnormal notching or elevated pulsatility index in the uterine artery at this early gestation predicts *early-onset pre-eclampsia* (before 34 weeks), which is the severe, placenta-mediated form associated with poor placentation.

Late-onset pre-eclampsia (>37 weeks) has a different pathophysiology (maternal constitutional factors) and is NOT predicted by first-trimester uterine artery Doppler.

Q977 — Uterine artery Doppler at 14 weeks

A woman with an obstetric score of G2P1 comes to the clinic at 14 weeks of gestation for her antenatal checkup. A uterine artery Doppler was suggested by the doctor. What would it detect?

- A. Early onset preeclampsia
- B. Late-onset preeclampsia
- C. Fetal growth restriction
- D. Placenta accreta

► Answer & Explanation:

****A. Early onset preeclampsia****

Clinical reasoning: Uterine artery Doppler at 11–14 weeks detects *impaired trophoblastic invasion* of the spiral arteries — the pathological basis of early-onset pre-eclampsia and early-onset IUGR. It does not reliably predict late-onset conditions (different mechanism) or structural abnormalities like placenta accreta.

8. The Non-Stress Test (NST): Understanding What "Reactive" Really Means

The NST measures fetal heart rate (FHR) accelerations in response to fetal movement. An acceleration is a sign of an intact fetal autonomic nervous system and adequate oxygenation. Think of it as a fetal "stress ball" — a healthy, well-oxygenated fetus will show brisk FHR responses to its own movements.

Reactive NST criteria (remember ALL THREE numbers — 15/15/20): - ≥ 2 accelerations - Each ≥ 15 bpm above baseline - Each lasting ≥ 15 seconds - Within a 20-minute window

If non-reactive in 20 minutes → extend to 40 minutes (may be fetal sleep cycle). If still non-reactive → vibroacoustic stimulation (VAS/BV buzzer) → if still non-reactive → proceed to BPP.



Examiner Note: Two distinct questions have been asked about NST: (1) "What are the criteria?" → 15/15 × 2 in 20 min; (2) "NST is reactive when?" → 2 accelerations in 20 min. The second framing drops the amplitude/duration, but from context it implies the standard criteria. Know the full criteria.

Q341 — Reactive NST criteria

A reactive non-stress test requires ≥ 2 accelerations in 20 minutes, each lasting:

- A. ≥ 5 bpm × 5 sec

- B. $\geq 10 \text{ bpm} \times 10 \text{ sec}$
- C. $\geq 15 \text{ bpm} \times 15 \text{ sec}$
- D. $\geq 20 \text{ bpm} \times 20 \text{ sec}$

► **Answer & Explanation:**

****C. $\geq 15 \text{ bpm} \times 15 \text{ sec}$ **** — Reactive NST: ≥ 2 accelerations of $\geq 15 \text{ bpm}$ lasting $\geq 15 \text{ sec}$ from baseline in 20-min tracing. Non-reactive → extend to 40 min, vibroacoustic stimulation, or BPP.

Q869 — NST reactive definition (2016 PYQ)

With reference to fetal heart rate, a nonstress test is considered reactive when?

- A. Two fetal heart rate accelerations are noted in 20 minutes
- B. One fetal heart rate acceleration is noted in 20 minutes
- C. Two fetal heart rate accelerations are noted in 10 minutes
- D. Three fetal heart rate accelerations are noted in 30 minutes

► **Answer & Explanation:**

****A. Two fetal heart rate accelerations are noted in 20 minutes**** — NST is reactive (reassuring) when ≥ 2 accelerations of $\geq 15 \text{ bpm}$ lasting $\geq 15 \text{ seconds}$ occur within 20 minutes, indicating intact fetal autonomic function and adequate oxygenation.

Q1054 — Non-reactive NST at 36 weeks

36 weeks, decreased fetal movements, NST: 1 acceleration, no deceleration in 20 min. Next step?

- A. Emergency C-section
- B. Induce labour
- C. Wait and observe
- D. Continue NST for 40 min

► **Answer & Explanation:**

****D. Continue NST for 40 min**** — A non-reactive NST in 20 minutes should be extended to 40 minutes (accounting for fetal sleep cycles) before declaring it non-reactive.

Clinical reasoning: One acceleration in 20 minutes = non-reactive, but this does NOT immediately mean fetal compromise. Fetuses have sleep cycles of 20–40 minutes. The protocol is: extend to 40 minutes first (the most common cause of non-reactive NST is fetal sleep). Emergency delivery is only warranted if the NST is persistently non-reactive with concerning decelerations or if BPP is also abnormal.

9. Biophysical Profile (BPP): The Fetal Report Card

The BPP scores five parameters (each scores 0 or 2, maximum 10):

Parameter	Score 2 (Normal)	Score 0 (Abnormal)
NST	≥2 accelerations in 20 min	< 2 accelerations
Fetal breathing movements	≥1 episode of ≥30 sec in 30 min	Absent
Fetal gross body movement	≥3 discrete movements in 30 min	< 3 movements
Fetal tone	≥1 episode of flexion/extension	Absent
Amniotic fluid volume (AFV)	Single deepest pocket ≥2 cm	< 2 cm

BPP interpretation: - 8–10: Normal (reassuring) - 6: Equivocal — repeat in 24 hours (or deliver if term) - 4: Abnormal — consider delivery - 0–2: Deliver immediately (fetal distress)

Critical exam concept: Which parameter is lost LAST in chronic hypoxia?

The BPP parameters are acquired in phylogenetically sequential order during fetal brain development (NST is the most recently acquired, tone is the oldest). In chronic hypoxia, central nervous system functions are lost in *reverse order of acquisition* — the most recently acquired functions fail first, the most primitive (oldest) functions fail last.

Order of loss in chronic hypoxia (first to last): 1. NST (most recently acquired) → fails FIRST 2. Fetal breathing movements 3. Gross body movements 4. **Fetal tone (most primitive) → fails LAST**

This is why a non-reactive NST in the setting of chronic IUGR is an early warning sign — it's the canary in the coal mine.

Q872 — BPP parameter lost last in hypoxia

If fetus is having hypoxia, which of the BPP parameter will be affected last?

- A. Fetal tone
- B. Fetal breathing movement
- C. Fetal movements
- D. NST

► Answer & Explanation:

****A. Fetal tone**** — In chronic fetal hypoxia, BPP parameters are lost in reverse order of phylogenetic development: NST first, then breathing movements, then gross movements, and fetal tone (earliest CNS function) is lost last.

Why students answer D (NST): NST is the most "sensitive" test — it becomes non-reactive earliest. Students confuse "affected first" with "affected last." The question asks "last," so the answer is fetal tone (the last parameter standing in a compromised fetus).

Memory anchor: "Tone is Tenacious — it holds on the Tillest end." Or: **NBBT** = NST → Breathing → Body movements → Tone (order of loss). Tone is always last.

10. Umbilical Artery Doppler and IUGR: The Severity Ladder

Doppler velocimetry of the umbilical artery measures *placental vascular resistance*. In normal pregnancy, blood flows forward throughout the cardiac cycle (systole AND diastole). As placental resistance increases with uteroplacental insufficiency, diastolic flow decreases, disappears, then reverses.

The severity ladder: 1. **Elevated S/D ratio (> 3 after 30 weeks):** Increased placental resistance — mild-moderate IUGR 2. **Absent end-diastolic flow (AEDF):** Severe placental insufficiency — deliver if ≥ 32 –34 weeks; intense surveillance if earlier 3. **Reversed end-diastolic flow (REDF):** Extreme placental failure — deliver immediately regardless of gestation

Adding MCA Doppler (brain-sparing): The middle cerebral artery (MCA) normally has *high* resistance. In IUGR, the fetal brain vasodilates to protect itself, making the MCA show *low* resistance. The ratio MCA PI / Umbilical artery PI = **cerebroplacental ratio (CPR)**. Normal CPR > 1; CPR < 1.08 indicates brain-sparing = significant uteroplacental insufficiency.

The ductus venosus (DV): the final alarm bell

When umbilical artery Doppler shows AEDF or REDF, the next step is DV Doppler. The ductus venosus carries oxygenated blood from the umbilical vein to the right atrium. A *reversed a-wave* in the DV indicates that right heart pressure is so high that flow reverses during atrial contraction — this reflects imminent cardiac failure. A reversed a-wave is the most ominous Doppler finding in IUGR and demands immediate delivery regardless of gestational age.

Q342 — Absent end-diastolic flow

Absent end-diastolic flow in umbilical artery Doppler implies:

- A. Normal fetus
- B. Mild placental insufficiency
- C. Severe placental insufficiency
- D. Polyhydramnios

► Answer & Explanation:

****C. Severe placental insufficiency**** — AEDF beyond 30–32 weeks → expedite delivery. Reversed end-diastolic flow → immediate delivery regardless of gestation.

Q567 — IUGR: investigation of choice

Investigation of choice for monitoring a fetus with suspected IUGR:

- A. Kick count chart
- B. Non-stress test (CTG)
- C. Umbilical artery Doppler
- D. Biophysical profile

► Answer & Explanation:

****C. Umbilical artery Doppler**** — RCOG/ISUOG guidelines: umbilical artery Doppler is the primary surveillance tool for suspected IUGR (SGA). Absent/reversed end-diastolic flow = urgent delivery consideration. MCA Doppler added for redistribution.

Q374 — IUGR with AEDF: ominous additional Doppler finding

A growth-restricted fetus at 30 weeks has absent end-diastolic flow (AEDF) on umbilical artery Doppler. Which additional Doppler finding would indicate imminent fetal compromise requiring immediate delivery?

- A. MCA pulsatility index < 5th centile
- B. Reversed end-diastolic flow (REDF) in umbilical artery
- C. DV pulsatility index > 95th centile

- D. Reversed a-wave in ductus venosus (DV)

► **Answer & Explanation:**

****D. Reversed a-wave in ductus venosus (DV)**** — Reversed a-wave in the ductus venosus indicates raised central venous pressure and right heart failure; it is the most ominous Doppler finding in IUGR and an indication for immediate delivery regardless of gestational age.

Why B (REDF) seems correct: REDF is also very serious and also an indication for immediate delivery. But the question specifies "additional Doppler finding" (i.e., something you add to the existing AEDF) and asks for "imminent fetal compromise" — the question is testing knowledge that DV reversed a-wave is the most extreme, terminal warning sign specifically. Both B and D indicate "deliver now," but D represents a more specific, mechanistically understood marker of cardiovascular decompensation.

Q375 — Brain-sparing: cerebroplacental ratio

In a fetus with suspected uteroplacental insufficiency, which Doppler ratio is calculated to assess "brain-sparing"?

- A. Umbilical artery PI / Uterine artery PI
- B. MCA PI / Umbilical artery PI (cerebroplacental ratio, CPR)
- C. Ductus venosus PI / MCA PI
- D. Aortic isthmus PI / MCA PI

► **Answer & Explanation:**

****B. MCA PI / Umbilical artery PI (cerebroplacental ratio, CPR)**** — A CPR < 1.08 (or MCA-PI below 5th centile with elevated UA-PI) indicates brain-sparing: vasodilation of cerebral vessels to maintain oxygen delivery in the context of placental insufficiency.

The clinical logic: In IUGR, placental resistance is HIGH (elevated UA-PI). The brain vasodilates as compensation → MCA resistance LOW (low MCA-PI). $CPR = MCA-PI \div UA-PI$. Low MCA-PI in numerator + high UA-PI in denominator = $CPR < 1$ = brain-sparing. A low CPR is the Doppler equivalent of the body's distress signal.

11. Chorionicity in Twins: The Critical Risk Stratifier

Twin pregnancies are risk-stratified primarily by chorionicity (not zygosity). The rule: earlier the division, the more shared structures, the higher the risk.

Division timing	Chorionicity	Risk level
Day 0–3	Dichorionic diamniotic (DCDA)	Lowest
Day 4–8	Monochorionic diamniotic (MCDA)	High (TTTS risk)
Day 8–13	Monochorionic monoamniotic (MCMA)	Highest
After day 13	Conjoined twins	Extreme

MCMA (monochorionic monoamniotic): Single placenta + single amniotic sac. Risks: cord entanglement (can happen from early in pregnancy), twin-twin transfusion syndrome (TTTS), discordant growth. Delivery: typically at 32–34 weeks by planned C-section due to cord entanglement risk.

Ultrasound signs: - DCDA: Two separate placentas or a thick inter-twin membrane ("lambda" or "twin peak" sign at the placental insertion) - MCDA: Single placenta, thin membrane ("T-sign" at membrane insertion) - MCMA: Single placenta, NO inter-twin membrane, cord entanglement often visible

Q281 — Chorionicity

Image shows a single placenta with no inter-twin membrane. Chorionicity?

- A. Dichorionic monoamniotic
- B. Monochorionic diamniotic
- C. Monochorionic monoamniotic
- D. Dichorionic diamniotic

► Answer & Explanation:

****C. Monochorionic monoamniotic**** — Single placenta + single amniotic sac. Highest-risk twin pregnancy (cord entanglement, TTTS). Delivery typically at 32–34 weeks by C-section.

 **Examiner Note:** "Dichorionic monoamniotic" (option A) does NOT exist in practice — if two chorions are present, there are always two amnions. This option is a fabricated distractor. Eliminate it immediately.

12. Bishop Score: Predicting Induction Success

The Bishop score assesses cervical readiness for induction of labour. It evaluates five parameters: dilation, effacement, station, consistency, and position. Maximum score = 13.

Parameter	0	1	2	3
Dilation (cm)	0	1–2	3–4	≥5
Effacement (%)	0–30	40–50	60–70	≥80
Station	-3	-2	-1/0	+1/+2
Consistency	Firm	Medium	Soft	—
Position	Posterior	Mid	Anterior	—

Interpretation: - Score ≥ 6 (some use ≥8): Favourable cervix → induction with oxytocin/ARM likely to succeed - Score < 6: Unfavourable cervix → cervical ripening first (PGE1 misoprostol, PGE2 dinoprostone, Foley catheter)

Q343 — Bishop score ≥ 6

A Bishop score ≥ 6 indicates:

- A. Unfavourable cervix needing ripening
- B. Favourable cervix — induction likely to succeed
- C. Cervical incompetence
- D. Established labour

► **Answer & Explanation:**

****B. Favourable cervix**** — Bishop ≥ 6 (some use ≥ 8): induction with oxytocin/ARM likely to succeed. Score $< 6 \rightarrow$ ripen cervix first (PGE1/PGE2/Foley).

13. Cervical Insufficiency and Cerclage

Cervical insufficiency (also called cervical incompetence) presents as recurrent *painless* second-trimester pregnancy losses — classically, the cervix silently dilates without contractions, and the fetus delivers prematurely.

History-indicated cerclage: - Indication: ≥ 3 prior painless mid-trimester losses OR previous cerclage with an indication - Timing: 12–14 weeks (after NT scan) - Technique: McDonald (purse-string suture) or Shirodkar (submucosal) - Removal: at 36–37 weeks, or immediately if preterm labour occurs

Q344 — Cervical cerclage indication

Classical indication for history-indicated (prophylactic) cervical cerclage:

- A. First-trimester miscarriage $\times 1$
- B. Recurrent painless second-trimester pregnancy loss (cervical insufficiency)
- C. Threatened preterm labour at 32 weeks
- D. Twin pregnancy

► **Answer & Explanation:**

****B. Recurrent painless second-trimester loss**** — Cervical insufficiency. McDonald (purse-string) at 12–14 weeks; Shirodkar variant. Remove at 36–37 weeks.

Why D (twin pregnancy) is wrong: Cerclage in twin pregnancies has been studied and is generally NOT recommended (some trials showed it may increase preterm birth in twins). The key word in option B is "recurrent" (≥ 3) + "painless" + "second trimester" — all three features must point to cervical insufficiency.

14. Preterm Labour: Tocolytics

Preterm labour (PTL) is defined as regular contractions causing cervical change before 37+0 weeks. Tocolysis (uterine relaxation) is used to gain time for corticosteroid administration and maternal transfer, not to prevent preterm birth indefinitely (tocolytics do not alter neonatal outcome if used beyond 48 hours).

First-line tocolytics: - **Nifedipine** (calcium channel blocker): most commonly recommended first-line worldwide by WHO and RCOG; oral, cheap, effective; inhibits L-type Ca^{2+} channels in myometrium - **Atosiban** (oxytocin antagonist): IV, preferred in Europe; fewer cardiovascular side effects than ritodrine; more expensive - **Ritodrine/salbutamol** (β_2 -agonists): older drugs, high cardiovascular side effects (tachycardia, pulmonary oedema); used less now - **Indomethacin** (COX inhibitor): used in very preterm (< 32 weeks), but risk of premature closure of ductus arteriosus; restricted to < 32 weeks

Q376 — Tocolytic of choice

A 28-year-old primigravida at 30 weeks presents with regular uterine contractions and cervical dilation of 2 cm. She has no contraindications to tocolysis. Cervical length on transvaginal ultrasound is 18 mm. Which tocolytic agent is currently considered FIRST-LINE by WHO and RCOG guidelines?

- A. Ritodrine (β_2 -agonist)
- B. Indomethacin (COX inhibitor)
- C. Nifedipine (calcium channel blocker)
- D. Atosiban (oxytocin antagonist)

► **Answer & Explanation:**

****C. Nifedipine (calcium channel blocker)**** — Nifedipine is the preferred first-line tocolytic in most guidelines due to its efficacy, oral route, low cost, and favourable side-effect profile versus ritodrine. Atosiban is equally effective but more expensive; ritodrine has more cardiovascular side effects.

Q377 — Atosiban mechanism

Atosiban is used as a tocolytic in preterm labour. What is its mechanism of action?

- A. Beta-2 adrenergic receptor agonist → myometrial relaxation
- B. Competitive oxytocin/vasopressin receptor antagonist → reduced myometrial contractions
- C. Inhibition of prostaglandin synthesis via COX-2
- D. L-type calcium channel blockade in myometrium

► **Answer & Explanation:**

****B. Competitive oxytocin/vasopressin receptor antagonist → reduced myometrial contractions**** — Atosiban competitively blocks oxytocin and vasopressin V1a receptors in the myometrium and decidua, reducing uterine contractility; fewer systemic side effects than ritodrine.

Key distinction: Nifedipine = calcium channel blocker (option D describes nifedipine's mechanism). Atosiban = oxytocin antagonist (option B). These mechanisms are cross-tested — the examiner may describe a mechanism and ask "which drug?" or name a drug and ask "which mechanism?"

15. Antenatal Corticosteroids: Betamethasone — The Lifesaver That Won't Stop PPH

Betamethasone (or dexamethasone) given to the mother when preterm birth is anticipated (24–34 weeks, with a single rescue course up to 37 weeks considered in some guidelines) crosses the placenta and accelerates fetal lung maturation:

Benefits of antenatal corticosteroids (ACS): 1. **Fetal lung maturity** — reduces respiratory distress syndrome (RDS) by inducing surfactant production 2. **Reduction in intraventricular haemorrhage (IVH)** — stabilises immature fetal blood vessels 3. **Prevention of periventricular leukomalacia (PVL)** — reduces periventricular white matter injury 4. **Reduction in necrotising enterocolitis (NEC)**

What betamethasone does NOT do: - **Prevent postpartum haemorrhage (PPH)** — no mechanism for this; PPH is a maternal uterine/coagulation problem, not related to fetal lungs

Q489 — Betamethasone: all EXCEPT

Betamethasone given to preterm patients is used for all EXCEPT:

- A. Fetal lung maturity
- B. Decrease intraventricular haemorrhage
- C. Prevention of periventricular leukomalacia
- D. Prevent PPH

► **Answer & Explanation:**

****D. Prevent PPH**** — Antenatal corticosteroids (betamethasone/dexamethasone) enhance fetal lung maturity and reduce neonatal IVH and periventricular leukomalacia. They have no role in preventing maternal PPH.

Why this question appears almost every year: Students learn that betamethasone has "many uses" and may be uncertain whether PPH is one of them. The key logic: betamethasone acts on the *fetal* system (crosses the placenta); PPH is a *maternal* complication of the third stage of labour. There is no biological connection.

Q847 — Betamethasone in preterm (2016 PYQ repeat)

Betamethasone given to preterm patient for all except:

- A. Fetal lung maturity
- B. Decrease intraventricular hemorrhage
- C. Prevents periventricular leukomalacia
- D. Prevent PPH

► **Answer & Explanation:**

****D. Prevent PPH**** — Antenatal betamethasone accelerates fetal lung maturity, reduces intraventricular hemorrhage and periventricular leukomalacia. It has no role in preventing postpartum hemorrhage, which is a maternal uterine/coagulation problem.

 **Examiner Note:** This question appeared in both Q489 and Q847 in the same 2016 paper in slightly different wording — confirming it is an examiner obsession. If you can only memorise one fact from Section 15, it is: betamethasone does NOT prevent PPH.

Q856 — Cardiovascular changes in pregnancy (2016 PYQ)

During pregnancy, the true statement about the cardiovascular system is:

- A. Cardiac output decreases
- B. Right axis deviation
- C. Increase in left ventricular end diastolic diameter
- D. All of the above

► **Answer & Explanation:**

****C. Increase in left ventricular end diastolic diameter**** — Pregnancy causes increased plasma volume and preload → increased LVEDD (physiological cardiac dilatation). Cardiac output increases (not decreases) and the ECG axis shifts left (not right) as the uterus elevates the diaphragm and rotates the heart.

Memory hook: In pregnancy, the heart is pressed UP and LEFT by the growing uterus → left axis deviation on ECG. Cardiac output increases from as early as 5 weeks. LVEDD increases from volume overload. None of the "all of the above" is correct because A and B are both wrong statements.

Q907 — Quadruple test timing (2018 PYQ repeat)

Best time to do quadruple test:

- A. 8–12 weeks
- B. 11–15 weeks
- C. 15–20 weeks
- D. 18–22 weeks

► Answer & Explanation:

****C. 15–20 weeks**** — The quad test is a second-trimester serum screening test. This question appeared identically in both 2017 and 2018 NEET-PG papers — confirming it is examiner-favourite material. Always place it in the second trimester (15–20 weeks), never the first.

16. Rhesus Incompatibility: The Three-Part Story

Rh incompatibility is one of the most clinically important and frequently tested topics in antenatal care. Understand it as three linked chapters: (1) how sensitisation happens, (2) how we measure the exposure, (3) how we prevent it.

Chapter A — Sensitisation: An Rh-negative mother carrying an Rh-positive fetus can be sensitised when fetal RBCs enter maternal circulation (fetomaternal haemorrhage — FMH). Once sensitised, maternal anti-D IgG crosses the placenta in subsequent pregnancies → haemolyses fetal RBCs → haemolytic disease of the fetus and newborn (HDFN).

High-risk events for FMH: delivery, miscarriage, CVS, amniocentesis, antepartum haemorrhage, external cephalic version, road traffic accidents (abdominal trauma).

Chapter B — Kleihauer-Betke test (measuring exposure): After a potentially sensitising event, the Kleihauer-Betke (acid elution) test determines how much fetal blood entered maternal circulation. Fetal RBCs (containing HbF) resist acid elution and stain darkly; adult RBCs (HbA) are decolourised. Count the proportion → calculate volume of FMH → determine if standard dose (500 IU anti-D) is sufficient or if additional doses are needed.

Chapter C — Anti-D prophylaxis (preventing sensitisation): - Standard postnatal dose: 500 IU anti-D IM within 72 hours of delivery (covers up to 4 mL of FMH) - Additional dose if Kleihauer-Betke shows large FMH (> 4 mL fetal RBCs) - Routine antenatal anti-D prophylaxis (RAADP): given to ALL non-sensitised Rh-negative mothers antenatally - Two-dose protocol: 500 IU at 28 weeks + 500 IU at 34 weeks - Single-dose protocol (NICE preferred): 1500 IU at 28 weeks

Q380 — Anti-D prophylaxis timing

An Rh-negative primigravida at 34 weeks with an Rh-positive partner is given anti-D immunoglobulin. What is the standard dose and timing for routine antenatal anti-D prophylaxis (RAADP) in the UK/India protocol?

- A. 500 IU at 28 weeks only
- B. 500 IU at 28 weeks AND 34 weeks
- C. 1500 IU as a single dose at 28 weeks

- D. 300 mcg (1500 IU) at 28 weeks and 34 weeks

► **Answer & Explanation:**

****C. 1500 IU as a single dose at 28 weeks**** — The NICE/RCOG single-dose RAADP protocol uses 1500 IU (300 mcg) at 28 weeks; the two-dose protocol uses 500 IU at 28 and 34 weeks. Both are acceptable; single-dose is now preferred in many centres for compliance.

Q381 — Kleihauer-Betke test

A Kleihauer-Betke (acid elution) test is performed after delivery in an Rh-negative mother. What does this test measure?

- A. Maternal antibody titre against fetal red cells
- B. Volume of fetal red cells in the maternal circulation (fetomaternal haemorrhage)
- C. Fetal blood group
- D. Presence of anti-D antibodies in fetal serum

► **Answer & Explanation:**

****B. Volume of fetal red cells in the maternal circulation (fetomaternal haemorrhage)**** — The Kleihauer-Betke test detects fetal (HbF-containing) red cells in the maternal blood by acid elution; it quantifies fetomaternal haemorrhage to determine if additional anti-D is needed beyond the standard 500 IU dose.

Why A is the trap: Maternal antibody titre (indirect Coombs test) is a *different test* used to monitor an already-sensitised Rh-negative mother. The Kleihauer-Betke test is done *after* an FMH event to quantify exposure — it counts fetal cells in maternal blood, not antibodies.

17. Day 20 of the Menstrual Cycle: Applying Cycle Phase Knowledge to ANC

Understanding the menstrual cycle phases is fundamental to understanding implantation timing, endometrial preparation, and the basis of luteal phase support in ANC/ART.

Day 20 of a 28-day cycle = luteal (secretory) phase. Ovulation occurs day 14. The corpus luteum produces progesterone days 15–28, peaking around days 20–22. The endometrium becomes secretory with corkscrew glands and oedematous stroma — optimally prepared for implantation.

Q497 — Day 20 of menstrual cycle

Day 20 of a regular 28-day menstrual cycle corresponds to:

- A. Menstrual phase
- B. Follicular phase
- C. Ovulation phase
- D. Luteal phase

► **Answer & Explanation:**

****D. Luteal phase**** — Ovulation on day 14; luteal phase days 15–28. Day 20 = secretory endometrium (peak progesterone, corkscrew glands, secretions).

Bias Removal Section

Common psychological traps in Chapter 2:

1. **The "more specific = better" fallacy in NST interpretation:** When NST shows 1 acceleration in 20 minutes, students want to act (extend to 40 min OR do BPP OR deliver). The protocol is clear: extend to 40 minutes first. Escalate only if still non-reactive. Do not jump to delivery.
2. **Confusing "first-line" between guidelines:** Atosiban is first-line in Europe; nifedipine is first-line in WHO/RCOG/India. Most NEET-PG questions ask about WHO/RCOG → answer nifedipine. If the question specifies "European guideline" → atosiban.
3. **Screening vs. diagnostic tests:** Double/triple/quadruple tests = SCREENING (risk estimation). CVS/amniocentesis = DIAGNOSTIC (definitive karyotype). When a patient has had an affected pregnancy and wants to "exclude" a condition → they need a diagnostic test, not a screening test.
4. **The "NT is only for Down syndrome" misconception:** NT elevation is most commonly Trisomy 21, but can also indicate ALL aneuploidies, Turner syndrome, cardiac defects, skeletal dysplasias, congenital diaphragmatic hernia. The question asks "most commonly" — answer Down syndrome.
5. **Reversed a-wave vs. REDF:** Both require delivery. The distinction tested is: "which is the MOST OMINOUS/FINAL sign?" → reversed DV a-wave. It represents right heart failure — the last step before fetal death.
6. **Anti-D dosing confusion:** 500 IU = standard postnatal dose / one arm of the two-dose antenatal regimen. 1500 IU = single-dose antenatal regimen. 300 mcg = 1500 IU (they are equivalent). These numbers get mixed up easily.

Variations & Twists

Twisted NST questions: - "A CTG shows 2 accelerations in 40 minutes. What does this mean?" → Still reactive (the 20-min window allows extension to 40 min) - "What is the next step after a non-reactive NST at 40 minutes?" → Biophysical profile (BPP) or vibroacoustic stimulation

Twisted BPP questions: - "A BPP score of 6 at 38 weeks. What is the next step?" → Deliver (score of 6 at term = deliver) - "Which BPP component, if absent, most likely indicates terminal fetal compromise?" → Absent tone (last to go)

Twisted Doppler questions: - "MCA shows low pulsatility with high umbilical artery pulsatility. What syndrome?" → Brain-sparing IUGR - "Umbilical artery shows absent diastolic flow at 34 weeks. Management?" → Give betamethasone (if not already given) + deliver at 34 weeks

Twisted betamethasone questions: - "Benefits of antenatal betamethasone for the NEONATE include all of the following EXCEPT..." → Look for non-neonatal option (PPH is maternal; this is the trap) - "Why is betamethasone preferred over hydrocortisone for fetal lung maturity?" → Hydrocortisone does not cross the placenta efficiently; betamethasone is not protein-bound

Memory Engineering

For NST reactive criteria — "15-15-2-20":

"Fifteen beats, fifteen seconds, two times, in twenty minutes" Mnemonic: **"1-5-1-5-2-2-0"** or think of it as two "15s" → just like a rugby conversion (15-yard line, twice) in 20 minutes.

For BPP parameters — FTBMA:

Fetal Tone, Breathing, Movements, Amniotic fluid + NST Lost in order: **NST first** → **Breathing** → **Movements** → **Tone last** "Not Breathing? Move Tone last" = NST → B → M → Tone

For Doppler severity ladder — "Normal → Elevated → Absent → Reversed":

"NEAR death": Normal → Elevated S/D → Absent EDF → Reversed EDF

For Kleihauer-Betke:

"Kleihauer Counts Fetal Cells" (counts HbF cells in maternal blood) Contrast with: Indirect Coombs test = counts antibodies (for monitoring sensitised mothers)

For anti-D dosing:

"Single dose = Six-week (28-week) injection: 1500 IU" **"Two doses = Twenty-eight and Thirty-four weeks: 500 IU each"**

For tocolytics — "NAIB":

Nifedipine → first-line; **Atosiban** → Europe/expensive; **Indomethacin** → < 32 weeks; **Beta-agonists** (ritodrine) → old, side effects

For the screening timeline:

"First trimester: NT and CVS (11–13 wks)" "Second trimester: Quad test and amnio (15–20 wks)" "24–28 weeks: GCT for Gestational diabetes"

Ultra High-Yield Revision Layer

"What the Examiner REALLY Wants"

This chapter tests three cognitive levels simultaneously: memorisation (NST criteria numbers, betamethasone indications), clinical reasoning (non-reactive NST → extend to 40 min, not emergency delivery), and conceptual understanding (why fetal tone is lost last, why CPR < 1 = brain-sparing). The questions that separate top scorers from average scorers are the clinical reasoning ones — memorise the numbers but also understand the clinical logic.

20-Second Revision Bullets

Scheduling and screening: - WHO 2016 ANC: minimum 8 contacts (replaced 4-visit model) - NT scan: 11+0 to 13+6 weeks; $> 3 \text{ mm}$ = raised - Most common cause raised NT: Trisomy 21 (Down syndrome) - Double marker (first-trimester combined): 11–13 weeks - Quad test: 15–20 weeks (second trimester) - CVS: 10–13 weeks (diagnostic, not screening) - Amniocentesis: 15–18 weeks (diagnostic) - CRL: most accurate dating in first trimester - Uterine artery Doppler at 11–14 weeks: predicts early-onset pre-eclampsia - Additional 300 kcal in 2nd trimester (not 1st or "throughout") - Anaemia: mild 10–10.9; moderate 7–9.9; severe $< 7 \text{ g/dL}$ - Thalassaemia screening: MCV $\leq 80 \text{ fL}$ first; then HPLC to confirm - GCT for GDM: 24–28 weeks

Fetal surveillance: - Reactive NST: ≥ 2 accelerations, $\geq 15 \text{ bpm}$, $\geq 15 \text{ sec}$, in 20 min - Non-reactive NST in 20 min → extend to 40 min (fetal sleep) - BPP parameters: NST, breathing, movements, tone, AFV (each 0 or 2; max 10) - BPP parameter lost LAST in hypoxia: fetal tone - BPP parameter lost FIRST: NST - Investigation of choice for IUGR: umbilical artery Doppler - Absent EDF: severe placental insufficiency → expedite delivery ≥ 32 –34 wks - Reversed EDF: immediate delivery regardless of gestation - Most ominous Doppler in IUGR: reversed a-wave in ductus venosus - Brain-sparing: $\text{CPR} = \text{MCA-PI} / \text{UA-PI} < 1.08$

Preterm and tocolysis: - First-line tocolytic (WHO/RCOG): nifedipine (calcium channel blocker) - Atosiban: oxytocin/V1a receptor antagonist; Europe preferred; IV - Betamethasone for: fetal lung maturity + IVH + PVL (NOT PPH) - Cervical insufficiency: recurrent painless mid-trimester loss → McDonald cerclage 12–14 wks - Bishop ≥ 6 : favourable cervix (ready for induction) - Bishop < 6 : ripen first (PGE1/PGE2/Foley)

Rhesus: - Kleihauer-Betke: measures fetal RBCs in maternal blood (quantifies FMH) - Indirect Coombs: measures maternal antibodies (for sensitised mothers) - Anti-D RAADP single dose: 1500 IU at 28 weeks - Anti-D two-dose: 500 IU at 28 + 34 weeks - Postnatal anti-D: 500 IU within 72 hours

Top Traps List

1. NST criteria: all THREE numbers ($15 \text{ bpm} \times 15 \text{ sec} \times 2$ times in 20 min)
2. Non-reactive NST → extend to 40 min FIRST (don't jump to C-section)
3. BPP tone lost LAST (not first)
4. Betamethasone does NOT prevent PPH
5. Kleihauer-Betke = counts fetal cells (not antibodies)
6. NT timing = 11–13 weeks (not 8–10, not 18–20)
7. Quad test = 15–20 weeks (second trimester)
8. CVS = diagnostic, not screening; earliest invasive test = 10–13 wks
9. Uterine artery Doppler at 14 wks → early-onset pre-eclampsia (not late-onset)
10. $\text{CPR} < 1$ = brain-sparing (MCA vasodilation + high UA resistance)
11. WHO ANC 2016 = 8 contacts (not 4)
12. Thalassaemia screening = MCV first, then HPLC (not HPLC first)

High-Yield Comparison Table: Invasive vs. Screening Tests

Test	Type	Timing	What it detects
NT scan	Screening (USG)	11–13 wks	Down syndrome, aneuploidies
Double marker	Screening (serum+USG)	11–13 wks	Down syndrome risk
Triple marker	Screening (serum)	15–20 wks	Down, Edwards, NTDs
Quad test	Screening (serum)	15–20 wks	Down syndrome (better sensitivity)
CVS	Diagnostic (invasive)	10–13 wks	Karyotype, gene mutations
Amniocentesis	Diagnostic (invasive)	15–18 wks	Karyotype, AFP, biochemistry
NIPT/cfDNA	Screening (blood)	≥ 10 wks	Aneuploidies (high sensitivity)

High-Yield Table: Doppler Findings Severity in IUGR

Doppler Finding	Significance	Management
Elevated UA S/D ratio	Mild-moderate insufficiency	Intensive surveillance
Absent EDF (AEDF)	Severe insufficiency	Deliver ≥ 34 wks; steroids if < 34 wks
Reversed EDF (REDF)	Extreme insufficiency	Immediate delivery regardless of gestation
MCA low PI (brain-sparing)	CNS redistribution	Alert; combined with UA Doppler (CPR)
CPR < 1	Brain-sparing present	High-risk, plan delivery
Reversed DV a-wave	Right heart failure / imminent fetal death	IMMEDIATE delivery

Most Volatile Facts

- The two-dose anti-D protocol (500 IU at 28+34 weeks) vs. single-dose (1500 IU at 28 weeks) — both correct per different protocols; know both
- The exact BPP score threshold for delivery: ≥ 6 at term = deliver; at < 34 weeks, score of 6 = repeat in 24 hours (gestational age matters)
- Nifedipine vs. atosiban: NEET-PG answers depend on which guideline is cited; when no guideline is specified, nifedipine (WHO/global) is safer
- Uterine artery Doppler also predicts early IUGR — the question specifically says "pre-eclampsia" when asked at 14 weeks



Linked Clinical Topics from the ANC Question Bank

The following questions appeared in the ANC question bank but test adjacent gynae and obstetric medicine topics. They are integrated here because they frequently appear alongside ANC questions in NEET-PG papers.

Ovarian Dermoid (Mature Cystic Teratoma) with Torsion

Mature cystic teratomas (dermoid cysts) contain all three germ layers. The *Rokitansky protuberance* (dermoid plug) is the ectodermal nodule visible on ultrasound — a pathognomonic finding. Dermoids are the most common ovarian tumour in reproductive-age women and the most commonly twisted ovarian cyst (due to their semi-solid composition making them heavier and prone to torsion).

Q282 — Mature cystic teratoma (Rokitansky protuberance)

A 30-year-old with Rokitansky's protuberance on imaging presents with acute abdomen. Diagnosis?

- A. Torsion of subserous fibroid
- B. Twisted ovarian cyst (dermoid)
- C. Ruptured ectopic pregnancy
- D. Ruptured ovarian cyst

► Answer & Explanation:

****B. Twisted ovarian cyst (mature cystic teratoma)**** — Rokitansky protuberance (dermoid plug) = ectodermal nodule in mature cystic teratoma. Torsion → acute abdomen; urgent laparoscopy.

Key link to ANC: Ovarian cysts discovered during pregnancy (often on the NT scan or anomaly scan) are frequently dermoids. Torsion risk increases in pregnancy due to displacement by the growing uterus. Management in pregnancy: laparoscopic cystectomy, ideally in second trimester.

Endometrioma: The "Ground Glass" Cyst

Endometriomas are ovarian cysts containing old ("chocolate") blood from ectopic endometrial tissue. On ultrasound, they show a characteristic *homogeneous ground-glass* echogenicity (due to hemosiderin-laden fluid). They are relevant to ANC because endometriosis is a common cause of infertility and subfertility — patients who conceive with endometriosis require monitoring.

Q1023 — Endometrioma

A 35-year-old woman presents with chronic lower abdominal pain that worsens during menstruation. An ultrasound shows a cystic ovarian mass with a "ground glass" appearance. Which condition is most likely?

- A. Intramural fibroid
- B. Adenomyoma
- C. Endometrioma
- D. Myomatous polyp

► Answer & Explanation:

****C. Endometrioma**** — Ground-glass echogenicity on USG = endometrioma (chocolate cyst). Cyclic pain worsening with menstruation is the classic history.

Genital Tuberculosis: A Mimic of PID

Genital tuberculosis (TB) is caused by *Mycobacterium tuberculosis* spreading haematogenously from a primary focus (usually lungs). It classically presents with *subacute/chronic* lower abdominal pain + intermenstrual bleeding + minimal or no discharge. The key diagnostic clue: despite a history of multiple sexual partners (which points toward

PID), TB causes more indolent, chronic symptoms with minimal purulent discharge. Gonorrhoea and chlamydia cause acute PID with purulent discharge.

Q995 — Genital TB vs. other lower abdominal infections

A 28-year-old woman with a history of multiple sexual partners presents with lower abdominal pain for 1 month, minimal discharge, and intermenstrual bleeding. Most likely cause?

- A. Herpes simplex
- B. Genital tuberculosis
- C. Candida
- D. Neisseria gonorrhoea

► Answer & Explanation:

****B. Genital tuberculosis**** — Chronic (1 month) course + minimal discharge + intermenstrual bleeding in India = genital TB until proven otherwise. Gonorrhoea causes acute purulent PID; candida causes discharge without pain; herpes causes vesicular ulcers.

Why D (gonorrhoea) is the trap: Multiple sexual partners = STI risk = PID from gonorrhoea is the immediate thought. But gonorrhoea causes *acute* PID (days, not months) with *copious purulent discharge*. The chronicity (1 month) and minimal discharge point to TB.

Anticoagulation in Pregnancy: Warfarin → LMWH Near Term

Warfarin crosses the placenta and is teratogenic in the first trimester (warfarin embryopathy) and causes fetal haemorrhage in the third trimester. For women with mechanical heart valves who require anticoagulation throughout pregnancy, the general approach is:

- **First trimester:** LMWH (to avoid warfarin embryopathy) or continue warfarin if dose < 5 mg/day
- **Second trimester:** Warfarin is acceptable (fetal risk lower)
- **After 36 weeks:** Switch to LMWH (to allow reversal before delivery and avoid fetal intracranial haemorrhage at delivery)

Q1028 — Mechanical heart valve at 37 weeks on warfarin

A 37-weeks pregnant woman with a mechanical prosthetic heart valve on warfarin presents for routine antenatal check-up. She is asymptomatic. Next step?

- A. Switch to LMWH
- B. Immediate induction of labour
- C. Immediate LSCS
- D. Continue the same

► Answer & Explanation:

****A. Switch to LMWH**** — At 36–37 weeks, warfarin should be switched to LMWH to allow safe neuraxial anaesthesia and to prevent fetal intracranial haemorrhage at the time of delivery. Delivery is planned, not emergent.

Heavy Menstrual Bleeding: When to Biopsy

In any woman with abnormal uterine bleeding, endometrial thickness on ultrasound guides the need for biopsy. The threshold varies by menopausal status: - **Postmenopausal**: > 4–5 mm → mandatory biopsy (to exclude endometrial cancer) - **Perimenopausal/premenopausal with abnormal bleeding**: > 10–12 mm or persistent/recurrent bleeding → biopsy indicated

Q1044 — Heavy menstrual bleeding with thick endometrium

46-year-old with HMB, endometrial thickness 16 mm on USG. Next best step?

- A. Hysterectomy
- B. Endometrial biopsy
- C. HPV testing
- D. Progesterone therapy

► Answer & Explanation:

****B. Endometrial biopsy**** — Endometrial thickness > 4–5 mm in postmenopausal or abnormal bleeding mandates biopsy to exclude endometrial carcinoma. Never treat (hysterectomy or hormones) before histological diagnosis.

Clinical logic: At 46 years (perimenopausal), with HMB and an endometrium of 16 mm, the priority is to *exclude endometrial cancer*. Jumping to hysterectomy (A) without a diagnosis is wrong — biopsy must come first. Progesterone therapy (D) without ruling out cancer is dangerous.

Menopause: Vasomotor Symptoms and HRT

Menopausal hot flushes are caused by oestrogen withdrawal acting on hypothalamic thermoregulatory centres. FSH > 40 IU/L confirms ovarian failure.

Q385 — Menopause: vasomotor symptoms

A 51-year-old woman with hot flushes and night sweats for 8 months after her last menstrual period. FSH is 60 IU/L. No contraindications. Most effective treatment?

- A. Low-dose SSRIs (venlafaxine)
- B. Tibolone
- C. Combined oestrogen-progestogen HRT (systemic)
- D. Black cohosh supplements

► Answer & Explanation:

****C. Combined oestrogen-progestogen HRT (systemic)**** — Systemic HRT is the most effective treatment for menopausal vasomotor symptoms. All other options (SSRIs, tibolone, phytoestrogens) are second-line for women who cannot take HRT.

Note: Women with an intact uterus must receive *combined* oestrogen + progestogen HRT (progestogen protects the endometrium from unopposed oestrogen → endometrial cancer risk). Women post-hysterectomy can take oestrogen alone.

Chapter 3: Hypertensive Disorders of Pregnancy, Medical Complications & Antepartum Haemorrhage

Big Picture First

If there is one chapter that has been the examiner's favourite across every single NEET-PG / INI-CET paper from 2015 to 2025, this is it. Hypertensive disorders of pregnancy (HDP), gestational diabetes, hepatitis, cholestasis, cardiac disease, and antepartum haemorrhage collectively account for roughly **12–18 questions per paper** — sometimes more. That is not a guess; look at any year's paper and you will count them. This chapter is your single highest-return investment in OBG.

The examiner in this topic is not testing whether you can recite definitions. The examiner is testing whether you can **navigate nuance under pressure**: Is this pre-eclampsia or eclampsia? Is this HELLP or AFLP? Is this Mg toxicity or a seizure? Which antihypertensive is safe and which kills the fetus? Why does smoking *protect* against pre-eclampsia? These are the exact question stems you will face.

The most misunderstood areas, based on a decade of PYQ analysis: (1) The Mg sulfate toxicity sequence — students confuse the order of signs; (2) GDM diagnostic cut-offs — IADPSG vs DIPSI vs WHO; students learn one set and mix up the others; (3) The APH differential — vasa praevia is the dark horse that kills the fetus within minutes, and questions exploit the fact that students don't instinctively distinguish it from placenta praevia; (4) Abruption — the concealed form is more dangerous but less visible, which is a perfect examiner trap.

The tricky differential that appears every year: **HELLP vs AFLP vs viral hepatitis vs obstetric cholestasis**. They all involve the liver, all occur in the third trimester, and the examiner knows you will panic and confuse them.

Question Pattern Map

What the examiner loves to ask: - "Drug of choice for eclampsia / chronic hypertension in pregnancy / acute severe hypertension" - "First sign of Mg toxicity" (DTR loss — this appears almost every year) - "Investigation of choice for placenta praevia" (TVUS — appears every year) - "Comparison: concealed vs revealed abruption" - "GDM diagnosis: which cut-off?" (IADPSG vs DIPSI) - "Which cardiac condition is absolutely contraindicated in pregnancy?" - "Which hepatitis has highest maternal mortality?" - "Drug of choice for ICP (cholestasis)" - "Smoking as a protective factor" (counterintuitive — classic trap)

Common examiner traps: - Giving a vignette of concealed abruption with *no bleeding* and asking for the diagnosis — students panic because they expect blood - Offering "emergency LSCS" for a patient with pre-eclampsia at 33 weeks who needs *medical stabilisation first* - Mixing up GDM screening thresholds (DIPSI = 140, IADPSG FPG = 92) - Asking about vasa praevia where the correct clue is bleeding *at the moment of membrane rupture* + immediate fetal distress - "Smoking protects against pre-eclampsia" — this is NOT a fabrication; it is a real, reproducible epidemiological finding

Core Concept Deep Dive

Section 1: Pre-eclampsia — Diagnosis, Severity, Management

The concept: Pre-eclampsia = new-onset hypertension (BP $\geq 140/90$ mmHg on two readings ≥ 4 hours apart) after 20 weeks, with either proteinuria (≥ 300 mg/24h or $\geq 2+$ dipstick) OR severe features without proteinuria (AKI, thrombocytopenia $< 100,000$, impaired liver function, pulmonary oedema, new severe headache, visual disturbance).

Pre-eclampsia with **severe features** = BP $\geq 160/110$ mmHg persistently, OR the above organ-specific criteria regardless of BP.

Eclampsia = pre-eclampsia + seizures. No pre-existing history of epilepsy. The most important immediate management question is: **what drug do you give?**

MCQ Encounter — Q524 (PYQ)

Drug of choice for management of eclamptic seizures is: - A. Oral magnesium sulphate - B. Intravenous magnesium sulphate - C. Oral phenytoin - D. Intravenous phenytoin

Answer: B — IV MgSO₄

Why students get trapped: "Magnesium" sounds nutritional and gentle. Phenytoin sounds like a "real" anticonvulsant. Students default to phenytoin out of habit from neurology.

Why IV MgSO₄ wins: The Magpie Trial (2002) showed MgSO₄ reduces eclampsia risk by 58% vs placebo and is superior to both phenytoin and diazepam. Its mechanism is unique — it is an NMDA receptor antagonist, not a traditional anticonvulsant. It also causes cerebral vasodilation, reducing cerebrovascular spasm.

Loading dose: 4 g IV over 5–10 min. Maintenance: 1–2 g/hr by infusion (Zuspan regimen) OR 10 g IM split dose + 5 g IM q4h (Pritchard regimen).

Key takeaway: IV MgSO₄ = drug of choice for both seizure treatment AND prophylaxis (Q560) in severe pre-eclampsia.

Examiner Note: Mg Toxicity — The Sequence You Must Know

This appears nearly every year, phrased as either "first sign" or "sequence of toxicity."

Serum Mg Level	Clinical Sign
4–7 mEq/L	Therapeutic range (seizure prophylaxis)
7–10 mEq/L	Loss of deep tendon reflexes (FIRST sign)
10–13 mEq/L	Respiratory depression
> 13 –15 mEq/L	Cardiac arrest

Monitor with the triad: Patellar reflex (present = safe), RR ≥ 12 /min, Urine output ≥ 25 mL/hr.

Antidote: Calcium gluconate 1 g IV (10 mL of 10% solution) — given SLOWLY over 10 min.

MCQ Encounter — Q268 & Q659 (PYQ)

Pregnant woman on MgSO₄ develops loss of DTRs and respiratory depression. Antidote?

Answer: Calcium gluconate IV — Calcium antagonises magnesium at the neuromuscular junction.

Why students get trapped: They say "calcium chloride" — this is wrong. Calcium chloride is for cardiac arrest/ACLS, not for Mg reversal in obstetrics. The OBG answer is always **calcium gluconate**.

Pre-eclampsia Management: Timing of Delivery

MCQ Encounter — Q514 (PYQ)

Primigravida with PIH at 32 weeks, BP 150/100 mmHg, controlled on medication, no complications.

Pregnancy should be terminated at: - A. 40 weeks B. 37 weeks C. 36 weeks D. 34 weeks

Answer: B — 37 completed weeks

The rule: Gestational hypertension or pre-eclampsia without severe features → deliver at **37 weeks**. With severe features at ≥34 weeks → deliver after stabilisation. With severe features at <34 weeks → steroids + close monitoring, aim for brief prolongation if possible.

Antihypertensive Therapy

MCQ Encounter — Q630 (PYQ)

Drug of choice for chronic hypertension in pregnancy:

Answer: D — Methyldopa

Methyldopa has the longest safety record (decades of data). It is the first-line choice.

Drug	Status in Pregnancy
Methyldopa	First-line (chronic hypertension)
Labetalol	First-line (acute severe hypertension, IV)
Nifedipine	Alternative (oral, acute and chronic)
Hydralazine	IV, acute severe hypertension
ACE inhibitors (enalapril, ramipril)	ABSOLUTELY CONTRAINDICATED — fetal renal dysplasia, oligohydramnios, death
Atenolol	Relatively contraindicated — IUGR
Sodium nitroprusside	Avoid if possible — cyanide toxicity in fetus

Examiner trap: Enalapril is offered as an option in a drug-for-PIH question. Students recognise it as an antihypertensive and choose it. Wrong. ACE inhibitors = Category D/X in 2nd and 3rd trimesters.

Not a Risk Factor for Pre-eclampsia:**MCQ Encounter — Q471 (PYQ 2015)**

Which is NOT a risk factor for gestational hypertension: Obesity / Smoking / Primigravida / Factor V Leiden?

Answer: B — Smoking

This is the paradigm counterintuitive fact of HDP. Smoking reduces placental oxidative stress and endothelial dysfunction via nicotine-mediated vasoconstriction and reduced trophoblast invasion — paradoxically reducing pre-eclampsia risk. This does not justify smoking; it simply predicts the "NOT a risk factor" answer type.

Genuine risk factors: Primigravida, multiple pregnancy, previous PE, chronic hypertension, diabetes, obesity, renal disease, SLE/APLA, extremes of age, Factor V Leiden (thrombophilia).

Complications of Pre-eclampsia:**MCQ Encounter — Q482 (PYQ 2016)**

Complications of pre-eclampsia are all EXCEPT: - A. Post-datism B. DIC C. Blindness D. None of the above

Answer: A — Post-datism (prolonged pregnancy is NOT a complication of pre-eclampsia).

Genuine complications: abruption, DIC, HELLP, eclampsia, cortical blindness (hypertensive encephalopathy/ occipital lobe oedema), renal failure, hepatic rupture, IUGR, IUFD.

Section 2: HELLP Syndrome — Diagnosis and Pitfalls

The concept: HELLP = Haemolysis + Elevated Liver enzymes + Low Platelets. It is a severe variant of pre-eclampsia (though 15% occur without classical HDP features). The triad is what the examiner tests.

MCQ Encounter — Q266 & Q660 (PYQ)

Pregnant woman with elevated LDH, low platelets, abnormal LFTs:

Answer: A — HELLP syndrome

Diagnostic criteria (Tennessee Classification): - LDH >600 IU/L (haemolysis marker) - AST/ALT >70 IU/L (liver) - Platelets <100,000/ μ L (thrombocytopenia)

The differentials:

Feature	HELLP	AFLP	ICP	Viral Hepatitis
Trimester	3rd	3rd (late)	3rd	Any
Jaundice	Mild/absent	Prominent	Mild/absent	Prominent
Pruritus	Absent	Absent	Hallmark	May be present
Platelets	Low	Low (late)	Normal	Normal
PT/APTT	Normal initially	Prolonged	Normal	Prolonged (severe)
Bile acids	Normal/slightly raised	Normal	Very high	Normal
Key fact	Pre-eclampsia context	LCHAD deficiency fetus	UDCA treatment	HEV = highest mortality

MCQ Encounter — Q269 (PYQ)

Acute fatty liver of pregnancy typically occurs in:

Answer: C — Third trimester (late, typically 35–37 weeks)

AFLP pathophysiology: fetus carries LCHAD (long-chain 3-hydroxyacyl CoA dehydrogenase) deficiency → long-chain fatty acids accumulate in maternal liver → microvesicular steatosis → liver failure.

Treatment: IMMEDIATE DELIVERY regardless of gestational age. Do not wait.

Section 3: Gestational Diabetes Mellitus (GDM)

Why students get confused: There are THREE sets of diagnostic criteria taught in Indian exams — IADPSG, DIPSI, and WHO/overt diabetes cut-offs. These are NOT interchangeable.

The framework:

Criterion	Test	Cut-off	Fasting?
IADPSG (gold standard, 75g OGTT)	FPG ≥ 92 mg/dL	Single abnormal value = GDM	Yes
	1-hr ≥ 180 mg/dL		
	2-hr ≥ 153 mg/dL		
DIPSI (India MoHFW)	2-hr PG ≥ 140 mg/dL	After 75g glucose, non-fasting	No fasting needed
Overt DM	FPG ≥ 126 mg/dL or RBG ≥ 200 or HbA1c $\geq 6.5\%$	At any gestation	—

MCQ Encounter — Q382 (PYQ)

DIPSI test: A 26-year-old at 24–28 weeks given 75g glucose non-fasting. GDM if 2-hour PG $\geq$ ____?

Answer: B — 140 mg/dL

Why students get trapped: They remember 153 from IADPSG (2-hr cut-off) and apply it to DIPSI. But DIPSI is a non-fasting test with a different threshold. Remember: **DIPSI = 140, IADPSG 2-hr = 153.**

MCQ Encounter — Q272 & Q545 (PYQ)

Overt diabetes threshold in pregnancy (fasting PG):

Answer: ≥ 126 mg/dL — This is the WHO criteria for diagnosing pre-existing (overt) DM at any gestation, distinct from GDM cut-offs.

Mechanism of GDM:**MCQ Encounter — Q562 (PYQ)**

Primary mechanism of gestational diabetes:

Answer: B — Placental lactogen-induced peripheral insulin resistance

hPL, cortisol, progesterone, and oestrogen are all insulin antagonists that rise progressively from mid-pregnancy. This "diabetogenic stress" unmasks GDM in women with limited beta-cell reserve. Note: GDM is NOT autoimmune (that's Type 1 DM).

Pre-gestational diabetes and fetal risk:**MCQ Encounter — Q633 (PYQ)**

Pre-gestational diabetes counselling about fetal anomaly risk:

Answer: C — Four-fold higher risk of congenital anomalies

This is proportional to periconceptional HbA1c. The highest-risk anomalies: sacral agenesis (pathognomonic), VSD, neural tube defects. GDM (onset in pregnancy) does NOT increase structural anomaly risk — the anomalies form in 1st trimester, before GDM typically manifests.

Section 4: Cardiac Disease in Pregnancy

The framework: The key question is always "which condition is most dangerous / absolutely contraindicated?"

MCQ Encounter — Q383 & Q699 (PYQ)

Which cardiac condition is an **ABSOLUTE** contraindication to pregnancy? / Which has highest maternal mortality?

Answer: Eisenmenger's syndrome — Maternal mortality 30–50%

Why: Pregnancy causes systemic vasodilation (progesterone effect), which in a right-to-left shunt patient worsens hypoxaemia and causes pulmonary vascular crisis. The shunt reverses further → refractory hypoxia → death. Advise **termination** if already pregnant.

NYHA Classification and Risk:

NYHA Class	Description	Maternal Mortality
I–II	Asymptomatic or mild symptoms	<1%
III	Symptomatic on ordinary activity	5–15%
IV	Symptomatic at rest	25–50%

Peripartum cardiomyopathy: New-onset heart failure in the last month of pregnancy or within 5 months postpartum. Treatment: diuretics, ACE inhibitors (postpartum only!), beta-blockers, anticoagulation if EF <35%. Can recur in subsequent pregnancies.

The mitral stenosis trap: Mitral stenosis is the most common rheumatic valvular lesion in pregnancy in India. The increased blood volume of pregnancy is particularly dangerous with MS (mitral stenosis) — the fixed cardiac output through a narrowed valve cannot accommodate the demand → acute pulmonary oedema.

Section 5: Hepatitis in Pregnancy**MCQ Encounter — Q700 (PYQ)**

Which hepatitis has highest maternal mortality?

Answer: D — Hepatitis E — Maternal mortality 15–25% in 3rd trimester; fulminant hepatic failure is the mechanism.

Hepatitis Type	Pregnancy Feature	Vertical Transmission	Key Fact
Hep E	Highest maternal mortality (15–25%)	Yes	Faeco-oral, epidemic in India
Hep B	Usually no increased maternal morbidity	High (if HBeAg+)	HBsAg = first serological marker (Q270)
Hep A	No increased risk	Rare	Resolves; no chronicity
Hep C	Low risk	~5%	No vaccine

MCQ Encounter — Q270 (PYQ)

Pregnant woman HBsAg positive at 14 weeks. True statement?

Answer: C — HBsAg is the first serological marker

HBsAg appears 1–10 weeks post-exposure. Neonatal vaccination + HBIG at birth prevents 90% of vertical transmission. The confusion: students think "first marker = anti-HBc IgM" — that is the first marker of acute infection immune response; HBsAg is the first serological antigen.

Section 6: Intrahepatic Cholestasis of Pregnancy (ICP)

The concept: ICP = pruritus in pregnancy (especially palms and soles), raised serum bile acids, elevated transaminases, NO rash, reverses postpartum.

MCQ Encounter — Q349 & Q547 (PYQ)

Drug of choice for ICP:

Answer: B — Ursodeoxycholic acid (UDCA)

UDCA dose: 500 mg BD or 10–15 mg/kg/day. It reduces bile acid levels, relieves pruritus, and reduces fetal risk.

Why not cholestyramine? Cholestyramine relieves pruritus but does NOT reduce bile acids or improve fetal outcomes. UDCA does both.

MCQ Encounter — Q536 (PYQ)

Best diagnostic marker for ICP:

Answer: C — Serum bile acids

ALP rises normally in pregnancy (placental ALP) and is NOT a reliable marker. Bile acids $>40 \mu\text{mol/L}$ = significant risk; $>100 \mu\text{mol/L}$ = very high risk of stillbirth.

Delivery timing: 36–37 weeks when bile acids >40 ; some centres deliver at 35–36 if $>100 \mu\text{mol/L}$.

Section 7: Antepartum Haemorrhage (APH) — The Big Three

APH = bleeding from the genital tract after 28 weeks (24 weeks in some guidelines) and before delivery of the baby.

The three causes you must master:

A. Placenta Praevia

The concept: Placenta implanted in the lower uterine segment, overlying or near the internal os.

Classification (traditional): | Type | Relation to OS | |---|---| | Type I (lateral) | Encroaches lower segment | | Type II (marginal) | Reaches edge of internal os | | Type III (partial/incomplete) | Partially covers internal os | | Type IV (complete/major) | Completely covers internal os |

Classic presentation: Painless, bright red, fresh bleeding, proportional to visible blood loss, soft non-tender uterus, abnormal lie/presentation common.

MCQ Encounter — Q345 (PYQ)

Investigation of choice for suspected placenta praevia:

Answer: C — Transvaginal ultrasound (TVUS)

TVUS is safe (probe stays >3 cm from cervix) and more accurate than transabdominal USG. **PV examination is absolutely contraindicated** until praevia is excluded — the "double set-up" examination (Q429) in OT is only for when you must examine but may need to proceed immediately to caesarean.

MCQ Encounter — Q477 & Q762 (PYQ 2015)

Management of complete placenta praevia at 38 weeks without bleeding:

Answer: C — Elective caesarean section

At ≥ 37 weeks, active delivery is mandatory. No bleeding = elective (planned) caesarean. Bleeding = emergency caesarean.

The Macafee–Johnson regimen (expectant, hospitalise, transfuse, wait until 37 weeks) is appropriate for the patient with praevia diagnosed before 37 weeks without active bleeding.

MCQ Encounter — Q625 (PYQ)

Placenta praevia found at 28 weeks — confirmatory scan at:

Answer: A — 32 weeks (then again at 36 weeks if persisting). Many cases "migrate" as the LUS develops.

Placenta Accreta Spectrum:

MCQ Encounter — Q814 (PYQ 2016) & Q655 (PYQ)

G3P2 with previous 2 LSCS + anterior placenta praevia — highest risk of:

Answer: A — Placenta accreta

Prior LSCS scar + anterior praevia = Nitabuch's layer deficiency → villi invade myometrium. Risk with 2 previous scars $\approx 40\%$. This patient needs multidisciplinary planning, possible caesarean hysterectomy.

B. Abruptio Placentae

The concept: Premature separation of a normally situated placenta. Pain (unlike praevia), retroplacental haematoma.

The classic trap — concealed abruption:

MCQ Encounter — Q404 (PYQ)

32-year-old multigravida at 36 weeks, sudden severe abdominal pain, "woody hard" uterus, NO visible bleeding, fetal distress. Type:

Answer: B — Type II (concealed/severe) — complete separation with retroplacental haematoma

This is the trap: "no bleeding" does NOT mean no abruption. In concealed abruption, blood is trapped behind the placenta → the apparent blood loss massively underestimates true loss → DIC. Woody-hard uterus = haematomyometrium (Couvelaire uterus).

Abruptio vs Placenta Praevia:

Feature	Placenta Praevia	Abruptio Placentae
Pain	Absent	Present (often severe)
Bleeding	Bright red, fresh	Dark red $\pm$ concealed
Uterus	Soft, non-tender	Tender, board-like
Shock	Proportional to blood	Disproportionate (concealed)
Cause	Abnormal implantation	Vascular disease, trauma, cocaine
Recurrence	Low	High (10–15%)
Fetal presentation	Abnormal (floating)	Normal lie

Risk factors for abruption: Hypertension (most common), smoking, cocaine, prior abruption, PPRM, trauma, rapid uterine decompression (polyhydramnios).

C. Vasa Praevia

The concept: Fetal vessels traverse the membranes over the internal os, unsupported by Wharton's jelly, associated with velamentous cord insertion.

MCQ Encounter — Q478 & Q768 (PYQ 2015)

Vasa praevia is associated with:

Answer: B — Velamentous placenta

MCQ Encounter — Q403 & Q493 (PYQ)

Classic scenario: painless bleeding AT the moment of membrane rupture + immediate fetal bradycardia. Diagnosis?

Answer: Vasa praevia with fetal vessel rupture

*The key distinguishing feature: bleeding is **FETAL blood** — the fetus exsanguinates within minutes. Apt test (Ogita test) confirms fetal blood in vaginal bleeding. Management: immediate emergency caesarean. Antenatal diagnosis by TVUS + colour Doppler → planned caesarean at 34–36 weeks with steroids.*

Section 8: Medical Complications — Drugs and Pregnancy

MCQ Encounter — Q408 & Q578 (PYQ)

Which drug does NOT cross the placenta?

Answer: A — Heparin (large, charged molecule; cannot cross). Warfarin, morphine, naloxone = all cross freely.

LMWH advantage (Q384): Does not cross placenta, predictable pharmacokinetics, less HIT, less osteoporosis vs UFH. Preferred for DVT prophylaxis in pregnancy.

MCQ Encounter — Q307 & Q701 (PYQ)

Recurrent pregnancy loss + prolonged APTT / APLA. Treatment:

Answer: Aspirin + LMWH — Low-dose aspirin (75 mg) from pre-conception + prophylactic LMWH once intrauterine pregnancy confirmed.

Why the APTT is prolonged in APS: Lupus anticoagulant interferes with phospholipid-dependent coagulation tests in vitro — paradoxically, patients are pro-thrombotic in vivo.

Bias Removal Section

Top traps in this chapter:

1. **Mg toxicity sequence:** Students write "respiratory depression is first." Wrong. Loss of DTRs is first (7–10 mEq/L). Respiratory depression is second. Cardiac arrest is last.
2. **GDM thresholds:** DIPSI uses 140 (non-fasting), IADPSG 2-hr uses 153 (fasting OGTT). These are NOT the same test. Don't mix them.
3. **UDCA vs cholestyramine for ICP:** Both relieve itch but only UDCA reduces bile acids and fetal risk. UDCA = drug of choice.
4. **Enalapril/ACE inhibitors:** Students see "antihypertensive" and choose it. In pregnancy = absolutely contraindicated. The answer is always methyldopa (chronic) or labetalol/hydralazine (acute).
5. **Placenta praevia investigation:** Students write "transabdominal USG." Wrong. TVUS is more accurate and safe.
6. **Smoking = protective:** Students assume this is a trick question and choose "smoking is a risk factor." Smoking genuinely reduces pre-eclampsia risk (though it causes IUGR, abruption, and other harms).

Variations and Twists

How examiners modify HDP questions: - The scenario gives you a 33-week patient with BP 180/110, headache, visual disturbance, and oliguria — then asks "NEXT step." Students jump to "LSCS." Wrong. **Stabilise first** (MgSO₄ + antihypertensive), THEN deliver. Q484 illustrates this perfectly. - "A woman develops DIC after abruption at 34 weeks. Most likely underlying mechanism?" → Thromboplastin release from retroplacental clot activates extrinsic pathway. - They may offer "phenytoin for seizure prophylaxis in severe pre-eclampsia" — this was the old practice and is WRONG by current guidelines.

GDM vignette twists: - They give HbA1c 6.7% at first prenatal visit → this is OVERT DM, not GDM (HbA1c $\geq 6.5\%$ = overt DM; GDM is diagnosed only 2nd trimester onwards and uses OGTT). - "GDM patient, best indicator of fetal compromise" → Non-stress test (NST) is the first-line test (Q657).

Memory Engineering

HELLP mnemonic: Haemolysis – Elevated liver enzymes – Low Platelets

Mg toxicity sequence — "DTR before RR before Arrest": - D = DTRs lost (first warning) - R = Respiration stops - A = Arrest (cardiac) - Antidote = **Ca gluconate** (Calcium Counters Magnesium)

APH causes — "The 3 Ps": - Placenta praevia (low-lying placenta) - Placental abruption (separation) - vasa Praevia (fetal vessel over os) — "tiny bleed, fatal for fetus"

Hepatitis mortality in pregnancy — "E for Emergency": Hepatitis E = Emergency = highest mortality (15–25%)

Antihypertensives — safe vs banned: - Safe: **M**ethyldopa, **L**abetalol, **N**ifedipine, **H**ydralazine → "M.L.N.H" = "My Labour Needs Help" - Banned: **ACE inhibitors** (enalapril, ramipril), **ARBs** — all nephrotoxic to fetus

GDM thresholds — "DIPSI is 140; IADPSG FPG is 92 (lowest)"

Ultra High-Yield Revision Layer

What the Examiner REALLY Wants

The examiner wants you to know three things cold: (1) the Mg toxicity sequence + antidote, (2) the APH differential (especially the vasa praevia trap), and (3) GDM diagnostic cut-offs.

20-Second Revision Bullets

- MgSO₄ = drug of choice for eclampsia (both treatment and prophylaxis)
- First sign of Mg toxicity = loss of deep tendon reflexes; antidote = calcium gluconate 1g IV
- Mg therapeutic range = 4–7 mEq/L; toxic >7; respiratory paralysis >10; cardiac arrest >13
- Methyldopa = drug of choice for chronic hypertension in pregnancy
- ACE inhibitors (enalapril) = absolutely contraindicated in pregnancy
- Smoking = protective against pre-eclampsia (NOT a risk factor)
- DIPSI GDM cut-off = 140 mg/dL (non-fasting 75g 2-hr PG)
- IADPSG: FPG ≥ 92 / 1-hr ≥ 180 / 2-hr ≥ 153 (any ONE = GDM)
- Overt DM in pregnancy: FPG ≥ 126 / RBG ≥ 200 / HbA1c $\geq 6.5\%$
- HELLP = Haemolysis + Elevated liver enzymes + Low platelets (<100,000)
- HELLP vs AFLP: AFLP = more liver failure, LCHAD deficiency fetus, treat by immediate delivery
- ICP treatment = UDCA (not cholestyramine); best marker = serum bile acids (not ALP)
- Hepatitis E = highest maternal mortality (15–25%); HBsAg = first serological marker
- Eisenmenger's = absolute contraindication to pregnancy (>50% mortality)
- TVUS = investigation of choice for placenta praevia (NOT per-vaginal exam)
- Complete praevia at term = elective LSCS; no bleeding = elective; bleeding = emergency

- Vasa praevia = bleeding at ROM + immediate fetal distress = fetal blood loss = emergency LSCS
- Concealed abruption = woody hard uterus + no PV bleeding + fetal distress + DIC
- Heparin does NOT cross placenta; warfarin DOES
- APS treatment = aspirin + LMWH (not steroids)
- Pre-gestational diabetes = 4-fold increased fetal anomaly risk
- GDM does NOT increase structural anomaly risk (anomalies form in 1st trimester, before GDM manifests)

Top 5 Examiner Traps in Chapter 3

1. "First sign of Mg toxicity" → DTR loss (NOT respiratory depression)
 2. "Vasa praevia" scenario → bleeding after ROM + immediate fetal distress (NOT placenta praevia)
 3. "Drug for PIH" options include enalapril → always wrong (ACE inhibitors = contraindicated)
 4. "Smoking and pre-eclampsia" → smoking is PROTECTIVE (counterintuitive but tested repeatedly)
 5. "Investigation for praevia" → TVUS (NOT transabdominal USG and NEVER PV examination)
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Chapter 4: Normal Labour, Intrapartum Care, Operative Delivery & Complications

Big Picture First

Labour and delivery is where obstetrics becomes life-or-death in real time. It is also where the NEET-PG examiner feels most at home. Roughly **10–15 questions per paper** come directly from this chapter — stages of labour, partogram interpretation, PPH management, shoulder dystocia, instrumental delivery, episiotomy, breech manoeuvres, cord prolapse, uterine inversion. No OBG preparation is complete without mastering this section.

What makes this chapter especially dangerous for the student is that it looks straightforward — "just know the stages, know the drugs, know the manoeuvres" — but the examiner deliberately writes questions that test the *order of steps*, the *exact criteria*, and the *exceptions to the rule*. A question on PPH does not test whether you know "uterine atony is the commonest cause" — you know that. It tests whether you know that carboprost is contraindicated in asthma, that its maximum dose is 2 mg (not 2000 µg, not 20 mg), and that ergometrine must NOT be given before the second stage.

The most misunderstood area in this chapter: **fetal presentations and their diameters**. Students know brow is bad and face can sometimes deliver vaginally, but they confuse the exact diameters, forget that mento-posterior face presentation = mandatory caesarean, and don't connect the examining-room findings (what landmarks are palpable) to the diagnosis.

The second most misunderstood area: the **partogram**. Specifically, the alert-action line gap (4 hours), the transition from latent to active phase (6 cm by ACOG 2014), and what crossing the action line means clinically.

Question Pattern Map

What the examiner loves to ask: - "Most common cause of primary PPH / secondary PPH" (atony vs retained tissue) - "Maximum dose of carboprost / which drug is contraindicated in PPH with asthma" - "Active management of third stage — which component is NOT included" - "Partogram alert-to-action line distance" (4 hours — this recurs every year) - "Drug contraindicated before second stage" (ergometrine) - "First manoeuvre for shoulder dystocia" (McRoberts) - "Last resort for shoulder dystocia" (Zavanelli) - "Forceps vs vacuum — preterm preference / face presentation contraindication" - "Johnson's formula for fetal weight calculation" - "Stages of labour — duration in primigravida vs multigravida" - "Mento-vertical diameter = which presentation?" - "Frank breech = most common type"

Common traps: - Carboprost dose: "2000 µg" looks like "2 mg" — students pick 2000 µg thinking it means 2 mg when written differently - Ergometrine given "before second stage" in a question — recognising it as dangerous - "McRoberts" described without name — "hyperflexion of maternal thighs onto abdomen" - Vasa praevia appearing in a labour chapter question (it is both an APH and intrapartum emergency) - Sheehan's syndrome presented as a labour complication consequence — severe PPH → pituitary necrosis → amenorrhoea (Q866, Q412)

Core Concept Deep Dive

Section 1: Stages of Labour — The Framework

Every question in this chapter is anchored to your understanding of the stages.

Stage	Definition	Duration (Primigravida)	Duration (Multigravida)
First stage — Latent phase	Onset of regular contractions → 6 cm dilatation	Up to 20 hours	Up to 14 hours
First stage — Active phase	6 cm → 10 cm (full dilatation)	≥0.5–1 cm/hr (min rate)	Faster
Second stage (propulsive)	Full dilatation → delivery of baby	Up to 2 hours (3 with epidural)	~20 minutes (Q736)
Third stage	Delivery of baby → delivery of placenta	Up to 30 minutes (active mgmt)	Same
Fourth stage	First 2 hours postpartum	—	—

MCQ Encounter — Q410 (PYQ)

Which is **NOT** true about the latent phase? - A. Starts with onset of regular contractions - B. Begins at the end of the active phase - C. Ends at 6 cm dilatation (ACOG 2014) - D. Patient may present with mild crampy pain

Answer: B — The latent phase **PRECEDES** the active phase; it does not begin at the end of it. The false statement is B.

Key update (ACOG 2014): Active phase now defined as starting at **6 cm** (previously 4 cm). This raised the threshold for diagnosing "failure to progress" — do not diagnose arrest of active phase before 6 cm dilatation.

Section 2: The Partogram — The Examiner's Favourite Tool

The partogram (WHO model) is a graphical record of labour progress. Three lines matter:

1. **Alert line:** Starts at 4 cm dilatation; based on the slowest 10th centile of normal labours (Friedman's curve).
2. **Action line:** Drawn **4 hours to the right** of the alert line. Crossing the action line = intervention warranted.
3. The gap between alert and action line = **4 hours** (not 2, not 1, not 5).

MCQ Encounter — Q492 & Q864 (PYQ 2016)

Distance between alert and action lines in WHO partogram:

Answer: C — 4 hours

This appears almost every single year. Four hours. Commit to memory and never hesitate.

What the action line means: Labour has deviated significantly from expected progress; the patient needs assessment for CPD, malposition, inadequate contractions → consider ARM, oxytocin augmentation, or caesarean.

Section 3: Fetal Presenting Diameters — The Anatomy You Must Know

This is the area where clinical vignettes are written. "On vaginal exam, the examiner palpates orbital ridges, nose, anterior fontanelle but NOT the mouth or chin — what is the presentation?"

Key diameters:

Presentation	Degree of Flexion	Presenting Diameter	Size	Can Deliver Vaginally?
Vertex (well-flexed)	Maximum flexion	Suboccipito-bregmatic	9.5 cm	Yes (ideal)
Vertex (partial deflexion)	Partial	Suboccipito-frontal	10.0 cm	Yes
Brow	Partial deflexion	Mento-vertical	13.5 cm	Almost never
Face (mento-anterior)	Full extension	Submento-bregmatic	9.5 cm	Yes
Face (mento-posterior)	Full extension	—	—	NEVER
Breech	—	—	After-coming head	With manoeuvres

MCQ Encounter — Q294 (PYQ)

Mento-vertical engaging diameter = which presentation?

Answer: B — Brow presentation

The brow is the partially deflexed head. Mento-vertical is the largest AP diameter (13.5 cm). This is undeliverable in a term fetus — the pelvis (average 11–12 cm AP) cannot accommodate it.

MCQ Encounter — Q295 (PYQ)

Presenting diameter of fully flexed (vertex) head:

Answer: A — Suboccipito-bregmatic (9.5 cm)

This is the smallest presenting diameter and why good flexion matters. Full flexion = smallest diameter = easiest delivery.

Section 4: Fetal Presentations — Recognition and Management

Brow Presentation

MCQ Encounter — Q370 (PYQ)

Term primigravida in active labour: orbital ridges, nose, and anterior fontanelle palpable; mouth and chin NOT felt; station -1. Diagnosis:

Answer: B — Brow presentation

Landmarks palpated in brow: Orbital ridges + anterior fontanelle (neither the occiput of vertex nor the chin/mouth of face). **Management:** almost always requires caesarean section in a term fetus. Small chance it converts to face (mento-anterior) and delivers vaginally, but never count on it.

Face Presentation

Mento-anterior: Can deliver vaginally (chin rotates to occiput anterior position; extension is maintained through the pelvis). **Mento-posterior:** CANNOT deliver vaginally — the fully extended head cannot negotiate the sacral curve.

MCQ Encounter — Q369 (PYQ)

Primigravida at term, active labour, face presentation with mentum directed posteriorly. Management:

Answer: C — Caesarean section

Persistent mento-posterior = mandatory LSCS. Any attempt at vaginal delivery risks fetal and maternal injury.

Breech Presentation

MCQ Encounter — Q473 (PYQ 2015)

Most common type of breech:

Answer: B — Frank breech (thighs flexed, knees extended = "pike position") — accounts for 65–70%.

Complete breech (both hips and knees flexed) = next most common. Footling = least common but highest cord prolapse risk.

Manoeuvres in breech delivery: - **Lovset manoeuvre** (Q297): For **extended/nuchal arms** — rotate baby 180° to deliver arms. - **Mauriceau-Smellie-Veit (MSV) manoeuvre:** For the **after-coming head** — jaw flexion and occiput pressure with one hand, traction with the other. - **Burns Marshall technique:** After-coming head, allows fetus to "hang" to allow head to flex, then swing up. - **Piper's forceps** (Q706): Applied to the after-coming head when it is arrested. - **Prague manoeuvre:** For delivery of head when occiput is posterior.

MCQ Encounter — Q297 (PYQ)

Lovset manoeuvre is used for:

Answer: D — Extended arms in breech delivery

Never confuse: MSV = head; Lovset = arms.

Transverse Lie / Shoulder Presentation

MCQ Encounter — Q624 (PYQ)

Transverse lie, shoulder presentation, absent membranes in second stage. Management:

Answer: D — Caesarean section (neglected shoulder presentation — do NOT attempt ECV or internal podalic version with absent membranes at term; risk of uterine rupture is high).

Section 5: Bishop Score — Cervical Favourability

The Bishop score determines whether the cervix is "ripe" for induction. Five parameters:

Parameter	0	1	2	3
Dilatation (cm)	Closed	1–2	3–4	≥5
Effacement (%)	0–30	40–50	60–70	≥80
Station	–3	–2	–1/0	+1/+2
Consistency	Firm	Medium	Soft	—
Position	Posterior	Mid	Anterior	—

Score ≥8 = favourable cervix → induction likely successful with oxytocin alone.

Score <6 = unfavourable → cervical ripening needed first.

MCQ Encounter — Q618 (PYQ)

Cervix 3 cm dilated, soft, anterior, station –1, length <0.5 cm. Bishop score?

Calculation: Dilatation 3 cm (+2) + Effacement >80% (+3) + Station –1 (+1) + Soft (+1) + Anterior (+1) = **8**

Answer: B — 8 — Favourable cervix; induction can proceed with oxytocin.

Section 6: Induction of Labour — Drugs and Contraindications

Cervical ripening agents: - PGE2 (dinoprostone) gel/tablet — pre-induction cervical ripening - PGE1 (misoprostol) — cervical ripening and induction - Mechanical: Foley's catheter balloon (mechanical dilation), hygroscopic dilators

Drug NOT used for induction:

MCQ Encounter — Q339 & Q921 (PYQ)

41-week pregnancy, unfavourable cervix. Which should NOT be used for induction?

Answer: D — PGF2α (carboprost)

PGF2α = carboprost = used for **PPH treatment** (uterotonic for atony). It causes bronchospasm and is too potent for induction. PGE1 (misoprostol), PGE2 (dinoprostone), and Foley's catheter are all appropriate.

Misoprostol pharmacology:**MCQ Encounter — Q330 (PYQ)**

Misoprostol is an analogue of:

Answer: A — PGE1 (prostaglandin E1)

Misoprostol is the methyl ester of PGE1. Used for: induction of labour, cervical ripening, medical TOP, missed abortion, and atonic PPH (off-label).

MCQ Encounter — Q515 (PYQ)

Misoprostol NOT effective for:

Answer: C — Menorrhagia — Misoprostol is a PG agonist; menorrhagia is treated with PG inhibitors (NSAIDs). This is the perfect "opposite mechanism" trap.

Ergometrine contraindication:**MCQ Encounter — Q298 (PYQ)**

Drug contraindicated before the second stage of labour:

Answer: D — Ergometrine

Ergometrine causes sustained tonic contraction. If given before delivery of the fetus, it can cause placental entrapment, tetanic contraction, and fetal death. It is given only after delivery of the anterior shoulder or in the third stage.

Contraindications to induction:**MCQ Encounter — Q280 & Q935 (PYQ)**

Which is NOT an absolute contraindication to induction?

Q280 Answer: C — IUGR (IUGR is actually an indication for induction, not a contraindication)

Q935 Answer: C — Heart disease (vaginal delivery is preferred in cardiac patients; induction is acceptable with monitoring)

Absolute contraindications: Complete placenta praevia, active genital herpes, pelvic obstruction (tumour), classical uterine scar, cord prolapse, transverse lie, fetal malpresentation preventing vaginal birth, multiple previous LSCS.

Section 7: Caesarean Section — Classical vs LSCS

Standard procedure: Lower uterine segment caesarean section (LSCS), Kerr incision (transverse) — preferred because: less blood loss, easier repair, lower rupture risk in subsequent pregnancy.

MCQ Encounter — Q365 (PYQ)

28 weeks, transverse lie with back-down, emergency caesarean required. Which incision?

Answer: C — Classical (upper segment vertical)

Classical LSCS indications: - Extreme prematurity (<28 weeks) — lower segment not formed - Transverse lie with back-down (lower segment inaccessible) - Anterior low-lying placenta/praevia with extensive vasculature - Carcinoma cervix coexisting

Disadvantage: Classical scar has much higher rupture risk in subsequent labour (5–10% vs 0.5% for LSCS scar) → future pregnancies must be LSCS.

Scar rupture risk:**MCQ Encounter — Q300 (PYQ)**

Risk of scar rupture in next pregnancy after ONE LSCS:

Answer: A — ~2% (0.5–2% for lower transverse scar). Rises with classical scar and multiple LSCS.

Section 8: Postpartum Haemorrhage (PPH) — The 4 Ts

Definition: Blood loss ≥ 500 mL after vaginal delivery or ≥ 1000 mL after caesarean, within 24 hours. Primary PPH = within 24 hours; secondary PPH = 24 hours to 12 weeks.

The 4 Ts framework:

Cause	Frequency	Examples
Tone (uterine atony)	70–80%	Most common cause of PRIMARY PPH
Trauma	10–20%	Cervical/vaginal tears, uterine rupture, episiotomy
Tissue (retained)	5–10%	Retained placenta, cotyledon, membranes — most common cause of SECONDARY PPH
Thrombin (coagulopathy)	1–5%	DIC, HELLP, AFLP, AFE

MCQ Encounter — Q276 & Q906 (PYQ)

Which is NOT a cause of PPH?

Answer: A — Placenta praevia

Placenta praevia causes antepartum haemorrhage, NOT PPH. Students trap themselves by thinking "placenta = postpartum."

MCQ Encounter — Q299 & Q476 (PYQ 2015)

Most common cause of secondary PPH:

Answer: B — Retained placental tissue ($\pm$ endometritis)

Secondary PPH = 24 hours to 12 weeks; most commonly 8–14 days. Cause: retained cotyledon/membrane fragment $\rightarrow$ failure of uterus to involute normally.

PPH-increasing risk factors:**MCQ Encounter — Q654 (PYQ)**

Which complication does NOT increase PPH risk?

Answer: A — Hypertension (pre-eclampsia)

Uterine overdistension = macrosomia, twins, polyhydramnios $\rightarrow$ all predispose to atony. Hypertension does not cause uterine distension; it is not in the 4 Ts.

Uterotonic Drugs — The Hierarchy:

Step 1: Oxytocin 10 IU IM or IV infusion (first-line; universal) **Step 2:** Ergometrine 0.5 mg IM (or Syntometrine = oxytocin + ergometrine combined) **Step 3:** Carboprost 250 μ g IM (if above fail) **Step 4:** Misoprostol 800 μ g sublingual/rectal (second-line PG) **Step 5:** Tranexamic acid 1 g IV (antifibrinolytic; add early in all cases of PPH)

MCQ Encounter — Q334 & Q561 & Q944 (PYQ)

Maximum cumulative dose of carboprost (15-methyl PGF₂ α) in PPH:

Answer: 2 mg (250 μ g IM every 15–90 min $\times$ maximum 8 doses = 2000 μ g = 2 mg)

The trap: Options offer 200 μ g, 2 mg, 20 mg, 2000 μ g. Candidates confuse 2000 μ g and 2 mg — they are the SAME. But "200 μ g" is wrong (one dose is 250 μ g) and "20 mg" is way too high.

MCQ Encounter — Q557 (PYQ)

Carboprost is contraindicated in:

Answer: B — Asthma — Carboprost (PGF₂ α analogue) causes bronchoconstriction. Absolutely contraindicated in asthma. Safe alternatives: oxytocin, ergometrine, misoprostol, tranexamic acid.

Active Management of Third Stage (AMTSL):

MCQ Encounter — Q606 (PYQ)

Active management of third stage includes all EXCEPT:

Answer: D — Ergometrine

Current WHO AMTSL components: 1. Oxytocin 10 IU IM (within 1 minute of delivery) 2. Uterine massage after placenta delivery 3. Controlled cord traction

Ergometrine has been removed from routine AMTSL in current WHO guidelines (though it can be used as a second uterotonic if oxytocin fails).

Section 9: Instrumental Delivery

Vacuum vs Forceps — When to Use Which

MCQ Encounter — Q363 (PYQ)

Which statement about vacuum vs forceps is correct?

Answer: A — Forceps preferred when fetal scalp must be protected (preterm <34 weeks)

Rules: - Vacuum contraindicated below **34 weeks** (risk of cephalhaematoma, intracranial haemorrhage in preterm neonates) - Vacuum contraindicated in **face presentation** - Vacuum contraindicated if **fetal bleeding disorder** - Forceps causes more **maternal perineal trauma** than vacuum - Vacuum requires more **maternal effort** (active pushing) than forceps

MCQ Encounter — Q364 (PYQ)

Absolute contraindication to outlet forceps:

Answer: B — Unengaged fetal head

For ANY instrumental delivery (forceps or vacuum): head must be **engaged** (at or below ischial spines = station 0 or positive). Applying forceps to an unengaged head is one of the most dangerous obstetric errors.

Forceps types (examiner likes to test recognition): - **Wrigley's:** Outlet forceps, low cavity, short blades - **Neville-Barnes:** Mid-cavity, long handles - **Kielland's:** Rotational forceps (for OP positions) — NO pelvic curve - **Piper's:** After-coming head in breech

Section 10: Shoulder Dystocia

The concept: After delivery of the fetal head, anterior shoulder is impacted behind the pubic symphysis. "Turtle sign" = head retracts back against perineum.

Management — HELPERR Mnemonic: - **H** = Call for Help - **E** = Evaluate for Episiotomy - **L** = Legs (McRoberts manoeuvre) - **P** = Suprapubic Pressure - **E** = Enter (internal rotational manoeuvres: Rubin II, Woods screw) - **R** = Remove the posterior arm - **R** = Roll the patient (Gaskin all-fours)

MCQ Encounter — Q358 (PYQ)

Turtle sign is noted, obstetrician flexes maternal thighs sharply onto abdomen. Manoeuvre:

Answer: B — McRoberts manoeuvre — *FIRST step in shoulder dystocia. Hyperflexion of thighs flattens lumbar lordosis, rotates symphysis superiorly, increases functional AP diameter. Success rate alone: ~40%.*

MCQ Encounter — Q359 (PYQ)

All other manoeuvres have failed. Obstetrician pushes head back into uterus and proceeds to caesarean. This is:

Answer: C — Zavanelli manoeuvre (cephalic replacement) — *Absolute LAST RESORT. Rotate head to OA, push back into vagina under tocolysis, perform emergency LSCS.*

Section 11: Cord Complications

Cord Prolapse

MCQ Encounter — Q360 (PYQ)

Cord visible at introitus, fetal bradycardia. Most important IMMEDIATE step:

Answer: C — Digital elevation of presenting part (*manually push presenting part upward to relieve cord compression*)

This buys time while preparing for emergency LSCS (the definitive treatment). The sequence: call for help → manually relieve compression → Trendelenburg/knee-chest position → fill bladder if possible (lifts head) → emergency LSCS.

Do NOT try to replace the cord into the uterus — *this is futile and dangerous.*

Vasa Praevia (Intrapartum Scenario)

This is covered in Chapter 3 but revisited here because it presents during labour. Bleeding at ROM + immediate fetal distress = vasa praevia = fetal exsanguination = emergency LSCS immediately.

Section 12: Uterine Inversion

Degrees: - 1st: Dimpling, fundus remains above cervix - 2nd (complete): Fundus passes through cervical os but not beyond introitus - **3rd (prolapsed):** Fundus protrudes beyond the introitus - 4th (total): Vaginal walls also inverted

MCQ Encounter — Q361 (PYQ)

Fundus not palpable abdominally, dark mass through introitus. Degree:

Answer: C — Third degree (prolapsed)

Management — critical sequence:

MCQ Encounter — Q362 (PYQ)

Manual reposition fails. Before Johnson's hydrostatic method, which drug to give?

Answer: C — Terbutaline (tocolytic)

The rule: To reposition the inverted uterus, you must RELAX it first. Tocolytics (terbutaline, nitroglycerin, MgSO₄) relax the cervical ring.

The trap: Students give oxytocin or ergometrine (uterotonics). This contracts the uterus further, trapping the inversion and making it impossible to replace. Uterotonics are given ONLY AFTER successful repositioning.

Section 13: Amniotic Fluid Embolism (AFE)

MCQ Encounter — Q367 (PYQ)

Multiparous woman: sudden respiratory distress + hypotension + cyanosis within minutes of delivery + DIC + cardiac arrest. Most likely diagnosis:

Answer: B — Amniotic fluid embolism

Classic triad: Hypoxia + Hypotension + Coagulopathy (DIC) in peripartum period. Mortality >80%.

Mechanism: amniotic fluid (containing fetal debris, prostaglandins, thromboplastin) enters maternal circulation via uterine/cervical veins → anaphylactoid reaction → cardiac collapse + DIC.

Section 14: Twin Pregnancy Complications

MCQ Encounter — Q371 (PYQ)

MCDA twins at 22 weeks: one sac polyhydramnios (DVP >8 cm), other sac oligohydramnios (DVP <2 cm), "stuck" twin. First-line treatment for stage II–III TTTS:

Answer: C — Fetoscopic laser photocoagulation of anastomotic vessels

TTTS (Twin-to-twin transfusion syndrome) = monochorionic placenta with arteriovenous anastomoses → donor twin (anaemic, IUGR, oligohydramnios) + recipient twin (polycythaemic, polyhydramnios).

Laser ablation addresses the root cause (pathological vascular anastomoses) and is superior to amnioreduction for stage II–IV TTTS before 26 weeks. Amnioreduction = palliation only.

MCQ Encounter — Q372 (PYQ)

What complication is UNIQUE to monoamniotic (MA) twins?

Answer: B — Cord entanglement — Monoamniotic twins share one sac → cords inevitably entangle → risk of cord compression/knotting → fetal death. TTTS requires monochorionic placenta but can occur in MCDA; cord entanglement is MA-only.

MCQ Encounter — Q338 (PYQ)

36-week pregnant, one previous delivery of twins. Gravida-parity:

Answer: A — G2P1 — Gravida = number of pregnancies. Parity = number of deliveries ≥20 weeks, regardless of number of fetuses. Twin delivery = ONE parity event.

Section 15: Perineal Tears — Classification and Repair

Degree	Structures Involved
1st	Skin only
2nd	Skin + perineal muscles (not sphincter)
3rd	Perineum + external anal sphincter (EAS)
— 3a	<50% EAS thickness
— 3b	>50% EAS thickness
— 3c	Complete EAS + internal anal sphincter (IAS)
4th	Perineum + EAS + IAS + anal/rectal mucosa

MCQ Encounter — Q772 (PYQ)

Tear involves perineum and external anal sphincter, intact mucosa. Grade:

Answer: C — Third degree (mucosa intact = not 4th degree)

Episiotomy comparison:

MCQ Encounter — Q274 (PYQ)

Advantage of mediolateral over midline episiotomy:

Answer: D — Less extension to 3rd/4th degree tears

Feature	Midline	Mediolateral
Blood loss	Less	More
Pain	Less	More
Healing	Faster	Slower
Extension to 3rd/4th degree	Higher risk	Lower risk
Common in	USA	UK/India

Delayed perineal tear repair:**MCQ Encounter — Q617 & Q964 (PYQ)**

Complete perineal tear, home delivery 2 weeks ago. Management:

Answer: D — Repair after 3 months — Primary repair window is within 24 hours. Presenting at 2 weeks = oedema, infection, fibrosis are present. Wait 3 months for infection/oedema to fully resolve, then do secondary repair.

Section 16: Johnson's Formula and Fetal Weight Estimation**MCQ Encounter — Q301 (PYQ)**

SFH = 40 cm, vertex at -1 station. Estimated fetal weight by Johnson's formula:

Formula: $EFW (g) = (SFH \text{ in cm} - n) \times 155$ - $n = 12$ if vertex is ABOVE the ischial spines - $n = 11$ if vertex is AT or BELOW the ischial spines

Calculation: Station -1 = above ischial spines $\rightarrow n = 12$ $(40 - 12) \times 155 = 28 \times 155 = 4,340 \text{ g} \approx 4.3 \text{ kg}$

Answer: D — ~4.3 kg

Section 17: Sheehan's Syndrome — A Labour Complication Consequence

MCQ Encounter — Q412 & Q866 (PYQ)

35-year-old, amenorrhoea + failure of lactation after second delivery + empty sella. Diagnosis:

Answer: C — Sheehan's syndrome

Mechanism: Massive postpartum haemorrhage → pituitary ischaemia (the pituitary is enlarged and hypervascular in pregnancy → sensitive to hypoperfusion) → anterior pituitary necrosis → panhypopituitarism.

Classic triad: Agalactia (failure of lactation — first sign, within days), secondary amenorrhoea, empty sella on imaging.

Why students confuse with pituitary adenoma: Adenoma = prolactinoma typically = galactorrhoea + amenorrhoea. Sheehan's = agalactia (no milk) + amenorrhoea. Opposite milk finding.

Bias Removal Section

Top traps in this chapter:

1. **Carboprost dose:** 2 mg = 2000 µg. Both are correct representations of the same maximum dose. "200 µg" (too low) and "20 mg" (way too high) are the wrong options.
2. **Ergometrine timing:** "Given before delivery of baby" = wrong and dangerous. Only after anterior shoulder delivery or in 3rd stage.
3. **Zavanelli vs McRoberts:** McRoberts = FIRST step (thigh flexion); Zavanelli = LAST resort (push head back).
4. **Active management of third stage:** Current WHO guidelines no longer include ergometrine as a routine component. Oxytocin + cord traction + uterine massage (after placenta delivery) = the three components.
5. **Mento-posterior face = always LSCS:** Students think "face presentations can deliver vaginally" → TRUE for mento-anterior. NOT for mento-posterior. Never confuse them.
6. **Partogram:** Students say "2 hours" for alert-action line gap. The answer is ALWAYS 4 hours for WHO partogram.
7. **Sheehan's vs prolactinoma:** Sheehan's = no milk (agalactia). Prolactinoma = too much milk (galactorrhoea). Key differentiator.

Variations and Twists

How examiners vary these questions:

- "A multiparous woman's placenta does not deliver for 40 minutes" → Manual removal under anaesthesia (Q366). The trap answer is "wait longer" or "give more oxytocin."

- "Bakri balloon inserted after LSCS for praevia" — examiner asks mechanism. Answer (Q368): direct intrauterine tamponade pressure exceeding arterial perfusion pressure. Not "stimulates contraction" and not "compresses uterine arteries."
- Sheehan's presented as a question about a patient who had severe PPH from placenta praevia: "Most likely outcome post delivery?" Answer = absence of menstrual cycle (Q866). The trap is "galactorrhoea" — but Sheehan's causes agalactia.
- Forceps type image questions (Q706): Piper's forceps = long perineal curve, used for after-coming head in breech. Kielland's = no pelvic curve, used for rotational delivery.

Memory Engineering

Stages of labour — "FADS": - First stage = Fully dilate (latent + active) - Active phase starts at **6 cm** (ACOG 2014) - Delivery = Second stage - Separation of placenta = Third stage

PPH 4 Ts — "Tone Trauma Tissue Thrombin": - Most common = **Tone** (atony, 70–80%) - Secondary PPH = **Tissue** (retained) - Coagulopathy = **Thrombin**

Uterotonic hierarchy — "O-E-C-M-T": - Oxytocin first - Ergometrine second - Carboprost third (not in asthma) - Misoprostol fourth - Tranexamic acid (add throughout)

Shoulder dystocia — HELPERR: Help, Episiotomy, Legs (McRoberts), Pressure (suprapubic), Enter (Rubin II/ Woods), Remove posterior arm, Roll (Gaskin)

Presenting diameters — "The best position is minimum deflexion": - Vertex well-flexed = suboccipito-bregmatic = **9.5 cm** (smallest = best) - Brow = mento-vertical = **13.5 cm** (largest = worst) - Face mento-anterior = submento-bregmatic = **9.5 cm** (can deliver)

Breech manoeuvres — "Arms = Lovset, Head = MSV": - Extended Arms → Lovset (**A** for Arms) - After-coming Head → Mauriceau-Smellie-Veit (**H** for Head)

Episiotomy — "Midline = More extension; Mediolateral = More blood": Midline easier to repair but risky for sphincter. Mediolateral safer for sphincter but more blood/pain.

Ultra High-Yield Revision Layer

What the Examiner REALLY Wants

The examiner wants you to sequence manoeuvres correctly (McRoberts before Zavanelli), dose drugs correctly (carboprost = 2 mg max), and diagnose by landmarks (mento-vertical diameter = brow presentation). These are precision questions, not recall questions.

20-Second Revision Bullets

- Latent phase ends at 6 cm (ACOG 2014); active phase minimum rate = 0.5 cm/hr
- WHO partogram alert-to-action line distance = **4 hours** (tested every year)
- Second stage = full dilatation to delivery; duration in multipara = ~20 minutes
- Suboccipito-bregmatic diameter = 9.5 cm = well-flexed vertex (smallest, best)
- Mento-vertical diameter = 13.5 cm = brow presentation (largest, worst)

- Brow presentation at term → almost always LSCS
- Face presentation: mento-anterior = can deliver vaginally; mento-posterior = always LSCS
- Frank breech = most common type (65–70%)
- Lovset manoeuvre = extended arms in breech; MSV = after-coming head
- Piper's forceps = after-coming head in breech
- Bishop score ≥ 8 = favourable cervix
- PGF2 α (carboprost) NOT used for induction; used for PPH only
- Misoprostol = PGE1 analogue; NOT for menorrhagia (pro-PG; menorrhagia needs anti-PG)
- Ergometrine = contraindicated before second stage
- PPH first-line drug = oxytocin; carboprost max = **2 mg** (8 doses $\times$ 250 μ g); carboprost contraindicated in asthma
- Active management of third stage = oxytocin + cord traction + uterine massage (ergometrine NOT routine in current WHO)
- Primary PPH = within 24 hours; most common cause = **uterine atony** (Tone)
- Secondary PPH = 24h to 12 weeks; most common cause = **retained placenta/tissue** (Tissue)
- Hypertension does NOT increase PPH risk (not a uterine distension factor)
- Shoulder dystocia: first step = McRoberts; last resort = Zavanelli (cephalic replacement)
- Cord prolapse: first step = digitally elevate presenting part; definitive = emergency LSCS
- Uterine inversion: TOCOLYTIC (terbutaline) BEFORE repositioning; uterotonic only AFTER replacement
- 3rd degree tear = EAS involved, mucosa intact; 4th degree = mucosa involved
- Delayed perineal tear at 2 weeks → repair after 3 months
- Sheehan's syndrome = massive PPH → agalactia + amenorrhoea + empty sella
- Johnson's formula: EFW = (SFH – 12) $\times$ 155 g (if above spines)
- Frank breech = G2P1 (parity counts deliveries, not fetuses)
- AMTSL does NOT include early cord clamping or fundal pressure; ergometrine removed from routine WHO protocol

Top 5 Examiner Traps in Chapter 4

1. "Partogram alert-to-action line gap" → **4 hours** (not 2, not 1, not 5)
 2. "Carboprost in a patient with asthma" → **contraindicated** (causes bronchospasm); choose misoprostol/oxytocin instead
 3. "Uterine inversion + manual replacement fails — which drug?" → **tocolytic** (terbutaline), NOT oxytocin/ergometrine (those make it worse)
 4. "McRoberts vs Zavanelli" → McRoberts = FIRST (described as 'flexing thighs onto abdomen'); Zavanelli = LAST (described as 'pushing head back')
 5. "Mento-posterior face presentation — can it deliver vaginally?" → **No, always caesarean** (mento-anterior = yes; mento-posterior = never)
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PART II — GYNAECOLOGY

Chapter 5 — Contraception

"The mark of a good exam candidate is knowing not just which method works, but WHY it works, when it fails, and what to do in the edge cases."

Big Picture First

Exam Weightage

Contraception consistently delivers 4–7 marks per NEET-PG paper. The questions cluster around four poles:

1. **Failure rates / Pearl Index** — pure number recall, but with sneaky option ordering
2. **Mechanism of action** — OCPs, emergency contraception, IUCDs: subtle "EXCEPT" traps
3. **Contraindications** — absolute vs. relative; situation-based (cardiac disease, lactation, postpartum)
4. **Surgical sterilisation** — named techniques for tubectomy/vasectomy, timing, complications

The Pearl Index: Get This Right First

The Pearl Index (PI) = number of pregnancies per 100 woman-years of use. Higher number = WORSE method.

The Golden Rule: When a question asks "most effective" or "lowest failure rate," you want the **LOWEST** Pearl Index.

The Reversibility Spectrum

PERMANENT	←	—————	→	TEMPORARY		
Tubectomy	Vasectomy	IUCD/Implant	DMPA	OCP/POP	Barrier	LAM
(irreversible)	(irreversible)	(LARC)		(6-9mo delay)	(immediate)	

Question Pattern Map

Recurring PYQ themes from 2015–2024:

Theme	Frequency
Emergency contraception — mechanism, drug of choice, timing	Very High
Progestin-only pill — mechanism, NOT a mechanism	High
Ideal contraceptive in lactation / cardiac disease	High
IUCD — absolute contraindications, side effects, generations	High
Vasectomy — false statement about sterility	High
Pearl Index / lowest failure rate	Medium
Tubectomy techniques (Pomeroy, Kroener, Uchida, Irving)	Medium
LAM criteria	Medium

Core Concept Deep Dive

1. Combined Oral Contraceptive Pill (OCP)

Mechanism (and the "EXCEPT" traps)

The combined OCP works through **four mechanisms**. Examiners love asking which is NOT a mechanism:

1. **Inhibition of LH surge** → **anovulation** (primary mechanism)
2. **Thickening of cervical mucus** → sperm cannot penetrate
3. **Endometrial atrophy** → impaired implantation
4. **Altered tubal motility**

PYQ 1 (Q540)

All of the following are mechanisms of action of combined OCPs EXCEPT: - A. Inhibition of ovulation - B. Spermicidal effect - C. Interference with implantation - D. Interference with placental function

Answer: D. Interference with placental function

The OCP has no effect once placentation has occurred. Option B (spermicidal effect) is technically not a classic mechanism either, but option D is definitively wrong. Options A, B, and C are accepted as partial mechanisms. Once the embryo has implanted and the placenta is forming, the OCP is completely ineffective — this is the conceptual boundary between contraception and abortion.

Non-Contraceptive Benefits of OCP

Examiners test this indirectly through "OCP is NOT indicated for ____" questions. OCPs are beneficial in: - Dysmenorrhoea (reduces PGF2α) - Endometriosis (decidualisation → atrophy) - Menorrhagia (thins endometrium) - PCOS (suppresses androgens) - PMS - Reduces risk of ovarian and endometrial cancer

OCP Failure Rate

PYQ 2 (Q313)

Which contraceptive has the LEAST pregnancy failure rate? - A. OCP - B. Cu-IUCD - C. Diaphragm - D. Condom

Answer: A. OCP (per source key — among these four options)

However, in reality, **subdermal implants** have a lower failure rate than OCPs (Pearl Index 0.05 vs. 0.3). When implants are in the options, implants win. When the question restricts to these four, OCP wins.

PYQ 3 (Q456/Q634)

A young woman wants the most effective reversible long-acting contraception with the lowest Pearl Index. The method with the LOWEST failure rate is: - A. Subdermal implants (etonogestrel) - B. Combined oral contraceptive pill - C. Condom - D. Copper IUD

Answer: A. Subdermal implants

Subdermal etonogestrel implants (Implanon/Nexplanon): Pearl Index 0.05 per 100 woman-years — the lowest of ALL reversible methods. Effective for 3 years, no user compliance needed, immediately reversible on removal.

OCP in Lactation — A Crucial Distinction

PYQ 4 (Q458)

A woman who is lactating comes for contraception 6 weeks after delivery. Which contraceptive is NOT advisable? - A. Combined oral contraceptive pill - B. Progestogen-only pill (mini-pill) - C. Copper IUD - D. DMPA (depot medroxyprogesterone acetate)

Answer: A. Combined oral contraceptive pill

The oestrogen component suppresses prolactin → reduces milk volume. This is a direct pharmacological effect, not just theoretical. All progestogen-only methods (POP, DMPA, LNG-IUS, implants) and IUDs are safe during breastfeeding.

2. Progestogen-Only Pill (Mini-Pill, POP)

Mechanisms

PYQ 5 (Q904)

Which of the following is NOT a mechanism of action of progesterone-only pills? - A. Thicken the cervical mucus - B. Prevent ovulation - C. Kill the sperms in the cervical canal - D. Inhibit penetration of sperm

Answer: C. Kill the sperms in the cervical canal

POPs are NOT spermicidal. Their three mechanisms are: (1) cervical mucus thickening, (2) suppression of ovulation (inconsistent — only ~50% with traditional POP; higher with desogestrel-POP), (3) endometrial atrophy reducing implantation.

PYQ 6 (Q918) (from 2018)

What is the effect of progesterone-only pills? - A. Completely suppresses ovulation - B. Thin lining of uterus - C. Thick cervical mucus - D. All of the above

Answer: D. All of the above

Note the apparent contradiction with the previous question — the 2018 paper accepted "all of the above." The nuance: traditional POPs suppress ovulation in ~50% of cycles; desogestrel-containing POPs (Cerazette) suppress in ~97%. For exam purposes, "all of the above" when all three are correct mechanisms is also valid.

POP Side Effects — What's NOT a Side Effect

PYQ 7 (Q757)

Which is not a side effect of POP (Progestin-only pill)? - A. Ovarian cysts - B. Venous thromboembolism - C. Increased risk of diabetes mellitus - D. Ectopic pregnancy

Answer: B. Venous thromboembolism

VTE is the side effect of OESTROGEN, not progestogen. POPs are the safe choice in women with: - Prior VTE / thrombophilia - Lactation - Cardiovascular disease risk factors - Smokers over 35 years

POPs can cause: ovarian cysts (follicles accumulate without rupturing), functional ectopic pregnancy risk (if pill fails, altered tubal motility increases ectopic risk), irregular bleeding, and mild carbohydrate intolerance.

Missed POP Dose

PYQ 8 (Q620)

A 26-year-old on a progestogen-only pill usually takes it at 10 am but took it yesterday at 5 pm. Advice: - A. No need to worry; continue same contraceptive - B. Use additional contraception for the next 48 hours - C. Change over to combined oral contraceptive pill - D. Take double dose next morning

Answer: B. Use additional contraception for the next 48 hours

The window of tolerance: Traditional POP (norethisterone) = 3-hour window. Desogestrel POP = 12-hour window. If taken late beyond these windows, the cervical mucus effect is lost — additional barrier contraception for 48 hours is needed. Never double dose.

3. Emergency Contraception

This is a perennial high-yield area with multiple PYQs testing mechanism and drug choice.

Drug of Choice

PYQ 9 (Q314/Q469)

Progestogen of choice for emergency contraception? - A. Norethisterone - B. Medroxyprogesterone - C. Oxytocin - D. Levonorgestrel

Answer: D. Levonorgestrel

LNG 1.5 mg as a single dose (or 0.75 mg $\times$ 2 doses 12 hours apart) within 72 hours. Effective up to 120 hours (5 days), but efficacy decreases significantly after 72 hours.

Ulipristal Acetate — The Better Option

PYQ 10 (Q315)

Dose of ulipristal acetate for emergency contraception? - A. 300 mg - B. 30 mg - C. 300 μ g - D. 30 μ g

Answer: B. 30 mg

Ulipristal acetate (UPA) is a selective progesterone receptor modulator (SPRM). Single 30 mg oral dose, effective up to 120 hours. It is MORE effective than LNG beyond 72 hours because it can delay ovulation even after the LH surge has begun (LNG cannot).

Mechanism of Emergency Contraception — The "NOT a Mechanism" Trap

PYQ 11 (Q7/Q505)

Emergency contraception prevents pregnancy by all mechanisms EXCEPT: - A. Delaying or inhibiting ovulation - B. Inhibiting fertilisation - C. Preventing implantation of the fertilised egg - D. Interrupting an early established pregnancy

Answer: D. Interrupting an early established pregnancy

This is a conceptual boundary question. Emergency contraception is CONTRACEPTION — it prevents pregnancy. Once implantation is complete and pregnancy is established, EC has zero efficacy. Any intervention at that stage constitutes termination. This question has appeared multiple times (Q7, Q505) with identical answer.

Emergency Contraception Comparison

Feature	LNG (Levonorgestrel)	UPA (Ulipristal)	Cu-T 380A
Dose	1.5 mg single	30 mg single	Device insertion
Window	Up to 72h (up to 120h)	Up to 120h	Up to 5 days (120h)
Mechanism	Delay ovulation; prevent fertilisation	Delay ovulation (even post-LH surge)	Spermicidal + anti-implantation
Efficacy	~85%	~98%	>99%
Contraindication	Pregnancy	UPA: avoid if on regular hormonal contraception	Active PID, STI risk
Most effective option	—	—	YES (most effective EC)

Key exam trap: Cu-T 380A is the most effective single EC method (>99%), can be left in for ongoing contraception for up to 10 years, AND is the only EC method that also provides future contraception.

4. IUCD Types — Exact Specifications Examiners Ask

Generations of Copper IUCDs

Generation	Device	Surface area
1st	Lippes Loop	None (inert)
2nd	Cu-T 200, Multiload 250	200–250 mm ²
3rd	Cu-T 380A, Multiload 375	375–380 mm ²
Frameless	GyneFix	Frameless, anchored

PYQ 12 (Q798)

Which of the following is false about intra-uterine devices (IUDs)? - A. Multiload Cu-375 is a third generation intra-uterine device - B. Copper devices are effective as post-coital contraceptives - C. LNG-20 (Mirena) has an effective life of 5 years - D. Pregnancy rates of Lippes Loop and T Cu-200 are similar

Answer: A. Multiload Cu-375 is a third generation intra-uterine device

Multiload Cu-375 is SECOND generation. Third-generation devices include Cu-T 380A and the frameless GyneFix. Option C (LNG-IUS/Mirena effective for 5 years) is technically true for original labelling, though regulatory approvals now extend to 8 years in some countries — the answer as set remains A.

Hormonal IUDs — Precise Data

Device	Hormone	Release rate	Duration	Primary use
Progestasert	Progesterone	65 µg/day	1 year	Historical
LNG-IUS (Mirena)	Levonorgestrel	20 µg/day	5–8 years	Contraception + menorrhagia
Kyleena	LNG	17.5 µg/day	5 years	Smaller device, nullipara
Jaydess/Skyla	LNG	14 µg/day	3 years	Nullipara

PYQ 13 (Q6)

The quantity of hormone released by the Progestasert IUCD per day is: - A. 25 µg - B. 45 µg - C. 65 µg - D. 85 µg

Answer: C. 65 µg

Progestasert contains 38 mg progesterone in a silicone oil reservoir. Released at 65 µg/day. Valid for only 1 year. Compare with Mirena (20 µg LNG/day, 5 years) — the numbers are frequently confused in MCQs. Mechanism: thickens cervical mucus, suppresses endometrium.

IUCD Mechanism — Copper Devices

The copper IUD works by: 1. **Spermicidal effect** — copper ions are directly toxic to sperm 2. **Inhibition of fertilisation** — copper alters tubal fluid 3. **Impaired implantation** — endometrial inflammatory reaction 4. **Increased tubal peristalsis** — faster egg transit

Copper IUDs do NOT suppress ovulation (unlike LNG-IUS which partially does).

IUCD Side Effects — Most Common

PYQ 14 (Q528)

The most common side effect of IUCD insertion is: - A. Bleeding (menorrhagia) - B. Pain/dysmenorrhoea - C. Pelvic infection - D. Ectopic pregnancy

Answer: A. Bleeding (menorrhagia)

Copper IUCDs increase menstrual blood loss by **30–50%** due to prostaglandin-mediated endometrial reaction. This is the most common reason for discontinuation. Compare: LNG-IUS (Mirena) reduces blood loss by 90% — it is used to TREAT menorrhagia.

IUCD Absolute Contraindications

PYQ 15 (Q475/Q750)

Absolute contraindications to IUCD are all EXCEPT: - A. Pregnancy - B. Undiagnosed vaginal bleeding - C. Pelvic inflammatory disease - D. Uterine malformation

Answer: D. Uterine malformation

Uterine malformation (distorted cavity) is only a RELATIVE contraindication. Absolute contraindications include: - Current pregnancy - Active PID / current STI - Undiagnosed abnormal uterine bleeding - Gestational trophoblastic disease (Q969) - Known/suspected genital malignancy - Wilsons disease (copper IUD only — contraindicated due to copper accumulation)

IUCD and Ectopic Pregnancy — A Conceptual Trap**PYQ 16 (Q682)**

A P2L2 patient using an IUCD presents to the emergency with vaginal bleeding and recurrent syncope. Most likely diagnosis? - A. Ectopic pregnancy - B. Recurrent abortion - C. Septicaemia - D. Coagulation disorder

Answer: A. Ectopic pregnancy

The IUCD prevents INTRAUTERINE pregnancy very effectively, but provides LESS protection against ectopic pregnancy. If a woman conceives with an IUCD in situ, the probability that pregnancy is ectopic is higher than in the general population. The IUCD does NOT increase the absolute risk of ectopic pregnancy in users compared with no contraception — but if breakthrough pregnancy occurs, the relative chance of it being ectopic increases.

LNG-IUS — What It Treats vs. What It Doesn't**PYQ 17 (Q422/Q594)**

Levonorgestrel-releasing IUD (LNG-IUS) is helpful in all of the following EXCEPT: - A. Menorrhagia - B. Dysmenorrhoea - C. Endometriosis - D. Pelvic inflammatory disease

Answer: D. Pelvic inflammatory disease

LNG-IUS is therapeutic for menorrhagia (reduces blood loss ~90%), primary and secondary dysmenorrhoea, and endometriosis (induces endometrial decidualisation/atrophy). It is CONTRAINDICATED in active PID — it is not protective against ascending infection and it should not be inserted in the setting of infection.

IUCD Displacement — Management**PYQ 18 (Q413/Q1063)**

A woman with a copper IUD inserted 2 years ago has absent threads on speculum examination; pelvic ultrasound shows the device partially displaced into the abdominal cavity. Management: - A. Hysteroscopy - B. No removal needed - C. IUCD hook retrieval - D. Laparoscopy

Answer: D. Laparoscopy

A displaced IUD in the abdominal/peritoneal cavity MUST be removed — it causes bowel adhesions, visceral perforation, and injury if left. Laparoscopic removal is standard. Hysteroscopy is for intracavitary devices with missing threads where the device is still uterine.

IUCD in Pregnancy — Visible Threads

PYQ 19 (Q1025)

A 28-year-old woman with a 10-week intrauterine gestation presents to the OPD with an IUCD in place. The IUCD thread is clearly visible. She expresses her desire to continue the pregnancy. What is the next best step? - A. Leave IUCD inside and continue pregnancy with prophylactic antibiotics - B. Leave IUCD inside and continue pregnancy - C. Remove IUCD immediately and continue with pregnancy - D. Termination of pregnancy due to risk of infection

Answer: C. Remove IUCD immediately and continue with pregnancy

If the strings are visible and the woman wants to continue the pregnancy, remove the IUCD to prevent septic abortion (risk ~50% if left), preterm labour, and PROM. If strings are not visible, do NOT remove blindly — proceed with ultrasound-guided management or leave it and counsel about risks.

5. Barrier Methods

Female Condom

PYQ 20 (Q1 — Q5)

A contraceptive image shows a polyurethane sheath with two flexible rings (one inserted, one external). Identify: - A. Male condom - B. Female condom - C. Diaphragm - D. Cervical cap

Answer: B. Female condom

Female condom: polyurethane sheath inserted into the vagina; the outer ring remains external. Unlike the male condom (latex), it is polyurethane. It provides STI protection (including HIV). Pearl Index ~5% with perfect use, ~21% typical use.

Contraceptive Sponge

PYQ 21 (Q604)

"Today" contraceptive sponge contains: - A. Prostaglandin F2α - B. Norethisterone - C. Nonoxynol-9 - D. Copper-releasing mesh

Answer: C. Nonoxynol-9

The "Today" sponge is a polyurethane disc impregnated with Nonoxynol-9 (a surfactant-spermicide). It provides both physical barrier and spermicidal action. Nonoxynol-9 also appears in spermicidal gels/creams. It is NOT recommended alone (high failure rate) but acceptable as adjunct.

Ideal Contraceptive in Cardiac Disease

PYQ 22 (Q529)

Ideal contraceptive for a patient with cardiac disease is: - A. Cu-IUCD - B. Depot medroxyprogesterone acetate (DMPA) - C. Barrier method (diaphragm/condom) - D. Combined oral contraceptive pill

Answer: C. Barrier method (diaphragm/condom)

Combined OCPs → oestrogen → VTE risk (absolutely contraindicated in Eisenmenger's, severe MS, etc.) DMPA → fluid retention → can worsen cardiac failure Cu-IUCD → vasovagal at insertion → haemodynamic risk, prostaglandin-mediated effects Barrier methods → no systemic effects, no haemodynamic consequences

6. DMPA (Depot Medroxyprogesterone Acetate)

PYQ 23 (Q316)

DMPA is given once every: - A. 3 months - B. 6 months - C. 9 months - D. 45 days

Answer: A. 3 months (12 weeks)

150 mg IM every 12 weeks. Key facts: - Delayed return to fertility: **6–12 months** after last injection - Causes amenorrhoea (50% after 1 year) - Safe in lactation (progestin only) - NOT ideal for women desiring future fertility soon (due to delay in return)

7. Sterilisation

Tubectomy — Named Techniques (High Yield)

Technique	Method
Pomeroy	Loop of tube ligated, segment excised — most common laparoscopic method
Modified Pomeroy	Same but with delayed absorbable suture
Uchida	Subserosal saline injection, tube elevated and excised
Irving	Both ends buried into myometrium and broad ligament — lowest failure rate
Madlener	Knuckle of tube crushed and ligated — NOT excised (higher failure rate, rarely used)
Kroener (Fimbriectomy)	Fimbrial end removed
Filshie clips	Titanium clips — increasingly used, most reversible

PYQ 24 (Q889)

Fimbriectomy procedure is known as: - A. Uchida method - B. Irving method - C. Madlener technique - D. Kroener method

Answer: D. Kroener method

Kroener = Fimbriectomy. Remember: "Kroener cuts the corner (fimbrial end)." The other three techniques all involve a mid-tubal approach. Irving buries both ends and has the **LOWEST** failure rate.

Site of Tubal Ligation

PYQ 25 (Q1019)

A woman is undergoing elective tubal ligation for permanent contraception. Which is the most common site selected for ligation, and what is the reason? - A. Ampulla – As it has a uniform thickness of muscle - B. Ampulla – As it is the widest part - C. Ampulla–Isthmus junction – As it is the site of fertilization - D. Isthmus – As it is the narrowest part

Answer: D. Isthmus – As it is the narrowest part

The isthmus is the narrowest segment, with the most uniform muscle wall thickness and the least blood supply — easiest to ligate, divide, and achieve effective occlusion. Fertilisation actually occurs at the ampullary-isthmic junction, but this is not the reason for choosing the isthmus for ligation.

Vasectomy — The "Sterility Is Immediate" Trap

PYQ 26 (Q459/Q638)

All of the following about non-scalpel vasectomy are correct EXCEPT: - A. Haematoma occurs in up to 5% of subjects - B. Sexual function following healing is rarely affected - C. Sterility is immediate after vasectomy - D. Recanalization is possible

Answer: C. Sterility is immediate after vasectomy

This has appeared TWICE (Q459, Q638) — it is a classic NEET-PG trap. After vasectomy, residual sperm remain in the vas deferens distal to the ligation. The patient remains fertile until these sperm are cleared (typically after 20 ejaculations or 12 weeks). Backup contraception is mandatory until azoospermia is confirmed on semen analysis.

8. Lactational Amenorrhoea Method (LAM)

Mechanism

PYQ 27 (Q697)

Lactational amenorrhoea is caused by: - A. Prolactin-induced inhibition of GnRH - B. Prolactin-induced inhibition of FSH - C. Oxytocin-induced inhibition of GnRH - D. Oxytocin-induced inhibition of FSH

Answer: A. Prolactin-induced inhibition of GnRH

The suckling reflex → pituitary prolactin surge → prolactin directly suppresses hypothalamic GnRH pulsatility → reduced FSH and LH → follicular development fails → anovulation and amenorrhoea. Note: Oxytocin is released by suckling too, but its role is milk let-down, NOT contraception.

LAM Criteria (ALL THREE Must Be Present)

The Triple Lock:

Criterion	Specification
Feeding	Exclusive breastfeeding (no supplemental feeds, bottles, or water)
Duration	Infant < 6 months old
Menstrual status	Postpartum amenorrhoea (no period since delivery)

If any ONE criterion fails, LAM is unreliable and another method should be added.

Pearl Index of LAM: 0.5–2% when all three criteria are met. Rises to 5–10% if criteria are not all met.

PYQ 28 (Q418/Q587)

Ideal contraceptive for a woman who is exclusively breastfeeding is: - A. Barrier method alone - B. Combined oral contraceptive pill - C. Lactational amenorrhoea alone - D. Progestogen-only pill

Answer: D. Progestogen-only pill

LAM alone fails when: infant >6 months, supplemental feeding starts, or periods return. At that point, the ideal additional contraceptive is the progestogen-only pill because it does not suppress prolactin or reduce milk supply.

Bias Removal Section

Common Misconceptions That Cost Marks

- "Higher Pearl Index = Better method"** WRONG. Pearl Index = pregnancies per 100 woman-years. Lower = better. If you see "Pearl Index 0.05 vs 9," the 0.05 method (implant) is vastly superior to the 9 method (condom).
- "Emergency contraception terminates early pregnancy"** WRONG. EC prevents pregnancy. Once the fertilised egg has implanted, EC has zero effect. This is not a legal or ethical nuance — it is a pharmacological fact. Q7 and Q505 both test this.
- "Vasectomy gives immediate sterility"** WRONG — Q459 and Q638 specifically test this. Backup contraception is needed for 12 weeks / 20 ejaculations.
- "Combined OCP is safe in breastfeeding"** WRONG — oestrogen reduces milk supply. Progestogen-only methods are safe.
- "Uterine malformation is an absolute contraindication to IUCD"** WRONG — it is RELATIVE, not absolute.
- "Multiload Cu-375 is a third-generation IUD"** WRONG — it is second-generation. Third = Cu-T 380A, GyneFix.

Variations and Twists

Scenario: Postpartum Contraception

Situation	Best method
Exclusively breastfeeding, < 6 months	LAM + POP (belt-and-suspenders)
Not breastfeeding	Any method from 3 weeks postpartum
Lactating at 6 months	Add POP/Cu-IUCD
Wants permanent contraception	Tubectomy (day 2–3 postpartum or interval)
Wants reversible long-term	Cu-IUCD (can insert immediately postpartum)
Cardiac disease, lactating	Barrier + POP

Scenario: Missed Periods While on OCP

If a woman on combined OCP misses her period, first step = urine pregnancy test. Do NOT assume OCP side effect without ruling out pregnancy.

Scenario: IUCD and Pregnancy (2022 actual PYQ Q988)

A 24-year-old lactating female with an 18-month-old child with irregular, heavy bleeding seeking contraceptive advice:

Answer: "Mala" — In the Indian National Family Planning Programme, "Mala-N" and "Mala-D" are the official brand names of combined OCPs distributed through government health system. With heavy irregular bleeding, a combined OCP helps regulate cycles and reduce blood loss. At 18 months, LAM criteria are no longer met (infant too old).

Memory Engineering

Pearl Index Table (Memorise This Sequence)

Method	Pearl Index (typical use)
Implants	0.05 ← LOWEST
LNG-IUS (Mirena)	0.1–0.2
Male sterilisation (vasectomy)	0.1
Female sterilisation (tubectomy)	0.5
Combined OCP	0.3–9
DMPA (injectable)	0.3–3
Copper IUD	0.6–0.8
Progestogen-only pill	0.3–9
Male condom	2–15
Female condom	5–21
Diaphragm + spermicide	6–12
LAM (all 3 criteria met)	0.5–2
Spermicide alone	18–29
No contraception	85

Memory hook for the bottom of the table: "SDNF" — Spermicide, Diaphragm, No method, Female condom — Pearl Indices go from bad to catastrophic.

LAM Criteria Anchor: "EAM"

- **E**xclusive breastfeeding
- **A**menorrhoea (no period)
- **M**onths < 6 (infant age) All three = effective. Miss one = add another method.

Emergency Contraception Window Mnemonic: "3-5-5"

- LNG: effective to **3** days (72h), works to **5** days with reduced efficacy
- UPA: works for full **5** days (120h), more effective than LNG
- Cu-T: works for **5** days, **most** effective option overall

Ultra High-Yield Revision Layer

Emergency Contraception Decision Table

Presentation	Preferred Choice
Within 72h, routine	LNG 1.5 mg single dose
72–120h post-coitus	UPA 30 mg (more effective)
Any time within 120h, all circumstances	Cu-T 380A (+ future contraception bonus)
On enzyme-inducing drugs (rifampicin, phenytoin)	Cu-T 380A (hormonal EC reduced efficacy)
BMI > 26, within 72h	Cu-T 380A preferred (LNG efficacy reduced)
Rape / sexual assault	Cu-T 380A or UPA (+ STI prophylaxis)

Top 5 Traps in Contraception MCQs

1. "Sterility immediate after vasectomy" = **FALSE** — the most-tested vasectomy fact
2. **Pearl Index confusion** — higher number = worse; subdermal implant = lowest
3. **Combined OCP in lactation** — contraindicated; not progestogen-only
4. **Multiload Cu-375 = 2nd generation** (not 3rd)
5. **IUCD and ectopic** — IUCD prevents IUP, if breakthrough pregnancy occurs, higher relative chance it is ectopic

All Chapter 5 PYQs Integrated

Q-Bank Practice Block

Q1 — Female condom identification

A contraceptive image shows a polyurethane sheath with two flexible rings (one inserted, one external). Identify:

- A. Male condom
- B. Female condom
- C. Diaphragm
- D. Cervical cap

► Answer & Explanation:

****B. Female condom**** — Polyurethane sheath; inserted into vagina; provides STI protection. Diaphragm covers cervix; cervical cap is smaller.

Q2 — Progestasert IUCD: daily hormone release

The quantity of hormone released by the Progestasert IUCD per day is:

- A. 25 µg

- B. 45 µg
- C. 65 µg
- D. 85 µg

► **Answer & Explanation:**

****C. 65 µg**** — Progestasert contains 38 mg progesterone in a silicone oil reservoir; releases 65 µg/day. Compare LNG-IUS/Mirena: 20 µg/day, valid 5 years.

Q3 — Emergency contraception: NOT a mechanism

Emergency contraception prevents pregnancy by all mechanisms EXCEPT:

- A. Delaying or inhibiting ovulation
- B. Inhibiting fertilisation
- C. Preventing implantation of the fertilised egg
- D. Interrupting an early established pregnancy

► **Answer & Explanation:**

****D. Interrupting an early established pregnancy**** — Once implantation is complete, EC is ineffective and any intervention constitutes termination, not contraception.

Q4 — Lowest pregnancy failure rate

Which contraceptive has the LEAST pregnancy failure rate?

- A. OCP
- B. Cu-IUCD
- C. Diaphragm
- D. Condom

► **Answer & Explanation:**

****A. OCP**** — Among these four options, OCP has the lowest Pearl Index. (Note: implants are even lower if that option is present.)

Q5 — Emergency contraception progestogen of choice

Progestogen of choice for emergency contraception?

- A. Norethisterone
- B. Medroxyprogesterone
- C. Oxytocin
- D. Levonorgestrel

► **Answer & Explanation:**

****D. Levonorgestrel**** — 1.5 mg single dose within 72h (up to 120h). Ulipristal 30 mg is more effective at 72–120h.

Q6 — Ulipristal acetate dose

Dose of ulipristal acetate for emergency contraception?

- A. 300 mg
- B. 30 mg
- C. 300 µg
- D. 30 µg

► **Answer & Explanation:**

****B. 30 mg**** — Selective progesterone receptor modulator; single 30 mg PO within 120h.

Q7 — DMPA dosing interval

DMPA is given once every:

- A. 3 months
- B. 6 months
- C. 9 months
- D. 45 days

► **Answer & Explanation:**

****A. 3 months (12 weeks)**** — 150 mg IM every 12 weeks. Delays return to fertility ~6–9 months.

Q8 — Subdermal implants: failure rate

A young woman wants the most effective reversible long-acting contraception with the lowest Pearl Index:

- A. Subdermal implants (etonogestrel)
- B. Combined oral contraceptive pill
- C. Condom
- D. Copper IUD

► **Answer & Explanation:**

****A. Subdermal implants**** — Pearl Index 0.05 per 100 woman-years — the lowest failure rate among all reversible methods. Effective for 3 years, no user compliance.

Q9 — Combined OCP: contraindicated in lactation

A lactating woman 6 weeks postpartum. Which contraceptive is NOT advisable?

- A. Combined oral contraceptive pill
- B. Progestogen-only pill
- C. Copper IUD
- D. DMPA

► **Answer & Explanation:**

****A. Combined oral contraceptive pill**** — Oestrogen suppresses prolactin → reduces breast milk volume. All progestogen-only methods and IUDs are safe during breastfeeding.

Q10 — Non-scalpel vasectomy: false statement

All of the following about non-scalpel vasectomy are correct EXCEPT:

- A. Haematoma occurs in up to 5% of subjects
- B. Sexual function following healing is rarely affected
- C. Sterility is immediate after vasectomy
- D. Recanalization is possible

► **Answer & Explanation:**

****C. Sterility is immediate after vasectomy**** — NOT immediate; residual sperm remain. Backup contraception until azoospermia confirmed (~12 weeks or 20 ejaculations).

Q11 — IUD absolute contraindication

Absolute contraindications to IUCD are all EXCEPT:

- A. Pregnancy
- B. Undiagnosed vaginal bleeding
- C. Pelvic inflammatory disease
- D. Uterine malformation

► **Answer & Explanation:**

****D. Uterine malformation**** — Relative, not absolute contraindication. The others (pregnancy, unexplained bleeding, active PID) are absolute.

Q12 — Most common side effect of IUCD

The most common side effect of IUCD insertion is:

- A. Bleeding (menorrhagia)
- B. Pain/dysmenorrhoea
- C. Pelvic infection
- D. Ectopic pregnancy

► **Answer & Explanation:**

****A. Bleeding (menorrhagia)**** — Copper IUCDs increase menstrual blood loss ~30–50%; most common reason for discontinuation. LNG-IUS reduces blood loss.

Q13 — Ideal contraceptive for cardiac disease

Ideal contraceptive for a patient with cardiac disease:

- A. Cu-IUCD
- B. DMPA
- C. Barrier method
- D. Combined OCP

► Answer & Explanation:

****C. Barrier method**** — No systemic haemodynamic or thrombotic effects. Combined OCP → VTE risk; DMPA → fluid retention; Cu-IUCD → vasovagal risk.

Q14 — LNG-IUS: NOT indicated for

Levonorgestrel-releasing IUD is helpful in all EXCEPT:

- A. Menorrhagia
- B. Dysmenorrhoea
- C. Endometriosis
- D. Pelvic inflammatory disease

► Answer & Explanation:

****D. Pelvic inflammatory disease**** — LNG-IUS is contraindicated in active PID; it does not treat or prevent PID.

Q15 — Lactational amenorrhoea mechanism

Lactational amenorrhoea is caused by:

- A. Prolactin-induced inhibition of GnRH
- B. Prolactin-induced inhibition of FSH
- C. Oxytocin-induced inhibition of GnRH
- D. Oxytocin-induced inhibition of FSH

► Answer & Explanation:

****A. Prolactin-induced inhibition of GnRH**** — Suckling → prolactin → suppresses hypothalamic GnRH → reduced FSH/LH → anovulation. Effective only with exclusive breastfeeding, < 6 months, and amenorrhoea.

Q16 — Ideal contraceptive for lactating mother

Ideal contraceptive for a lactating mother:

- A. Barrier method
- B. Combined OCP
- C. Lactational amenorrhoea alone
- D. Progestogen-only pill

► Answer & Explanation:

****D. Progestogen-only pill**** — Does not suppress prolactin or milk production. LAM alone is unreliable beyond 6 months.

Q17 — OCP mechanism: FALSE

All of the following are mechanisms of action of combined OCPs EXCEPT:

- A. Inhibition of ovulation
- B. Spermicidal effect
- C. Interference with implantation
- D. Interference with placental function

► Answer & Explanation:

****D. Interference with placental function**** — Once placentation occurs, OCPs have no effect.

Q18 — POP: not a side effect

Which is not a side effect of POP (progestogen-only pill)?

- A. Ovarian cysts
- B. Venous thromboembolism
- C. Increased risk of diabetes mellitus
- D. Ectopic pregnancy

► Answer & Explanation:

****B. Venous thromboembolism**** — VTE is an oestrogen effect. POPs are safe in women with VTE risk.

Q19 — Progestogen-only pill: NOT a mechanism

Which of the following is NOT a mechanism of action of progesterone-only pills?

- A. Thicken the cervical mucus
- B. Prevent ovulation
- C. Kill the sperms in the cervical canal
- D. Inhibit penetration of sperm

► Answer & Explanation:

****C. Kill the sperms in the cervical canal**** — POPs are not spermicidal.

Q20 — Progesterone effect (2018)

What is the effect of progesterone-only pills?

- A. Completely suppresses ovulation
- B. Thin lining of uterus

- C. Thick cervical mucus
- D. All of the above

► **Answer & Explanation:**

****D. All of the above**** — All three are accepted effects.

Q21 — Progestogen-only pill — missed dose advice

A 26-year-old on a POP usually takes it at 10 am but took it yesterday at 5 pm. Advice:

- A. No need to worry; continue same contraceptive
- B. Use additional contraception for the next 48 hours
- C. Change over to combined OCP
- D. Take double dose next morning

► **Answer & Explanation:**

****B. Use additional contraception for the next 48 hours**** — Traditional POP: 3-hour window; desogestrel: 12-hour window. If taken outside window, add barrier for 48 hours.

Q22 — Fimbriectomy method

Fimbriectomy procedure is known as:

- A. Uchida method
- B. Irving method
- C. Madlener technique
- D. Kroener method

► **Answer & Explanation:**

****D. Kroener method**** — Removes fimbrial end. Uchida = saline injection; Irving = bury both ends; Madlener = crush a loop.

Q23 — Site of tubal ligation

Most common site for tubal ligation:

- A. Ampulla – uniform muscle thickness
- B. Ampulla – widest part
- C. Ampulla–Isthmus junction – site of fertilization
- D. Isthmus – narrowest part

► **Answer & Explanation:**

****D. Isthmus – As it is the narrowest part**** — Easiest to ligate, least blood supply, uniform muscle wall.

Q24 — IUCD and ectopic pregnancy

A P2L2 patient using an IUCD presents with vaginal bleeding and recurrent syncope. Most likely diagnosis?

- A. Ectopic pregnancy
- B. Recurrent abortion
- C. Septicaemia
- D. Coagulation disorder

► Answer & Explanation:

****A. Ectopic pregnancy**** — IUCD protects against intrauterine pregnancy but not ectopic. Syncope + vaginal bleeding in reproductive-age woman = ectopic until proven otherwise.

Q25 — IUCD: false statement (2016)

Which of the following is false about IUDs?

- A. Multiload Cu-375 is a third-generation IUD
- B. Copper devices are effective as post-coital contraceptives
- C. LNG-20 (Mirena) has an effective life of 5 years
- D. Pregnancy rates of Lippes Loop and T Cu-200 are similar

► Answer & Explanation:

****A. Multiload Cu-375 is a third-generation IUD**** — It is second-generation. Third-generation = Cu-T 380A and frameless GyneFix.

Q26 — IUCD absolute contraindication (2022)

Which is an absolute contraindication for insertion of the IUCD shown?

- A. Menstruation
- B. Trophoblastic disease
- C. Immediately after delivery
- D. Ruptured condom during intercourse

► Answer & Explanation:

****B. Trophoblastic disease**** — GTD is absolute contraindication. IUCD can be inserted during menstruation, immediately postpartum, and as emergency contraception after unprotected sex.

Q27 — IUCD displaced into abdomen

Copper IUD with absent threads; ultrasound shows partial displacement into abdominal cavity. Management:

- A. Hysteroscopy
- B. No removal needed
- C. IUCD hook retrieval
- D. Laparoscopy

► Answer & Explanation:

****D. Laparoscopy**** — Abdominal/peritoneal displacement must be removed laparoscopically to prevent bowel adhesions and visceral injury.

Q28 — Progestin IUCD and upper genital tract infection

All explain reduced risk of upper genital tract infection with progestin IUDs EXCEPT:

- A. Reduced retrograde menstruation
- B. Decreased ovulation
- C. Thickened cervical mucus
- D. Decidual changes in the endometrium

► Answer & Explanation:

****B. Decreased ovulation**** — The protective mechanisms against ascending infection are: thickened cervical mucus, reduced retrograde menstruation, decidualisation. Ovulation suppression is not the mechanism.

Q29 — "Today" contraceptive sponge

"Today" contraceptive sponge contains:

- A. Prostaglandin F_{2α}
- B. Norethisterone
- C. Nonoxynol-9
- D. Copper-releasing mesh

► Answer & Explanation:

****C. Nonoxynol-9**** — Polyurethane sponge impregnated with Nonoxynol-9 spermicide; provides barrier + spermicidal action.

Q30 — IUCD in pregnancy, visible threads (2024)

A 28-year-old at 10 weeks with IUCD in place, thread visible, wants to continue pregnancy. Next step?

- A. Leave IUCD inside with prophylactic antibiotics
- B. Leave IUCD inside and continue pregnancy
- C. Remove IUCD immediately and continue pregnancy
- D. Termination of pregnancy

► Answer & Explanation:

****C. Remove IUCD immediately and continue with pregnancy**** — Removing IUCD (when threads visible) reduces risk of septic abortion (~50% risk if left in), preterm labour, and PROM.

Q31 — Lactating female with heavy bleeding seeking contraception (2022)

A 24-year-old lactating female with 18-month-old child, irregular heavy bleeding, seeks contraception:

- A. Progestasert
- B. Copper-T 380A
- C. Mala
- D. Norethisterone enanthate depot injection

► **Answer & Explanation:**

****C. Mala**** — "Mala-D" and "Mala-N" are combined OCP brands in India's National Family Planning Programme. With heavy irregular bleeding, combined OCP regulates cycles and reduces blood loss. At 18 months, LAM criteria are not met (infant too old).

Chapter 6 — Infertility and Assisted Reproductive Technology (ART)

"Infertility questions reward the candidate who understands sequence — what to investigate first, what each test tells you, and when to escalate to ART. Follow the logic and the marks follow you."

Big Picture First

WHO Definitions (Current, Exam-Ready)

- **Infertility:** Failure to achieve a clinical pregnancy after **12 months** of regular, unprotected sexual intercourse in women < **35 years**
- **Infertility in women ≥ 35 :** Evaluate after **6 months** (accelerated timeline due to declining ovarian reserve)
- **Primary infertility:** No previous pregnancy ever
- **Secondary infertility:** Previous pregnancy (including ectopic, miscarriage), currently unable to conceive
- **Subfertility:** Reduced capacity to conceive — used interchangeably with infertility in most exams

Prevalence and Causes

In India: ~10–15% of couples affected.

Distribution of causes: | Cause | Approximate proportion | |---|---| | Male factor | 30–40% | | Female factor (ovulatory) | 25–30% | | Female factor (tubal/pelvic) | 25–35% | | Unexplained | 10–15% | | Combined (both partners) | 20–30% |

Exam insight: Questions frequently test male factor first because it is simple, non-invasive, and cheap to investigate (semen analysis). Always investigate BOTH partners simultaneously.

Investigation Sequence — The Logical Order

- Step 1: Semen analysis (male) – cheap, non-invasive, rules in/out male factor
- Step 2: Document ovulation (female)
- Day 21 serum progesterone (> 5 ng/mL = ovulatory)
 - LH urinary kit (LH surge)
 - BBT charting (biphasic)
 - TVS follicle monitoring
- Step 3: Tubal patency (female)
- HSG first-line
 - Laparoscopy + chromopertubation if HSG abnormal
- Step 4: Uterine cavity assessment
- HSG (basic)
 - Hysteroscopy (definitive)
 - Saline sonohysterography
- Step 5: Ovarian reserve
- AMH (best single marker)
 - Day 3 FSH + estradiol
 - Antral follicle count (AFC) on TVS

Question Pattern Map

Recurring PYQ themes from 2015–2024:

Theme	Frequency
Bilateral cornual block on HSG — management	Very High
WHO 2021 semen analysis reference values	High
Semen terminology (aspermia vs azoospermia vs asthenospermia)	High
Hydrosalpinx on HSG — diagnosis and IVF implication	High
OHSS — diagnosis, severity, management	High
Asherman syndrome — post-curettage amenorrhoea	High
Ovulation induction: clomiphene vs letrozole	High
Obstructive azoospermia — clinical features	Medium
Genital TB — tubal involvement, HSG features	Medium
IVF sequence, PGT (preimplantation genetic testing)	Medium
Endometriosis and infertility	Medium

Core Concept Deep Dive

1. Semen Analysis — WHO 2021 Reference Values

This is the single most frequently tested ART topic. The WHO 6th Edition (2021) significantly revised normal values downward from previous editions.

WHO 2021 Lower Reference Limits (5th percentile)

Parameter	WHO 2021 (6th edition)	WHO 2010 (5th edition — OLD)
Volume	≥ 1.4 mL	≥ 1.5 mL
Sperm concentration	≥ 16 million/mL	≥ 15 million/mL
Total sperm count	≥ 39 million/ejaculate	≥ 39 million
Total motility (PR + NP)	≥ 42%	≥ 40%
Progressive motility	≥ 30%	≥ 32%
Morphology (strict Kruger)	≥ 4%	≥ 4%
Vitality	≥ 54%	≥ 58%

PYQ 1 (Q350)

WHO 2021 lower reference limit for sperm concentration: - A. 5 million/mL - B. 10 million/mL - C. 16 million/mL - D. 20 million/mL

Answer: C. 16 million/mL

The 2021 WHO 6th edition number is 16 million/mL. Option A (5 million) = severe oligospermia. Option D (20 million) = the pre-2010 threshold. This is a direct recall question — the number must be committed to memory.

2. Semen Terminology — Extremely Exam Friendly

These definitions are tested repeatedly and frequently confused:

Term	Definition
Aspermia	COMPLETE absence of ejaculate (no semen volume at all)
Azoospermia	No sperm in the ejaculate (but semen present)
Oligospermia	Sperm count < 16 million/mL (WHO 2021)
Asthenozoospermia	Reduced sperm motility
Teratozoospermia	Abnormal sperm morphology (< 4% normal forms)
Oligoastheno-teratozoospermia (OAT)	All three abnormalities combined
Hypospermia	Low ejaculate volume (< 1.4 mL)
Hyperspermia	High ejaculate volume (> 6 mL)

PYQ 2 (Q532)

Aspermia refers to: - A. Complete absence of semen (ejaculate) - B. Absence of sperm in the ejaculate (azoospermia) - C. Absence of sperm motility (asthenozoospermia) - D. Occurrence of abnormal sperm morphology (teratozoospermia)

Answer: A. Complete absence of semen

Aspermia = no ejaculate at all (retrograde ejaculation into the bladder, or ejaculatory duct obstruction). The ejaculate volume is zero. Contrast with azoospermia where ejaculate is present but contains no sperm. This distinction is a classic NEET trap.

3. Male Factor Infertility — Clinical Investigation Logic

Obstructive vs. Non-Obstructive Azoospermia

PYQ 3 (Q671)

A 25-year-old man presents with azoospermia. Testes are normal sized; FSH and testosterone are normal. Most likely cause? - A. Obstructed vas deferens - B. Kallmann syndrome - C. Klinefelter syndrome - D. Noonan syndrome

Answer: A. Obstructed vas deferens

This clinical vignette demonstrates the diagnostic algorithm: - **Normal FSH** = pituitary is working normally = testes are receiving gonadotropin signal - **Normal testicular size** = testes are producing sperm normally - **Azoospermia** = sperm are produced but cannot reach the ejaculate

This is **obstructive azoospermia (OA)**. The obstruction is in the vas, epididymis, or ejaculatory ducts. Treatment: surgical sperm retrieval (PESA, TESA, MESA) + ICSI.

Compare: - **Klinefelter (47,XXY)** = small testes + elevated FSH + azoospermia (non-obstructive) - **Kallmann syndrome** = small testes + LOW FSH/LH + anosmia (hypogonadotropic hypogonadism) - **Noonan syndrome** = elevated FSH + variable testicular function

Key Differentiator: FSH in Azoospermia

FSH level	Testes size	Interpretation
Normal/Low	Normal	Obstructive azoospermia (OA)
Elevated	Small	Non-obstructive (spermatogenic failure)
Very Low	Small	Hypogonadotropic hypogonadism (Kallmann, pituitary failure)

4. Female Factor — Ovulation Assessment

When to Measure Progesterone

PYQ 4 (Q607)

Best day for serum progesterone evaluation in a 28-day cycle to confirm ovulation: - A. Day 3 - B. Day 8 - C. Day 14 - D. Day 21

Answer: D. Day 21

Mid-luteal progesterone is measured 7 days before the next expected period. In a 28-day cycle, that is day 21 (midluteal = approximately day 14 + 7). A level > 5 ng/mL (some sources say > 10 ng/mL) confirms ovulation. Day 3 is for FSH/estradiol (ovarian reserve). Day 14 is for LH surge detection.

Indicators of Anovulation (and Their MCQ Relevance)

Anovulatory infertility classically presents as: - Irregular/oligomenorrhoea + difficulty conceiving - Low mid-luteal progesterone (< 3 ng/mL) - No biphasic BBT chart - No dominant follicle on TVS monitoring

PYQ 5 (Q388)

A 35-year-old obese woman (BMI 34) presents with irregular, heavy, unpredictable periods. Endometrial biopsy shows simple hyperplasia without atypia; serum progesterone in luteal phase < 3 ng/mL. The mechanism of her AUB is: - A. AUB-L (leiomyoma) - B. AUB-O (anovulatory cycles with unopposed oestrogen) - C. AUB-C (coagulopathy) - D. AUB-P (endometrial polyp)

Answer: B. AUB-O

Obesity → peripheral aromatisation of androgens to oestrone → unopposed oestrogen stimulates endometrium → no progesterone opposition (low luteal progesterone confirms anovulation) → irregular heavy bleeding + endometrial hyperplasia. This patient also has infertility potential (PCOS likely).

5. Ovarian Reserve

Markers of Ovarian Reserve

PYQ 6 (Q322)

All are markers of ovarian reserve EXCEPT: - A. Inhibin A - B. Estradiol (day 3) - C. Inhibin B - D. Antral follicle count / ovarian volume

Answer: A. Inhibin A

Ovarian reserve markers: AMH (best), Inhibin B, Day-3 FSH, Day-3 estradiol, Antral Follicle Count (AFC), ovarian volume.

Inhibin A is produced mainly by: (1) dominant follicle in follicular phase, and (2) corpus luteum in luteal phase → most importantly, by the PLACENTA during pregnancy. It is used in the Down syndrome screening "quad test," NOT as an ovarian reserve marker.

Premature Ovarian Insufficiency (POI)

PYQ 7 (Q321)

Which hormone is elevated in premature ovarian failure? - A. Serum inhibin - B. Serum FSH - C. Serum estradiol - D. Both A and B

Answer: B. Serum FSH

POI (formerly POF): FSH > 25–40 IU/L on two occasions (at least 1 month apart), age < 40 years. Pattern: elevated FSH, elevated LH, low estradiol, very low AMH, low inhibin B. Inhibin DECREASES in POI (not increases) because the granulosa cells are depleted.

PYQ 8 (Q11 — clinical vignette)

36-year-old P2L2 with 9 months amenorrhoea, FSH 36, LH 56, AMH 0.5. Diagnosis?

Answer: B. Premature ovarian failure (POI)

The combination of elevated FSH/LH + very low AMH + amenorrhoea in a woman < 40 years = POI. Normal range for FSH is 2–10 IU/L in the follicular phase; FSH > 25 on two separate occasions = diagnostic of POI.

6. Tubal Factor Infertility — HSG and Laparoscopy

HSG: When to Perform

PYQ 9 (Q351)

Best timing for HSG in the menstrual cycle: - A. Day 2–5 - B. Day 6–12 - C. Day 14–16 - D. Day 21–28

Answer: B. Day 6–12

Reasons: Post-menstrual (avoids pregnant state if cycle fails), pre-ovulatory (endometrium thin → better visualisation), before ovulation occurs (no risk of exposing a fertilised egg to radiation or contrast). Days 2–5 = menses may be ongoing. Days 14+ = risk of disrupting ovulation or early pregnancy.

PID and Infertility — First Investigation

PYQ 10 (Q614)

A 22-year-old with history of PID presents with 12 months of infertility, normal menses, normal examination. Next investigation: - A. Hysterosalpingography (HSG) - B. AMH - C. Hysteroscopy - D. CA-125

Answer: A. HSG

History of PID specifically raises concern for tubal damage (salpingitis → peritubal adhesions → hydrosalpinx → blocked tubes). HSG is the first-line test for tubal patency after PID. AMH assesses ovarian reserve (not the priority here). Hysteroscopy is for intrauterine pathology. CA-125 is a tumour marker for ovarian cancer (not useful for routine infertility).

Bilateral Cornual Block — Management Decision

This question has appeared in multiple years (Q664, Q704, Q784, Q1059) with consistent answer. Master it.

PYQ 11 (Q664/Q784/Q1059)

A woman presents with infertility and bilateral tubal block at the cornua on HSG. Best management? - A. Laparoscopy and hysteroscopy - B. Hydrotubation - C. IVF - D. Tuboplasty

Answer: A. Laparoscopy and hysteroscopy

The Clinical Reasoning: Cornual (proximal) block on HSG is very frequently a FALSE POSITIVE — due to: 1. Tubal spasm at the time of contrast injection 2. Mucous plugs at the cornua 3. Endometrial debris occluding the ostia
Laparoscopy + chromopertubation (methylene blue dye) confirms true tubal status under direct vision. Hysteroscopy simultaneously allows cannulation of the cornual ostia.

If laparoscopy confirms TRUE bilateral cornual block → then IVF is the treatment of choice (bypass the tubes entirely).

Hydrotubation (injection of fluid under pressure) is obsolete and may cause iatrogenic infection.

Why NOT IVF first? Because up to 40–60% of cornual blocks on HSG are spasm artefacts — no point committing to expensive ART without confirming true occlusion.

Hydrosalpinx and IVF

PYQ 12 (Q668)

An HSG of a 26-year-old infertile woman shows blocked fallopian tubes with fluid-filled dilatation at the fimbrial end. Diagnosis? - A. Hydrosalpinx - B. Retroverted fixed uterus - C. Uterine fibroids - D. Cervical fibroid

Answer: A. Hydrosalpinx

Hydrosalpinx = distal tubal occlusion (fimbrial end) + accumulation of serous fluid → clubs the tube into a sausage-shaped structure. HSG shows a clubbed, blind-ending tube.

Critical ART implication: Hydrosalpinx fluid contains inflammatory cytokines, endotoxins, and free radicals that are TOXIC to IVF embryos if they reflux into the uterine cavity. Hydrosalpinx reduces IVF success rates by ~50%. Before IVF, the hydrosalpinx must be dealt with — either: - Laparoscopic **salpingectomy** (removes the tube entirely) — preferred - Laparoscopic **proximal occlusion** (if salpingectomy not feasible)

PYQ 13 (Q1026) — 2024 actual paper

A 29-year-old woman with infertility and chronic pelvic pain. TVS reveals adnexal mass with incomplete septations and a cogwheel appearance. Most likely diagnosis? - A. Hydrosalpinx - B. Follicular ovarian cyst - C. Endometriosis - D. Haemorrhagic ovarian cyst

Answer: A. Hydrosalpinx

The "cogwheel sign" on TVS (cross-sectional view showing mural nodules/mucosal folds) is PATHOGNOMONIC of hydrosalpinx. Incomplete septations (actually tubal folds appearing as septa) are another feature.

7. Genital Tuberculosis and Infertility

PYQ 14 (Q323/Q621)

Most common site of female genital TB? - A. Fallopian tubes - B. Endometrium - C. Ovaries - D. Vulva/vagina

Answer: A. Fallopian tubes (~100%)

The hierarchy of genital TB involvement: 1. Fallopian tubes — virtually 100% (hallmark organ) 2. Endometrium — 60–80% 3. Ovaries — 20–30% 4. Cervix — 5% 5. Vagina/vulva — <1%

Genital TB is a common cause of tubal factor infertility in India. Always consider in a young woman presenting with: - Infertility + amenorrhoea + constitutional features (fever, weight loss) - Bilateral tubal block on HSG (especially pipe-stem or beaded pattern) - Asherman syndrome-like picture

PYQ 15 (Q686)

32-year-old with 6-month secondary amenorrhoea, chronic pelvic pain, intermittent fever, weight loss, and adnexal mass. Most likely diagnosis? - A. PCOS - B. Endometriosis - C. Genital tuberculosis - D. Ectopic pregnancy

Answer: C. Genital tuberculosis

Constitutional features (fever, weight loss) + secondary amenorrhoea + adnexal mass in an endemic region = genital TB. Diagnostic gold standard: endometrial biopsy during menstruation (culture, PCR, histology for caseating granulomas with Langhans giant cells).

HSG Features of Genital TB

PYQ 16 (Q844)

HSG findings suggestive of genital TB? - A. Beaded tubes - B. Honeycomb uterus - C. Golf-club tube - D. All of the above

Answer: D. All of the above

Genital TB on HSG shows: - **Beaded/pipestem tubes** (TB salpingitis with fibrosis) - **Golf-club appearance** (irregular narrowing at ampullary end) - **Honeycomb uterus** (multiple filling defects = intrauterine synechiae) - Feathering of the mucosal pattern - Calcification in healed lesions

8. Asherman Syndrome

PYQ 17 (Q667)

A woman presents with secondary amenorrhoea following curettage for abortion. FSH is 7 IU/L. Most likely diagnosis? - A. Pituitary failure - B. Ovarian failure - C. Uterine synechiae (Asherman syndrome) - D. Pregnancy

Answer: C. Uterine synechiae (Asherman syndrome)

The diagnostic key is **normal FSH** (7 IU/L is normal). This excludes: - Pituitary failure (FSH would be low) - Ovarian failure (FSH would be high, >25) - Pregnancy (would need beta-hCG, but FSH being normal + curettage history → Asherman is the answer)

Asherman syndrome: Intrauterine adhesions (synechiae) destroy the functional endometrium post-curettage.

Diagnosis by **hysteroscopy** (gold standard) or HSG (filling defects). Treatment: hysteroscopic adhesiolysis + high-dose oestrogen (2–4 mg/day for 4–6 weeks) to re-epithelialise.

PYQ 18 (Q992) — 2023 actual paper

For which procedure can hysteroscopy preferably be done? - A. Tubal ligation - B. Asherman syndrome - C. Endocervical polyp - D. Subserosal fibroid

Answer: B. Asherman syndrome

Tubal ligation requires laparoscopy. Subserosal fibroids are outside the uterine cavity — laparoscopic myomectomy. Endocervical polyp may be managed with simple avulsion or hysteroscopic resection (both possible, but hysteroscopy is more definitive). Asherman syndrome: hysteroscopic adhesiolysis is the treatment — definitive and diagnostic at the same time.

9. Endometriosis and Infertility**PYQ 19 (Q672)**

23-year-old with lower abdominal pain and dyspareunia; retroverted uterus with adnexal and cervical motion tenderness. Most important long-term sequela? - A. Recurrent pregnancy loss - B. Infertility - C. Pelvic organ prolapse - D. Amenorrhoea

Answer: B. Infertility

Endometriosis causes infertility through multiple mechanisms: 1. **Peritubal/periovarian adhesions** → mechanical obstruction 2. **Peritoneal toxins** (prostaglandins, cytokines) → hostile peritoneal environment for gametes 3.

Endometriomas → compresses ovarian reserve, surgical cystectomy further reduces AMH 4. **Impaired folliculogenesis** and oocyte quality 5. **Uterine receptivity** — altered endometrial gene expression

~30–50% of infertile women have endometriosis. ~50% of women with endometriosis have some degree of infertility.

PYQ 20 (Q833) — 2016 actual paper

Examination of a 26-year-old obese infertile female: Fixed retroverted uterus and nodularity of the uterosacral ligaments. Most likely diagnosis? - A. PCOS - B. Endometriosis - C. Adenomyosis - D. TB

Answer: B. Endometriosis

Fixed retroverted uterus (adhesions pulling uterus posteriorly) + nodular uterosacral ligaments = pathognomonic combination for endometriosis. The posterior cul-de-sac (Pouch of Douglas) becomes obliterated by adhesions and endometriotic deposits. Gold standard diagnosis: laparoscopy with histological confirmation.

PYQ 21 (Q673) — Severe endometriosis: management

29-year-old nulliparous woman with endometriosis and infertility: laparoscopy shows severe pelvic adhesions, "kissing ovaries" (each 6–8 cm cystic), bowel stuck to fundus. Best next management? - A. GnRH analogues - B. Combined OCP - C. Hysterectomy with bilateral oophorectomy - D. Bilateral cystectomy, adhesiolysis, and prepare for IVF

Answer: D. Bilateral cystectomy, adhesiolysis, and prepare for IVF

In a young nulliparous woman desiring fertility with severe endometriosis: - GnRH analogues alone → reduce pain but do NOT restore fertility; 50% recurrence off treatment - Combined OCP → for pain management, not fertility - Hysterectomy/oophorectomy → radical surgery, destroys fertility (last resort in non-fertility-desiring patients) - Conservative surgery → maximises natural and IVF-assisted fertility

Standard approach: conservative laparoscopic surgery (cystectomy, adhesiolysis, excision of deposits) → attempt natural conception for 6 months → proceed to IVF.

10. Ovulation Induction

Clomiphene Citrate

Clomiphene citrate (CC) = selective oestrogen receptor modulator (SERM).

Mechanism: - Blocks hypothalamic oestrogen receptors → hypothalamus "thinks" oestrogen is low → increases GnRH → increases FSH/LH → stimulates follicular development

Anti-oestrogenic effects (side effects from mechanism): - **Thin endometrium** (oestrogen receptors blocked in uterus) - **Thick cervical mucus** (less favourable for sperm) - **Hot flushes** - Ovarian cysts - Multiple pregnancy risk (5–10%)

Dose: 50 mg/day, days 2–6 or 3–7 of cycle. Can increase to 100–150 mg if no response.

Limitation: Despite increasing ovulation rate to ~70%, pregnancy rate is only ~15–40% per cycle — partly due to anti-oestrogenic endometrial and cervical effects.

Letrozole — Current First-Line in PCOS

PYQ 22 (Q352)

First-line ovulation induction agent in PCOS infertility (2018 ESHRE / 2023 international guideline): - A. Clomiphene citrate - B. Letrozole - C. Gonadotropins - D. Metformin

Answer: B. Letrozole

Letrozole is an **aromatase inhibitor** — blocks conversion of androgens to oestrogen. This causes: - Oestrogen ↓ → FSH ↑ via negative feedback → follicular stimulation - No peripheral anti-oestrogenic effects (no endometrial

thinning, no mucus thickening) - Result: Better endometrial receptivity, higher live-birth rate in PCOS vs. clomiphene

The 2014 NEJM landmark trial (Legro et al.) confirmed letrozole superiority. Current guidelines (ESHRE 2018, WHO 2023): Letrozole is FIRST-LINE for anovulatory infertility in PCOS.

Note: Letrozole is technically **off-label** for ovulation induction in many countries (licensed for breast cancer) but widely used and guideline-recommended.

Clomiphene vs. Letrozole Comparison

Feature	Clomiphene	Letrozole
Class	SERM	Aromatase inhibitor
Mechanism	Blocks oestrogen receptors → ↑ FSH	Blocks aromatase → ↓ oestrogen → ↑ FSH
Endometrial effect	THIN (anti-oestrogenic)	NORMAL (no anti-oestrogenic effect)
Cervical mucus	Thick (sperm hostile)	Normal
Multiple pregnancy risk	5–10%	~2–3%
Current guideline status	Second-line for PCOS	FIRST-LINE for PCOS
Ovulation induction rate	~70%	~70–80%
Live birth rate (PCOS)	~19%	~27%

11. PCOS and Infertility

PYQ 23 (Q696)

The cysts in PCOS are formed by: - A. Failure of atretic follicles to undergo apoptosis - B. Oocyte proliferation - C. Multiple corpora lutea - D. Cystic degeneration of the ovarian cortex

Answer: A. Failure of atretic follicles to undergo apoptosis

In PCOS: elevated androgens (from the theca cells, driven by excess LH and insulin resistance) inhibit follicular atresia. Multiple small follicles (2–9 mm) accumulate in a "pearl necklace" pattern at the ovarian periphery without one achieving dominance or ovulating. They are "arrested" antral follicles, not true cysts with epithelial linings.

Imaging criteria for PCOS (Rotterdam 2003): ≥12 follicles of 2–9 mm per ovary OR ovarian volume > 10 mL on TVS.

12. The IVF Process — Step by Step

Understanding the IVF sequence is essential for both process questions and complication questions.

IVF Protocol Steps

1. CONTROLLED OVARIAN STIMULATION (COS)
 - GnRH agonist (long protocol) or GnRH antagonist (short protocol)
 - Gonadotropins (FSH/LH) to stimulate multiple follicle growth
 - Monitoring: Serial TVS + serum estradiol every 1–2 days
2. TRIGGER SHOT
 - hCG 5000–10,000 IU when leading follicle ≥ 18 mm
 - Initiates final oocyte maturation
 - Or GnRH agonist trigger (in high OHSS-risk patients)
3. OOCYTE RETRIEVAL
 - 36 hours after trigger
 - Trans-vaginal ultrasound-guided needle aspiration
 - Follicular fluid collected → oocytes identified in lab
4. FERTILISATION
 - Conventional IVF: oocytes + sperm co-incubated
 - ICSI: Single sperm injected directly into oocyte
5. EMBRYO CULTURE
 - Day 3 (8-cell cleavage stage)
 - Day 5 (blastocyst stage – higher implantation rates)
6. PREIMPLANTATION GENETIC TESTING (PGT) – if indicated
 - Biopsy of blastomere (day 3) or trophectoderm (day 5)
7. EMBRYO TRANSFER
 - Fresh cycle (day 3 or day 5) OR
 - Frozen embryo transfer (FET)

PYQ 24 (Q873) — 2016 actual paper

Test used to detect genetic abnormality in embryo before transferring it to the uterus in IVF is: - A. Embryo cell biopsy - B. CVS - C. ICSI - D. All of the above

Answer: A. Embryo cell biopsy

Preimplantation Genetic Testing (PGT) = biopsy of embryonic cells before transfer. Types: - **PGT-A (aneuploidy)** — screens for chromosomal number abnormalities - **PGT-M (monogenic)** — for single-gene disorders (CFTR, BRCA, etc.) - **PGT-SR (structural rearrangements)** — for translocations

CVS (chorionic villus sampling) is an invasive prenatal diagnostic test AFTER implantation — not preimplantation. ICSI is a fertilisation technique, not a genetic test.

13. OHSS — Ovarian Hyperstimulation Syndrome

OHSS is the most serious iatrogenic complication of ART. High yield for both NEET-PG and INI-CET.

Pathophysiology

Exogenous hCG trigger → massive VEGF (vascular endothelial growth factor) release from stimulated follicles → increased vascular permeability → fluid shifts from intravascular to third space (ascites, pleural effusion) → hypovolaemia + haemoconcentration → VTE risk.

Classification (Golan / RCOG)

Grade	Features
Mild	Abdominal bloating, mild discomfort, ovaries 5–10 cm
Moderate	Nausea, vomiting, ascites on USG, ovaries 5–12 cm
Severe	Massive ascites, hydrothorax, Hct > 45%, oliguria, ovaries > 12 cm
Critical	Arterial thromboembolism, ARF, ARDS, torsion

Risk factors for OHSS: - PCOS (most important) - Young age - Low BMI - Previous OHSS - High AFC / AMH - Large number of follicles recruited - High serum estradiol (> 2500–3000 pg/mL at trigger)

PYQ 25 (Q400)

A 29-year-old woman with PCOS undergoing IVF develops severe abdominal distension, nausea, shortness of breath 5 days after oocyte retrieval. Ultrasound shows bilateral enlarged ovaries (each > 10 cm) with massive ascites. Haematocrit 52%. This complication is: - A. Ruptured ovarian cyst - B. Severe ovarian hyperstimulation syndrome - C. Ovarian torsion - D. PID post-retrieval

Answer: B. Severe OHSS

Features of severe OHSS: ovaries > 10 cm, massive ascites, Hct > 45% (52% here = haemoconcentration), respiratory distress, oliguria. The PCOS background + IVF context + timing (5 days post-retrieval) = classic late-onset OHSS (triggered by endogenous hCG from early pregnancy if an embryo was transferred, or by exogenous trigger).

PYQ 26 (Q336)

Patient on infertility treatment with gonadotropins develops abdominal pain, ascites, dyspnoea; USG shows enlarged multicystic ovaries. Diagnosis? - A. PCOS - B. OHSS - C. Mucinous cystadenoma - D. Theca lutein cysts

Answer: B. OHSS

The iatrogenic context (currently on gonadotropins for infertility treatment) + bilateral enlarged multicystic ovaries + ascites = OHSS. Theca lutein cysts are also bilateral and multicystic but occur in gestational trophoblastic disease context (very high hCG). Mucinous cystadenoma is typically unilateral and slowly growing.

PYQ 27 (Q666)

A 25-year-old nulliparous woman undergoing controlled ovarian stimulation has 8 growing follicles and serum estradiol 800 pg/mL. Best management? - A. Retrieve all follicles immediately - B. Give cabergoline - C. Cancel the cycle - D. Withhold HCG trigger

Answer: D. Withhold HCG trigger

8 follicles growing = high OHSS risk. Withholding or switching the trigger is a risk-reduction strategy. Options include: - **Withhold hCG** ("coasting" — continue stimulation but hold trigger until estradiol drops) - **Switch to GnRH agonist trigger** (instead of hCG) — dramatically reduces OHSS risk - **Freeze all embryos** (freeze-all strategy — prevents early pregnancy hCG from worsening OHSS)

Cabergoline (dopamine agonist) can reduce OHSS severity by blocking VEGF-mediated permeability, but it is not first-line here.

14. OHSS Prevention Strategy

Strategy	Mechanism
GnRH antagonist protocol	Allows GnRH agonist trigger (not hCG)
GnRH agonist trigger (instead of hCG)	Short LH surge, less sustained stimulus
Freeze all embryos	No endogenous hCG from early pregnancy
Cabergoline	Reduces VEGF receptor signalling
Albumin infusion	Expands intravascular volume
Coasting	Hold gonadotropins until estradiol stabilises

Memory hook for OHSS classification: "M-M-S-C" Mild → Moderate → Severe → Critical

Severe OHSS numbers to remember: - Ovaries > 10 cm - Hct > 45% - WBC > 25,000 (infection risk/haemoconcentration marker)

15. ICSI — When to Use

ICSI (Intracytoplasmic Sperm Injection) injects a single sperm directly into the oocyte cytoplasm. It bypasses: - Sperm swimming/motility - Zona pellucida penetration - Acrosome reaction

Indications for ICSI (over conventional IVF): - Severe oligospermia - Asthenozoospermia (poor motility) - Teratozoospermia - Obstructive azoospermia (surgically retrieved sperm) - Non-obstructive azoospermia (testicular sperm extraction) - Previous failed fertilisation on conventional IVF - Low oocyte numbers - Frozen-thawed sperm

16. Infertility Workup — Initial Investigations

PYQ 28 (Q542)

A 29-year-old couple presents with 3 years of infertility, regular cycles, normal coitus. What initial investigations are indicated? - A. Semen analysis, CXR, Mantoux - B. Semen analysis, tubal patency test, ovulation assessment - C. Ovulation assessment, tubal patency test, Mantoux - D. Testicular biopsy, scrotal USG, sperm penetration assay

Answer: B. Semen analysis, tubal patency test, ovulation assessment

The basic infertility workup triad: 1. **Semen analysis** — investigate male factor first (cheap, non-invasive) 2. **Ovulation documentation** — Day 21 progesterone/TVS 3. **Tubal patency** — HSG

Testicular biopsy and sperm penetration assay come only if initial semen analysis shows azoospermia or IVF failure. Mantoux/CXR would be indicated if genital TB is suspected (constitutional features, tubal disease pattern).

17. Uterine Causes of Infertility and Miscarriage

Submucosal Fibroid

PYQ 29 (Q665)

28-year-old with menorrhagia, primary infertility, and Hb 8 g/dL; hysteroscopy shows a round firm mass on a stalk in the uterine cavity. Diagnosis? - A. Pedunculated subserosal fibroid - B. Submucosal fibroid - C. Carcinoma endometrium - D. Gestational sac

Answer: B. Submucosal fibroid

Submucosal fibroids protrude into the uterine cavity → distort endometrium → cause menorrhagia + infertility + recurrent pregnancy loss. FIGO classification: Type 0 (entirely intracavitary, on a stalk), Type 1 (< 50% intramural), Type 2 (≥ 50% intramural). Types 0 and 1 are treated hysteroscopically.

Incompetent Cervix and Recurrent Miscarriage

PYQ 30 (Q669)

A woman with two previous abortions; the Hegar dilator test shows size-8 dilator passes easily. Diagnosis? - A. Septate uterus - B. Unicornuate uterus - C. Uterus didelphys - D. Incompetent cervix

Answer: D. Incompetent cervix

Cervical insufficiency: painless dilatation of the cervix in the second trimester → pregnancy loss. Classical feature: Hegar dilator No. 8 passes without resistance outside pregnancy. Management in subsequent pregnancy: **McDonald cerclage** at 12–14 weeks gestation.

18. Tubal Anatomy and Function

Kartagener Syndrome and Tubal Motility

PYQ 31 (Q535)

Decreased motility/ciliary dysfunction of the fallopian tube is associated with: - A. Noonan syndrome - B. PCOS - C. Churg-Strauss syndrome - D. Kartagener's syndrome

Answer: D. Kartagener's syndrome

Primary ciliary dyskinesia (PCD): absent dynein arms → immotile cilia. Features: bronchiectasis, recurrent sinusitis, situs inversus (50%), male infertility (immotile sperm), and female subfertility (impaired tubal egg transport). The egg cannot be transported toward the uterus without ciliary action.

19. Embryology Concepts Relevant to ART

Implantation Timing

PYQ 32 (Q687/Q411/Q580)

After ovulation, embryo implantation occurs after approximately: - A. 3–5 days - B. 7–9 days - C. 10–12 days - D. 13–15 days

Answer: B. 7–9 days

Timeline: - Day 0: Ovulation / fertilisation - Day 3–4: 8-cell cleavage → morula → enters uterine cavity - Day 4–5: Blastocyst formation (cavitation of morula) - Day 5–6: Hatching from zona pellucida - Day 7–9: Implantation into secretory endometrium

This question appears THREE TIMES in the database — the answer is consistently B (7–9 days).

Blastocyst Formation Timing

PYQ 33 (Q548)

Blastocyst formation (cavitation of morula) occurs after fertilisation on approximately: - A. Day 4–7 - B. Day 10–12 - C. Day 12–15 - D. Day 15–20

Answer: A. Day 4–7

Morula (Day 3–4) → blastocyst (Day 4–5 by cavitation) → hatching from zona pellucida → implantation Day 6–7. The uterine cavity is reached by Day 4 as a morula.

Secondary Oocyte — Meiotic Arrest

PYQ 34 (Q1013) — 2024 actual paper

After ovulation, the egg is released from the ovary. At what stage of meiosis is the released egg arrested? - A. Secondary oocyte arrested in meiosis I - B. Secondary oocyte arrested in meiosis II - C. Primary oocyte arrested in meiosis I - D. Primary oocyte arrested in meiosis II

Answer: B. Secondary oocyte arrested in meiosis II

The story of meiotic arrests: - **Primary oocyte**: arrested in Prophase of Meiosis I → from fetal life until puberty (sometimes 12–50 years) - **At ovulation**: LH surge → meiosis resumes → first polar body extruded → becomes secondary oocyte → arrested in Metaphase II - **After fertilisation**: sperm entry triggers completion of Meiosis II → second polar body extruded → female pronucleus formed

So at the moment of ovulation and release, the egg is a **secondary oocyte in Metaphase II arrest**.

Bias Removal Section

Common Traps in Infertility MCQs

1. **"Cornual block on HSG = IVF immediately"** WRONG. Cornual block is often spasm. First step = laparoscopy + hysteroscopy to confirm. IVF is the answer only if true blockage is confirmed.
2. **"Hydrosalpinx: proceed directly to IVF"** Partially right, but INCOMPLETE. Hydrosalpinx MUST be dealt with (salpingectomy or proximal occlusion) BEFORE IVF, otherwise IVF success rates are halved.
3. **"FSH elevated = ovarian failure"** Correct direction, but context matters. Elevated FSH is also seen in Klinefelter syndrome (in males). In females < 40 years: POI.
4. **"Inhibin A is an ovarian reserve marker"** WRONG. Inhibin B is the reserve marker. Inhibin A is placental.
5. **"Sterility after vasectomy is immediate"** WRONG (also tested in Chapter 5). Azoospermia confirmed ~12 weeks later.
6. **"WHO 2010 values are current"** WRONG. WHO 2021 (6th edition) is current. Key change: concentration is 16 million/mL (was 15 million). Volume is 1.4 mL (was 1.5 mL).

Variations and Twists

Scenario: Normal HSG but Still Infertile

If semen analysis is normal, HSG is normal, and ovulation is confirmed → **unexplained infertility** OR consider: - Mild endometriosis (not visible on HSG) - Sperm-cervical mucus incompatibility - Immunological factors - Luteinised unruptured follicle (LUF) syndrome - Poor oocyte quality

Next step: **Diagnostic laparoscopy** — can find endometriosis invisible on HSG.

Scenario: Male with Bilateral Absence of Vas Deferens

CBAVD (congenital bilateral absence of vas deferens): associated with **cystic fibrosis** (CFTR gene mutation). Semen analysis shows azoospermia + very low volume + acidic semen (seminal vesicle also absent). Treatment: testicular sperm extraction + ICSI + genetic counselling.

Scenario: Woman with Elevated Prolactin + Infertility

Hyperprolactinaemia → suppresses GnRH → anovulation → infertility. Treatment: **cabergoline** (dopamine agonist) → normalises prolactin → restores ovulation. First exclude pituitary macroadenoma (MRI).

Memory Engineering

WHO 2021 Semen Analysis — Memory Hook: "16-42-30-4-54"

"16 men, 42 total, 30 progressive, 4 shape, 54 alive" - Concentration: **16** million/mL - Total motility: **42%** - Progressive motility: **30%** - Morphology: **4%** (strict Kruger) - Vitality: **54%** - Volume: **1.4** mL

OHSS Severity — Memory Hook: "5-10-45"

- Mild/moderate: ovaries 5–10 cm
- Severe: ovaries > **10 cm**, Hct > **45%**

Semen Terminology — Memory Hook: "AZOAT"

- Aspermia = absent ejaculate
- Azoospermia = absent sperm (but ejaculate present)
- Oligospermia = low count
- Asthenospermia = poor motility
- Teratospermia = abnormal morphology

IVF Indications Mnemonic: "TUBES MALE UNEXPLAINED"

- **T** — Tubal disease (bilateral block, hydrosalpinx)
- **U** — Unexplained infertility (failed other treatments)
- **B** — Both partners contributing (mixed factor)
- **E** — Endometriosis (severe, surgery failed)
- **S** — Semen abnormality (severe OAT)
- **MALE** — Male factor (azoospermia requiring ICSI)

Ultra High-Yield Revision Layer

Semen Analysis Normal Values (WHO 2021) — Direct Recall

Parameter	Normal Lower Limit
Volume	≥ 1.4 mL
Concentration	≥ 16 million/mL
Total motility	≥ 42%
Progressive motility	≥ 30%
Morphology (strict)	≥ 4%
Vitality	≥ 54%

Investigation Sequence Summary

Suspected cause	First-line investigation
Male factor	Semen analysis
Anovulation	Day 21 progesterone (cycle day 21 in 28-day cycle)
Tubal factor (history of PID/TB)	HSG (days 6–12)
Cornual block on HSG	Laparoscopy + hysteroscopy
Uterine adhesions (post-curettage)	Hysteroscopy
Ovarian reserve	AMH
PCOS	TVS + LH:FSH ratio + androgen levels

Top 5 Traps in Infertility/ART MCQs

1. **Cornual block on HSG** → **lap + hysteroscopy first** (not straight to IVF)
2. **Aspermia (no semen) ≠ azospermia (no sperm in semen)** — definitions swapped in distractors
3. **Inhibin A ≠ ovarian reserve** — Inhibin B is the reserve marker
4. **Hydrosalpinx must be treated before IVF** — toxic to embryo
5. **Letrozole is now first-line, not clomiphene** in PCOS infertility

All Chapter 6 PYQs Integrated

Q-Bank Practice Block

Q1 — WHO 2021 sperm concentration

WHO 2021 lower reference limit for sperm concentration:

- A. 5 million/mL
- B. 10 million/mL
- C. 16 million/mL
- D. 20 million/mL

► Answer & Explanation:

****C. 16 million/mL**** — WHO 6th edition: concentration ≥ 16 M/mL, total motility $\geq 42\%$, progressive $\geq 30\%$, morphology $\geq 4\%$ (strict), volume ≥ 1.4 mL.

Q2 — Best timing for HSG

Best timing for HSG in the menstrual cycle:

- A. Day 2–5
- B. Day 6–12

- C. Day 14–16
- D. Day 21–28

► **Answer & Explanation:**

****B. Day 6–12**** — Post-menses, pre-ovulatory. Assesses tubal patency and uterine cavity without risk of pregnancy exposure.

Q3 — First-line ovulation induction in PCOS

First-line ovulation induction agent in PCOS infertility:

- A. Clomiphene citrate
- B. Letrozole
- C. Gonadotropins
- D. Metformin

► **Answer & Explanation:**

****B. Letrozole**** — Aromatase inhibitor; higher live-birth rate than clomiphene in PCOS (ESHRE 2018, WHO 2023 guidelines).

Q4 — IVF: severe OHSS

29-year-old PCOS woman, 5 days after oocyte retrieval: severe abdominal distension, nausea, shortness of breath. USG: ovaries > 10 cm, massive ascites. Hct 52%. Diagnosis:

- A. Ruptured ovarian cyst
- B. Severe OHSS
- C. Ovarian torsion
- D. PID post-retrieval

► **Answer & Explanation:**

****B. Severe OHSS**** — Ovaries > 10 cm, massive ascites, haemoconcentration (Hct > 45%), triggered by hCG in PCOS patient after oocyte retrieval.

Q5 — Cornual tubal block: management

A woman with infertility and bilateral tubal block at the cornua on HSG. Best management:

- A. Laparoscopy and hysteroscopy
- B. Hydrotubation
- C. IVF
- D. Tuboplasty

► **Answer & Explanation:**

****A. Laparoscopy and hysteroscopy**** — Cornual block is often a spasm artefact; confirms true occlusion + treats simultaneously. IVF reserved for confirmed true obstruction.

Q6 — Aspermia definition

Aspermia refers to:

- A. Complete absence of semen (ejaculate)
- B. Absence of sperm in the ejaculate (azoospermia)
- C. Absence of sperm motility (asthenozoospermia)
- D. Occurrence of abnormal sperm morphology (teratozoospermia)

► Answer & Explanation:

****A. Complete absence of semen**** — Aspermia = no ejaculate. Azoospermia = ejaculate present but no sperm. Classic MCQ distinction.

Q7 — Bilateral cornual block management (2016)

An infertile woman has bilateral tubal block at the cornua on HSG. Next treatment of choice:

- A. IVF
- B. Laparoscopy and hysteroscopy
- C. Tuboplasty
- D. Hydrotubation

► Answer & Explanation:

****B. Laparoscopy and hysteroscopy**** — HSG cornual block is often spasm; laparoscopy + chromopertubation confirms and hysteroscopy treats.

Q8 — Male azoospermia, normal testes and FSH

25-year-old man with azoospermia, normal-sized testes, normal FSH and testosterone. Most likely cause:

- A. Obstructed vas deferens
- B. Kallmann syndrome
- C. Klinefelter syndrome
- D. Noonan syndrome

► Answer & Explanation:

****A. Obstructed vas deferens**** — Normal FSH + normal testes = spermatogenesis intact; azoospermia due to obstruction. Treatment: surgical sperm retrieval + ICSI.

Q9 — Asherman's syndrome diagnosis

Woman with secondary amenorrhoea following curettage for abortion. FSH is 7 IU/L. Most likely diagnosis:

- A. Pituitary failure
- B. Ovarian failure
- C. Uterine synechiae (Asherman's syndrome)

- D. Pregnancy

► **Answer & Explanation:**

****C. Uterine synechiae (Asherman's syndrome)**** — Normal FSH excludes pituitary/ovarian failure. Post-curettage + secondary amenorrhoea = Asherman's. Diagnosis: hysteroscopy. Treatment: adhesiolysis + oestrogen.

Q10 — Hydrosalpinx on HSG

HSG of a 26-year-old infertile woman shows blocked fallopian tubes with fluid-filled dilatation at the fimbrial end. Diagnosis:

- A. Hydrosalpinx
- B. Retroverted fixed uterus
- C. Uterine fibroids
- D. Cervical fibroid

► **Answer & Explanation:**

****A. Hydrosalpinx**** — Distal tubal occlusion + serous fluid accumulation. HSG: clubbed blind-ending tubes. Reduces IVF success by ~50%; salpingectomy/proximal occlusion before IVF is recommended.

Q11 — Hydrosalpinx: cogwheel appearance (2024)

29-year-old woman, infertility + chronic pelvic pain. TVS: adnexal mass with incomplete septations and cogwheel appearance. Most likely diagnosis:

- A. Hydrosalpinx
- B. Follicular ovarian cyst
- C. Endometriosis
- D. Haemorrhagic ovarian cyst

► **Answer & Explanation:**

****A. Hydrosalpinx**** — Cogwheel sign on TVS = cross-sectional view of fallopian tube wall folds; pathognomonic of hydrosalpinx.

Q12 — Infertility workup: initial investigations

29-year-old couple, 3 years infertility, regular cycles, normal coitus. Initial investigations:

- A. Semen analysis, CXR, Mantoux
- B. Semen analysis, tubal patency test, ovulation assessment
- C. Ovulation assessment, tubal patency test, Mantoux
- D. Testicular biopsy, scrotal USG, sperm penetration assay

► Answer & Explanation:

****B.** Semen analysis, tubal patency test, ovulation assessment** — Basic triad. Testicular biopsy is invasive and comes later.

Q13 — Severe endometriosis + infertility management

29-year-old nulliparous with severe endometriosis: laparoscopy shows severe adhesions, kissing ovaries (6–8 cm), bowel involvement. Best management:

- A. GnRH analogues
- B. Combined OCP
- C. Hysterectomy with bilateral oophorectomy
- D. Bilateral cystectomy, adhesiolysis, prepare for IVF

► Answer & Explanation:

****D.** Bilateral cystectomy, adhesiolysis, and prepare for IVF** — Conservative surgery first; IVF for residual infertility. GnRH and OCP manage symptoms but do not restore fertility.

Q14 — Genital TB: most common site

Most common site of female genital TB:

- A. Fallopian tubes
- B. Endometrium
- C. Ovaries
- D. Vulva/vagina

► Answer & Explanation:

****A.** Fallopian tubes (~100%)** — Always involved. Endometrium 60–80%, ovaries 20–30%. Common cause of tubal infertility in India.

Q15 — HSG findings in genital TB (2016)

HSG findings suggestive of genital TB:

- A. Beaded tubes
- B. Honeycomb uterus
- C. Golf-club tube
- D. All of the above

► Answer & Explanation:

****D.** All of the above** — All three are pathognomonic HSG features of genital tuberculosis.

Q16 — Genital TB: clinical presentation

32-year-old with 6-month secondary amenorrhoea, chronic pelvic pain, intermittent fever, weight loss, adnexal mass. Most likely diagnosis:

- A. PCOS
- B. Endometriosis
- C. Genital tuberculosis
- D. Ectopic pregnancy

► Answer & Explanation:

****C. Genital tuberculosis**** — Constitutional features + secondary amenorrhoea + adnexal mass in endemic region. Endometrial biopsy (culture, PCR, histology) is the diagnostic gold standard.

Q17 — PID and infertility investigation

22-year-old with history of PID, 12 months infertility, normal examination. Next investigation:

- A. HSG
- B. AMH
- C. Hysteroscopy
- D. CA-125

► Answer & Explanation:

****A. HSG**** — Post-PID infertility = first assess tubal patency. HSG detects tubal blockage or hydrosalpinx.

Q18 — Ovarian reserve markers — NOT

All are markers of ovarian reserve EXCEPT:

- A. Inhibin A
- B. Estradiol (day 3)
- C. Inhibin B
- D. Antral follicle count / ovarian volume

► Answer & Explanation:

****A. Inhibin A**** — Inhibin A is mainly placental; inhibin B is the granulosa-cell reserve marker.

Q19 — Day 21 progesterone

Best day for serum progesterone evaluation in a 28-day cycle to confirm ovulation:

- A. Day 3
- B. Day 8
- C. Day 14
- D. Day 21

► Answer & Explanation:

****D. Day 21**** — Mid-luteal (7 days before next period). Level > 5 ng/mL confirms ovulation.

Q20 — PCOS: cyst formation mechanism

The cysts in PCOS are formed by:

- A. Failure of atretic follicles to undergo apoptosis
- B. Oocyte proliferation
- C. Multiple corpora lutea
- D. Cystic degeneration of the ovarian cortex

► Answer & Explanation:

****A. Failure of atretic follicles to undergo apoptosis**** — Elevated androgens inhibit follicular atresia; multiple small follicles persist in peripheral necklace pattern.

Q21 — Ovulation-to-implantation interval

After ovulation, embryo implantation occurs after approximately:

- A. 3–5 days
- B. 7–9 days
- C. 10–12 days
- D. 13–15 days

► Answer & Explanation:

****B. 7–9 days**** — Blastocyst hatches from zona (day 5–6) and implants into secretory endometrium day 7–9 post-ovulation.

Q22 — Secondary oocyte arrested in (2024)

After ovulation, the egg released is arrested at:

- A. Secondary oocyte arrested in meiosis I
- B. Secondary oocyte arrested in meiosis II
- C. Primary oocyte arrested in meiosis I
- D. Primary oocyte arrested in meiosis II

► Answer & Explanation:

****B. Secondary oocyte arrested in meiosis II**** — LH surge triggers completion of meiosis I (first polar body extruded) → secondary oocyte formed → arrested in metaphase II until fertilisation.

Q23 — Preimplantation genetic testing (2016)

Test used to detect genetic abnormality in embryo before transferring to uterus in IVF:

- A. Embryo cell biopsy
- B. CVS
- C. ICSI
- D. All of the above

► **Answer & Explanation:**

****A. Embryo cell biopsy**** — Preimplantation genetic testing (PGT) = trophectoderm biopsy (day 5) or blastomere biopsy (day 3). CVS is postnatal; ICSI is a fertilisation technique.

Q24 — OHSS: HCG withholding

25-year-old undergoing COS with 8 growing follicles and estradiol 800 pg/mL. Best management:

- A. Retrieve all follicles immediately
- B. Give cabergoline
- C. Cancel the cycle
- D. Withhold HCG trigger

► **Answer & Explanation:**

****D. Withhold HCG trigger**** — 8 follicles = high OHSS risk; withholding hCG (or switching to GnRH agonist trigger) prevents severe OHSS.

Q25 — Endometriosis: long-term sequela

23-year-old with dyspareunia, retroverted fixed uterus, adnexal + cervical motion tenderness. Most important long-term sequela:

- A. Recurrent pregnancy loss
- B. Infertility
- C. Pelvic organ prolapse
- D. Amenorrhoea

► **Answer & Explanation:**

****B. Infertility**** — Endometriosis causes infertility through adhesions, peritoneal toxins, and endometriomas. ~30–50% of infertile women have endometriosis.

Q26 — Endometriosis + infertility on exam (2016)

26-year-old obese infertile female: fixed retroverted uterus, nodular uterosacral ligaments. Most likely diagnosis:

- A. PCOS
- B. Endometriosis
- C. Adenomyosis

- D. TB

► **Answer & Explanation:**

****B. Endometriosis**** — Fixed retroverted uterus + nodular uterosacral ligaments = pathognomonic. Pouch of Douglas obliterated by adhesions.

Q27 — Hysteroscopy for Asherman (2023)

For which procedure can hysteroscopy preferably be done?

- A. Tubal ligation
- B. Asherman syndrome
- C. Endocervical polyp
- D. Subserosal fibroid

► **Answer & Explanation:**

****B. Asherman syndrome**** — Hysteroscopic adhesiolysis + oestrogen. Tubal ligation = laparoscopy; subserosal fibroid = laparoscopic myomectomy.

Q28 — Decreased tubal motility syndrome

Decreased motility/ciliary dysfunction of the fallopian tube is associated with:

- A. Noonan syndrome
- B. PCOS
- C. Churg-Strauss syndrome
- D. Kartagener's syndrome

► **Answer & Explanation:**

****D. Kartagener's syndrome**** — Primary ciliary dyskinesia; absent dynein arms; immotile cilia → bronchiectasis, situs inversus, male + female subfertility.

Q29 — Submucosal fibroid on hysteroscopy

28-year-old with menorrhagia, primary infertility, Hb 8; hysteroscopy shows round firm mass on stalk in uterine cavity. Diagnosis:

- A. Pedunculated subserosal fibroid
- B. Submucosal fibroid
- C. Carcinoma endometrium
- D. Gestational sac

► **Answer & Explanation:**

****B. Submucosal fibroid**** — Protrudes into uterine cavity; causes menorrhagia + infertility. FIGO types 0 and 1 treated hysteroscopically.

Q30 — Blastocyst formation timing

Blastocyst formation occurs after fertilisation on approximately:

- A. Day 4–7
- B. Day 10–12
- C. Day 12–15
- D. Day 15–20

► **Answer & Explanation:**

****A. Day 4–7**** — Morula (Day 3–4) → blastocyst by cavitation (Day 4–5) → hatching → implantation Day 6–7.

Q31 — AUB-O in obese anovulatory woman

35-year-old obese (BMI 34) with irregular heavy bleeding, simple endometrial hyperplasia without atypia, luteal progesterone < 3 ng/mL. AUB mechanism:

- A. AUB-L (leiomyoma)
- B. AUB-O (anovulatory cycles with unopposed oestrogen)
- C. AUB-C (coagulopathy)
- D. AUB-P (endometrial polyp)

► **Answer & Explanation:**

****B. AUB-O**** — Obesity → peripheral aromatisation → unopposed oestrogen → anovulatory bleeding + endometrial hyperplasia. Low luteal progesterone confirms anovulation.

Q32 — Premature ovarian failure: hormone profile

Which hormone is elevated in premature ovarian failure?

- A. Serum inhibin
- B. Serum FSH
- C. Serum estradiol
- D. Both A and B

► **Answer & Explanation:**

****B. Serum FSH**** — POI: FSH > 25–40 IU/L on two occasions; low estradiol, low AMH, low inhibin B.

Chapter 7 — AUB, Menstrual Disorders & Endocrine Gynaecology

"The uterus does not bleed in a vacuum — every abnormal bleed has a cause. Your job is to find the cause before prescribing a drug."

Big Picture First

Abnormal uterine bleeding (AUB) is one of the top three reasons women present to a gynaecology clinic. For NEET-PG and INI-CET, the examiner is testing one fundamental concept repeatedly: **before you label it DUB, you must rule out everything structural**. DUB (now subsumed under the PALM-COEIN framework) is a diagnosis of exclusion.

The 2011 FIGO classification — PALM-COEIN — organised AUB into structural causes (PALM: Polyp, Adenomyosis, Leiomyoma, Malignancy/hyperplasia) and non-structural causes (COEIN: Coagulopathy, Ovulatory, Endometrial, Iatrogenic, Not yet classified). Understanding where DUB sits in this framework is the single most important conceptual anchor for this chapter.

Equally central to this chapter is the endocrine axis governing menstruation — the HPO axis. When it misfires at any level (hypothalamus, pituitary, ovary, or outflow tract), bleeding and amenorrhoea result. The chapter takes you from the top of the axis to the uterine cavity, hitting every examiner favourite along the way.

Question Pattern Map

From past NEET-PG papers, AUB and endocrine gynaecology questions cluster around these themes:

Tested Concept	What the Examiner Wants
DUB definition	That it does NOT occur postmenopausally
Metropathia haemorrhagica	Amenorrhoea → heavy bleed; anovulatory mechanism
PALM-COEIN	Classification recognition; which category
Primary amenorrhoea	Most common cause = ovarian dysgenesis (Turner)
Secondary amenorrhoea	First investigation after pregnancy test = prolactin
Sheehan's syndrome	PPH → pituitary necrosis → panhypopituitarism
AIS (testicular feminisation)	46,XY + female phenotype + absent pubic hair
MRKH	46,XX + absent uterus + normal ovaries
Imperforate hymen	Bluish bulge at introitus; cyclical pain with no menarche
Granulosa cell tumour	Oestrogen-secreting → menorrhagia
Menorrhagia management	Perimenopausal → D&C first to exclude pathology



Core Concept Deep Dive

1. Normal Menstrual Parameters — The Baseline You Must Know

Before diagnosing any disorder, you need the normal range burned into memory.

Parameter	Normal Range	Abnormal Label
Cycle length	21–35 days	Polymenorrhoea (<21d); Oligomenorrhoea (>35d)
Duration of flow	3–7 days	Prolonged (>7d)
Blood loss	20–80 mL per cycle	Menorrhagia (>80 mL)
Pain	Minimal dysmenorrhoea acceptable	Dysmenorrhoea if disabling

Memory peg: "21–35 / 3–7 / 20–80" — three number pairs. Anything outside is abnormal.

Menorrhagia is objectively >80 mL but is clinically assessed by passage of clots >1 cm, soaking more than one pad per hour, or iron-deficiency anaemia.

2. PALM-COEIN Classification (FIGO 2011) — The Examiner's Favourite

The classification organises AUB causes into two groups:

PALM = **Structural** (visible on imaging or histology)

Letter	Cause	Key Feature
P	Polyp	Endometrial or cervical; premenopausal most common benign structural cause
A	Adenomyosis	Uterus uniformly enlarged; dysmenorrhoea + menorrhagia
L	Leiomyoma	Fibroids; submucosal fibroids most responsible for AUB
M	Malignancy/Hyperplasia	Must always exclude in perimenopausal/postmenopausal

COEIN = Non-structural (no imaging abnormality)

Letter	Cause	Key Feature
C	Coagulopathy	Von Willebrand disease commonest; always ask in teenagers
O	Ovulatory dysfunction	Classic DUB — anovulation; includes PCOS, thyroid, hyperprolactinaemia
E	Endometrial	Local endometrial defect — primary (no systemic cause found)
I	Iatrogenic	Anticoagulants, copper IUD, progestogen-only methods
N	Not classified	Arteriovenous malformation, myometrial hypertrophy

Memory Technique: "PALM trees COIN the diagnosis" - PALM = structural (you can see it) - COIN = non-structural (you can't see it)

3. DUB — The Diagnosis of Exclusion

DUB (dysfunctional uterine bleeding) = AUB with no structural, systemic, or iatrogenic cause. It is now classified under AUB-O (ovulatory dysfunction) in FIGO terminology but the old term appears extensively in exam questions.

Two critical rules for the exam: 1. DUB is a **diagnosis of exclusion** — only after ruling out PALM causes. 2. DUB does **NOT** occur in postmenopausal women. Postmenopausal bleeding = cancer until proven otherwise.

MCQ Alert — This exact question appears multiple times in NEET-PG:

Q415 / Q584 — DUB: Postmenopausal Exclusion

Dysfunctional uterine bleeding (DUB) is commonly seen in all EXCEPT:

- A. Adolescence
- B. Following childbirth
- C. Premenopausal period
- D. Postmenopausal period

► Answer & Explanation:

****D. Postmenopausal period**** — DUB is anovulatory by definition and does not occur after menopause; postmenopausal bleeding always requires investigation to exclude endometrial carcinoma.

Why this keeps appearing: The trap is that "following childbirth" sounds pathological — but the newly postpartum uterus is oestrogen-dominant, anovulatory, and can bleed dysfunctionally during the transition back to cycles. This is legitimate DUB. Postmenopausal bleeding is never DUB.

Q836 — Perimenopausal Menorrhagia: First Step

A 46-year-old P3L3 complains of menorrhagia for 3 months. Next line of management?

- A. D&C
- B. Progesterone x 6 months
- C. OC pills x 6 months
- D. Hysterectomy

► **Answer & Explanation:**

****A. D&C**** — In a perimenopausal woman with menorrhagia, D&C (dilatation and curettage) is performed first to exclude endometrial pathology (hyperplasia/carcinoma) before any medical therapy is started.

Examiner's logic: Age 46 is perimenopausal. The risk of endometrial hyperplasia and carcinoma is significant. You cannot prescribe progestogens or OCPs until malignancy is excluded. D&C provides both diagnosis and temporary treatment. Note: In modern practice, pipelle endometrial biopsy + hysteroscopy is increasingly preferred, but D&C remains the classic exam answer for this scenario.

4. Metropathia Haemorrhagica — The Classic Anovulatory Bleed

This is a specific clinical pattern that examiners love because it encapsulates the entire physiology of anovulation.

Pathophysiology chain: 1. Anovulatory cycle → no corpus luteum → no progesterone 2. Prolonged unopposed oestrogen → continuous endometrial proliferation 3. Endometrium becomes thickened, cystically hyperplastic ("Swiss cheese" pattern on histology) 4. Oestrogen level eventually falls (follicle atresia) 5. No organised shedding → irregular, heavy, prolonged bleeding after a period of amenorrhoea

Classic presentation: A premenopausal or perimenopausal woman with a period of weeks of amenorrhoea followed by sudden, profuse, irregular bleeding. This is the hallmark.

Histology: Cystic glandular hyperplasia (dilated glands in a sea of stroma — "Swiss cheese endometrium"). This is benign but distinguishable from complex atypical hyperplasia (which is premalignant).

Q420 / Q590 — Metropathia Haemorrhagica (appears twice)

Period of amenorrhoea followed by massive bleeding in a premenopausal woman is seen in:

- A. Irregular ripening
- B. Irregular shedding
- C. Metropathia haemorrhagica
- D. All of the above

► **Answer & Explanation:**

****C. Metropathia haemorrhagica**** — Prolonged oestrogenic stimulation without progesterone (anovulation) leads to cystic glandular hyperplasia; when the oestrogen level falls, the thickened hyperplastic endometrium sheds irregularly causing heavy breakthrough bleeding after amenorrhoea.

Option analysis: - Irregular ripening: defective corpus luteum → short luteal phase → shortened cycles with light bleeding. No preceding amenorrhoea. - Irregular shedding: prolonged corpus luteum activity → bleeding continues

days after expected cessation. The pattern is prolonged bleeding, not amenorrhoea then flood. - Metropathia: the only pattern with the signature amenorrhoea-then-flood sequence.

5. AUB: Structural Cause on Hysteroscopy

Q899 — AUB on Hysteroscopy

In a woman complaining of AUB, the following image was seen on endoscopic examination of the uterus. Diagnosis?

- A. Leiomyoma
- B. Adenomyosis
- C. Ovarian neoplasm
- D. Carcinoma of uterus

► Answer & Explanation:

****A. Leiomyoma**** — Hysteroscopy in AUB revealing a smooth, rounded, well-circumscribed intracavitary or submucosal mass is diagnostic of a leiomyoma (fibroid); submucosal fibroids are the most common structural cause of AUB identified on hysteroscopy.

Why adenomyosis is wrong: Adenomyosis is a myometrial disease — it cannot be seen directly at hysteroscopy (only irregular openings on endometrial surface, not a discrete mass). It is diagnosed by MRI or transvaginal ultrasound.

6. Ovarian Tumour Causing Menorrhagia

Q824 — Granulosa Cell Tumour

Which ovarian tumour can present with menorrhagia?

- A. Dermoid cyst
- B. Epithelial ovarian cancer
- C. Granulosa cell tumour
- D. Yolk sac tumour

► Answer & Explanation:

****C. Granulosa cell tumour**** — Granulosa cell tumour is an oestrogen-secreting sex-cord stromal tumour; excess oestrogen stimulates the endometrium causing menorrhagia in premenopausal women or postmenopausal bleeding in older women. The classic associated marker is inhibin B.

Complete picture: Granulosa cell tumours also cause endometrial hyperplasia and even endometrial carcinoma due to chronic unopposed oestrogen. In prepubertal girls, they cause isosexual precocious puberty. Call sign = inhibin B elevated on tumour marker testing.

7. Amenorrhoea — Framework and Investigation

Definition: - Primary amenorrhoea: No menstruation by age 14 with absent secondary sexual characteristics, OR by age 16 regardless of secondary sexual characteristics - **Secondary amenorrhoea:** Absence of periods for ≥ 3 months in a woman who previously had normal cycles (or ≥ 6 months if cycles were irregular)

Investigation ladder: 1. Pregnancy test (always first — never skip this) 2. Serum prolactin (hyperprolactinaemia = commonest correctable cause) 3. TSH (hypothyroidism raises TRH → elevates prolactin) 4. FSH/LH (to determine if hypo- or hypergonadotrophic) 5. Pelvic ultrasound (uterus, ovarian morphology) 6. Karyotype (if primary amenorrhoea with absent secondary characteristics)

Hypergonadotrophic hypogonadism (high FSH, high LH): - Ovarian failure: Turner syndrome, premature ovarian failure (POF), radiation - Streak gonads present in Turner

Hypogonadotrophic hypogonadism (low FSH, low LH): - Hypothalamic: weight loss, exercise, stress, Kallmann syndrome - Pituitary: Sheehan's syndrome, prolactinoma, haemochromatosis

Q480 — Primary Amenorrhoea Most Common Cause

Most common cause of primary amenorrhoea is:

- A. Constitutional/idiopathic
- B. MRKH syndrome
- C. Ovarian dysgenesis
- D. None of the above

► Answer & Explanation:

****C. Ovarian dysgenesis (Turner syndrome)**** — Ovarian dysgenesis (45,X Turner syndrome) is the most common single cause of primary amenorrhoea, accounting for ~30–40% of cases.

Q649 — Secondary Amenorrhoea: Initial Investigation

A 32-year-old with 6-month secondary amenorrhoea and a history of hypothyroidism; pregnancy test negative. Initial diagnostic test?

- A. Serum prolactin
- B. Serum FSH and LH
- C. Pelvic ultrasound
- D. Serum estradiol

► Answer & Explanation:

****A. Serum prolactin**** — Hypothyroidism raises TRH, which stimulates prolactin; hyperprolactinaemia inhibits GnRH → secondary amenorrhoea. Prolactin is the first-line test after excluding pregnancy.

The clinical logic: The history of hypothyroidism is a deliberate clue — it points you toward the TRH-prolactin-GnRH axis. But even without that clue, prolactin is the answer because hyperprolactinaemia is the commonest correctable cause of secondary amenorrhoea overall.

8. Turner Syndrome (45,X0) — The High-Yield Triad

Clinical triad: Short stature + primary amenorrhoea + webbed neck

Additional features: Shield chest, widely spaced nipples, cubitus valgus, horseshoe kidney, bicuspid aortic valve (cardiovascular complication), streak ovaries

Hormonal profile: High FSH, high LH (hypergonadotrophic hypogonadism) — the ovaries are streaks, producing no oestrogen, so no negative feedback

Q391 — Turner Syndrome Investigation

A 16-year-old girl presents with primary amenorrhoea, short stature, and webbed neck. Karyotype is 45,X0. Which investigation would show streak gonads on imaging?

- A. MRI of pituitary
- B. Pelvic ultrasound
- C. Laparoscopy
- D. CT scan of abdomen

► Answer & Explanation:

****B. Pelvic ultrasound**** — Pelvic ultrasound in Turner syndrome typically shows streak ovaries (fibrous bands replacing gonads), an infantile uterus, and absent follicles; serum FSH and LH are markedly elevated (hypergonadotrophic hypogonadism). Laparoscopy is definitive but ultrasound is first-line imaging.

9. MRKH Syndrome vs AIS vs Turner — The Three-Way Differential

This is a classic NEET-PG trap question scenario. Three conditions present with primary amenorrhoea — but the clues distinguish them clearly.

Feature	Turner (45,X0)	MRKH (46,XX)	AIS — CAIS (46,XY)
Karyotype	45,X0	46,XX	46,XY
Secondary sexual characteristics	Absent/poor	Normal female	Advanced (Tanner V breast)
Pubic/axillary hair	Reduced	Normal	Absent/sparse
Uterus	Absent (streak)	Absent (Müllerian aplasia)	Absent
Ovaries	Streak gonads	Normal ovaries	Testes (inguinal/intra-abdominal)
FSH/LH	High (hypergonadotrophic)	Normal	High after gonadectomy
Short stature	Yes	No	No

Q392 — MRKH Syndrome

A 17-year-old presents with primary amenorrhoea, normal secondary sexual characteristics, and a blind vaginal pouch of 2 cm. Ultrasound shows absent uterus with bilateral normal ovaries. Karyotype is 46,XX. This is:

- A. Complete androgen insensitivity syndrome (CAIS)
- B. Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome
- C. Swyer syndrome (46,XY pure gonadal dysgenesis)
- D. Imperforate hymen

► **Answer & Explanation:**

****B. Mayer-Rokitansky-Küster-Hauser (MRKH) syndrome**** — MRKH: absent uterus + upper vagina (Müllerian aplasia), 46,XX karyotype, normal ovaries, normal female secondary sexual characteristics. Distinguished from CAIS by karyotype (46,XX vs 46,XY) and from imperforate hymen by absent uterus.

Q393 — AIS: Long-term Management

A phenotypically female 18-year-old presents with primary amenorrhoea and absence of pubic/axillary hair. Karyotype is 46,XY. Testes are found in the inguinal canals. Most important long-term management concern?

- A. Cyclical oestrogen-progestogen HRT to induce menstruation
- B. Gonadectomy to prevent malignant transformation of intra-abdominal/inguinal testes
- C. Testosterone supplementation to masculinise
- D. Clomiphene to stimulate gonadotrophin release

► **Answer & Explanation:**

****B. Gonadectomy to prevent malignant transformation of intra-abdominal/inguinal testes**** — In complete AIS (46,XY), the testes have a 2–5% lifetime risk of malignant transformation (gonadoblastoma, dysgerminoma); gonadectomy is recommended after puberty (allowing puberty to complete with endogenous oestrogen from testosterone aromatisation), followed by oestrogen-only HRT.

Critical details: - Gonadectomy is done AFTER puberty completes — the endogenous oestrogen from aromatisation of testosterone in the testes drives feminisation (breast development, feminine habitus) - Post-gonadectomy: oestrogen-only HRT (no uterus present, so no progestogen needed) - Option A is wrong: there is no uterus to menstruate - Option C is the opposite of desired management

Q582 — Testicular Feminisation Syndrome

A girl with primary amenorrhoea, advanced breast development (Tanner V), minimal pubic hair (Tanner II), and absent axillary hair. Most likely diagnosis:

- A. Testicular feminisation syndrome (AIS)
- B. Müllerian agenesis (MRKH)
- C. Turner syndrome
- D. Gonadal dysgenesis

► **Answer & Explanation:**

****A. Testicular feminisation syndrome (AIS)**** — Complete AIS: 46,XY karyotype, normal female habitus with excellent breast development (Tanner V), absent/sparse pubic and axillary hair (the pathognomonic sign), blind vagina, no uterus.

The pathognomonic clue: Absent/sparse pubic and axillary hair in a well-feminised girl = AIS until proven otherwise. The hair follicles are androgen-dependent; without androgen receptors, pubic/axillary hair does not grow despite normal androgen levels.

10. Imperforate Hymen — Cryptomenorrhoea

Q513 / Q1001 — Imperforate Hymen

A 13-year-old girl presents with cyclical lower abdominal pain for 6 months and primary amenorrhoea. Examination reveals a tense bluish bulge at the introitus. Diagnosis?

- A. MRKH syndrome
- B. Imperforate hymen
- C. Asherman's syndrome
- D. Congenital adrenal hyperplasia

► Answer & Explanation:

****B. Imperforate hymen**** — Cryptomenorrhoea: haematocolpos behind an imperforate membrane. Bulging bluish hymen on inspection is pathognomonic. Treatment: cruciate incision and drainage.

Clinical logic — why not MRKH: - MRKH has no bulge at introitus (no haematocolpos forms because there is no outflow from the uterus at all — the uterus is absent) - Imperforate hymen: uterus PRESENT, outflow BLOCKED → blood accumulates (haematocolpos) → bluish bulge - The cyclical pain is the uterus contracting to expel blood with nowhere to go

Second 2023 version (Q1001): Suprapubic bulge + rectal bulge on PR exam — this is haematometra (blood in uterus) + haematocolpos, confirmed by clinical and PR findings.

11. Lactational Amenorrhoea

Q504 — Mechanism

Lactational amenorrhoea is due to:

- A. Prolactin-induced inhibition of GnRH
- B. Prolactin-induced inhibition of FSH
- C. Oxytocin-induced inhibition of GnRH
- D. Oxytocin-induced inhibition of FSH

► Answer & Explanation:

****A. Prolactin-induced inhibition of GnRH**** — Suckling → prolactin surge → suppression of hypothalamic GnRH → ↓ LH/FSH → anovulation. Effective contraception only if: <6 months postpartum, exclusively breastfeeding, amenorrhoeic (LAM method).

Why not oxytocin: Oxytocin is the milk ejection hormone — it acts on myoepithelial cells. It has no significant role in suppressing GnRH. Prolactin is the anti-reproductive hormone of lactation.

12. Ovarian Hyperstimulation Syndrome (OHSS)

Q623 — OHSS Clinical Scenario

A 26-year-old with delayed cycles triggered with hCG for follicular rupture develops, on day 19, dyspnoea, reduced urine output, abdominal bloating, and pain. Diagnosis:

- A. Theca lutein cysts

- B. Ovarian hyperstimulation syndrome (OHSS)
- C. Ruptured ectopic pregnancy
- D. Ruptured corpus luteum cyst

► **Answer & Explanation:**

****B. Ovarian hyperstimulation syndrome (OHSS)**** — OHSS typically appears 5–10 days after hCG trigger; features include massive ovarian enlargement, ascites, pleural effusion, oliguria, and haemoconcentration.

Distinguishing features: The timing (post-hCG trigger, delayed onset) and the combination of abdominal symptoms + respiratory symptoms + oliguria are the OHSS signature. Ruptured ectopic would present acutely with haemoperitoneum; corpus luteum rupture does not cause pleural effusion.

13. Menorrhagia: Drug Without Evidence

Q509 — Drugs for Menorrhagia

Use of which drug for menorrhagia is NOT supported by clinical evidence?

- A. Tranexamic acid
- B. Combined oral contraceptive pill
- C. Ethamsylate
- D. Progesterone-only pills (cyclical)

► **Answer & Explanation:**

****C. Ethamsylate**** — Studies have failed to demonstrate efficacy of ethamsylate in reducing menstrual blood loss. Tranexamic acid (antifibrinolytic), OCPs, and cyclical progestogens all have documented efficacy.

Memory aid — drugs that WORK for menorrhagia: - **Tranexamic acid** — antifibrinolytic; 30–58% reduction in blood loss - **NSAIDs (mefenamic acid)** — reduce prostaglandins; 25–35% reduction - **Cyclical progestogens** — for anovulatory DUB - **LNG-IUS (Mirena)** — most effective long-term (80% reduction) - **OCP** — reduces flow by 40–50% - **Ethamsylate** — theoretically reduces capillary fragility; NO good trial evidence

14. Risk Factors for Carcinoma Cervix

Q592 — Cervical Cancer Risk Factors

Risk factors for carcinoma cervix include: a) HPV b) Smoking c) Late menarche d) Nulliparity e) Early sexual intercourse

- A. a, b, e
- B. b, c, d
- C. c, d, e
- D. a, c, d

► **Answer & Explanation:**

****A. a, b, e (HPV, smoking, early sexual intercourse)**** — HPV infection (especially types 16 and 18), smoking, and early sexual activity are major risk factors. Late menarche and nulliparity are not risk factors.

Mnemonic — cervix cancer risk: "HEMP-S" - H — HPV (16, 18, 31, 33) - **E** — Early sexual intercourse (< age 16) - **M** — Multiple sexual partners - **P** — Parity (multiparous, multiple partners through deliveries) - **S** — Smoking (doubles risk independently of HPV)

Late menarche is protective (less cervical metaplasia during adolescent high-risk period). Nulliparity is actually not a risk factor — it is a protective factor for cervical cancer.

15. Chadwick's Sign — A Misplaced Question

Q705 — Chadwick's Sign Classification

2-month amenorrhoea, bluish vaginal discoloration (Chadwick's sign) — sign type?

- A. Presumptive signs
- B. Probable signs
- C. Positive signs
- D. Diagnostic signs

► Answer & Explanation:

****B. Probable signs**** — Chadwick's sign (bluish discoloration of vagina/cervix due to vascular congestion) is a probable sign of pregnancy. Presumptive = missed period, nausea. Positive = FHR heard, US visualisation.

This question appears in amenorrhoea/endocrine sections because Chadwick's sign is a CAUSE of secondary amenorrhoea (pregnancy). Remember the hierarchy: Presumptive (subjective) → Probable (objective but not conclusive) → Positive (conclusive).

Bias Removal Section

Common Confusions

Metropathia haemorrhagica vs Endometrial Hyperplasia vs Carcinoma: | Feature | Metropathia | Simple Hyperplasia | Atypical Hyperplasia | Carcinoma | ---|---|---|---|---| | Pattern | Cystic glandular hyperplasia | Glandular crowding without atypia | Atypia present | Invasion | | Age | Perimenopausal | Any reproductive age | Perimenopausal | Postmenopausal usually | | Malignant risk | Nil (benign) | <1% | 29% → cancer | Already malignant | | Treatment | Progestogens, D&C | Progestogens | Hysterectomy | Surgery ± RT |

DUB vs PMB: - DUB = reproductive age + anovulatory + no structural cause - PMB (postmenopausal bleeding) = ANY bleeding >12 months after last period → endometrial cancer until proven otherwise → ALWAYS do endometrial sampling

Hypothalamic vs Pituitary amenorrhoea: - Both are hypogonadotrophic (low FSH/LH) - Distinguish with GnRH stimulation test: pituitary responds to GnRH (levels rise); hypothalamic cause = pituitary responds to exogenous GnRH (pituitary is intact, the problem is upstream)

Variations and Twists

Scenario: Teenager with menorrhagia from day 1 of menarche Think coagulopathy (AUB-C category) — Von Willebrand disease, thrombocytopaenia. The PALM causes are unlikely in a young teen with no history of procedures. Always ask about bleeding from other sites (gums, easy bruising).

Scenario: Athlete with secondary amenorrhoea, low BMI This is the female athlete triad: low energy availability + bone stress injury + hypothalamic amenorrhoea. FSH and LH will both be low (hypogonadotrophic). Treatment = weight gain and reduced training intensity; not hormonal replacement alone.

PCOS trap in amenorrhoea: Rotterdam criteria for PCOS diagnosis (need 2 of 3): 1. Oligo/anovulation (irregular cycles) 2. Clinical or biochemical hyperandrogenism (acne, hirsutism; elevated testosterone) 3. Polycystic ovarian morphology on ultrasound (≥ 12 follicles 2–9mm in one ovary OR ovarian volume > 10 mL)

Trap question: LH:FSH ratio elevated in PCOS is a classic exam point (typically $> 2:1$ or $> 3:1$ in classic PCOS), BUT it is NOT part of the Rotterdam criteria and is NOT required for diagnosis. Many exams try to make you choose LH:FSH as a diagnostic criterion — it is not.

Sheehan's syndrome: - Cause: Massive PPH → anterior pituitary infarction (pituitary enlarges during pregnancy, making it vulnerable to ischaemia) - Sequence of loss: Growth hormone (first, most sensitive) → LH/FSH → TSH → ACTH (last) - Clinical triad: Failure to lactate + no return of menstruation + cold intolerance/fatigue - Prolactin is low (not high) — this distinguishes it from prolactinoma

Memory Engineering

PALM-COEIN: "PALM trees need COINS to grow" - PALM = structural (you can PALM them — feel them) - COEIN = non-structural (coin-sized problems you cannot feel)

Amenorrhoea investigation order: "P-P-T-G-K" 1. **P**regnancy test 2. **P**rolactin 3. **T**SH 4. **G**onadotrophins (FSH/LH) 5. **K**aryotype (if primary, with absent secondary characteristics)

Turner vs AIS memory: - Turner: "TurNER — No Egg Reserves" → hypergonadotrophic - AIS: "AIS — Absent Inguinal Sensitivity" → testes in inguinal canal, no androgen response

Imperforate hymen: "Bluish Bulge Before Bleeding" — blood backs up, bulges blue at introitus

Metropathia haemorrhagica: "No Progesterone → No Stop Signal → Floods after Drought"

Ultra High-Yield Revision Layer

Normal Menstrual Parameters Quick Reference

- Cycle: 21–35 days | Duration: 3–7 days | Loss: 20–80 mL

PALM-COEIN Quick Reference

- Polyp | Adenomyosis | Leiomyoma | Malignancy
- Coagulopathy | Ovulatory | Endometrial | Iatrogenic | Not classified

Primary Amenorrhoea — Top 3 Causes

1. Ovarian dysgenesis (Turner) — 30–40% (most common)
2. MRKH syndrome — 15%
3. Imperforate hymen — common outflow tract cause

Secondary Amenorrhoea — Investigation Priority

1. Pregnancy test → 2. Prolactin → 3. TSH → 4. FSH/LH → 5. USS

AUB-O (DUB) Key Rules

- Diagnosis of exclusion
- Does NOT occur postmenopausally
- Metropathia = amenorrhoea then flood
- Perimenopausal menorrhagia → D&C first

Drugs for Menorrhagia — Evidence Table

Drug	Evidence	Mechanism
Tranexamic acid	Strong	Antifibrinolytic
LNG-IUS	Strongest (long-term)	Local progestogen
NSAIDs	Good	↓ Prostaglandins
OCP	Good	↓ Endometrial growth
Cyclical progestogens	Moderate	Oppose oestrogen
Ethamsylate	NONE	—

Chapter 7 covers 20 questions from the NEET-PG question bank. All concepts tested above have appeared in previous papers from 2015–2024.

Chapter 8 — Menopause, HRT & Pelvic Floor Disorders

"The pelvis supports not just the uterus but the entire quality of a woman's later life. When it fails — whether through hormonal withdrawal or structural collapse — understanding the mechanism tells you exactly how to fix it."

Big Picture First

This chapter covers two seemingly distinct areas — the hormonal transition of menopause and the mechanical failures of the pelvic floor — but they are linked by one physiological thread: **oestrogen withdrawal**. Oestrogen maintains the epithelium, the bone, the vascular wall, and critically, the collagen and fascial support of the pelvic floor. When oestrogen falls, everything falls.

For NEET-PG and INI-CET, this chapter generates a reliable cluster of questions every year:

Menopause/HRT questions almost always test the same three facts: 1. HRT is NOT indicated for coronary artery disease prevention (WHI trial — the most high-yield clinical trial in this chapter) 2. Oestrogen-only HRT is used only after hysterectomy; intact uterus always requires combined therapy 3. Raloxifene does NOT treat hot flushes (it may worsen them)

Pelvic floor questions cluster around: 1. Incontinence type identification from clinical description 2. Prolapse grade and its surgical management 3. Fistula aetiology in India vs developed countries 4. Asherman's syndrome — cause, investigation, treatment



Question Pattern Map

Tested Concept	Exam Pattern
HRT not for CAD	"HRT useful in all EXCEPT..." (coronary heart disease) — appears 3+ times
Menopause hormone profile	FSH↑↑, LH↑, oestrogen↓ (high FSH + high LH)
Hot flush mechanism	Increased noradrenaline + decreased oestrogen
Raloxifene and hot flushes	Does NOT help; may worsen
Prolapse complete family	Vaginal hysterectomy + pelvic floor repair
Ring pessary verification	Remains in situ after Valsalva
Mackenrodt's ligament	Primary support damaged in prolapse
Asherman's syndrome	Amenorrhoea after D&C; hysteroscopy gold standard
Asherman's highest risk	D&C for postpartum haemorrhage
Cystocele location	Upper 2/3 anterior vaginal wall
Rectovaginal fistula initial	Diversion colostomy first
VVF post-op advice	Sexual abstinence 3 months + avoid pregnancy 1 year



Core Concept Deep Dive

1. Menopause — Physiology and Definition

Definition: Menopause is the **permanent cessation of menstruation** resulting from loss of ovarian follicular activity. It is diagnosed clinically as **12 consecutive months of amenorrhoea** with no other pathological or physiological cause (e.g., not pregnancy, not breastfeeding).

Key numbers to commit: - Average age of menopause: 51 years (Indian data: 45–47 years) - Premature ovarian failure (POF)/premature ovarian insufficiency (POI): menopause before age 40 - Premature menopause: before age 45

Perimenopause: The transition period beginning with irregular cycles and ending 12 months after the final menstrual period. Can last 2–8 years. Vasomotor symptoms are most prominent during this phase.

The hormonal cascade of menopause:

The ovary runs out of responsive follicles. Here is the chain: 1. Follicular depletion → ↓ Inhibin B (first to fall, years before menopause) 2. ↓ Inhibin B → ↑ FSH (FSH starts rising as early as 5–10 years before menopause — FSH is the earliest hormonal marker) 3. Eventually ↓ oestradiol → ↑ LH as well 4. Final state: FSH >30–40 IU/L, LH elevated, oestradiol <30 pmol/L

FSH is the most sensitive single test — it rises earliest and most markedly. LH also rises but less dramatically.

Q662 — Menopause: Hormone Profile

A 52-year-old woman presents with 16 months of amenorrhoea. Expected hormone profile?

- A. High LH and high FSH
- B. Low FSH and low LH
- C. Low FSH and normal LH
- D. High FSH and low LH

► **Answer & Explanation:**

****A. High LH and high FSH**** — Ovarian failure → loss of oestrogen/inhibin feedback → pituitary releases high FSH (>30–40 IU/L) and LH. FSH is the most sensitive single marker of menopause.

Why D is wrong: High FSH with LOW LH would suggest a selective problem — but in true ovarian failure, both gonadotrophins are elevated because both receive inhibitory feedback from oestrogen and inhibin. If only FSH were high with low LH, think inhibin B deficiency in isolation (not a standard clinical scenario tested at NEET-PG).

Q754 — Premature Ovarian Failure

Which is increased in premature ovarian failure?

- A. Serum Inhibin
- B. Serum FSH
- C. Serum Estradiol
- D. Both A and B

► **Answer & Explanation:**

****B. Serum FSH**** — POF shows hypergonadotrophic hypogonadism: FSH rises (>40 IU/L) due to loss of negative feedback from depleted follicles producing low estradiol and inhibin.

Pattern recognition: Whenever you see "premature ovarian failure" in a question — think hypergonadotrophic hypogonadism, elevated FSH, the hormonal profile identical to surgical menopause or radiation-induced menopause but occurring before age 40.

2. Symptoms of Menopause

Vasomotor symptoms (hot flushes, night sweats): - Affect 75–80% of menopausal women - Most severe in the first 1–2 years after menopause - Mechanism: oestrogen withdrawal → decreased hypothalamic endorphins → central norepinephrine and serotonin dysregulation → narrowed thermoregulatory set point → small temperature rises trigger disproportionate heat dissipation

Q709 — Hot Flush Mechanism

Hot flushes are experienced as a result of:

- A. Increased noradrenaline alone
- B. Decreased oestrogen alone
- C. Increased noradrenaline and decreased oestrogen
- D. Increased noradrenaline and oestrogen

► Answer & Explanation:

****C. Increased noradrenaline and decreased oestrogen**** — Oestrogen deficiency reduces hypothalamic endorphins, increasing norepinephrine and serotonin, which disturb the hypothalamic thermoregulatory set-point.

Why not B alone: Decreased oestrogen is the initial trigger, but the immediate pathophysiology involves increased noradrenaline acting on the thermoregulatory centre. Both components are part of the mechanism. Examiner is specifically testing whether you know the neurotransmitter pathway.

Complete symptom overview for examination:

Domain	Symptoms
Vasomotor	Hot flushes, night sweats, palpitations
Urogenital	Vaginal dryness, dyspareunia, urinary frequency, recurrent UTIs
Psychological	Mood changes, sleep disturbance, poor concentration
Skeletal	Accelerated bone loss → osteoporosis (most rapid in first 5 years post-menopause)
Cardiovascular	Increased CVD risk (after menopause)
Skin/hair	Skin thinning, hair loss

3. Hormone Replacement Therapy — The Most Tested Topic

HRT is oestrogen administered to replace the ovarian oestrogen lost at menopause. The choice of regimen depends on whether the uterus is present.

Critical decision rule: - **Intact uterus** → **Combined oestrogen + progestogen HRT** (progestogen protects endometrium from oestrogen-driven hyperplasia/carcinoma) - **Post-hysterectomy** → **Oestrogen-only HRT** (no endometrium to protect; adding unnecessary progestogen increases breast cancer risk)

Q386 — HRT and Endometrial Protection

A 53-year-old woman with an intact uterus requires HRT for severe menopausal symptoms. Why must progestogen be added to oestrogen therapy?

- A. To prevent cardiovascular events
- B. To protect against oestrogen-induced endometrial hyperplasia and cancer
- C. To prevent breast cancer from oestrogen
- D. To improve mood and libido

► Answer & Explanation:

****B. To protect against oestrogen-induced endometrial hyperplasia and cancer**** — Unopposed oestrogen stimulates endometrial proliferation, increasing endometrial hyperplasia and carcinoma risk; progestogen (continuous or cyclical) is mandatory in women with an intact uterus. Women after hysterectomy can take oestrogen-only HRT.

HRT Indications — Confirmed by Evidence: 1. Vasomotor symptoms (hot flushes, night sweats) — MOST effective treatment 2. Genitourinary syndrome of menopause (vaginal atrophy, urinary symptoms) 3. Osteoporosis prevention (especially in women <60 or within 10 years of menopause) 4. Premature ovarian failure/premature menopause (HRT until age 51 as physiological replacement)

WHI Trial (Women's Health Initiative) — The Most Important Clinical Trial Here: The WHI trial (2002) changed practice globally. Key findings: - Combined E+P HRT → increased breast cancer, MI, stroke, VTE - Oestrogen-only HRT → decreased breast cancer (but still increased stroke/VTE) - **HRT does NOT prevent coronary artery disease** — this is the most repeatedly examined WHI finding

The "Timing Hypothesis" (nuance for advanced questions): - WHI studied women who were >10 years post-menopause when starting HRT - In younger women (<60 years, within 10 years of menopause), HRT may reduce CVD risk - This is the basis for the "window of opportunity" — but for exams, the tested fact remains: HRT is NOT recommended for CAD prevention

Q409 / Q541 / Q579 / Q663 — HRT Not for CAD (appears 4 times)

HRT is useful in all of the following EXCEPT:

- A. Vasomotor symptoms (hot flushes)
- B. Osteoporosis prevention
- C. Vaginal atrophy
- D. Coronary heart disease prevention

► Answer & Explanation:

****D. Coronary heart disease prevention**** — Current evidence (WHI trial) shows HRT does NOT reduce cardiovascular events in older postmenopausal women. HRT relieves vasomotor symptoms, protects bone, and reverses urogenital atrophy.

Examiner's pattern: This exact question with these exact options has appeared at least four times across NEET-PG 2015, 2016, PrepLadder, and PG Blazer. Answer is always D. Do not overthink it. The WHI trial data is definitive for the exam context.

Q498 — Vulvovaginal Atrophy Treatment

Drug of choice for symptomatic vulvovaginal atrophy (atrophic vaginitis) in menopause?

- A. Topical oestrogen cream
- B. Systemic antihistamines
- C. Tamoxifen
- D. Systemic HRT only

► Answer & Explanation:

****A. Topical oestrogen cream**** — Local oestrogen (cream/pessary/ring) relieves vulvovaginal atrophy with minimal systemic absorption. Tamoxifen is anti-oestrogenic and worsens vaginal dryness.

Why topical, not systemic: Topical oestrogen acts locally on the urogenital epithelium. It has minimal systemic absorption, meaning it can be used even in women where systemic HRT is contraindicated (e.g., those with previous breast cancer, in consultation with oncology). Tamoxifen is a selective oestrogen receptor modulator (SERM) — it is anti-oestrogenic in breast and uterine tissue but paradoxically worsens vaginal dryness.

4. Raloxifene — The Hot Flush Trap

Q527 — Treatment of Hot Flushes: Raloxifene

Which does NOT help decrease hot flushes in menopausal women?

- A. Androgens
- B. Raloxifene
- C. Isoflavones (phytoestrogens)
- D. Tibolone

► Answer & Explanation:

****B. Raloxifene**** — Raloxifene (SERM) protects bone and breast/endometrium but does NOT relieve vasomotor symptoms and may worsen hot flushes. Tibolone, isoflavones, androgens, and HRT all reduce flushes.

Raloxifene's profile: - Acts as oestrogen agonist in bone → prevents osteoporosis - Acts as oestrogen antagonist in breast and endometrium → reduces breast cancer risk (STAR trial) - Acts as oestrogen antagonist in hypothalamus → WORSENS hot flushes - Does NOT relieve vaginal dryness

This is why raloxifene is used for osteoporosis prevention (not vasomotor symptoms) in postmenopausal women who cannot take HRT. It is NOT HRT.

Non-hormonal options for hot flushes: Venlafaxine (SNRI), clonidine, gabapentin, phytoestrogens (isoflavones), cognitive behavioural therapy.

5. Lichen Sclerosus — Postmenopausal Vulval Condition

Q395 — Lichen Sclerosus Treatment

A 60-year-old post-menopausal woman presents with severe vulval pruritus and white, parchment-like atrophic skin of the vulva in a "figure-of-8" distribution around the vulva and perianal area. Biopsy shows epidermal thinning and homogenised dermis. Treatment of choice?

- A. Topical oestrogen cream
- B. High-potency topical corticosteroid (clobetasol propionate 0.05%)
- C. Topical antifungal (clotrimazole)
- D. Vulvectomy

► Answer & Explanation:

****B. High-potency topical corticosteroid (clobetasol propionate 0.05%)**** — Lichen sclerosus is treated with ultra-potent topical corticosteroids (clobetasol propionate); topical oestrogen is ineffective as it is NOT an oestrogenic condition. Long-term follow-up is needed as LS carries ~5% risk of squamous cell carcinoma of the vulva.

Critical trap: Lichen sclerosus looks like it should respond to oestrogen (white atrophic postmenopausal skin) — but it is an autoimmune condition, NOT caused by oestrogen deficiency. Topical oestrogen does nothing for lichen sclerosus. The treatment is immunosuppressive — ultra-potent topical steroid.

Figure-of-8 distribution (involving both vulva and perianal area) is pathognomonic of lichen sclerosus. Candida vulvovaginitis would have a cheesy discharge; atrophic vaginitis responds to topical oestrogen.

Long-term risk: ~5% risk of vulval squamous cell carcinoma — these patients need indefinite surveillance.

6. Urinary Incontinence — Clinical Differentiation

Incontinence questions in NEET-PG require you to identify the TYPE from a clinical scenario. The types are:

Type	Mechanism	Clinical Feature	Treatment
Stress (GSI)	Urethral sphincter failure / pelvic floor laxity	Leak on coughing, sneezing, exercise; no urgency	Pelvic floor exercises, TVT/TOT surgery
Urge (OAB)	Detrusor overactivity	Strong urge, cannot defer, frequency, nocturia	Anticholinergics (oxybutynin), bladder training
Overflow	Detrusor underactivity / outflow obstruction	Dribbling, incomplete emptying, distended bladder	Catheterisation, treat obstruction
Fistula	Anatomical defect	Continuous leakage day and night	Surgical repair
Mixed	Combined stress + urge	Features of both	Treat dominant component first

The key clinical differentiators: - Stress: leak WITH physical stress, NO urgency, NO frequency - Urge: leak PRECEDED by urgency, increased frequency, nocturia - Overflow: constant dribble, large post-void residual, no urge component - Fistula: CONTINUOUS leakage, even lying down — this is the distinguishing feature

Q398 — Stress Urinary Incontinence Mechanism

A 45-year-old multiparous woman leaks urine on coughing, sneezing, and exercise. She denies urgency or frequency. Urodynamics shows involuntary urine loss with increased intra-abdominal pressure without detrusor contraction. The pathophysiological mechanism is:

- A. Detrusor overactivity
- B. Inadequate urethral closure mechanism (urethral sphincter deficiency/hypermobility)
- C. Vesicovaginal fistula
- D. Overflow incontinence from detrusor underactivity

► Answer & Explanation:

****B. Inadequate urethral closure mechanism (urethral sphincter deficiency/hypermobility)**** — Stress urinary incontinence results from failure of the urethral closure pressure to exceed bladder pressure during physical stress; caused by pelvic floor weakness, urethral hypermobility, or intrinsic sphincter deficiency (ISD) — not detrusor overactivity (which causes urge incontinence).

Urodynamics insight: The gold standard for diagnosing incontinence type is urodynamic study. Stress incontinence shows urine loss during provocation WITHOUT a detrusor contraction. Urge incontinence (detrusor overactivity) shows an involuntary detrusor contraction coinciding with the urge.

Q399 — Urgency Incontinence: First-Line Drug

A 55-year-old woman complains of sudden, compelling urge to void followed by involuntary urine leakage. She voids 12 times per day and twice at night. Urodynamics confirms detrusor overactivity. First-line pharmacological treatment?

- A. Duloxetine (SNRI)
- B. Oxybutynin (anticholinergic)
- C. Mirabegron (beta-3 adrenergic agonist)
- D. Topical oestrogen vaginal cream

► **Answer & Explanation:**

****B. Oxybutynin (anticholinergic)**** — Anticholinergics (oxybutynin, solifenacin, tolterodine) are first-line pharmacological treatment for overactive bladder/detrusor overactivity; they reduce detrusor contractions by blocking muscarinic receptors. Mirabegron is an alternative with fewer anticholinergic side effects.

Drug mechanism quick reference: - Oxybutynin: M2/M3 muscarinic receptor blocker → relaxes detrusor - Duloxetine: SNRI → used for STRESS incontinence (enhances urethral sphincter tone via pudendal nerve), NOT urge - Mirabegron: β_3 agonist → detrusor relaxation; used when anticholinergics are contraindicated (glaucoma, cognitive impairment in elderly)

7. Uterovaginal Prolapse

Anatomy of pelvic support:

Three levels of pelvic support (DeLancey levels): - **Level I:** Apex/uterus — cardinal and uterosacral ligaments - **Level II:** Mid-vagina — pubocervical fascia (anterior), rectovaginal fascia (posterior) - **Level III:** Distal — urogenital diaphragm, perineal body

Primary supporting ligament of the uterus: Mackenrodt's (cardinal/transverse cervical) ligament

Q959 — Primary Damaged Ligament in Prolapse

P2L2 with uterine prolapse — which ligament is primarily damaged?

- A. Sacrospinous ligament
- B. Mackenrodt's (cardinal/transverse cervical) ligament
- C. Broad ligament
- D. Round ligament

► **Answer & Explanation:**

****B. Mackenrodt's (cardinal) ligament**** — Primary support of uterus at cervical level. Combined with uterosacral ligaments. Damage in multiparity → uterovaginal prolapse.

Ligament function summary: - Cardinal = primary cervical support (horizontal) - Uterosacral = posterior suspension (keeps cervix pulled posteriorly) - Round ligament = maintains anteversion (NOT a support ligament for prolapse) - Broad ligament = NOT a true ligament; it is a peritoneal fold

Classification of Uterovaginal Prolapse:

Classical (UK) grading: - 1st degree: Cervix descends but remains within vagina - 2nd degree: Cervix reaches introitus - 3rd degree (Procidentia): Uterus entirely outside vagina (the entire uterus protrudes)

Compartments of prolapse: - Anterior wall prolapse: Cystocele (upper 2/3) + Urethrocele (lower 1/3) - Posterior wall prolapse: Rectocele (lower 2/3) + Enterocele (upper posterior) - Apical: Uterine prolapse / vault prolapse post-hysterectomy

Q878 — Cystocele Location

Upper two-thirds anterior vaginal wall prolapse is?

- A. Cystocele
- B. Urethrocele
- C. Rectocele
- D. Enterocele

► Answer & Explanation:

****A. Cystocele**** — The upper two-thirds of the anterior vaginal wall is supported by the pubocervical fascia overlying the bladder base; weakness here allows the bladder to herniate, forming a cystocele.

Anatomical aid: "Cystocele = upper anterior (bladder herniates). Urethrocele = lower anterior (urethra protrudes)." In clinical practice, they often coexist as cystourethrocele.

Q804 — Cystocele Treatment Options

Which of the following are treatment options for cystocele?

- A. Anterior colporrhaphy
- B. Transvaginal tape
- C. Transobturator tape
- D. All of the above

► Answer & Explanation:

****D. All of the above**** — Cystocele is treated by anterior colporrhaphy (surgical repair), transvaginal tape (TVT), or transobturator tape (TOT) depending on associated stress urinary incontinence.

Surgical management of complete prolapse:

Q474 / Q745 / Q922 — Complete Prolapse in Completed Family (appears 3 times)

Preferred treatment of complete uterine prolapse in a female with completed family is:

- A. Sling surgery
- B. Vaginal hysterectomy
- C. Le Forte's repair
- D. Pessary

► **Answer & Explanation:**

****B. Vaginal hysterectomy**** — Vaginal hysterectomy (usually with anterior and posterior colporrhaphy) is the definitive surgical option for major degree uterine prolapse in a parous woman with a completed family.

Management matrix:

Patient	Management
Completed family, fit for surgery	Vaginal hysterectomy + pelvic floor repair
Completed family, elderly, fit	Vaginal hysterectomy + repair (same)
Completed family, unfit for surgery	Ring pessary
Desirous of fertility	Pelvic floor repair (Manchester repair) — uterus preserved
Very elderly, medically unfit	Le Fort's colpocleisis (obliterates vagina) — only if no desire for intercourse

Le Fort's repair: A colpocleisis — vagina is partially closed off. Used only in elderly medically unfit women who are not sexually active. The answer "Le Fort's" is a trap — it sounds surgical and definitive but is NOT the answer for a completed-family standard patient.

Q891 / Q909 — Ring Pessary Verification (appears twice)

In uterine prolapse, how do you know the ring pessary is in place?

- A. If not expelled after increased abdominal pressure
- B. If bleeding does not occur
- C. If the patient feels discomfort
- D. None of the above

► **Answer & Explanation:**

****A. If not expelled after increased abdominal pressure**** — A correctly placed ring pessary should remain in situ even with Valsalva manoeuvre/increased intra-abdominal pressure.

8. Vesicovaginal and Rectovaginal Fistulae

Aetiology in India vs Developed Countries:

Country	Most Common Cause of VVF
India / Sub-Saharan Africa	Obstetric — prolonged obstructed labour causing pressure necrosis
Developed countries	Surgical — post-hysterectomy (most common), post-pelvic radiation

For NEET-PG in India: **obstetric VVF** is the most common cause of vesicovaginal fistula.

Clinical presentation of VVF: Continuous, painless, uncontrollable leakage of urine through the vagina — even while sleeping (this distinguishes it from incontinence, which has some voluntary control component).

Q968 / Q975 — VVF Post-operative Advice (appears twice)

While discharging a patient after vesicovaginal fistula repair, what would you recommend?

- A. Sexual abstinence for 3 months and avoid pregnancy for a year

- B. Sexual abstinence for 3 weeks and avoid pregnancy for 6 months
- C. Sexual abstinence for 6 weeks and avoid pregnancy for a year
- D. Sexual abstinence for 6 months and avoid pregnancy for 6 years

► **Answer & Explanation:**

****A. Sexual abstinence for 3 months and avoid pregnancy for a year**** — After VVF repair, prolonged abstinence and delay of pregnancy allow adequate healing and reduce recurrence risk.

Memory peg: "3 months REST, 1 year BABY-WAIT" - 3 months abstinence (coital pressure on repair site) - 1 year before next pregnancy (uterine scar healing + VVF repair healing)

Q526 — Rectovaginal Fistula: Initial Management

Patients with rectovaginal fistula should be initially treated with:

- A. Colostomy
- B. Primary repair
- C. Posterior colporrhaphy
- D. Anterior resection

► **Answer & Explanation:**

****A. Colostomy**** — Initial diversion colostomy reduces inflammation and faecal contamination, improving surgical outcomes of definitive repair. Direct repair without diversion has higher failure rates for most rectovaginal fistulas.

Principle: Faecal contamination at a fresh fistula site causes infection and repair breakdown. The sequence is: 1. Diversion colostomy → allows fistula track to become clean, inflammation to settle 2. Definitive repair of rectovaginal fistula (after 3–6 months) 3. Closure of colostomy

For VVF: urinary diversion (catheter) serves the analogous purpose — Foley catheter drainage allows some small VVFs to close spontaneously, and larger ones to be repaired in optimal conditions.

Boari flap: A bladder flap mobilisation technique used to bridge a defect in the lower third of the ureter when direct uretero-ureteral anastomosis is not possible. It is used in uretero-vaginal fistula repair or ureteric injuries.

Remember: Boari = bladder mobilised to reach ureter. Not used for VVF directly.

9. Asherman's Syndrome — Intrauterine Adhesions

Definition: Intrauterine adhesions (synechiae) caused by trauma to the endometrium, resulting in partial or complete obliteration of the uterine cavity.

Q490 — Asherman Syndrome Characteristic

Asherman syndrome is characterised by:

- A. Amenorrhoea
- B. Menorrhagia
- C. Polymenorrhoea
- D. All of the above

► **Answer & Explanation:**

****A. Amenorrhoea**** — Asherman syndrome (intrauterine adhesions/synechiae) following uterine trauma (D&C, infection) causes amenorrhoea or hypomenorrhoea due to obliteration of the uterine cavity.

Full clinical picture: - Secondary amenorrhoea or markedly reduced menstrual flow (hypomenorrhoea) after a uterine procedure - Cyclical pelvic pain (the uterus contracts but blood cannot escape — similar mechanism to imperforate hymen but acquired) - Infertility / recurrent miscarriage (the endometrium cannot support implantation) - Diagnosis: Hysteroscopy (GOLD STANDARD — directly visualises adhesions) - Saline infusion sonography and HSG can show filling defects but hysteroscopy is definitive

Q695 — Asherman Syndrome: Highest Risk Procedure

The risk of Asherman syndrome is highest when dilatation and curettage is performed for:

- A. Medical termination of pregnancy
- B. Missed abortion
- C. Dysfunctional uterine bleeding
- D. Postpartum haemorrhage

► **Answer & Explanation:**

****D. Postpartum haemorrhage**** — D&C on an infected/necrotic postpartum uterus carries the highest risk of Asherman syndrome. Curettage at 2–4 weeks postpartum (fibrotic uterus) is particularly dangerous. Missed abortion is also a significant risk.

Why postpartum D&C is most dangerous: The postpartum uterus is highly vascular, the decidua is thin, and the myometrium is friable. D&C at 2–4 weeks postpartum — when the uterus is involuting and the endometrium is regenerating — leads to raw opposing endometrial surfaces that adhere as they heal. Combined with any superimposed infection (endometritis), scar formation is rapid and severe.

Q390 — Asherman Syndrome: Definitive Treatment

A 28-year-old woman with secondary amenorrhoea following suction evacuation undergoes hysteroscopy, revealing intrauterine synechiae obliterating the cavity. Definitive treatment?

- A. Combined oral contraceptive pill for 3 months
- B. GnRH agonist followed by add-back therapy
- C. Hysteroscopic adhesiolysis followed by high-dose oestrogen and intrauterine device/balloon
- D. IVF with donor embryos

► **Answer & Explanation:**

****C. Hysteroscopic adhesiolysis followed by high-dose oestrogen and intrauterine device/balloon**** — Treatment of Asherman's syndrome is hysteroscopic division of adhesions; post-operative high-dose oestrogen ($\pm$ intrauterine device or Foley catheter balloon) prevents re-formation of adhesions and promotes endometrial regeneration.

Why each part of the treatment is necessary: - Hysteroscopic adhesiolysis: removes the adhesions (mechanical) - High-dose oestrogen post-operatively: stimulates endometrial regeneration, prevents re-adhesion - Intrauterine device / Foley balloon: physical separator — prevents the raw surfaces from touching and re-adhering while they

heal under oestrogen stimulus - After 1–3 cycles of cyclical oestrogen-progestogen (to shed regenerated endometrium), the IUD is removed

Bias Removal Section

HRT Contraindications — Absolute vs Relative

Absolute contraindications to HRT: - Oestrogen-dependent malignancy (breast cancer, endometrial cancer) - Undiagnosed vaginal bleeding - Active venous thromboembolism (DVT/PE) - Active cardiovascular disease (recent MI, angina) - Liver disease (active hepatitis, cirrhosis) - Pregnancy

Relative contraindications: - Past VTE (can consider with careful risk assessment ± anticoagulation) - Hypertension (poorly controlled) — though transdermal oestrogen avoids hepatic first pass and has lower VTE risk - Migraine with aura (increased stroke risk) - Fibroids or endometriosis (relative — may be exacerbated by oestrogen)

Common exam trap: "Can a woman with a previous hysterectomy for endometrial cancer take HRT for menopausal symptoms?" — This is debated, but exam answer = NO (absolute contraindication to oestrogen in oestrogen-sensitive tumour).

Confusions to Eliminate

Confusion	Resolution
Raloxifene = HRT?	NO — raloxifene is a SERM. It does NOT relieve hot flushes
Topical oestrogen for lichen sclerosis?	NO — lichen sclerosis is autoimmune, not oestrogenic
VVF presentation = stress incontinence?	NO — VVF is continuous, day and night; stress incontinence is only with physical exertion
Le Fort's for all prolapse?	NO — only for elderly sexually inactive women unfit for major surgery
Asherman's treated with D&C?	NO — D&C CAUSES Asherman's; treatment is hysteroscopic adhesiolysis

Variations and Twists

Scenario — Incontinence type identification:

Woman leaks when she coughs → Stress incontinence (no urgency) Woman leaks before she reaches the toilet → Urge incontinence (urgency-driven) Woman has constant wet underwear → Fistula or overflow Elderly woman with dementia, no awareness of leaking → Functional incontinence (separate category)

Scenario — PCOS and menopause: PCOS women often have late menopause (more follicular reserve due to polycystic ovaries). They may still have cycles in their mid-50s. Their FSH will eventually rise but later than average.

Scenario — Premature ovarian insufficiency: A 32-year-old with hot flushes, elevated FSH, and secondary amenorrhoea. This is NOT menopause (by definition) — it is premature ovarian insufficiency (POI). Management differs: HRT is given as physiological replacement (not just symptom relief) and continued until the expected age of natural menopause (~51 years) to protect bone and cardiovascular health.

HRT: Cyclical vs Continuous combined regimen: - Cyclical (sequential): Oestrogen continuously + progestogen added in second half of cycle. A monthly bleed occurs (planned withdrawal bleed). Used in perimenopausal women (accept a bleed). - Continuous combined: Oestrogen + progestogen every day. No bleed (amenorrhoea). Used in postmenopausal women (>1 year post-menopause). If used too early, causes irregular spotting.

Memory Engineering

Menopause hormone changes: "FSH Goes First and Fastest" - FSH ↑↑ (first and highest rise, most sensitive marker) - LH ↑ (rises too but later and less) - Oestrogen ↓ - Inhibin ↓ (the actual trigger for FSH rise)

HRT rule: "Uterus Present → Protect It (add Progestogen)" - P for Progestogen, P for Protection, P for Presence of uterus

WHI trial memory: "WHI Warned Heart Is Not Improved" - HRT ≠ heart protection — WHI proved it

Raloxifene: "RAL protects RAL bones, but RAL runs from hot flushes" - Bone agonist, hot flush antagonist

Incontinence types: "SU-FO" - Stress — physical exertion triggers - Urge — cannot hold, must void now - Fistula — constant, cannot control at all - Overflow — dribbles continuously, distended bladder

Asherman's: "D&C Does Damage → Hysteroscopy Heals" - Caused by D&C; investigated and treated by hysteroscopy

VVF post-op: "3 months no sex, 1 year no baby"

Prolapse treatment: "If family complete = Vaginal Hysterectomy + Floor Repair"

Ultra High-Yield Revision Layer

Menopause Quick Facts

- Definition: 12 months amenorrhoea (natural, no pathological cause)
- Average age India: 45–47 years; Global: 51 years
- POF: menopause <40 years
- Hormone profile: FSH ↑↑, LH ↑, Oestradiol ↓, Inhibin ↓

HRT Decision Table

Uterus Present?	HRT Choice
Yes (intact)	Combined oestrogen + progestogen
No (post-hysterectomy)	Oestrogen only
Breast cancer contraindication	Topical oestrogen for urogenital only (consult oncology)
For hot flushes, not HRT candidate	Venlafaxine, clonidine, gabapentin, isoflavones
For osteoporosis, not HRT candidate	Raloxifene, bisphosphonates

HRT Indications vs NOT-Indications

Indication	HRT Status
Hot flushes / vasomotor symptoms	YES — first-line
Osteoporosis prevention	YES
Vaginal atrophy	YES (topical preferred)
Coronary artery disease prevention	NO — WHI trial

Incontinence Quick Comparison

Feature	Stress	Urge	Overflow	Fistula
Trigger	Exertion	Urgency	Obstruction/atony	None
Urgency	No	Yes	No	No
Frequency	No	Yes	No	No
Timing	With exertion only	After urge	Constant dribble	Constant
Detrusor on urodynamics	Normal	Overactive	Underactive	Normal
Drug treatment	Duloxetine	Oxybutynin	Bethanecol (if detrusor)	Surgery
Surgery	TVT/TOT	Sacral neuromodulation	Catheter ± surgery	Fistula repair

Prolapse Management Summary

Scenario	Management
Grade 3 prolapse, completed family, fit	Vaginal hysterectomy + pelvic floor repair
Grade 3 prolapse, elderly, unfit	Ring pessary
Grade 3 prolapse, elderly, sexually inactive	Le Fort's colpocleisis
Grade 1-2 prolapse, young, desires fertility	Pelvic floor physiotherapy + Manchester repair

Fistula Key Points

Fistula	India Cause	Developed World Cause	Initial Rx	Post-op
VVF	Obstetric (obstructed labour)	Post-hysterectomy	Catheter drainage	Abstinence 3m, no pregnancy 1yr
RVF	Obstetric tear / infection	Post-radiation / Crohn's	Diversion colostomy	Repair after 3–6 months

Asherman's Syndrome at a Glance

Feature	Detail
Cause	D&C (esp. postpartum haemorrhage, missed abortion)
Highest risk D&C	For postpartum haemorrhage (fibrotic, infected uterus)
Presentation	Secondary amenorrhoea / hypomenorrhoea / infertility
Investigation	Hysteroscopy (gold standard), HSG (filling defect)
Treatment	Hysteroscopic adhesiolysis + high-dose oestrogen + IUD/balloon

Chapter 8 covers 30 questions from the NEET-PG question bank (11 menopause/HRT + 19 pelvic floor). All concepts tested above have appeared in previous papers from 2015–2024, with several questions appearing 2–4 times across different years.

Chapter 9: Benign Gynaecological Conditions

Fibroids · Endometriosis · Ovarian Cysts · PCOS · Adenomyosis

Big Picture First

Benign gynaecological conditions dominate the gynaecology section of NEET-PG. Year after year, roughly **25–30% of all gynaecology MCQs** come from just four topics covered here: uterine fibroids, endometriosis, PCOS, and ovarian cysts. These are not pure recall topics — the examiner tests whether you can apply concepts to clinical vignettes. A stem that says "35-year-old, menorrhagia, uterus 16-week size, wants fertility" expects you to reason through fibroid type, fertility implications, and surgical choice — not just recite definitions.

The three patterns the examiner loves: 1. Identifying the correct subtype from a clinical feature or complication (e.g., which fibroid causes menorrhagia vs. obstructive symptoms) 2. Choosing the right management step at the right life stage (young woman wanting fertility vs. 45-year-old with completed family) 3. Knowing which investigation is gold standard vs. which is useful-but-not-definitive (laparoscopy vs. USS for endometriosis; MRI vs. CT for fibroids)



Question Pattern Map

Trap	What Students Write	What Examiners Want
Most common fibroid degeneration	Red degeneration (because it's dramatic)	Hyaline degeneration (~65% of all)
Red degeneration: when in pregnancy	Any trimester	2nd trimester (rapid uterine growth)
Fibroid: malignant transformation risk	"It's common"	Extremely rare, < 0.1%
Endometriosis gold standard	CA-125	Laparoscopy + biopsy
Endometriosis vs adenomyosis	Confused in clinical vignettes	Adenomyosis = glands inside myometrium; globular uterus
PCOS USG description	"cysts"	String of pearls = 12+ follicles (2–9 mm) at periphery
PCOS Rotterdam: how many criteria needed	3 of 3	2 of 3 criteria
CT vs MRI for fibroid mapping	Both acceptable	CT is NOT an adequate alternative; MRI is gold standard
Ovarian torsion: which side	Either side equally	Right side more common (sigmoid limits left)



Core Concept Deep Dive

PART 1: Uterine Fibroids (Leiomyomas)

What They Are

Fibroids are benign monoclonal tumours of uterine smooth muscle, dependent on oestrogen and progesterone for growth. They are the most common solid pelvic tumours in women. Incidence: 20–40% of women of reproductive age; more common in African women. They are oestrogen-sensitive — hence they grow during pregnancy and regress after menopause.

Types and Their Clinical Signatures

The type of fibroid determines the dominant symptom. This is a favourite MCQ angle.

Type	Location	Dominant Symptom	Key Points
Submucosal	Protrudes into uterine cavity	Menorrhagia (most severe)	Even a small submucosal fibroid causes heavy bleeding; type 0 = fully intracavitary
Intramural	Within myometrium	Menorrhagia + pelvic pressure	Most common type overall
Subserosal	Outer surface of uterus	Pressure symptoms (urinary, bowel)	Least likely to cause menorrhagia
Pedunculated	On a stalk (submucosal or subserosal)	Torsion; submucosal type can prolapse through cervix	Pedunculated subserosal = risk of torsion
Broad ligament	Between leaves of broad ligament	Ureteric compression, flank pain	Can mimic adnexal mass
Cervical	In cervical tissue	Obstructive dystocia, retention of urine	Rare

Exam pearl: Submucosal fibroids cause the most menorrhagia per unit size because they distort the endometrial cavity. A 1 cm submucosal fibroid causes more bleeding than a 5 cm intramural fibroid.

PYQ Integrated: Fibroid Subtypes and Imaging

Question from database:

Regarding imaging of uterine fibroids, which statement is FALSE? - A. Ultrasound is ideal to confirm the diagnosis - B. Saline infusion sonography (SIS) better detects submucous fibroids - C. MRI is superior to ultrasound for identifying exact fibroid location - D. CT scanning is an adequate alternative to MRI ← CORRECT ANSWER (FALSE statement)

Why this matters: MRI is the gold standard for fibroid mapping pre-myomectomy or uterine artery embolisation (UAE). It identifies exact location (submucosal vs. intramural), number, vascularity, and relationship to the endometrial cavity. CT has poor soft-tissue contrast for the uterus and cannot substitute for MRI. USS is the first-line investigation; SIS (saline infusion sonography) enhances detection of submucosal fibroids by filling the cavity with saline.

Fibroid Degeneration — The Most Tested Complication

Types of degeneration (remember: "Hyaline Monsters Cry Red Calcified Sarcomas"): - **Hyaline** — most common (65%), acellular collagen replacement - **Myxomatous** — gelatinous, mucoid change - **Cystic** — liquefaction of hyaline; large cavities - **Red (Carneous)** — haemorrhagic infarction; most common in **pregnancy** (2nd trimester) - **Calcification** — "womb stones"; post-menopausal; dystrophic - **Sarcomatous** — malignant transformation; extremely rare (< 0.1%)

PYQ — Red degeneration:

Red (carneous) degeneration of a uterine fibroid most commonly occurs in: - A. 1st trimester - B. 2nd trimester ← ANSWER - C. 3rd trimester - D. Puerperium

Mechanism: Rapid uterine growth → relative ischaemia of fibroid → haemorrhagic infarction. The haemoglobin leeches into the fibroid, giving the cut surface a raw meat (carneous/red) appearance. Clinically: acute abdominal pain, low-grade fever, focal uterine tenderness, no peritoneal signs. **Management is conservative** — analgesics and hydration. Do NOT operate; myomectomy in pregnancy carries catastrophic risk.

PYQ — Fibroid in pregnancy:

28 weeks pregnancy with abdominal pain, fever, and ultrasound showing a fibroid. Next management: - A. Hysterectomy - B. Conservative management (analgesics, hydration) ← ANSWER - C. Termination of pregnancy - D. Myomectomy

PYQ — Least likely change:

An asymptomatic woman with a uterine fibroid declines treatment. Which change is LEAST likely? - A. Calcium deposition (calcification) - B. Malignant transformation (sarcoma) ← ANSWER - C. Infection and haemorrhage - D. Torsion (pedunculated fibroid)

Do not confuse leiomyosarcoma (a separate entity, not transformation of a fibroid) with sarcomatous degeneration of a fibroid. Sarcomatous degeneration in a pre-existing fibroid is < 0.1%.

Fibroid Management — Decision Framework

This is the single most important table for clinical scenario MCQs:

Clinical Scenario	Preferred Management
Asymptomatic fibroid, premenopausal	Expectant — reassure, annual review
Symptomatic, completed family, perimenopausal	Hysterectomy (definitive)
Symptomatic, young woman, wants fertility, submucosal	Hysteroscopic myomectomy
Symptomatic, intramural/subserosal, wants fertility	Laparoscopic or open myomectomy
Pre-operative shrinkage or perimenopausal bridge	GnRH agonist (3–6 months; not long-term)
UAE (Uterine Artery Embolisation)	Non-surgical; good for multiple fibroids; not if fertility desired
Fibroids in pregnancy (red degeneration)	Conservative (analgesics)
45-year-old, menorrhagia, submucosal fibroid	Hysterectomy (most definitive)

PYQ — Management of menorrhagia with submucosal fibroid (45-year-old):

A 45-year-old woman with 3 months of menorrhagia and a 2–3 cm submucosal fibroid. Best management? - A. OCP for 3 months - B. Cyclical progesterone for 3 months - C. Endometrial sampling alone - D. Hysterectomy ← ANSWER

Why D? Age 45, completed family, submucosal fibroid is the structural cause. Medical options (OCP, progestins) are temporising and do not address the structural cause. Hysteroscopic myomectomy would be reasonable if fertility preservation was desired, but with completed family and persistent heavy bleeding, hysterectomy is definitive.

GnRH Agonists for Fibroids — Why Only Short-Term?

GnRH agonists (leuprolide, goserelin) suppress oestrogen to near-castrate levels → fibroid volume decreases 30–50% over 3 months. **Why not use long-term?** - Hypo-oestrogenic side effects: hot flushes, vaginal atrophy, bone loss (osteoporosis) - Fibroids return to pre-treatment size within 3–6 months of stopping - Use: pre-operative shrinkage (reduces intraoperative blood loss), correction of anaemia, or as a bridge to menopause in perimenopausal women

Hysterectomy Types — The Examiner Always Asks

Type	What Is Removed	When Used
Subtotal (supracervical)	Uterine body; cervix kept	Simple benign conditions
Total hysterectomy	Uterus + cervix	Standard for benign disease
Wertheim's (radical) hysterectomy	Uterus + cervix + parametrium + upper vaginal cuff + pelvic nodes	Cervical cancer (Stage IA2–IIA)
Schauta's operation	Vaginal radical hysterectomy	Same indications as Wertheim's, vaginal approach

PART 2: Endometriosis

Definition and Pathogenesis

Endometriosis = presence of functional endometrial glands and stroma **outside the uterus**. The dominant theory is **Sampson's retrograde menstruation** — menstrual blood flows retrograde through the tubes into the peritoneal cavity, where shed endometrial cells implant and grow.

Do not confuse with adenomyosis: - Adenomyosis = endometrial glands/stroma > 2.5 mm deep within the **myometrium** (internal endometriosis) - Endometriosis = ectopic implants **outside** the uterus (external endometriosis)

PYQ — Definition:

Endometriosis is: - A. Endometrium within the myometrium - B. Functional endometrium outside the uterus
 ← **ANSWER** - C. Myometrium within the endometrium - D. Rare squamous variety of carcinoma endometrium

Option A describes adenomyosis. The word "functional" is key — the ectopic tissue responds to cyclical hormonal changes, bleeds monthly, and creates inflammation and fibrosis.

Sites — In Descending Frequency

1. Ovaries (most common — "chocolate cyst" / endometrioma)
2. Uterosacral ligaments
3. Pouch of Douglas (Rectovaginal septum)
4. Fallopian tubes
5. Bladder / ureter
6. Distant sites: lung (catamenial haemothorax), scar endometriosis (umbilicus, laparotomy scars)

Clinical Features — The Classic Triad

1. **Dysmenorrhoea** — severe, secondary, progressive (gets worse with each cycle)
2. **Dyspareunia** — deep, due to involvement of uterosacral ligaments and pouch of Douglas
3. **Infertility** — from adhesions, tubal blockage, altered peritoneal fluid environment

Other features: dyschezia (pain on defaecation), haematuria (bladder endo), cyclical haemoptysis (pulmonary endo), pelvic mass (endometrioma).

Important: Severity of symptoms does not correlate with stage. A woman with minimal-stage disease may have crippling dysmenorrhoea.

Diagnosis

Investigation	Role	Notes
Laparoscopy + biopsy	Gold standard	Powder-burn/blue-black spots, chocolate cysts, adhesions
Transvaginal USS	First-line	Identifies endometriomas (chocolate cysts) as homogeneous low-level echo mass
CA-125	Elevated in ~80%	Elevated in many conditions (PID, pregnancy); NOT diagnostic; used for monitoring
MRI	Good for deep infiltrating endo	Rectovaginal, bladder involvement

Laparoscopy findings: Powder-burn spots (blue-black deposits), red vesicles (early active lesions), white fibrotic plaques, chocolate cysts, "kissing ovaries," violin-string adhesions in advanced disease.

Management

Medical management (suppresses cyclical stimulation of ectopic tissue):

Drug	Mechanism	Duration/Notes
COCP (continuous)	Suppress ovulation; anovulatory amenorrhoea	First-line for mild-moderate symptoms
Progestins (norethisterone, medroxyprogesterone)	Decidualisation of ectopic tissue	Injectable DMPA every 3 months
GnRH agonists (leuprolide, goserelin)	Surgical menopause; profound oestrogen suppression	Max 6 months (bone loss); add-back HRT if longer
Danazol	Androgenic; creates pseudomenopause	Androgenic side effects; rarely used now
Gestrinone	19-nortestosterone derivative	2.5–5 mg TWICE WEEKLY × 6–9 months (not daily — common MCQ trap)
LNG-IUS (Mirena)	Local progestin; reduces dysmenorrhoea	Good for endometriosis-associated dysmenorrhoea

PYQ — False statement about gestrinone:

All are TRUE regarding medical treatment of endometriosis EXCEPT: - A. Oral progestins cause decidualisation of endometrial tissue - B. Injectable progestagen is used once every 3 months - C. LNG-IUS reduces dysmenorrhoea in endometriosis - D. Gestrinone is given orally 2.5–5 mg per day ← ANSWER (FALSE)

Gestrinone is given **twice weekly**, not daily. This is a consistent PYQ trap.

Surgical management: - **Laparoscopy:** ablation/excision of implants, adhesiolysis, cystectomy (for endometrioma > 3 cm) — preferred when fertility desired - **Laparotomy:** for extensive disease, bowel/bladder involvement -

Definitive: Hysterectomy + bilateral salpingo-oophorectomy — only for severe, completed family

PYQ — Endometriosis failing GnRH, next step:

A 30-year-old with moderate-to-severe pelvic pain remains symptomatic after 3 months on leuprolide. Next step? - A. Increase GnRH agonist dose - B. Expectant management - C. Diagnostic and operative laparoscopy ← ANSWER - D. Add progestins

GnRH agonists are a bridge, not a cure. When symptoms persist or recur, laparoscopy is the next step — it confirms the diagnosis histologically and allows concurrent surgical treatment.

Adenomyosis — The Great Masquerader

Adenomyosis = endometrial glands > 2.5 mm deep into the myometrium. - Uterus is **bulky, globular, and tender** (not irregular as in fibroids) - Classic patient: 40-year-old multiparous woman with progressive dysmenorrhoea and menorrhagia - **Diagnosis:** MRI (best non-invasive); definitive = histology on hysterectomy specimen - **Treatment:** Hysterectomy is curative (GnRH agonist helps temporarily)

PYQ — Adenomyosis:

A 30-year-old with dysmenorrhoea, dyspareunia, chronic pelvic pain. Hysterectomy specimen shows endometrial glands deep within myometrium. Diagnosis? - A. Endometriosis - B. Adenomyosis ← ANSWER - C. Leiomyoma - D. Endometrial polyp

Key: "within myometrium" = adenomyosis. Endometriosis is outside the uterus.

PART 3: Ovarian Cysts**Functional vs. Neoplastic — First Distinction**

Functional cysts (physiological, arise from follicular development): - **Follicular cysts:** unruptured dominant follicle; thin-walled, unilocular, < 8 cm; resolves spontaneously in 4–8 weeks - **Corpus luteum cyst:** from failed regression of corpus luteum; haemorrhagic; may cause pain if it bleeds

Management of simple ovarian cyst in premenopausal woman: - < 5 cm, unilocular, no septae/solid components → **conservative: repeat USS in 6 weeks** - Persisting or > 5 cm → consider laparoscopic cystectomy

Dermoid Cyst (Mature Cystic Teratoma)

The single most important benign ovarian tumour to know:

Key facts: - Most common ovarian tumour in **women under 30** and in **pregnancy** - Contains elements from all three germ cell layers: hair, sebum, teeth, bone, neural tissue, thyroid - **Rokitansky protuberance (dermoid plug):** solid protuberance projecting into the cyst lumen — the dense spot on USS/X-ray - **USS:** hyperechoic nodule with acoustic shadowing; "tip of iceberg" sign - **X-ray:** teeth visible (50% calcified components) - Most common cause of **ovarian torsion** in young women (right side > left) - **Malignant transformation:** < 2%; squamous cell carcinoma arising from dermoid (in older women with large, long-standing cysts) - **Management:** Cystectomy (preserve ovary), ideally laparoscopic; spillage risk during surgery (chemical peritonitis from sebaceous material)

Chocolate Cyst (Endometrioma)

- Ovarian endometriosis → repeated cyclical bleeding inside the ovary → thick, dark brown "chocolate" fluid (old blood, haemosiderin)
- USS: homogeneous low-level echoes ("ground glass" appearance); thick-walled unilocular cyst
- CA-125 is elevated but not specific
- **Gold standard: laparoscopy + histology**
- **Management:**
 - Medical: GnRH agonists (temporary; cyst returns on stopping)
 - Surgical: Laparoscopic cystectomy (preferred for cysts > 3–4 cm, especially if symptomatic or fertility investigation planned)
 - Drainage alone = high recurrence rate

Ovarian Torsion

PYQ:

*Which side ovarian torsion is more common? - A. Left side - **B. Right side** ← ANSWER - C. Equal on both sides - D. Bilateral simultaneous*

Why right? The sigmoid colon (left side of pelvis) acts as a physical barrier limiting left ovarian movement. The right ovary has more freedom to rotate. Also, in pregnant women, the right side is more affected due to uterine dextrorotation.

Presentation: Sudden-onset severe unilateral pelvic pain, nausea/vomiting; may be intermittent (partial/intermittent torsion). USS shows enlarged ovary, absence of blood flow on Doppler. **Management: laparoscopic detorsion** (untwist before adnexectomy; even if the ovary looks dusky, it is usually viable). Reserve salpingo-oophorectomy for clearly necrotic tissue.

PART 4: Polycystic Ovary Syndrome (PCOS)

Diagnostic Criteria (Rotterdam 2003)

Diagnosis requires **2 of 3** criteria: 1. Oligo/anovulation (irregular cycles, amenorrhoea) 2. Clinical or biochemical hyperandrogenism (hirsutism, acne, elevated testosterone/DHEA-S) 3. Polycystic ovarian morphology on USS

PYQ — NOT a Rotterdam criterion:

*All of the following are Rotterdam criteria for diagnosing PCOS EXCEPT: - A. Hyperandrogenism (clinical or biochemical) - B. Oligomenorrhoea / anovulation - C. **Elevated oestrogen levels** ← ANSWER - D. Polycystic ovarian morphology on ultrasound*

Oestrogen is NOT part of Rotterdam criteria. In PCOS, oestrone (not oestradiol) is elevated due to peripheral aromatisation of androgens, but oestrogen level is not a diagnostic criterion.

USG Findings in PCOS — "String of Pearls"

PYQ:

*Pearl necklace appearance on ultrasound is characteristic of: - A. Ectopic pregnancy - B. **PCOS** ← ANSWER - C. Endometriosis - D. PID*

Specific USS criteria (AES or Rotterdam): At least one ovary with ≥ 12 follicles measuring 2–9 mm in diameter, arranged at the periphery ("pearl necklace" / "string of pearls"), and/or increased ovarian volume (> 10 mL). Increased ovarian stromal echogenicity.

Why not true cysts? These follicles have not undergone atresia due to androgen-mediated failure of apoptosis.

PYQ — Cyst formation mechanism:

*Cysts in polycystic ovarian syndrome are formed by: - A. **Failure of atretic follicles to undergo apoptosis** ← ANSWER - B. Oocyte proliferation - C. Multiple corpus lutea - D. Cystic degeneration of ovarian cortex*

Hormonal Profile

Hormone	PCOS Level	Explanation
LH	Elevated	Increased pulse frequency from GnRH dysregulation
FSH	Normal or low	Relatively low compared to LH
LH/FSH ratio	$> 2:1$ (classically $> 3:1$)	Characteristic but not in Rotterdam criteria
Androgens (Testosterone, DHEA-S)	Elevated	Theca cell overproduction under LH
Insulin	Elevated	60–80% have insulin resistance
Prolactin	Normal (mild elevation occasionally)	Hyperprolactinaemia is a SEPARATE diagnosis
Oestradiol	Normal or slightly low	
Oestrone	Elevated	Peripheral aromatisation of androgens in adipose tissue

PYQ — False regarding hormone levels:

Which is **FALSE** regarding hormone levels in PCOS? - A. Increased androgen - **B. Increased prolactin** ← **ANSWER (FALSE)** - C. Increased LH - D. Increased LH/FSH ratio > 2:1

Prolactin is typically **normal** in PCOS. If prolactin is elevated, investigate separately for prolactinoma.

PCOS Management

PYQ — Drug NOT used in PCOS:

Which of the following is **NOT** used in PCOS? - A. OC pills - B. Cyclical progestogens - C. Myoinositol - **D. Danazol** ← **ANSWER**

Danazol is an androgen; it would worsen the androgen excess in PCOS. It has no role here.

Goal	Drug/Intervention
Regulate cycles	OC pills (preferred), cyclical progestogens
Hirsutism / acne	COC + anti-androgen (spironolactone, cyproterone acetate)
Insulin resistance	Metformin (first-line insulin sensitiser); myoinositol also used
Ovulation induction for fertility	Clomifene (first-line); letrozole (emerging preferred); FSH injections if clomifene fails
Weight loss	5–10% weight reduction restores ovulation in many cases

PART 5: AMH and Ovarian Reserve

PYQ:

In low ovarian reserve, anti-Müllerian hormone (AMH) level is: - **A. < 1 ng/mL** ← **ANSWER** - B. 1–4 ng/mL - C. > 7 ng/mL - D. > 10 ng/mL

AMH Level	Interpretation
< 1 ng/mL	Poor/diminished ovarian reserve
1–3.5 ng/mL	Normal range
> 3.5–7 ng/mL	Good reserve
> 7 ng/mL	High reserve (PCOS, OHSS risk)

AMH is secreted by granulosa cells of **preantral and small antral follicles**. It is **cycle-independent** (can be measured on any day of cycle) — this makes it superior to Day-3 FSH for ovarian reserve assessment.

PYQ — AMH vs FSH:

AMH is a better marker of ovarian reserve than day-3 FSH because: - **A. AMH is cycle-independent** ← **ANSWER** - B. AMH is produced by corpus luteum - C. AMH is elevated in premature ovarian failure - D. AMH requires fasting for accurate measurement

! Bias Removal: Classic Confusions

1. Fibroid vs. Adenomyosis

Feature	Fibroid	Adenomyosis
Uterine contour	Irregular, nodular	Smooth, globular
Uterine size	Variable enlargement	Symmetrically enlarged (like 8–10 week size)
Tenderness	Non-tender	Tender (especially pre-menstrually)
USS	Distinct focal masses, acoustic shadowing	Asymmetric myometrial thickening, loss of junctional zone
MRI	Well-circumscribed masses	Junctional zone > 12 mm
Diagnosis	Clinical + USS	Definitive on hysterectomy histology
Curative treatment	Myomectomy (if fertility desired) or hysterectomy	Hysterectomy (no myomectomy option)

2. Endometriosis vs. PID — Differentiating Features

Feature	Endometriosis	PID
Onset	Gradual, progressive	Acute
Fever	Absent	Present
Discharge	Absent	Purulent
Dysmenorrhoea	Secondary, progressive	Cyclic pain but not typical secondary dysmenorrhoea
CA-125	Elevated	Also elevated (less helpful)
WBC	Normal	Elevated
Laparoscopy	Powder-burn implants, endometriomas	Hyperaemia, pyosalpinx, tubo-ovarian abscess
Treatment	Hormonal / laparoscopy	Antibiotics

3. Dermoid (Teratoma) vs. Endometrioma

Feature	Dermoid	Endometrioma
Age	Reproductive age, < 30 common	Reproductive age, 25–40
USS	Hyperechoic with acoustic shadow; Rokitansky protuberance	Homogeneous low-level echoes ("ground glass")
Contents	Hair, teeth, sebum	Old blood (chocolate fluid)
CA-125	Normal or mildly elevated	Elevated
Associated with	Torsion risk	Endometriosis, infertility



Variations and Twists

Scenario 1: Young Woman with Recurrent Miscarriage and Fibroid

A 28-year-old has had 3 first-trimester miscarriages. Hysterosalpingography shows a filling defect in the uterine cavity.

Think: Submucosal fibroid (or endometrial polyp, or septate uterus). The filling defect indicates an intracavitary pathology. Next step: hysteroscopy (diagnostic and therapeutic). If submucosal fibroid confirmed → **hysteroscopic myomectomy** → improves pregnancy outcomes.

Scenario 2: Pelvic Pain Not Responding to NSAIDs

A 29-year-old has progressive dysmenorrhoea, deep dyspareunia, and subfertility for 2 years. USS shows a 4 cm left ovarian cyst with ground-glass echoes.

Think: Endometrioma. Next step: **laparoscopy** — confirms endometriosis, biopsies implants, performs ovarian cystectomy. If she wants fertility: conservative surgery (ovarian cystectomy, not ablation — lower recurrence).

Scenario 3: Postmenopausal Woman with Uterine Enlargement

A 55-year-old presents with an irregular uterus of 12-week size. No bleeding.

Think: Fibroids post-menopause should regress (oestrogen withdrawal). If enlarging or new, consider **leiomyosarcoma** (very rare but important). Investigate with MRI; offer hysterectomy for histological diagnosis.



Memory Engineering

Fibroid Degeneration (HMCRCs — "Hungry Monsters Can't Run Carefully, Sadly")

- Hyaline (most common, 65%)
- Myxomatous
- Cystic

- **Red/Carneous** (pregnancy — 2nd trimester)
- **Calcification** (postmenopausal)
- **Sarcomatous** (< 0.1%, rare)

PCOS Rotterdam Criteria — "HOT"

- **H**yperandrogenism (clinical or biochemical)
- **O**ligo/anovulation
- **T**ransabdominal or transvaginal USS showing polycystic ovaries

Need 2 of 3. Oestrogen is NOT in HOT.

Dermoid Cyst Contents — "Hair To Skin"

- **H**air, teeth (ectoderm)
- **T**hyroid (endoderm — struma ovarii when > 50% thyroid tissue)
- **S**kin, sebaceous glands (ectoderm)

Ultra High-Yield Revision Layer

Fibroid Management Decision Table

Patient	Family Status	Symptom	Choice
25-year-old	Wants children	Menorrhagia, submucosal	Hysteroscopic myomectomy
32-year-old	Wants children	Intramural 6 cm	Laparoscopic/open myomectomy
40-year-old	Completed	Menorrhagia, multiple fibroids	Hysterectomy
48-year-old	Completed	Perimenopausal, mild symptoms	GnRH agonist 3–6 months → observe → may reach menopause
Any age	—	Asymptomatic	Expectant management
Fibroid in pregnancy (red degeneration)	—	Pain, fever	Conservative (analgesics)

Tumour Markers Quick Reference

Marker	Tumour
CA-125	Epithelial ovarian cancer, endometriosis (non-specific)
AFP	Yolk sac (endodermal sinus) tumour
LDH + Placental ALP	Dysgerminoma
hCG	Choriocarcinoma, gestational trophoblastic disease
Inhibin	Granulosa cell tumour
CEA	Mucinous ovarian tumours

PYQ — Dysgerminoma marker:

In a case of dysgerminoma of the ovary, which tumour marker is most likely raised? - A. Serum HCG - B. Serum alpha-fetoprotein - C. Serum LDH ← ANSWER - D. Serum inhibin

PCOS Hormone Summary

LH high, FSH relatively low, LH/FSH > 2, androgens high, insulin high, prolactin NORMAL, oestradiol normal/low, oestrone HIGH (peripheral aromatisation).

Endometriosis: Medical Drugs and Key Details

Drug	Key Fact
GnRH agonist	Max 6 months; add-back HRT if longer; 30–50% fibroid shrinkage
Gestrinone	Twice weekly (not daily)
Danazol	Androgenic; rarely used; causes acne, hirsutism, voice changes
DMPA (depot medroxyprogesterone)	Every 3 months
LNG-IUS	Reduces dysmenorrhoea; local effect

Chapter 9 covers the highest-density benign gynaecological topics. Revisit the Decision Tables before exam day — scenario questions can be solved directly from these frameworks.

Chapter 10: Gynaecological Infections, Developmental Anomalies & Miscellaneous

Müllerian Anomalies · Developmental Disorders · STDs · PID · Vaginal Physiology

Big Picture First

This chapter is a tale of two halves. The developmental anomaly half (Müllerian anomalies, imperforate hymen, AIS) is concept-dense and directly tested — examiners love the "most common anomaly" question, the "septate vs. bicornuate" management distinction, and the "primary amenorrhoea with cyclic pain" clinical scenario. The infections half (BV, trichomonas, gonorrhoea, chlamydia, PID) is classic association territory — every organism has a pathognomonic sign, every treatment has a key rule.

The two unbreakable exam rules for this chapter: 1. **Septate uterus** is the most common Müllerian anomaly AND the most common cause of recurrent pregnancy loss from uterine anomalies → treated by **hysteroscopic septoplasty**. 2. **Clue cells = BV (Gardnerella)**, **Strawberry cervix = Trichomonas**, **Gram-negative diplococci = Gonorrhoea** — never mix these.

Question Pattern Map

Trap	What Students Write	What Examiners Want
Most common Müllerian anomaly	Bicornuate uterus	Septate uterus
Septate uterus management	Laparotomy metroplasty	Hysteroscopic septoplasty
Bicornuate uterus management	"Same as septate"	No surgery needed (different embryology)
BV clue — wet mount	Flagellate organism	Clue cells (stippled epithelial cells) = Gardnerella
Trichomonas treatment	Vaginal clotrimazole	Oral metronidazole (both partners)
PID antibiotic	Single drug	Triple therapy: ceftriaxone + doxycycline + metronidazole
Fitz-Hugh-Curtis	Liver disease	PID complication (perihepatitis; violin string adhesions)
AIS karyotype	46,XX	46,XY (male karyotype, female phenotype)
MRKH: ovarian function	Absent	Ovaries normal (46,XX; only uterus/upper vagina absent)

Core Concept Deep Dive

PART 1: Müllerian Anomalies (Uterovaginal Anomalies)

Embryological Background

The uterus, cervix, fallopian tubes, and upper two-thirds of the vagina all develop from the **Müllerian (paramesonephric) ducts**. The two Müllerian ducts fuse in the midline in a caudal-to-cranial direction during weeks 8–12 of embryogenesis. The fused segment becomes the uterus and cervix; the midline septum must be resorbed. Failure of any of these steps produces the array of Müllerian anomalies.

Classification (AFS/ASRM):

Class	Anomaly	Embryological Failure	Clinical Notes
I	Uterine agenesis/hypoplasia (MRKH)	Müllerian duct aplasia	No uterus, no upper vagina; normal ovaries
II	Unicornuate uterus	One duct develops partially	Banana-shaped; 65% have rudimentary horn
III	Uterus didelphys	Failure of fusion	Two separate uteri and cervices; two vaginae
IV	Bicornuate uterus	Partial fusion failure	Two horns; single cervix; Y-shaped cavity
V	Septate uterus	Failure of septal resorption	Most common anomaly ; single external uterus; internal septum
VI	Arcuate uterus	Near-complete resorption; minor indentation	Near-normal; minimal clinical significance
VII	DES-related anomalies	Exogenous drug exposure	T-shaped uterus

Septate Uterus — The High-Yield Anomaly

Why is it the most important? - Most common Müllerian anomaly overall - Most common cause of recurrent pregnancy loss (RPL) from uterine anomalies - Has a surgical cure: **hysteroscopic septoplasty (septum resection)**

The septum is avascular fibrous tissue. When an embryo implants on the septum, it gets poor blood supply → miscarriage (usually first trimester). The septum can be: - **Complete:** extends to the cervical os - **Partial (incomplete):** extends partway into the cavity

Diagnosis: - Hysterosalpingography (HSG): shows divided cavity; cannot distinguish septate from bicornuate - **MRI:** gold standard to distinguish septate vs. bicornuate (external uterine contour is key) - 3D ultrasonography - **Hysteroscopy + laparoscopy:** combined approach is definitive

Management:

PYQ:

Septate uterus with recurrent pregnancy loss. Management? - A. Hysterectomy - B. Laparoscopy and metroplasty - C. Hysteroscopy and septoplasty (resection) ← ANSWER - D. Laparotomy and metroplasty

PYQ:

*A 24-year-old with recurrent abortions found to have a septate uterus. Which surgery has the best reproductive outcome? - A. Strassmann metroplasty - B. Tompkins metroplasty - C. **Transcervical hysteroscopic resection of septum** ← ANSWER - D. Jones metroplasty*

Strassmann, Tompkins, and Jones are all **open metroplasty procedures** — they are now obsolete for septate uterus. Hysteroscopic resection is minimally invasive, does not require uterine incision (no weakened scar for future pregnancy), and has superior outcomes.

Bicornuate Uterus vs. Septate — The Critical Distinction

This is the most tested distinction in this chapter. Students consistently confuse these two.

Feature	Septate Uterus	Bicornuate Uterus
Cause	Failure of septal resorption	Failure of ductal fusion
External uterine contour	Normal (fundus convex or flat)	Concave fundal notch (heart-shaped)
Internal division	Fibrovascular septum	Muscular septum (myometrium)
MRI/USS	Intact outer contour	External fundal indentation > 1 cm
Most common anomaly	YES	No
RPL association	Strong (avascular septum)	Less severe
Surgical treatment	Hysteroscopic septoplasty	Usually NOT treated surgically
Open surgery	Not indicated	Strassmann metroplasty (rarely done)

Key rule: On HSG, both look like a divided cavity. MRI is needed to see the **external contour** — this distinguishes them. If the external uterine fundus is convex/flat = septate. If it has a cleft/notch = bicornuate.

Uterus Didelphys

PYQ:

*Uterus didelphys is NOT typically complicated by: - A. Recurrent abortion - B. Endometriosis - C. Preterm labour - D. **Transverse lie** ← ANSWER (NOT a complication)*

Two completely separate uterine horns = each horn can accommodate a fetus in longitudinal lie. Transverse lie is NOT a specific complication. Recognised complications: recurrent miscarriage (small cavity), preterm labour, endometriosis (retrograde menstruation with two horns), ectopic pregnancy (in rudimentary horn).

Unicornuate Uterus and Rudimentary Horn

- Single Müllerian duct develops → banana-shaped uterus
- 65% have a contralateral **rudimentary horn**

- If the rudimentary horn has a functioning endometrium and is non-communicating → **haematometra** and cyclic pain
- **Risk:** ectopic pregnancy in the rudimentary horn — can rupture catastrophically in the 2nd trimester (presents like uterine rupture); management = **excision of the rudimentary horn**

PART 2: Developmental Disorders

Imperforate Hymen

- Hymen fails to perforate at puberty → **obstructed outflow tract**
- **Clinical presentation:** Primary amenorrhoea + cyclic pelvic pain (cyclical haematocolpos)
- Examination: **bluish, bulging membrane** at the introitus
- USS/MRI: haematocolpos (blood-filled vagina), haematometra (blood in uterus)
- **Treatment: cruciate incision** (hymenotomy) — drainage of accumulated blood; excellent prognosis

The classic scenario: Adolescent girl with cyclical lower abdominal pain every month but no visible menstrual flow. Do not miss the bulging bluish membrane on examination.

MRKH Syndrome (Mayer-Rokitansky-Küster-Hauser)

- Müllerian aplasia → **absent uterus and upper vagina**
- Karyotype: **46,XX** (normal female)
- **Ovaries: normal and functional** → secondary sexual characteristics develop normally (unlike Turner's)
- Presentation: Primary amenorrhoea; no uterus on USS; normal female phenotype; normal breast development
- Associated: renal anomalies (horseshoe kidney, renal agenesis), skeletal anomalies
- Treatment: Progressive vaginal dilators (Frank's technique) to create a functional neovagina; surgical (McIndoe procedure) if dilators fail

PYQ:

*In which condition do the ovaries function normally? - A. Turner's syndrome - **B. Rokitansky-Küster-Hauser (MRKH) syndrome** ← ANSWER - C. Androgen insensitivity syndrome - D. Swyer's syndrome*

Turner = streak gonads (dysgenetic); AIS = testes (not ovaries); Swyer = streak gonads (46,XY). MRKH is the only one with **normal functional ovaries** — hence normal pubertal development, normal FSH/LH, but no menstruation (no uterus).

Androgen Insensitivity Syndrome (AIS) — Complete (CAIS)

- Karyotype: **46,XY** (genetically male)
- Phenotype: **female** (complete female external genitalia)
- Cause: Defect/absence of androgen receptors → testosterone cannot act on external genitalia or Wolffian ducts → female phenotype develops by default
- **Testes:** present in inguinal canals or labia majora (produce testosterone and MIF/AMH)
- **MIF (Müllerian inhibiting factor):** still produced → Müllerian structures **absent** (no uterus, no tubes, no upper vagina)

- **Features:**
- Primary amenorrhoea
- Female phenotype with normal breast development (peripheral aromatisation of testosterone → oestrogen)
- **Absent/sparse axillary and pubic hair** (no androgen effect on hair follicles)
- **Blind-ending vagina** (short, no cervix)
- **No uterus** on USS
- **Treatment:** Orchidectomy (gonads are testes; increased risk of malignant transformation — gonadoblastoma, seminoma) — usually after puberty is complete; HRT thereafter

AIS vs. MRKH comparison:

Feature	AIS (Complete)	MRKH
Karyotype	46,XY	46,XX
Gonads	Testes (inguinal canal)	Normal ovaries (in pelvis)
Uterus	Absent (MIF from testes)	Absent (Müllerian aplasia)
Pubic/axillary hair	Absent/sparse	Normal
Breast development	Normal (oestrogen from aromatisation)	Normal
FSH/LH	Normal or elevated	Normal

Virilisation of Female Fetus — Urogenital Sinus Concept

PYQ:

Excess androgen exposure of a female fetus in early embryogenesis leads to virilisation of: - A. Müllerian ducts - B. Ovary - C. Urogenital sinus ← ANSWER - D. Mesonephric ducts

Mechanism: Androgens (from fetal adrenal CAH, maternal tumour, or exogenous) cause labioscrotal fusion and clitoral hypertrophy (urogenital sinus derivatives). The Müllerian-derived internal genitalia (uterus, tubes, upper vagina) develop normally because: - Androgens don't suppress Müllerian development - Wolffian ducts don't develop (no testosterone-mediated SRY pathway)

Clinical result: A 46,XX infant born with ambiguous genitalia, virilised external genitalia, but **normal uterus and ovaries** — typical of **congenital adrenal hyperplasia (21α-hydroxylase deficiency)**.

PART 3: Gynaecological Infections

Bacterial Vaginosis (BV)

BV is not a true infection but a **dysbiosis** — replacement of normal Lactobacillus-dominant flora by anaerobes (Gardnerella vaginalis, Mobiluncus, Prevotella).

Amsel Criteria — **diagnose BV if 3 of 4 criteria present:**

Criterion	Detail
1. Thin, greyish-white homogeneous discharge	Adherent to vaginal walls
2. Vaginal pH > 4.5	(Normal = 3.5–4.5)
3. Positive whiff (amine) test	Add 10% KOH → fishy amine odour (amines released from anaerobes)
4. Clue cells on wet mount	Vaginal squamous epithelial cells studded with bacteria (Gardnerella)

PYQ — Clue cells:

"Clue cells" on vaginal wet mount are pathognomonic of: - A. Bacterial vaginosis ← ANSWER - B. Vaginal candidiasis - C. Chlamydial vaginitis - D. Trichomoniasis

Clue cells = squamous cells with stippled/granular appearance due to Gardnerella bacteria coating the cell surface. Pathognomonic.

PYQ — Organism and treatment:

A woman presents with fishy-smelling greyish discharge; pH 5.5, whiff test positive, clue cells on wet mount. Organism and drug? - A. Candida albicans — fluconazole - B. Trichomonas vaginalis — metronidazole - C. Gardnerella vaginalis — metronidazole ← ANSWER - D. Neisseria gonorrhoeae — doxycycline

Treatment: Metronidazole 500 mg BD × 7 days (oral) OR 0.75% vaginal gel × 5 days. Alternative: clindamycin 2% vaginal cream × 7 days. Partner treatment is **NOT** routine in BV (it is not primarily sexually transmitted, though it is associated with sexual activity).

PYQ — Malodorous discharge cause:

Malodorous vaginal discharge is due to: - A. Bacterial vaginosis ← ANSWER - B. Chlamydia trachomatis - C. Trichomonas vaginalis - D. Neisseria gonorrhoeae

The amine odour in BV is from metabolic products of anaerobes, not from Gardnerella itself.

Trichomonas Vaginalis

- **Organism:** Trichomonas vaginalis (protozoan flagellate, sexually transmitted)
- **Symptoms:** Intense pruritus, frothy yellow-green discharge, vulvovaginal irritation, dysuria
- **Sign: Strawberry cervix** (colpitis macularis) — punctate haemorrhagic spots on cervix; visible on colposcopy in ~50%, naked eye in 10%
- **Vaginal pH:** > 4.5 (alkaline, unlike candida)
- **Wet mount:** motile flagellated trichomonads (pear-shaped, motile)

PYQ — Strawberry cervix:

Strawberry cervix (punctate haemorrhages) is caused by: - A. Trichomonas vaginalis ← ANSWER - B. Gardnerella vaginalis - C. Neisseria gonorrhoeae - D. Candida albicans

PYQ — Green frothy discharge:

Green frothy vaginal discharge is produced by: - A. Herpes simplex virus - B. Candida albicans - C. Trichomonas vaginalis ← ANSWER - D. Normal vaginal flora

PYQ — True statement about Trichomonas:

Regarding Trichomonas vaginalis infection, which statement is TRUE? - A. Vaginal pH is typically < 4.5 - B. Male sexual partners must be treated simultaneously ← ANSWER - C. Pruritus is usually absent - D. Treated by intravaginal clotrimazole

Treatment: Metronidazole 2 g single dose (oral) OR 500 mg BD × 7 days. **Both partners must be treated simultaneously** — Trichomonas is sexually transmitted and ping-pong reinfection occurs without partner treatment. Clotrimazole is antifungal — completely wrong for Trichomonas (this option appears in PYQs to trap students).

Vaginal Candidiasis

- **Organism:** Candida albicans (90%)
- **Risk factors:** Antibiotics, uncontrolled diabetes, pregnancy, immunosuppression
- **Symptoms:** Intense pruritus, thick white "cottage cheese" discharge, dysuria, vulval burning
- **pH:** < 4.5 (acidic — normal pH)
- **Wet mount (KOH prep):** pseudohyphae and budding yeast cells
- **Treatment:** Clotrimazole vaginal pessary (single dose or 3–7 days) OR oral fluconazole 150 mg single dose
- Partner treatment: NOT routine (not primarily STI)

STD Quick Reference Table

Organism	Characteristic Finding	Treatment
Gardnerella vaginalis	Clue cells, fishy odour, grey discharge	Metronidazole
Trichomonas vaginalis	Strawberry cervix, frothy green discharge, motile flagellates	Metronidazole (both partners)
Candida albicans	Cottage cheese discharge, pseudohyphae on KOH	Clotrimazole / fluconazole
Neisseria gonorrhoeae	Gram-negative intracellular diplococci; urethritis, cervicitis	Ceftriaxone
Chlamydia trachomatis	Silent salpingitis → tubal infertility; intracellular; serovars D–K	Doxycycline / azithromycin
HSV-2	Painful genital ulcers, active lesions → LSCS	Aciclovir
Treponema pallidum	Primary: painless chancre; FTA-ABS specific test	Benzathine penicillin

Chlamydia Trachomatis — The Silent Epidemic

PYQ:

Chlamydia trachomatis infection commonly causes: - A. Infertility ← ANSWER - B. Post-coital bleeding - C. Amenorrhoea - D. Malignancy

Chlamydia causes **silent salpingitis** — most patients are asymptomatic. The tubal mucosa is progressively damaged → tubal scarring and peritubal adhesions → **tubal factor infertility**. It is the most common STI-related cause of infertility worldwide. It can also cause ectopic pregnancy (damaged ciliary function + scarring). The absence of symptoms means diagnosis is often delayed.

Pelvic Inflammatory Disease (PID)

Definition: Ascending infection from the lower genital tract (cervix/vagina) to the upper genital tract (endometrium, fallopian tubes, ovaries, peritoneum).

Causative organisms: Polymicrobial — *Neisseria gonorrhoeae*, *Chlamydia trachomatis*, and vaginal anaerobes. Both may be responsible.

Diagnosis (CDC criteria — need all 3 minimum criteria): - Uterine tenderness - Adnexal tenderness - **Cervical motion tenderness (cervical excitation)** — the most specific sign

Additional features supporting diagnosis: fever $> 38.3^{\circ}\text{C}$, mucopurulent cervical discharge, elevated CRP/ESR, elevated WBC, laparoscopy (gold standard confirmation).

PYQ — Outpatient antibiotic regimen:

A 24-year-old sexually active woman diagnosed with PID, suitable for outpatient treatment. Appropriate regimen? - A. Azithromycin 1 g + metronidazole $\times 14$ days - B. Ceftriaxone 500 mg IM single dose + doxycycline 100 mg BD $\times 14$ days + metronidazole 500 mg BD $\times 14$ days ← ANSWER - C. Levofloxacin 500 mg + metronidazole $\times 14$ days - D. Amoxicillin-clavulanate + doxycycline $\times 14$ days

Rationale: - **Ceftriaxone:** covers *Neisseria gonorrhoeae* (3rd-gen cephalosporin) - **Doxycycline:** covers *Chlamydia trachomatis* - **Metronidazole:** covers anaerobes and BV-associated organisms

Inpatient indications: Tubo-ovarian abscess, surgical emergency cannot be excluded, failed outpatient treatment, pregnancy, severe illness (high fever, vomiting, peritoneal signs).

Fitz-Hugh-Curtis Syndrome (Perihepatitis)

PYQ:

A 24-year-old sexually active woman presents with right upper quadrant pain, fever, and cervical motion tenderness. Laparoscopy shows "violin string" adhesions between the liver capsule and anterior abdominal wall. This is: - A. Acute cholecystitis - B. Fitz-Hugh-Curtis syndrome (perihepatitis complicating PID) ← ANSWER - C. Hepatic abscess - D. Budd-Chiari syndrome

Key features: - Complication of PID (*Neisseria gonorrhoeae* or *Chlamydia* reach the liver capsule via peritoneal spread or lymphatics) - **Violin string adhesions** (fibrinous then fibrous bands) between liver capsule and anterior abdominal wall/diaphragm - RUQ pain can mimic cholecystitis or pleurisy - Confirmed on laparoscopy; mildly elevated LFTs - **Treatment: same as PID** (antibiotics, NOT surgical)

Genital Herpes (HSV) in Labour

PYQ:

A pregnant woman presents in labour with active genital ulcers likely due to HSV. Management? - A. LSCS (Lower Segment Caesarean Section) ← ANSWER - B. Classic Caesarean Section - C. Vaginal Delivery - D. Induction of Labour

Active genital herpes at onset of labour = indication for LSCS to prevent vertical transmission (neonatal herpes is a devastating infection). Vaginal delivery is only acceptable when there are no active lesions and no prodromal symptoms.

Tubercular Salpingitis — Special Points

PYQ:

All of the following about tubercular salpingitis are correct EXCEPT: - A. Abdominal ostium may be patent with eversion of fimbriae - B. Early lesions may be confused with adenocarcinoma on histology - C. Genital TB is always secondary, with tubes as primary genital site - D. Salpingitis isthmica nodosa is exclusive to tuberculosis ← ANSWER (FALSE)

Salpingitis isthmica nodosa (SIN) = diverticular outpouching of tubal epithelium into the muscularis at the isthmus. It occurs in TB but is **not exclusive** to TB — also seen in gonorrhoeal and other chronic salpingitis. This is a consistent "NOT exclusive" trap.

Other truths about genital TB: - Always secondary (primary site = lungs or lymph nodes; spreads haematogenously) - Fallopian tubes are the **primary genital site** (100% of genital TB cases) - "Tobacco pouch" appearance of tubes (fibrous, thickened fimbriae) - Caseous granulomas on histology; early stages mimic adenocarcinoma - **Asherman's syndrome** (intrauterine adhesions) is a known complication - Treatment: standard 6-month anti-TB regimen (HRZE × 2 months → HR × 4 months)

PART 4: Vaginal Physiology and Cervical Examination

Vaginal Changes in Pregnancy

PYQ:

Vaginal changes during pregnancy include: - A. Decreased Lactobacilli - B. Increased glycogen metabolism ← ANSWER - C. pH > 6 - D. Epithelial thinning

Mechanism: Oestrogen → **epithelial thickening** + glycogen accumulation in squamous cells → Lactobacilli metabolise glycogen → **lactic acid** → acidic pH (3.5–4.5) → protective against infection. In pregnancy, Lactobacilli **increase** (not decrease) because substrate (glycogen) is abundant. Vaginal pH remains acidic, not alkaline.

Eosinophilic (Karyopyknotic) Index

PYQ:

The eosinophilic (karyopyknotic) index of lateral vaginal wall cells indicates: - A. Effect of LH - B. Oestrogenic effect ← ANSWER - C. Progesterone effect - D. Prolactin effect

The eosinophilic index measures the percentage of superficial vaginal cells with eosinophilic cytoplasm and pyknotic nuclei. This reflects **oestrogen stimulation** — oestrogen causes superficial cell maturation (pink, small nucleus). It peaks at ovulation and is used in vaginal cytology to assess hormonal status. Progesterone causes intermediate cell predominance (cyanophilic, folded edges).

Cervical Ectropion (Erosion)

PYQ:

"Congenital" cervical erosion (ectropion) regresses after birth but may reappear at: - A. One year - B. Two years - C. Five years - D. Puberty ← ANSWER

Congenital ectropion: columnar epithelium extends onto the ectocervix (visible as red velvety area) due to maternal oestrogen stimulation. After birth, as maternal oestrogen declines, it regresses back behind the squamocolumnar junction. At **puberty**, rising oestrogens cause cervical eversion again. It also recurs with pregnancy and OCP use.

Physiological Leucorrhoea

PYQ:

Normal physiological vaginal discharge (leucorrhoea) is caused by: - A. Cervical mucus + Bartholin gland secretions - B. Transudation of plasma through vaginal epithelium ← ANSWER - C. Endometrial secretions - D. Peritoneal fluid

Normal discharge = plasma transudation through vaginal epithelium + cervical mucus. Bartholin glands provide only minor secretions for lubrication at the introitus. Leucorrhoea increases at mid-cycle (oestrogen peak, abundant clear cervical mucus), in pregnancy, and with arousal.

Bias Removal: Key Confusions

1. Septate vs. Bicornuate — Never Confuse Again

The external contour of the uterus distinguishes them: - **Septate**: External = **normal** (round/flat fundus); Internal = divided by septum; treat with hysteroscopy - **Bicornuate**: External = **cleft fundus** (heart-shaped); Internal = muscular separation; surgery generally NOT needed

HSG shows both as divided cavity — you cannot tell them apart on HSG. MRI shows the external contour. 3D USS also helpful.

The exam will give you an HSG or MRI finding + history of RPL and ask "management." If it says "septate" → hysteroscopic septoplasty. If it says "bicornuate" → NO surgery (Strassmann metroplasty is an option but not the expected answer unless specifically asked).

2. BV vs. Trichomonas — The Discharge Mixup

Feature	BV	Trichomonas
Discharge	Thin, grey, homogeneous	Frothy , yellow-green
Odour	Fishy (amine)	Mild odour
pH	> 4.5	> 4.5
Wet mount	Clue cells	Motile flagellates
Cervix	Normal	Strawberry cervix
Treatment	Metronidazole (not partner)	Metronidazole (both partners)
STI?	Associated but not classic STI	Yes, STI

3. AIS vs. MRKH — Same Presentation, Different Cause

Both present with **primary amenorrhoea and absent uterus**. The critical distinguishing features:

AIS	MRKH
46,XY	46,XX
Testes in inguinal canal	Normal ovaries in pelvis
Absent/sparse pubic and axillary hair	Normal hair
Short blind vagina	Short or absent vagina
Normal breast development	Normal breast development
High testosterone (but no effect)	Normal female hormone levels



Variations and Twists

Scenario 1: Primary Amenorrhoea with Cyclic Pelvic Pain

A 16-year-old presents with crampy lower abdominal pain every month for 6 months, no menstrual bleeding. Examination shows a bulging bluish membrane at the introitus.

Diagnosis: Imperforate hymen with haematocolpos

Management: Cruciate incision (hymenotomy) under anaesthesia. Pre-operative USS/MRI confirms haematocolpos and haematometra. Excellent prognosis if managed early (before secondary tubal damage from retrograde blood flow).

Do not mistake this for a vaginal septum (no visible bulging membrane) or MRKH (no cyclic pain — no uterus to cause pain).

Scenario 2: Recurrent Pregnancy Loss — Investigation Pathway

A 27-year-old has had 3 first-trimester pregnancy losses. Investigations show normal parental karyotype, normal thrombophilia screen, normal hormones.

Next investigation: 3D USS or HSG to assess uterine cavity

If a filling defect is found → **hysteroscopy** (diagnostic and therapeutic). If septate uterus found → **hysteroscopic septoplasty**.

Complete work-up for RPL: 1. Parental karyotypes (chromosomal cause, 5%) 2. Antiphospholipid syndrome (lupus anticoagulant + anti-cardiolipin antibodies — dRVVT test) 3. Uterine anatomy (3D USS, HSG, hysteroscopy) 4. Thyroid function, diabetes screen 5. Thrombophilia screen (factor V Leiden, protein C/S)

PYQ — Antiphospholipid syndrome test:

A 26-year-old with recurrent abortions. Which investigation confirms antiphospholipid syndrome? - A. PT - B. BT - C. Anti-Russell viper venom (anti-RVV) antibodies ← ANSWER - D. Clot solubility test

Lupus anticoagulant is detected by the **dilute Russell viper venom time (dRVVT)**. Together with anti-cardiolipin antibodies, it confirms APS.

Scenario 3: Right Upper Quadrant Pain in a Young Sexually Active Woman

A 22-year-old with RUQ pain, mild fever, mildly elevated ALT, and pelvic pain. On pelvic examination: cervical motion tenderness.

Think Fitz-Hugh-Curtis syndrome. The combination of RUQ pain + cervical motion tenderness is classic. Laparoscopy would show "violin string" adhesions. Treat as PID.

Memory Engineering

Müllerian Anomaly Classification — "Absent / Uni / Didel / Bi / Sep / Arc"

Number	Memory	Anomaly
I	Absent	MRKH (agenesis)
II	Unicorn	Unicornuate
III	Didel	Didelphys
IV	Bicorn	Bicornuate
V	Septate	Septate (most common)
VI	Arcuate	Arcuate (near-normal)

Amsel Criteria for BV — "PCDW"

- PH > 4.5
- Clue cells on wet mount
- Discharge (thin, grey, homogeneous)
- Whiff test positive (KOH → fishy amine)

3 of 4 = BV diagnosed.

STD Pathognomonic Signs — "Clue-Straw-Gram"

- **Clue** cells = BV (Gardnerella)
- **Strawberry** cervix = Trichomonas
- **Gram**-negative diplococci = Gonorrhoea
- **Pseudohyphae** = Candida
- **Schiller-Duval** bodies = Yolk sac tumour (not STD — bonus memory)

AIS Memory — "X-Y, No Y-es"

AIS = 46,**XY** karyotype, **No** uterus, **No** axillary/pubis hair, **Yes** breast development (aromatisation), **Yes** testes (in inguinal canal).

Ultra High-Yield Revision Layer

Müllerian Anomaly Summary Table

Anomaly	Most Common	RPL Risk	Surgery	Special Feature
Septate	YES	Highest	Hysteroscopic septoplasty	External contour normal
Bicornuate	—	Moderate	Not usually	Fundal cleft on MRI
Didelphys	—	Moderate	Not usually	Two cervices, two vaginae
Unicornuate	—	Moderate	Rudimentary horn excision	Risk of ectopic in horn
MRKH	—	—	Vaginal dilators (Frank's)	Normal ovaries; 46,XX
Arcuate	—	Minimal	None	Minor fundal indentation

Vaginal Discharge Differential — One-Glance Table

Feature	BV	Trichomonas	Candida	Normal
Colour	Grey-white	Yellow-green	White "cheesy"	Clear/white
Consistency	Thin, homogeneous	Frothy	Thick, curdy	Variable
Odour	Fishy	Mild	None	None
pH	> 4.5	> 4.5	< 4.5	3.5–4.5
Wet mount	Clue cells	Motile flagellates	Pseudohyphae	Lactobacilli
Itch	Mild	Severe	Severe	None
Cervix	Normal	Strawberry	Normal	Normal
Treatment	Metronidazole	Metronidazole (partners)	Clotrimazole/fluconazole	—

PID Management Summary

Severity	Setting	Regimen
Mild-moderate	Outpatient	Ceftriaxone 500 mg IM × 1 dose + Doxycycline 100 mg BD × 14d + Metronidazole 500 mg BD × 14d
Severe / TOA	Inpatient IV	Cefoxitin 2 g IV 6-hourly OR clindamycin + gentamicin; switch to oral after 24h improvement
Both partners	—	Screen and treat male partner for gonorrhoea/chlamydia

Developmental Anomaly — Primary Amenorrhoea Decision

Features	Diagnosis
Cyclic pain + bulging blue membrane	Imperforate hymen
No pain + absent uterus on USS + 46,XX + normal ovaries	MRKH
Female phenotype + absent uterus + no pubic hair + 46,XY	AIS (CAIS)
Short stature + webbed neck + streak gonads + 45,XO	Turner syndrome
Female body + no secondary sex characteristics + delayed bone age	Primary ovarian insufficiency / hypogonadotropic hypogonadism

The infectious/anomaly pairing in this chapter looks broad, but the exam tests the same 6–8 specific facts repeatedly. Mastering the tables in the Ultra High-Yield Revision Layer and the bias-removal section will cover the vast majority of MCQs on these topics.

Chapter 11: Cervical Cancer & Endometrial Cancer

Big Picture First

Gynaecological oncology is the single heaviest exam topic in OBG — expect 8–12 marks across cervical, endometrial, ovarian, and GTD questions combined. Cervical cancer is the **number one female cancer in India** (second globally after breast cancer). Examiners love three things: (1) FIGO 2018 staging questions — especially the hydronephrosis trap, (2) Pap smear / Bethesda system interpretation, and (3) treatment selection by stage. Endometrial cancer questions focus on Type 1 vs Type 2 distinction, hyperplasia risk table, and Lynch syndrome. Both topics are highly rewarding per hour of study because the same facts appear year after year.

Question Pattern Map — What Examiners Actually Ask

Cervical cancer traps (appearing in multiple NEET-PG years): - Hydronephrosis = Stage **IIIB** NOT IVA (IVA = bladder/rectal invasion — a different finding entirely) - FIGO 2018 changed the parametrium rule: Stage IIB now includes any parametrial extension; Stage IIIC added pelvic and para-aortic lymph node metastases (IIIC1 = pelvic LN, IIIC2 = para-aortic LN) — old question banks may use pre-2018 staging - Microinvasion dimensions: IA1 = depth ≤ 3 mm; IA2 = depth > 3 mm and ≤ 5 mm (width ≤ 7 mm, now removed in 2018) - Wertheim's hysterectomy is for Stage IB1 to IIA — never for Stage IIB or beyond - Point A (brachytherapy) = 2 cm above external os AND 2 cm lateral to uterine canal (NOT 5 cm — classic distractors use wrong lateral distance) - HPV 16 = most common in squamous cell carcinoma; HPV 18 = more associated with adenocarcinoma - Squamous cell carcinoma = 70–80% of cervical cancers; adenocarcinoma = 20–25% (rising)

Endometrial cancer traps: - Type 2 endometrial cancer arises on atrophic endometrium, NOT on hyperplastic endometrium — the inverse of what you might expect - Tamoxifen for breast cancer → oestrogen agonist on uterus → increased endometrial cancer risk - Lynch syndrome (HNPCC) = endometrial cancer is the SENTINEL cancer in women (not colon — colon comes first in men; in women, endometrial cancer may precede colon cancer) - Progestin is both the treatment for hyperplasia AND the protective factor against endometrial cancer from unopposed oestrogen - EIN (Endometrial Intraepithelial Neoplasia) = old terminology "complex atypical hyperplasia" = highest risk of progression (~29%)

Core Concept Deep Dive

11.1 HPV and Cervical Carcinogenesis

Human papillomavirus is a double-stranded DNA virus transmitted sexually. Of 200+ HPV types, ~14 are classified as high-risk (oncogenic).

Key HPV types to memorise:

HPV Type	Association
16	Squamous cell carcinoma of cervix (40–70%)
18	Adenocarcinoma of cervix; aggressive SCC
6, 11	Condyloma acuminata (genital warts) — LOW-RISK
31, 33, 45	Also oncogenic, less commonly tested

The oncogenic mechanism: E6 protein of HPV 16/18 degrades p53 (tumour suppressor); E7 protein inactivates pRb (retinoblastoma protein). Both drive uncontrolled cell proliferation. The transformation zone (squamocolumnar junction) of the cervix is the vulnerable site — columnar cells undergoing squamous metaplasia here are most susceptible to HPV integration.

HPV Vaccines — the exam now asks about schedule:

- **Cervarix (bivalent):** covers HPV 16 + 18 (oncogenic only)
- **Gardasil 4 (quadrivalent):** covers HPV 6, 11, 16, 18 (warts + cancer)
- **Gardasil 9 (nonavalent):** covers 9 types including 31, 33, 45, 52, 58

Vaccination schedule (WHO / India MoHFW): - Age 9–14 years: **2 doses**, 6 months apart - Age ≥ 15 years or immunocompromised: **3 doses** (0, 1–2, 6 months) - Note: WHO now endorses single-dose efficacy for routine national programmes — this is tested in recent NEET-PG questions

Exam Question — HPV Type (NEET-PG 2018-style)

Q116 — HPV type most commonly associated with carcinoma cervix:

- A. HPV 35
- B. HPV 33
- C. HPV 18
- D. HPV 16

► Answer & Explanation:

****D. HPV 16**** — HPV 16 causes 40–70% of invasive squamous cell carcinomas of the cervix. HPV 18 is more strongly linked to adenocarcinoma and accounts for an additional 10–15% of all cervical cancers. Together, HPV 16 + 18 account for approximately 70% of all cervical cancers — which is why the bivalent vaccine offers significant protection. Options A (HPV 35) and B (HPV 33) are high-risk types but far less common.

11.2 Risk Factors for Cervical Cancer

Risk factors (memory: "SIMPLE"): - **S** — Sexual activity: early age of first intercourse (< 16 years), multiple partners, partner with multiple partners - **I** — Immunosuppression (HIV, transplant patients) - **M** — Multiple pregnancies (multiparity) - **P** — Partner: high-risk male (history of penile cancer, multiple partners) - **L** — Long-term OCP use (slight increase in adenocarcinoma risk) - **E** — Environment: **smoking** (doubles the risk), low socioeconomic status, HPV exposure

NOT risk factors: Late menarche, nulliparity, obesity (these are endometrial cancer risk factors — the examiner deliberately places them in cervical cancer options to trap you).

Exam Question — Risk Factors (NEET-PG pattern)

Q126 — Risk factors for carcinoma cervix include: (a) HPV (b) Smoking (c) Late menarche (d) Nulliparity (e) Early sexual intercourse

- A. a, b, e
- B. b, c, d
- C. c, d, e
- D. a, c, d

► Answer & Explanation:

****A. a, b, e (HPV, Smoking, Early sexual intercourse)**** — These are the three established major risk factors. Late menarche (option c) is actually a PROTECTIVE factor for endometrial cancer (fewer anovulatory cycles, less oestrogen exposure). Nulliparity (option d) is a risk factor for endometrial cancer, not cervical cancer — in fact, high parity increases cervical cancer risk, while nulliparity does not. The examiner specifically places "late menarche" and "nulliparity" here because candidates confuse cervical and endometrial risk factor profiles.

11.3 Cervical Cancer Screening — The Full Picture

Pap Smear (Papanicolaou Smear): - Collects exfoliated cells from the transformation zone (using Ayre's spatula + endocervical brush or combined brush) - Interpreted using the **Bethesda System (2001)**:

Bethesda Classification	Old CIN Terminology	Clinical Meaning
NILM	Normal	No lesion
ASC-US	Atypical cells, undetermined significance	Most common abnormal result; reflex HPV testing
ASC-H	Atypical cells, cannot exclude HSIL	Urgent colposcopy
LSIL	CIN I + HPV effect (koilocytosis)	Low-grade; often regresses spontaneously
HSIL	CIN II + CIN III	High-grade; immediate colposcopy + biopsy
AGC	Atypical glandular cells	Investigate for endocervical/endometrial pathology
AIS	Adenocarcinoma in situ	Cone biopsy required
Carcinoma	Invasive cancer	Urgent referral

Key Bethesda fact for NEET-PG: LSIL = CIN I; HSIL = CIN II + CIN III. This is tested almost every year.

Koilocytes = hallmark cytological feature of HPV infection: cells with perinuclear clearing (halo) and wrinkled/raisinoid nucleus. Their presence = active HPV cytopathic effect.

CIN Grading (histological): - **CIN I (mild dysplasia):** Dysplastic cells confined to lower 1/3 of epithelium; high rate of spontaneous regression (~60%) - **CIN II (moderate dysplasia):** Dysplasia in lower 2/3; intermediate behaviour - **CIN III (severe dysplasia / Carcinoma in situ):** Full-thickness or near-full-thickness dysplasia; ~30% progress to invasive cancer if untreated

Exam Question — Bethesda System (NEET-PG 2018-style)

Q41 — LSIL in the Bethesda classification corresponds to:

- A. CIN I (mild dysplasia)
- B. CIN II (moderate dysplasia)
- C. CIN III (severe dysplasia/CIS)
- D. Squamous metaplasia

► Answer & Explanation:

****A. CIN I**** — Bethesda 2001: LSIL = CIN I + HPV cytopathic effect (koilocytosis). HSIL encompasses both CIN II and CIN III. Option D (squamous metaplasia) is a physiological process, not a dysplastic lesion — it is the normal process at the transformation zone where columnar cells are replaced by squamous cells, and it appears pink/mature on cytology, not dysplastic. ASC-US is the single most common abnormal Pap result (not LSIL — another frequent confusion).

VIA (Visual Inspection with Acetic Acid): - Acetic acid 3–5% applied to cervix; acetowhite lesions = positive - Detects high-grade CIN (CIN II/III) by causing temporary opacity in dysplastic cells due to protein coagulation - Acetowhite lesions at the transformation zone are colposcopically suspicious - Sensitivity ~79%, specificity ~85% for CIN II+ - Used in screen-and-treat programmes in low-resource settings (India's national programme)

VILI (Visual Inspection with Lugol's Iodine / Schiller's Test): - Normal glycogen-rich squamous cells stain mahogany brown (Schiller-positive = iodine-positive = normal) - Dysplastic cells lack glycogen → **iodine-negative (Schiller-negative)** = abnormal area = positive VILI test - Note the confusing double-negative: Schiller-negative means abnormal. "Schiller-negative" lesion = suspicious. - False positives (Schiller-negative but benign): columnar epithelium, metaplastic areas, atrophic epithelium, erosions

Colposcopy: - Magnified visualisation of cervix after acetic acid application - Colposcopic features of high-grade CIN: acetowhite epithelium with coarse punctuation, mosaic pattern, atypical vessels, abnormal transformation zone - Colposcopy-directed (punch) biopsy is the diagnostic gold standard for confirming CIN grade before treatment - Step in management: Abnormal Pap → Colposcopy → Directed biopsy → Histological confirmation → Treatment

Exam Question — Next Step after Abnormal Pap (NEET-PG 2018)

Q175 — A 45-year-old woman has post-coital bleeding and a positive Pap smear. Next line of management?

- A. Colposcopy-directed biopsy
- B. Cone biopsy
- C. Repeat Pap smear
- D. Hysterectomy

► Answer & Explanation:

****A. Colposcopy-directed biopsy**** — A positive Pap smear (abnormal cytology) requires HISTOLOGICAL confirmation before any treatment decision. Colposcopy identifies the most abnormal-appearing area; directed (punch) biopsy provides the tissue. Option B (cone biopsy) is both diagnostic AND therapeutic but is reserved for: (1) unsatisfactory colposcopy (transformation zone not fully visible), (2) suspected microinvasive disease, (3) positive endocervical curettage. Option C (repeat Pap) only wastes time when a positive result is already in hand. Option D (hysterectomy) without histological staging would be inappropriate — you cannot stage or select treatment without biopsy confirmation.

Exam Question — CIN on Pap Smear (NEET-PG 2020-style)

Q691 — PAP smear shows carcinoma-in-situ (CIN III / HSIL). Best next logical procedure:

- A. Cold-knife conisation
- B. Colposcopy + directed biopsy
- C. Hysterectomy
- D. HPV viral DNA typing

► Answer & Explanation:

****B. Colposcopy + directed biopsy**** — Even for CIN III on cytology, histological confirmation is mandatory before definitive treatment. Colposcopy will confirm the lesion; directed biopsy will provide the tissue diagnosis. Cold-knife conisation (option A) is used when colposcopy is unsatisfactory or microinvasion is suspected — not as the first step after an abnormal Pap. Hysterectomy (option C) is never done based on cytology alone without histology. HPV typing (option D) adds no management value when HSIL is already demonstrated.

11.4 Treatment of CIN

Grade	Treatment
CIN I	Observe (regression in 60%); treat at 24 months if persistent
CIN II	LLETZ / LEEP (loop excision); cryotherapy in resource-limited settings
CIN III	LLETZ / LEEP as first line; cold-knife cone if lesion extends to endocervical canal
CIN III + pregnancy	Defer treatment until after delivery; re-biopsy at 6 weeks postpartum

LLETZ (Large Loop Excision of the Transformation Zone) = LEEP in Indian terminology: - Uses low-voltage diathermy loop to excise the transformation zone - Advantages: quick, outpatient, provides specimen for histology (unlike ablative methods) - Loop size < 2 cm gives a better specimen with less thermal artefact at margins - Faster than laser ablation

Exam Question — LLETZ False Statement

Q436 — Which is NOT true regarding LLETZ?

- A. Uses low-voltage diathermy

- B. Applicable only to the cervix
- C. Loop size > 2 cm gives a better cone specimen
- D. Shorter procedure time than laser ablation

► **Answer & Explanation:**

****C. Loop size > 2 cm gives a better cone specimen**** — FALSE. A loop size LESS THAN 2 cm provides a better, more precise specimen with minimal thermal damage at the margins. The larger the loop, the greater the peripheral thermal artefact, which can compromise histological assessment of resection margins. All other options are true: LLETZ uses low-voltage diathermy (A), is cervix-specific (B), and is indeed faster than laser ablation (D) — making LLETZ the preferred modality in most settings.

11.5 FIGO 2018 Cervical Cancer Staging

FIGO 2018 introduced two major changes from the older system: (1) imaging and pathological findings can now be used to stage (previously clinical examination only), and (2) lymph node involvement is now included as Stage IIIC.

Complete FIGO 2018 Staging Table:

Stage	Definition
Stage I	Tumour confined to cervix
IA	Microscopic invasion only (diagnosed by microscopy)
IA1	Stromal invasion ≤ 3.0 mm
IA2	Stromal invasion > 3.0 mm and ≤ 5.0 mm
IB	Clinically visible or microscopic lesion $>$ Stage IA
IB1	Tumour ≤ 2 cm (previously ≤ 4 cm before 2018)
IB2	Tumour > 2 cm and ≤ 4 cm
IB3	Tumour > 4 cm
Stage II	Beyond cervix but not to pelvic wall / lower 1/3 vagina
IIA	No parametrial involvement
IIA1	Tumour ≤ 4 cm
IIA2	Tumour > 4 cm
IIB	Parametrial involvement (any)
Stage III	Tumour to pelvic wall / lower 1/3 vagina / hydronephrosis / LN mets
IIIA	Lower 1/3 vaginal involvement (no pelvic wall)
IIIB	Extension to pelvic wall AND/OR hydronephrosis / non-functioning kidney
IIIC1	Pelvic lymph node metastasis (NEW in FIGO 2018)
IIIC2	Para-aortic lymph node metastasis (NEW in FIGO 2018)
Stage IV	Spread beyond true pelvis / involvement of bladder or rectum
IVA	Invasion of bladder or rectal mucosa
IVB	Distant metastasis (lung, liver, bone)

Visual Staircase Memory Aid:

Stage I → Cervix only
 Stage II → Beyond cervix (IIA = vagina upper 2/3; IIB = parametrium)
 Stage III → Pelvic wall / lower vagina / HYDRONEPHROSIS(IIIB) / LYMPH NODES(IIIC)
 Stage IV → BLADDER/RECTUM (IVA) / Distant (IVB)

The Two Biggest NEET-PG Traps:

Trap 1 — Hydronephrosis: "Cervical cancer patient found to have hydronephrosis on IVP" = Stage **IIIB**, not IVA. IVA = bladder/rectal *mucosal* invasion (confirmed on cystoscopy/sigmoidoscopy, not just displacement). Hydronephrosis = ureteric obstruction by parametrial tumour = Stage **IIIB**.

Trap 2 — Bilateral hydronephrosis: Still Stage **IIIB**. The staging does not jump to Stage IV just because both kidneys are involved.

Exam Question — Hydronephrosis Staging (NEET-PG multiple years)

Q17 — Cervical cancer with hydronephrosis (non-functional kidney). FIGO stage?

- A. IIA
- B. IIIA
- C. IIIB
- D. IV

► Answer & Explanation:

****C. IIIB**** — FIGO 2018 (and even the older FIGO 2009): hydronephrosis or non-functioning kidney caused by tumour extension = Stage IIIB, regardless of primary tumour size or how mild the parametrial extension appears on examination. This is one of the highest-frequency staging traps on NEET-PG. Option A (IIA) = upper vaginal extension without parametrial involvement; Option B (IIIA) = lower 1/3 vaginal involvement; Option D (IV) requires either bladder/rectal mucosal invasion (IVA) or distant metastasis (IVB). Hydronephrosis = Stage IIIB.

11.6 Treatment of Cervical Cancer by Stage

Stage	Treatment of Choice
IA1 (no LVSI)	Conisation (fertility-sparing) or simple hysterectomy
IA1 (with LVSI)	Radical hysterectomy + pelvic LND, OR chemoradiation
IA2	Modified radical hysterectomy (Wertheim's Type II) + pelvic LND
IB1	Radical hysterectomy (Wertheim's Type III) + pelvic LND, OR chemoradiation
IB2–IIA	Radical hysterectomy OR concurrent chemoradiation (equivalent outcomes)
IIB and beyond	Concurrent chemoradiation (cisplatin-based) + brachytherapy
IVB	Palliative chemotherapy ± palliative RT

Wertheim's Hysterectomy: Radical (type III) hysterectomy removing uterus, upper 2–3 cm of vagina, parametrium to pelvic wall, and pelvic lymph nodes. Used for stages IA2 to IIA. **NOT** used for stage IIB or beyond (too much local disease for resection with clear margins).

Schauta's Operation: Radical vaginal hysterectomy (equivalent to Wertheim's but vaginal approach). Rarely used but may appear in older question banks.

Chemoradiation: - External beam radiotherapy (EBRT) to pelvis + concurrent cisplatin (weekly, 40 mg/m²) + intracavitary brachytherapy (ICB) - Standard for Stage IIB, III, IVA - **Point A** (brachytherapy reference point): 2 cm above external os AND **2 cm** lateral to uterine canal = receives ~7000–8500 cGy total - **Point B:** 3 cm lateral to Point A (5 cm from midline) = receives ~5000–6000 cGy (covers pelvic lymph nodes) - Note: Point A is 2 cm lateral, NOT 5 cm lateral — the 5 cm figure is the total distance of Point B from midline

Exam Question — Stage IIB Treatment (NEET-PG 2023)

Q245 — A 50-year-old woman with foul-smelling bloody discharge and lateral parametrium involvement. Management?

- A. Chemotherapy
- B. Brachytherapy
- C. Chemoradiation
- D. Surgery

► Answer & Explanation:

****C. Chemoradiation**** — Parametrial involvement = Stage IIB cervical cancer. Standard treatment is concurrent cisplatin-based chemoradiation (external beam RT + weekly cisplatin + intracavitary brachytherapy). Option A (chemotherapy alone) is never used as definitive treatment for localised cervical cancer. Option B (brachytherapy alone) is inadequate as primary therapy without EBRT for a stage IIB lesion. Option D (surgery — Wertheim's hysterectomy) is appropriate only up to Stage IIA; Stage IIB has parametrial involvement that cannot be cleared surgically with adequate margins.

Exam Question — Stage IIIB Treatment (NEET-PG pattern)

Q127 — Treatment for stage IIIB carcinoma cervix:

- A. Wertheim's hysterectomy
- B. Schauta's procedure
- C. Chemotherapy alone
- D. Intracavitary brachytherapy + external beam RT (with concurrent chemotherapy)

► Answer & Explanation:

****D. Brachytherapy + external beam RT (concurrent chemoradiation)**** — Stage IIIB (pelvic wall extension or hydronephrosis) is not surgically resectable. Standard of care: EBRT (45–50 Gy to the pelvis) + concurrent weekly cisplatin + intracavitary brachytherapy (boost to Point A). Options A and B are surgical operations appropriate only up to Stage IIA. Chemotherapy alone (option C) has insufficient local control for Stage III disease.

11.7 Endometrial Cancer — Type 1 vs Type 2

Endometrial carcinoma is the most common gynaecological malignancy in developed countries (in India, cervical cancer is commoner). Understanding the fundamental duality of Type 1 vs Type 2 is the foundation of exam performance.

Type 1 vs Type 2 Endometrial Cancer:

Feature	Type 1 (Endometrioid)	Type 2 (Serous/Clear Cell)
Frequency	~80%	~15–20%
Histology	Endometrioid adenocarcinoma	Serous or clear cell carcinoma
Oestrogen	Oestrogen-DEPENDENT	Oestrogen-INDEPENDENT
Background	Endometrial hyperplasia (EIN)	Atrophic endometrium
Typical patient	Obese, perimenopausal, younger	Thin, older, postmenopausal
Grade	Low-grade (G1, G2)	High-grade (G3)
Prognosis	Good (5-yr survival ~85%)	Poor (aggressive, early metastasis)
Molecular	PTEN, microsatellite instability, KRAS	TP53 mutation, HER2 amplification
Lynch syndrome	Associated (MLH1, MSH2 mismatch repair)	Not associated

Exam Question — Type 1 vs Type 2 (NEET-PG repeated question)

Q19 — Which statement about Type 1 versus Type 2 endometrial carcinoma is CORRECT?

- A. Type 1 is more aggressive with poorer prognosis than Type 2
- B. Type 1 is oestrogen-driven, arising in younger obese women on a background of hyperplasia
- C. Type 2 is low-grade and non-metastasising
- D. Type 2 is oestrogen-dependent and associated with endometrial hyperplasia

► Answer & Explanation:

****B. Type 1 is oestrogen-driven, arising in younger obese women on a background of hyperplasia**** — This is the correct statement. Option A reverses the prognosis: Type 2 is the aggressive one with poor prognosis. Option C reverses the grade: Type 2 is high-grade and metastasises early (peritoneal spread in serous carcinoma). Option D reverses the oestrogen dependence: Type 2 is oestrogen-INDEPENDENT — it arises on atrophic, not hyperplastic, endometrium, through TP53 mutation pathways. This question appears almost verbatim in multiple NEET-PG years because the Type 1/Type 2 reversal is the classic trap.

11.8 Risk Factors for Endometrial Cancer — Unopposed Oestrogen

The central concept: **unopposed oestrogen stimulates endometrial proliferation without progesterone opposition** → hyperplasia → carcinoma.

Risk factors (all increase oestrogen exposure): - Obesity (adipose tissue converts androstenedione to oestrone via aromatase) - Nulliparity (more anovulatory cycles, prolonged unopposed oestrogen) - Late menopause (prolonged oestrogen exposure) - Early menarche (same reason) - Polycystic ovary syndrome (chronic anovulation) - Oestrogen-secreting tumours (granulosa cell tumour) - Exogenous unopposed oestrogen (HRT without progestogen in women with intact uterus) - **Tamoxifen use** (for breast cancer — acts as oestrogen agonist on uterus) - Diabetes mellitus, hypertension (associated but likely via obesity linkage) - Lynch syndrome / HNPCC (hereditary mismatch repair gene defect)

Protective factors (add progesterone opposition or reduce oestrogen): - Combined OCP use (progestogen component opposes oestrogen) - Smoking (paradoxically anti-oestrogenic via increased 2-hydroxylation) - Multiparity - Physical activity - Progestogen-only IUD (LNG-IUS)

Lynch Syndrome (HNPCC) and Endometrial Cancer: - Autosomal dominant; mutations in mismatch repair genes: MLH1, MSH2, MSH6, PMS2 - Lifetime risk of endometrial cancer: 40–60% (equal to or greater than colon cancer risk in women) - Endometrial cancer is the SENTINEL cancer in women with Lynch syndrome (may precede colon cancer) - Screening: annual endometrial sampling from age 35; prophylactic hysterectomy + BSO after childbearing completed - Association: endometrioid subtype of both endometrial AND ovarian carcinoma

11.9 Endometrial Hyperplasia — The Cancer Risk Progression Table

Histological Type	Risk of Progression to Carcinoma
Simple hyperplasia without atypia	~1%
Simple hyperplasia with atypia	~8%
Complex hyperplasia without atypia	~3%
Complex hyperplasia WITH atypia (EIN)	~29%

EIN (Endometrial Intraepithelial Neoplasia) = modern terminology for complex atypical hyperplasia. The "atypia" is the key determinant — architecture alone (simple vs complex) matters much less than cytological atypia.

Exam Question — Endometrial Hyperplasia Cancer Risk (NEET-PG pattern)

Q437 — Which histological pattern of endometrial hyperplasia carries the HIGHEST risk of progression to endometrial carcinoma?

- A. Simple hyperplasia without atypia (~1%)
- B. Simple hyperplasia with atypia (~8%)
- C. Complex hyperplasia without atypia (~3%)
- D. Complex hyperplasia with atypia (~29%)

► Answer & Explanation:

****D. Complex hyperplasia with atypia (~29%)**** — The critical variable is cytological atypia (nuclear enlargement, prominent nucleoli, irregular chromatin), NOT the architectural pattern. Simple atypical hyperplasia (B, ~8%) already carries higher risk than complex hyperplasia without atypia (C, ~3%), because atypia outweighs architecture in predicting malignant potential. Complex atypical hyperplasia / EIN at ~29% risk is managed by: hysterectomy in women who have completed childbearing; continuous progestogen (medroxyprogesterone acetate 10 mg/day or LNG-IUS) in young women desiring fertility, with 3–6 monthly repeat biopsies until regression is confirmed.

11.10 Endometrial Cancer Staging — FIGO (2009, updated 2023)

Stage	Definition
IA	Tumour limited to endometrium or invades < 50% myometrium
IB	Tumour invades ≥ 50% myometrium
II	Tumour invades cervical stroma (but not beyond uterus)
IIIA	Tumour invades serosa or adnexa
IIIB	Vaginal or parametrial involvement
IIIC1	Pelvic lymph node metastasis
IIIC2	Para-aortic lymph node metastasis
IVA	Bladder or bowel mucosal involvement
IVB	Distant metastasis

Key differences from cervical staging: Endometrial staging is **surgical** (not clinical) — performed at time of hysterectomy. The lymph node categories (IIIC1/IIIC2) mirror the FIGO 2018 cervical system.

11.11 Endometrial Cancer Workup

Presenting symptom: Postmenopausal bleeding in 90% of cases (PMB = endometrial carcinoma until proven otherwise)

Investigation sequence: 1. Pelvic examination + Pap smear (Pap smear is often normal — it screens cervix, not endometrium) 2. **Transvaginal ultrasound (TVS):** Endometrial thickness (ET) - Postmenopausal, no HRT: ET > 4 mm → biopsy required - Postmenopausal on HRT: ET > 8 mm → biopsy - Premenopausal with AUB: ET ≥ 12 mm → biopsy 3. **Endometrial biopsy** (Pipelle sampler — outpatient): First-line tissue sampling; sensitivity ~95% for carcinoma 4. **D&C (Dilatation and Curettage):** If Pipelle sample inadequate; traditional gold standard for endometrial sampling 5. **Hysteroscopy + directed biopsy:** Best for polyps, focal lesions; gold standard for identifying and biopsying specific lesions 6. **MRI pelvis:** Assesses myometrial invasion depth and cervical involvement (staging pre-operatively)

Exam Question — Postmenopausal Bleeding (NEET-PG pattern)

Q107 — A 55-year-old postmenopausal woman with blood-stained vaginal discharge — next step?

- A. Reassure (postmenopausal discharge is normal)
- B. Immediate cardiology referral
- C. Pelvic exam, TVS, Pap smear
- D. Antihypertensives and observation

► Answer & Explanation:

****C. Pelvic exam, TVS, Pap smear**** — The axiom: postmenopausal bleeding = endometrial cancer until proven otherwise. Option A is dangerous — postmenopausal bleeding is NEVER normal and should never be dismissed. The initial workup includes pelvic examination (to rule out vulval/vaginal causes, visualise cervix), TVS (endometrial thickness), and Pap smear (to exclude cervical pathology). If ET > 4 mm on TVS, proceed to endometrial biopsy. The most important next step is NOT to reassure and NOT to ignore.

Exam Question — Perimenopausal AUB with Thick Endometrium

Q108 — A 46-year-old nulliparous woman with heavy irregular bleeding and endometrial thickness 18 mm — next step?

- A. Combined oral contraceptive pill
- B. Hysterectomy
- C. Wait and watch 6 months
- D. Endometrial biopsy

► Answer & Explanation:

****D. Endometrial biopsy**** — Endometrial thickness ≥ 12 mm in a perimenopausal woman with abnormal uterine bleeding mandates histological evaluation before any treatment. At 18 mm, the risk of hyperplasia or carcinoma is significant, and treatment cannot be selected without a tissue diagnosis. Starting hormonal therapy (option A) without excluding carcinoma could suppress symptoms while masking a developing malignancy. Hysterectomy (option B) without histological diagnosis is inappropriate. Waiting 6 months (option C) when ET is 18 mm and bleeding is present is unacceptable — the risk threshold has already been exceeded.

11.12 Treatment of Endometrial Cancer

Standard treatment (Stage I–II): - Total abdominal hysterectomy (TAH) + bilateral salpingo-oophorectomy (BSO) + pelvic lymph node assessment - Laparoscopic or robotic approach increasingly standard - Vaginal brachytherapy for Stage IA Grade 3 (vault radiotherapy) - External beam RT + chemotherapy for Stage III–IV

Fertility-sparing treatment (selected Stage IA, Grade 1, young woman): - Continuous high-dose progestogen (MPA 200 mg/day or LNG-IUS) - 3-monthly endometrial biopsies to confirm regression - IVF/pregnancy once remission documented - Definitive hysterectomy after childbearing

Bias Removal — Avoiding the Classic Cervical/Endometrial Confusion

The Four Most Common Exam Errors:

1. **Staging confusion — hydronephrosis:** Many candidates write Stage IIIC (LN involvement) for hydronephrosis. Wrong — hydronephrosis is Stage IIIB. IIIC is lymph nodes.
2. **Wertheim's for IIB:** Candidates correctly recall "Wertheim's = radical hysterectomy" but apply it to Stage IIB. Wrong — IIB has parametrial involvement; Wertheim's cannot achieve clear parametrial margins at Stage IIB. Chemoradiation is the answer.
3. **Point A = 5 cm lateral:** The distractors consistently give "5 cm lateral" for Point A to mislead. Point A = 2 cm lateral. Point B = 5 cm from midline (= 3 cm lateral to Point A). The question always specifies "Point A" to test this exact distinction.
4. **Schiller's test interpretation:** "Schiller-negative" means the iodine did NOT stain the tissue (abnormal, glycogen-poor). "Schiller-positive" (iodine-brown) = normal tissue. The double-negative confuses many candidates. A Schiller-negative lesion = suspicious = biopsy.

Ultra High-Yield Revision Layer

Top 10 Exam Facts — Cervical & Endometrial Cancer:

1. HPV 16 = most common in SCC; HPV 18 = adenocarcinoma; HPV 6/11 = warts (not cancer)
 2. LSIL = CIN I; HSIL = CIN II + CIN III (Bethesda 2001)
 3. Hydroureter = Stage **IIIB** (not IVA; IVA = bladder/rectal mucosa invasion)
 4. Wertheim's hysterectomy = up to Stage IIA only; IIB onward = chemoradiation
 5. Point A = 2 cm above external os + **2 cm** lateral; Point B = 3 cm further lateral (5 cm from midline)
 6. Type 1 endometrial cancer = oestrogen-driven, good prognosis; Type 2 = TP53, poor prognosis
 7. Lynch syndrome = endometrial cancer is sentinel cancer in women (40–60% lifetime risk)
 8. Complex atypical hyperplasia (EIN) = highest progression risk (~29%) → treat with hysterectomy or continuous progestin
 9. Postmenopausal bleeding: $ET > 4 \text{ mm}$ → endometrial biopsy; $ET \leq 4 \text{ mm}$ → 95% benign
 10. Tamoxifen = anti-oestrogenic in breast but pro-oestrogenic in uterus → endometrial cancer risk (annual endometrial surveillance for all women on tamoxifen)
-

Chapter 12: Ovarian Cancer & Gestational Trophoblastic Disease

Big Picture First

Ovarian cancer is called the "silent killer" because 70–75% of patients present at advanced stage (Stage III–IV), having had no specific early symptoms. The exam tests tumour markers (which marker with which tumour type), FIGO staging, and the distinction between epithelial, germ cell, and sex cord-stromal tumours. Get the marker table into your memory completely — it generates 2–3 questions every year.

Gestational Trophoblastic Disease (GTD) is a unique group of tumours arising from abnormal trophoblastic tissue. It includes molar pregnancy (complete and partial), invasive mole, choriocarcinoma, and placental site trophoblastic tumour. The critical exam facts are: (1) complete vs partial mole genetics, (2) post-molar beta-hCG surveillance protocol, (3) WHO/FIGO risk scoring for GTN, and (4) EMA-CO chemotherapy for high-risk disease. GTD is also remarkable because it can follow a completely normal pregnancy — an exam scenario favourite.

Question Pattern Map

Ovarian cancer traps: - CA-125 is NOT specific to ovarian cancer — it is elevated in endometriosis, PID, fibroids, pregnancy, and other cancers. The question will specify "primary tumour marker for ovarian cancer" — answer CA-125, but know its limitations. - Serous carcinoma = most common (70%) and most often bilateral (50%); mucinous = largest tumour; endometrioid = associated with endometriosis; Brenner = transitional (looks like urothelium) - Germ cell marker table: AFP = yolk sac tumour; HCG = choriocarcinoma (non-gestational); LDH + PLAP = dysgerminoma; AFP + HCG = embryonal carcinoma - Clear cell carcinoma is an EPITHELIAL tumour — NOT a germ cell tumour (despite the name "mesonephroid/clear cell" being confusingly similar to mesonephric/germ cell terminology) - Granulosa cell tumour = Call-Exner bodies + inhibin + oestrogen production (feminising) - Sertoli-Leydig cell tumour = androgen production (virilising/masculinising) - Meigs syndrome = ovarian FIBROMA + ascites + right hydrothorax

GTD traps: - Complete mole = 46XX (all paternal chromosomes), NO fetal parts, very high HCG → snowstorm on USS - Partial mole = 69XXX triploid, fetal parts present, lower HCG, Swiss-cheese placenta - "Complete mole → 15–20% risk of GTN; Partial mole → 0.5–1% risk of GTN" - Post-molar surveillance: weekly beta-hCG until normal, then monthly for 6 months (for complete mole). NO contraception for 6–12 months — but this means use CONTRACEPTION during surveillance (to avoid pregnancy confusing HCG results — the negative is the trap: "no contraception" = wrong) - GTN criteria: rising HCG or plateau for ≥ 3 weeks after evacuation, regardless of HCG value - WHO/FIGO score ≥ 7 = high-risk GTN → EMA-CO (not methotrexate alone) - Choriocarcinoma can follow ANY pregnancy — normal delivery, ectopic, abortion, or mole

Core Concept Deep Dive

12.1 Classification of Ovarian Tumours

Ovarian tumours are classified by cell of origin:

OVARIAN TUMOURS

- └─ EPITHELIAL (65–70% of all ovarian neoplasms; 90% of malignant ovarian tumours)
 - └─ Serous (most common)
 - └─ Mucinous (largest tumours)
 - └─ Endometrioid (associated with endometriosis)
 - └─ Clear cell / Mesonephroid (aggressive; associated with endometriosis)
 - └─ Brenner (transitional cell type; mostly benign)
 - └─ Mixed
- └─ GERM CELL (15–20%; most malignant germ cell tumours occur in young women)
 - └─ Dysgerminoma (most common MALIGNANT germ cell tumour)
 - └─ Mature cystic teratoma / Dermoid (most common BENIGN germ cell tumour)
 - └─ Immature teratoma
 - └─ Endodermal sinus / Yolk sac tumour
 - └─ Embryonal carcinoma
 - └─ Non-gestational choriocarcinoma
- └─ SEX CORD-STROMAL (5–8%)
 - └─ Granulosa cell tumour (oestrogen-producing, feminising)
 - └─ Thecoma (oestrogen-producing)
 - └─ Fibroma (associated with Meigs syndrome)
 - └─ Sertoli-Leydig cell tumour / Arrhenoblastoma (androgen-producing, virilising)
 - └─ Gonadoblastoma (in dysgenetic gonads)

12.2 Epithelial Ovarian Cancer — Subtypes

Serous cystadenocarcinoma: - Most common epithelial ovarian malignancy (~70%) - Bilateral in ~50% of cases - **Psammoma bodies** (calcified concentric lamellated structures) on histology — pathognomonic - Strongly CA-125 positive - Associated with **BRCA1/2** mutations (hereditary high-grade serous carcinoma) - BRCA1 mutations → 40–60% lifetime ovarian cancer risk; BRCA2 → 15–20%

Mucinous cystadenocarcinoma: - 15–20% of epithelial ovarian cancers - **Largest ovarian tumours** (can reach enormous sizes filling the abdomen) - Usually **unilateral** (bilateral mucinous = think metastatic Krukenberg) - Rupture → pseudomyxoma peritonei (mucin-secreting cells seed the peritoneum) - CEA may be elevated (mucinous tumours can produce CEA)

Endometrioid carcinoma: - ~20% of ovarian cancers - Associated with **ovarian endometriosis** (co-existent in ~15%) - Associated with **Lynch syndrome** (HNPCC — PTEN, ARID1A mutations) - Resembles endometrial glandular epithelium histologically - May coexist with a synchronous endometrial carcinoma (15% of cases)

Clear cell (Mesonephroid) carcinoma: - 5–10%; arises in endometriosis background - High-grade; poor prognosis relative to other epithelial subtypes - Characterised by "hobnail" cells and clear glycogen-rich cytoplasm - CA-125 variably positive; relatively chemoresistant

Brenner tumour: - Transitional cell type (histologically resembles urothelium = "coffee bean" grooved nuclei) - Usually benign (>95%); small, solid, unilateral - Often an incidental finding; may be associated with mucinous cystadenoma

Exam Question — Most Common Epithelial Ovarian Cancer (NEET-PG repeated)

Q18 — A 45-year-old with bilateral cystic-solid pelvic masses and elevated CA-125. Most likely epithelial ovarian cancer subtype:

- A. Mucinous cystadenocarcinoma
- B. Serous cystadenocarcinoma
- C. Endometrioid carcinoma
- D. Brenner tumour

► Answer & Explanation:

****B. Serous cystadenocarcinoma**** — High-grade serous carcinoma accounts for ~70% of epithelial ovarian malignancies and is bilateral in 50% — both features present in this scenario. Psammoma bodies on histology, strong CA-125 elevation, and BRCA1/2 association are its fingerprints. Mucinous (option A) tumours are typically unilateral and much larger (filling the abdomen). Endometrioid (option C) accounts for 20% and is associated with endometriosis — not characteristically bilateral. Brenner (option D) is almost always benign, small, and solid — not a large cystic-solid bilateral mass. The bilateral nature + CA-125 elevation points directly to serous carcinoma.

Exam Question — Endometrioid Ovarian Tumour Frequency

Q434 — Which ovarian tumour accounts for approximately 20% of all ovarian cancers and often co-exists with ovarian endometriosis?

- A. Serous cystadenocarcinoma
- B. Mucinous cystadenocarcinoma
- C. Endometrioid carcinoma
- D. Brenner tumour

► Answer & Explanation:

****C. Endometrioid carcinoma**** — The 20% figure and the endometriosis association are both the hallmarks of endometrioid ovarian carcinoma. This tumour shows glandular differentiation resembling endometrial glands and is associated with PTEN and ARID1A mutations. Option A (serous) accounts for ~70%; option B (mucinous) is the largest tumour, associated with pseudomyxoma peritonei if it ruptures; option D (Brenner) is mostly benign and very small. The endometriosis co-existence (option C) is the key distinguishing phrase — note that clear cell carcinoma also arises in endometriosis, but it does NOT account for 20% of ovarian cancers.

Exam Question — Pseudomyxoma Peritonei (NEET-PG pattern)

Q435 — Pseudomyxoma peritonei most commonly occurs as a complication of rupture of:

- A. Serous cystadenoma

- B. Mucinous cystadenoma / low-grade appendiceal mucinous neoplasm
- C. Dysgerminoma
- D. Gonadoblastoma

► Answer & Explanation:

****B. Mucinous cystadenoma / low-grade appendiceal mucinous neoplasm**** — Rupture of a mucinous tumour (ovarian or appendiceal) seeds the peritoneum with mucin-secreting epithelial cells, producing massive gelatinous peritoneal deposits — the jelly belly of pseudomyxoma peritonei. Modern evidence shows the appendix is the primary site in most true pseudomyxoma peritonei cases (with secondary ovarian involvement). The appendix should always be removed when pseudomyxoma peritonei is encountered. Dysgerminoma (option C) and gonadoblastoma (option D) are solid germ cell tumours with no mucin production capacity.

12.3 Tumour Markers — The Master Table (Memorise This)

This table generates 2–4 questions in every NEET-PG paper. Practise it until automatic.

Tumour	Primary Marker	Secondary/Other Markers
Epithelial ovarian cancer (serous)	CA-125	HE4, CEA (mucinous)
Dysgerminoma	LDH (+ PLAP)	beta-hCG (if syncytiotrophoblast elements)
Yolk sac tumour (Endodermal sinus)	AFP	—
Non-gestational choriocarcinoma	beta-hCG	—
Immature teratoma	AFP	LDH
Embryonal carcinoma	AFP + beta-hCG	Both elevated
Granulosa cell tumour	Inhibin (B most specific)	Oestradiol
Sertoli-Leydig cell tumour	Testosterone / androstenedione	AFP (if heterologous)
Mucinous tumours (ovary/appendix)	CEA	CA 19-9, CA-125
GTD / Gestational choriocarcinoma	beta-hCG	—
Krukenberg tumour (gastric met)	CEA	CA 19-9

Memory device for germ cell markers: - "AYS" = AFP → Yolk Sac - "HCG → H is for trophoblast (choriocarcinoma)" - "LDH → Like Dysgerminoma's Hash" - Granulosa = **inhibin** (makes sense — granulosa cells normally produce inhibin)

Exam Question — CA-125 Marker (NEET-PG 2020-style)

Q692 — CA-125 is a marker antigen primarily used for the diagnosis of:

- A. Colorectal carcinoma
- B. Breast carcinoma
- C. Brain tumour
- D. Ovarian carcinoma

► Answer & Explanation:

****D. Ovarian carcinoma**** — CA-125 is elevated in ~80% of advanced epithelial ovarian cancers, especially high-grade serous carcinoma. However, it is NOT specific: also elevated in endometriosis, PID, uterine fibroids, early pregnancy, liver disease, and other cancers (endometrial, pancreatic). CA-125 is most useful for monitoring treatment response and detecting recurrence in known ovarian cancer, NOT as a primary diagnostic test. For screening, CA-125 combined with TVS is used in high-risk women (BRCA mutation carriers), but alone has poor sensitivity for early-stage disease.

Exam Question — Dysgerminoma Marker — Secreted Marker (NEET-PG pattern)

Q431 — Which product is secreted by dysgerminoma?

- A. Alpha-fetoprotein (AFP)
- B. Human chorionic gonadotropin (hCG)
- C. Placental alkaline phosphatase (PLAP)
- D. Serum inhibin

► Answer & Explanation:

****C. Placental alkaline phosphatase (PLAP)**** — Pure dysgerminoma (the ovarian counterpart of testicular seminoma) secretes PLAP and LDH. It does NOT secrete AFP (that is yolk sac tumour) or hCG (that is choriocarcinoma). If a dysgerminoma has beta-hCG elevation, it suggests a mixed germ cell tumour with choriocarcinoma elements — important distinction. Inhibin (option D) is secreted by granulosa cell tumours and Sertoli-Leydig tumours, not germ cell tumours.

Exam Question — Dysgerminoma Monitoring Marker (NEET-PG repeated)

Q432 — In a confirmed case of dysgerminoma, which serum marker is monitored for treatment response and recurrence?

- A. Beta-hCG
- B. Alpha-fetoprotein
- C. Lactate dehydrogenase (LDH)
- D. Serum inhibin

► Answer & Explanation:

****C. Lactate dehydrogenase (LDH)**** — LDH (especially isoenzyme 1) is the preferred serum marker for monitoring dysgerminoma. While PLAP is secreted, LDH is the practically used monitoring marker. Beta-hCG (option A) is monitored in gestational trophoblastic disease and non-gestational choriocarcinoma. AFP (option B) is monitored in yolk sac tumours and immature teratomas. Inhibin (option D) is the monitoring marker for granulosa cell tumour. This question appears in a slightly different form multiple times — the answer is always LDH for dysgerminoma.

Exam Question — Clear Cell Tumour Classification Trap

Q433 — All of the following are germ cell tumours of the ovary EXCEPT:

- A. Mesonephroid (clear cell) tumour
- B. Mature cystic teratoma (dermoid)
- C. Dysgerminoma
- D. Endodermal sinus tumour (yolk sac tumour)

► Answer & Explanation:

****A. Mesonephroid (clear cell) tumour**** — Clear cell carcinoma is an EPITHELIAL (surface epithelial-stromal) ovarian tumour, NOT a germ cell tumour. The name "mesonephroid" (referring to mesonephric/Wolffian duct-like appearance) is deliberately confusing when placed next to germ cell options. Germ cell tumours arise from primordial germ cells: dysgerminoma (option C), mature and immature teratoma (option B), yolk sac tumour (option D), embryonal carcinoma, and non-gestational choriocarcinoma. Clear cell carcinoma is epithelial, arises in endometriosis, and is not related to germ cells at all.

12.4 Sex Cord-Stromal Tumours — Hormonal Behaviour

Granulosa Cell Tumour (GCT): - Produces **oestrogen** → FEMINISING effects - Postmenopausal women: postmenopausal bleeding (from oestrogen stimulation of endometrium) - Premenopausal women: menstrual irregularity; in prepubertal girls: precocious puberty - Histological hallmark: **Call-Exner bodies** (small follicle-like spaces filled with eosinophilic material, surrounded by granulosa cells) - Tumour marker: **Inhibin** (B most specific); also AMH (anti-Müllerian hormone) - Menopausal women with GCT: endometrial carcinoma in 5–10% (unopposed oestrogen effect) - Low-grade malignancy; late recurrence (10–20 years after initial treatment)

Sertoli-Leydig Cell Tumour (Arrhenoblastoma): - Produces **androgens** (testosterone) → VIRILISING effects - Clinical features: hirsutism, clitoromegaly, voice deepening, amenorrhoea, temporal balding, breast atrophy - Most common in women aged 20–40 years - Low-grade malignancy; good prognosis - AFP may be elevated if heterologous elements present

Fibroma (Ovarian): - Benign, solid, fibrous - **Meigs syndrome:** Ovarian fibroma + ascites + right-sided hydrothorax - Ascites and pleural effusion resolve completely after tumour removal - "Pseudo-Meigs syndrome" = same triad with OTHER ovarian tumours (serous cystadenoma, struma ovarii, etc.)

Exam Question — Call-Exner Bodies (NEET-PG pattern)

Q133 — Call-Exner bodies are seen in:

- A. Dysgerminoma
- B. Theca cell tumour
- C. Granulosa cell tumour
- D. Polyembryoma

► **Answer & Explanation:**

****C. Granulosa cell tumour**** — Call-Exner bodies are small follicle-like spaces filled with eosinophilic material, surrounded by granulosa cells arranged in rosette-like patterns. They are pathognomonic for granulosa cell tumours. The granulosa cell tumour also produces inhibin and oestrogen. Option B (theca cell tumour) is closely related and also produces oestrogen, but theca cells form luteinised clusters without the follicle-like Call-Exner spaces. Option A (dysgerminoma) shows large vesicular cells with lymphocytic infiltrate — no follicle-like structures. Inhibin and Call-Exner bodies = granulosa cell tumour: remember this pairing.

Exam Question — Schiller-Duval Bodies (NEET-PG pattern)

Q137 — Schiller-Duval bodies are pathognomonic of:

- A. Endodermal sinus (yolk sac) tumour
- B. Granulosa cell tumour
- C. Dysgerminoma
- D. Dermoid cyst

► **Answer & Explanation:**

****A. Endodermal sinus (yolk sac) tumour**** — Schiller-Duval bodies are perivascular glomeruloid structures: a central capillary surrounded by a layer of tumour cells within a cystic space — resembling a primitive glomerulus. AFP is markedly elevated. Yolk sac tumours (endodermal sinus tumours) occur most commonly in girls and young women (median age 19 years), grow rapidly, and require BEP chemotherapy. Granulosa cell tumours (option B) have Call-Exner bodies (different!). Dysgerminoma (option C) has large cells with central nuclei and lymphocytic infiltrate. Dermoid (option D) has squamous epithelium with skin appendages. Call-Exner = GCT; Schiller-Duval = Yolk Sac: always pair these.

Exam Question — Meigs Syndrome (NEET-PG pattern)

Q356 — Meigs syndrome consists of:

- A. Ovarian fibroma + ascites + right hydrothorax
- B. Granulosa cell tumour + virilisation + ascites
- C. Dysgerminoma + hypercalcaemia + effusion
- D. Mucinous tumour + pseudomyxoma peritonei

► **Answer & Explanation:**

****A. Ovarian fibroma + ascites + right hydrothorax**** — Meigs syndrome is the classic triad. Importantly, (1) the hydrothorax is right-sided in 70%, bilateral in 10%, and left-sided in 20%; (2) complete resolution of both ascites and pleural effusion occurs after surgical removal of the fibroma; (3) CA-125 may be markedly elevated, mimicking malignancy — a diagnostic trap. Option B describes features of a virilising sex cord-stromal tumour, not Meigs. Option C (dysgerminoma + hypercalcaemia) is a feature of small cell carcinoma of the ovary (hypercalcaemic type), not Meigs. Option D is pseudomyxoma peritonei from mucinous rupture, not Meigs.

Exam Question — Krukenberg Tumour (NEET-PG pattern)

Q355 — Krukenberg tumour most commonly metastasises from:

- A. Lung
- B. Stomach (signet-ring cells)
- C. Breast
- D. Colorectal

► Answer & Explanation:

****B. Gastric carcinoma (signet-ring cells)**** — Krukenberg tumours are bilateral ovarian metastases characterised by signet-ring cells (mucin displaces the nucleus peripherally) embedded within a cellular spindle-cell stroma. The stomach is the most common primary site in Asian patients, while breast and colon are next. Lung (option A) very rarely metastasises to the ovary. The key histological feature is signet-ring cells in cellular stroma — not seen in primary ovarian tumours. Krukenberg = Bilateral ovarian mass in a woman with GI symptoms → always ask about the stomach and colon.

12.5 FIGO Staging of Ovarian Cancer

Stage	Definition
IA	One ovary, capsule intact, no surface growth, negative washings
IB	Both ovaries, capsule intact
IC	As above + capsule rupture / surface involvement / positive washings
IIA	Extension to uterus or tubes
IIB	Extension to other pelvic structures
IIIA	Microscopic peritoneal metastases beyond pelvis
IIIB	Macroscopic peritoneal metastases ≤ 2 cm
IIIC	Peritoneal metastases > 2 cm $\pm$ retroperitoneal LN
IVA	Pleural effusion with cytologically positive fluid
IVB	Parenchymal liver/spleen metastases; extra-abdominal LN

Note: Ovarian cancer Stage IIIB and cervical cancer Stage IIIB are COMPLETELY DIFFERENT. Ovarian IIIB = peritoneal metastases ≤ 2 cm. Cervical IIIB = pelvic wall extension or hydronephrosis. This cross-chapter confusion is an exam trap.

Most common stage at diagnosis: Stage IIIC (~50% of ovarian cancers are Stage III at diagnosis — this is why it is the "silent killer").

Treatment: Surgery (TAH + BSO + omentectomy + peritoneal biopsies + LND = comprehensive staging/debulking) + platinum-taxane chemotherapy (carboplatin + paclitaxel).

12.6 Gestational Trophoblastic Disease — Classification

GTD spectrum:

GTD

- ├─ BENIGN (Non-neoplastic)
 - ├─ Complete hydatidiform mole
 - └─ Partial hydatidiform mole
- └─ MALIGNANT (GTN = Gestational Trophoblastic Neoplasia)
 - ├─ Invasive mole (locally invasive, high HCG)
 - ├─ Choriocarcinoma (highly malignant, haematogenous spread)
 - ├─ Placental site trophoblastic tumour (PSTT)
 - └─ Epithelioid trophoblastic tumour (ETT)

12.7 Complete vs Partial Hydatidiform Mole — The Critical Comparison

This table is the single most tested GTD concept:

Feature	Complete Mole	Partial Mole
Karyotype	46XX (diploid, all paternal)	69XXX or 69XXY (triploid)
Fetal tissue	ABSENT	PRESENT (abnormal fetus/fetal RBCs)
Trophoblastic hyperplasia	Diffuse, marked	Focal, mild
Villous hydrops	Diffuse	Focal ("Swiss-cheese placenta")
Uterine size	Large for dates (in 50%)	Small or appropriate for dates
HCG level	Very high (> 100,000 mIU/mL)	Lower than complete
Ultrasound	Snowstorm, NO fetus	Swiss-cheese placenta + small fetus
Theca-lutein cysts	Present in 25–30%	Rare
Risk of GTN	15–20%	0.5–1%
Pre-eclampsia risk	High (before 20 weeks — a classic finding)	Low
Hyperemesis	Severe	Mild

Genetics detail (important for NEET-PG): - Complete mole: typically formed by fertilisation of an empty ovum by **one sperm that duplicates itself** (monospermic, 46XX) OR by **two sperm** (dispermic, 46XX or 46XY — all paternal). Nuclear material is entirely paternal; maternal mitochondrial DNA is present but nuclear chromosomes are all paternal. - Partial mole: fertilisation of a normal ovum by **two sperm** simultaneously → triploid 69XXX, 69XXY, or 69XYY.

Exam Question — Complete Mole Ultrasound (NEET-PG repeated)

Q20 — Classic ultrasound appearance of a complete hydatidiform mole:

- A. Snowstorm pattern with echogenic foci filling the uterus
- B. Swiss-cheese placenta with a small fetus
- C. Anechoic intrauterine cyst
- D. Snowstorm pattern with a fetus

► Answer & Explanation:

****A. Snowstorm pattern with echogenic foci filling the uterus**** — Complete mole shows multiple echogenic/hydropsic villi filling the entire uterine cavity with NO fetal parts visible — the "snowstorm." Option B (Swiss-cheese placenta with a small fetus) describes a PARTIAL mole — the fetus is abnormal and triploid, the placenta shows focal hydropsic change. Option D is the critical distractor: a snowstorm WITH a fetus = partial mole scenario, not complete mole. The complete mole has NO fetus whatsoever. Option C (anechoic cyst) suggests a simple ovarian cyst or early failed pregnancy — not a mole.

12.8 Presentation of Molar Pregnancy

Classic clinical picture: - Amenorrhoea (missed periods) → patient thinks she is pregnant - Vaginal bleeding (brownish, continuous) — most common presenting symptom - Uterus larger than dates for gestational age (though may be normal or small in partial mole) - **Passage of grape-like vesicles** — pathognomonic when present (not always seen) - Hyperemesis gravidarum (worse than normal pregnancy due to very high HCG) - Signs of hyperthyroidism (HCG has weak TSH-like activity → can cause biochemical hyperthyroidism at very high levels) - Pre-eclampsia before 20 weeks — a classic sign; PIH before 20 weeks should always raise suspicion for molar pregnancy - **Theca-lutein cysts** (bilateral large multi-locular ovarian cysts): HCG overstimulates ovaries → bilateral ovarian enlargement; **not to be treated** (they regress with falling HCG after mole evacuation)

Scenario Question — Grape-like Vesicles Passage

Scenario: A 26-year-old G2P1 at 16 weeks presents with brownish vaginal bleeding and passage of grape-like vesicles. Ultrasound shows snowstorm pattern, no fetal parts. HCG = 250,000 mIU/mL. **Diagnosis, next step, and follow-up?**

Diagnosis: Complete hydatidiform mole (46XX, all paternal, no fetal parts, snowstorm USS, very high HCG, grape-like vesicles).

Next step: Suction evacuation (suction curettage) — the first-line treatment at any gestational age. NOT oxytocin induction, NOT prostaglandins (these cause trophoblastic dissemination). Hysterectomy is an alternative only in older women who desire sterilisation.

Pre-evacuation workup: HCG quantitative, blood group + crossmatch, CXR (pulmonary trophoblastic embolism?), coagulation screen, thyroid function.

During evacuation: Oxytocin infusion after suction starts (not before — to avoid trophoblastic dissemination), Anti-D immunoglobulin if Rh-negative.

Post-evacuation follow-up (most tested aspect): - **Serial beta-hCG:** Weekly until normal → then monthly for 6 months (complete mole); 3 months (partial mole) - **Use contraception** during the entire surveillance period (OCP is first choice — also reduces false HCG signals) — this is often incorrectly remembered as "no contraception" - **No pregnancy** during surveillance (rising HCG from new pregnancy would be confused with GTN) - **Normal HCG values:** Should normalise within 56–70 days of evacuation for complete mole

GTN criteria (when to start chemotherapy): 1. HCG plateau (< 10% change) for ≥ 3 consecutive weekly measurements 2. Rising HCG (≥ 10% rise) over 2 consecutive weekly measurements 3. HCG still elevated at 6 months after evacuation 4. Histological diagnosis of choriocarcinoma 5. Metastatic disease (lung, brain, liver)

Exam Question — Molar Pregnancy Treatment (NEET-PG pattern)

Q145 — A 29-year-old at 16 weeks with bleeding and snowstorm USS — most feasible treatment?

- A. Misoprostol alone
- B. Hysterectomy
- C. Mifepristone
- D. Suction evacuation

► Answer & Explanation:

****D. Suction evacuation**** — Standard definitive treatment for complete hydatidiform mole at any gestational age. Misoprostol (option A) is a prostaglandin used for termination of normal pregnancies — it causes uterine contractions that can disseminate trophoblastic tissue haematogenously, potentially promoting metastatic GTN. Mifepristone (option C) similarly induces contractions and is not appropriate for molar evacuation. Hysterectomy (option B) is reserved for women who have completed their family and wish simultaneous sterilisation; it eliminates the risk of invasive mole but is not first-line for a 29-year-old who may want future pregnancies.

Exam Question — GTD Incidence Geography (NEET-PG)

Q49 — The highest incidence of gestational trophoblastic disease (complete mole) is in:

- A. Australia
- B. Southeast Asia
- C. North America
- D. Western Europe

► Answer & Explanation:

****B. Southeast Asia**** — Complete mole incidence: ~1 in 600–1000 pregnancies in Southeast Asia, Philippines, and Taiwan, compared to ~1 in 1500–2000 in Western countries. Risk factors include dietary deficiency of vitamin A/carotene, Asian ethnicity, extremes of reproductive age (< 15 or > 40 years), history of previous molar pregnancy (1–2% recurrence risk), and blood group A in husband with group O wife (proposed association). The high incidence in Asia makes GTD a relatively common clinical entity in Indian practice.

12.9 Choriocarcinoma — The Unique Trophoblastic Malignancy

Key clinical facts: - Highly malignant tumour of trophoblastic cells (cytotrophoblast + syncytiotrophoblast) - **Can follow ANY pregnancy:** Normal delivery (30–40%), hydatidiform mole (50%), spontaneous abortion (20%), ectopic pregnancy (rare) - The interval between antecedent pregnancy and choriocarcinoma diagnosis can be months to years - Classic presentation: irregular bleeding after any pregnancy + rising HCG + lung metastases (cannonball lesions) - **No chorionic villi** histologically (distinguishes from invasive mole, which has villi) - Spreads haematogenously (not lymphatically): lungs most common, then brain, liver - **Extraordinarily chemosensitive:** 90–95% cure rate even with metastatic disease

Scenario: Post-hysterectomy patient with rising HCG - Any woman with persistently or rising HCG after hysterectomy should be evaluated for GTN (PSTT or choriocarcinoma) - Investigate: serum HCG, CXR, CT chest/abdomen/pelvis, brain MRI (if neuro symptoms) - Treat: WHO/FIGO risk score → single-agent or multi-agent chemotherapy

Exam Question — Choriocarcinoma Following Normal Pregnancy (NEET-PG 2021)

Q217 — A woman underwent D&C for molar pregnancy 6 months ago and now has abdominal pain, persistently elevated beta-hCG, and cannonball lung lesions. Appropriate management?

- A. Methotrexate with folinic acid
- B. Methotrexate alone
- C. EMA-CO regimen
- D. Hysterectomy

► Answer & Explanation:

****A. Methotrexate with folinic acid**** — This question requires WHO/FIGO risk scoring before choosing the answer. Post-molar GTN with lung metastases needs risk scoring (see section 12.10). If the WHO score is < 7 (low-risk), single-agent methotrexate with folinic acid (leucovorin) rescue is the standard. EMA-CO (option C) is reserved for HIGH-risk GTN (WHO score ≥ 7). The distinguishing feature here is the risk score — cannonball lesions alone (1 metastatic site = lung) may not necessarily equal high-risk if other score parameters are low. Hysterectomy (option D) is not the primary treatment for metastatic GTN; chemotherapy cures even metastatic disease in the vast majority. Note: if the question specified WHO score ≥ 7 OR more than one metastatic site OR prior chemotherapy failure, the answer would be EMA-CO.

Exam Question — EMA-CO for Resistant Choriocarcinoma (NEET-PG 2021)

Q15 — Choriocarcinoma failing single-agent methotrexate. Next regimen?

- A. Methotrexate + folinic acid
- B. Methotrexate alone (continue)
- C. EMACO regimen
- D. Hysterectomy

► Answer & Explanation:

****C. EMA-CO regimen**** — When single-agent chemotherapy fails (rising or plateau HCG despite adequate methotrexate courses), the patient is reclassified as high-risk and requires multi-agent EMA-CO: ****E**toposide, **M**ethotrexate, **A**ctinomycin D (given alternating weeks) + **C**yclophosphamide + **O**ncovin (Vincristine)**. EMA-CO achieves cure in ~75–80% of high-risk GTN. Continuing methotrexate alone (option B) when resistance is documented is incorrect and delays effective treatment. Hysterectomy (option D) may be used as an adjunct in PSTT or to manage haemorrhage but is not the chemotherapy choice.

12.10 WHO/FIGO Risk Scoring for GTN — The Score That Determines Treatment

The WHO/FIGO risk score predicts response to single-agent chemotherapy. Score ≥ 7 = high-risk = multi-agent EMA-CO required.

Prognostic Factor	Score 0	Score 1	Score 2	Score 4
Age	< 40	≥ 40	—	—
Antecedent pregnancy	Mole	Abortion	Term pregnancy	—
Interval (months)	< 4	4–6	7–12	> 12
Pre-treatment HCG (mIU/mL)	< 1000	1000–9999	10,000–99,999	≥ 100,000
Largest tumour size (cm)	< 3	3–4	≥ 5	—
Sites of metastasis	Lung	Spleen, kidney	GI tract	Brain, liver
Number of metastases	0	1–4	5–8	> 8
Previous failed chemotherapy	—	—	Single drug	≥ 2 drugs

Score interpretation: - **0–6 = Low-risk:** Single-agent chemotherapy (methotrexate 50 mg/m² IM every 48h + folinic acid 15 mg PO 30h after each MTX, for 8 days; or actinomycin D) - **≥ 7 = High-risk:** Multi-agent EMA-CO (or EMA-EP in resistant cases)

Memory aid for WHO score parameters (8 rows): "A² I² HT SM P" = Age, Antecedent pregnancy, Interval, HCG, Tumour size, Sites, Number of Metastases, Prior treatment

Key scoring facts for NEET-PG: - Term pregnancy as antecedent = score 2 (highest) → more aggressive disease expected - Brain and liver metastases = score 4 (highest for metastatic sites) → high-risk almost automatically - HCG ≥ 100,000 mIU/mL = score 4 → high-risk - Prior failed chemotherapy = most dire situation → score 2–4

Exam Question — GTN Marker for Monitoring (NEET-PG pattern)

Q87 — In GTN, the most reliable marker for monitoring treatment response and detecting recurrence:

- A. CA-125
- B. AFP
- C. β-hCG
- D. LDH

► Answer & Explanation:

****C. β-hCG**** — Beta-hCG is secreted by all types of trophoblastic tissue (normal and malignant); it is the **ONLY** reliable marker for GTN of all types. Serial beta-hCG measurements document: (a) complete remission (three consecutive weekly normal values), (b) resistance to current therapy (plateau or rise during chemotherapy), and (c) recurrence after treatment. CA-125 (option A) is an ovarian cancer marker — it has no reliable role in GTN monitoring. AFP (option B) is elevated in yolk sac tumours but not GTN. LDH (option D) monitors dysgerminoma. Beta-hCG is the gold standard — any examination question about GTN monitoring will have beta-hCG as the answer.

Exam Question — Beta-hCG Measurement Method (NEET-PG)

Q36 — The gold standard test for measuring serum β-hCG levels (in GTD) is:

- A. ELISA
- B. Bioassay

- C. Radioimmunoassay (RIA)
- D. Latex agglutination inhibition

► **Answer & Explanation:**

****C. Radioimmunoassay (RIA)**** — RIA is the most sensitive and specific quantitative technique for measuring serum beta-hCG, detecting levels as low as 1–5 mIU/mL. This precision is essential for GTD monitoring (where decisions to continue or stop chemotherapy depend on small changes in HCG). ELISA (option A) is used in home pregnancy tests and is adequate for qualitative detection but less precise for low-level quantification. Latex agglutination inhibition (option D) is a qualitative urine pregnancy test — wholly inadequate for GTN monitoring. Bioassay (option B) is historical and no longer used clinically.

12.11 Dysgerminoma — The Most Important Germ Cell Tumour

Why it gets its own section: Dysgerminoma is tested on almost every NEET-PG — typically in 2–3 questions per year across marker identification, histology, and treatment.

Clinical features: - Most common **malignant** germ cell tumour of the ovary (~50% of malignant GCT) - Ovarian counterpart of **testicular seminoma** - Age: typically 10–30 years (young women, adolescents) - Bilateral in 10–15% (more bilateral than other GCT) - Solid tumour on USS; moderate size - Associated with **gonadal dysgenesis** (45,X / 46,XY mosaics): dysgenetic gonads → gonadoblastoma → dysgerminoma

Histology: - Large, vesicular cells with clear cytoplasm and central nuclei (similar to primordial germ cells) - Lymphocytic infiltrate in the stroma (T-cell mediated immunity) - Fibrous septa dividing tumour into lobules - Syncytiotrophoblast giant cells may be present → HCG mildly elevated (NOT AFP)

Tumour markers: - **LDH** (especially isoenzyme 1) = primary monitoring marker - **PLAP** (Placental Alkaline Phosphatase) = secreted marker - Beta-hCG: only mildly elevated if syncytiotrophoblast elements present; AFP = negative (absence of AFP distinguishes pure dysgerminoma from mixed GCT)

Treatment: - **Radiosensitive AND chemosensitive** (unique advantage) - Stage IA: Unilateral salpingo-oophorectomy (fertility-sparing) — excellent prognosis - Advanced/bilateral: BEP chemotherapy (Bleomycin, Etoposide, Cisplatin) — 5-year survival > 90% - Historically treated with radiotherapy; now chemotherapy preferred to preserve fertility

Prognosis: Excellent — even with metastatic disease, > 90% cure rate with chemotherapy.

Exam Question — Most Common Pure Germ Cell Tumour (NEET-PG)

Q121 — Most common pure germ cell tumour of the ovary:

- A. Choriocarcinoma
- B. Dysgerminoma
- C. Embryonal cell tumour
- D. Malignant teratoma

► Answer & Explanation:

****B. Dysgerminoma**** — Dysgerminoma accounts for ~50% of all malignant ovarian germ cell tumours. Important to distinguish: the most common germ cell tumour **OVERALL** is the mature cystic teratoma (dermoid) — a **BENIGN** tumour. The most common **MALIGNANT** pure germ cell tumour = dysgerminoma. This distinction is an exam trap. Choriocarcinoma (option A) is rare in pure form; it usually occurs as a component of mixed GCT. Embryonal carcinoma (option C) is very rare. Malignant teratoma (option D) = immature teratoma, which is less common than dysgerminoma as a pure malignant GCT.

Bias Removal — Avoiding the High-Frequency Exam Errors

The Five Most Dangerous GTD/Ovarian Confusions:

- 1. Complete mole HCG vs Partial mole HCG:** - "Falling HCG after suction evacuation = GOOD news" (trophoblastic tissue being eliminated) - "Rising HCG after evacuation = BAD news" (GTN developing — invasive mole or choriocarcinoma) - Complete mole: very high HCG (> 100,000 mIU/mL usually) - Partial mole: lower HCG (often < 100,000 mIU/mL) - Theca-lutein cysts = seen with complete mole (HCG overstimulates ovaries); do NOT aspirate them
- 2. Contraception during mole surveillance:** - The correct instruction is: "USE contraception during the entire surveillance period" - The exam distractors often say "no contraception is required" — **WRONG** - OCP is preferred (also suppresses LH which could interfere with HCG assays) - IUD is contraindicated until HCG normalises (insertion risk of perforation in soft involuting uterus)
- 3. Ovarian vs Cervical IIIB:** - Cervical IIIB = hydronephrosis / pelvic wall extension - Ovarian IIIB = peritoneal metastases ≤ 2 cm - Never mix these staging definitions
- 4. Granulosa cell tumour hormone vs Sertoli-Leydig hormone:** - Granulosa = oestrogen → feminising (endometrial hyperplasia/cancer in post-menopausal woman) - Sertoli-Leydig = androgen → virilising (hirsutism, clitoromegaly in young woman) - Examiner favourite: "Young woman with irregular periods and hirsutism → which ovarian tumour?" = Sertoli-Leydig
- 5. Clear cell carcinoma is NOT a germ cell tumour:** - Clear cell (mesonephroid) carcinoma = **EPITHELIAL** tumour - The name "mesonephroid" evokes mesonephric/germ cell thinking — resist this - Germ cell = dysgerminoma, teratoma, yolk sac, embryonal, choriocarcinoma (non-gestational)

Memory Engineering

GTD Risk Score — AAIHTSNP (8 items, same order as the table): - Age (< 40 = 0; ≥ 40 = 1) - Antecedent pregnancy (Mole=0, Abortion=1, Term=2) - Interval months (< 4 = 0; 4–6 = 1; 7–12 = 2; > 12 = 4) - **HCG** level (< 1000 = 0 → ≥ 100,000 = 4) - Tumour size (< 3 cm = 0; 3–4 cm = 1; ≥ 5 cm = 2) - Sites (Lung = 1; Spleen/kidney = 2; GI = 2; Brain/liver = 4) - Number of mets (0=0; 1–4=1; 5–8=2; >8=4) - Prior treatment (Single drug=2; 2 drugs=4)

Score ≥ 7 → **High Risk** → EMA-CO

Ovarian tumour marker mnemonic — "DYGAL": - Dysgerminoma = **LDH** (L for LDH) - Yolk Sac = **AFP** (A for AFP) - Granulosa = **Inhibin** - Adenocarcinoma (epithelial) = **CA-125** - Choriocarcinoma = **HCG**

Ultra High-Yield Revision Layer

Ovarian Tumour Marker Table (Final Revision)

Tumour Type	Marker	Extra Point
Serous epithelial (most common)	CA-125	BRCA1/2 association; psammoma bodies
Granulosa cell	Inhibin B	Call-Exner bodies; oestrogen production; LMP malignancy
Sertoli-Leydig	Testosterone	Virilisation; AFP if heterologous elements
Dysgerminoma	LDH (+ PLAP)	Most common malignant GCT; radiosensitive; bilateral 10%
Yolk sac tumour	AFP	Schiller-Duval bodies; in young girls
Choriocarcinoma (non-gestational)	beta-hCG	No chorionic villi; hematogenous spread
Embryonal carcinoma	AFP + beta-hCG	Both markers elevated
Mucinous	CEA + CA 19-9	Largest tumour; pseudomyxoma if ruptured
Krukenberg (secondary)	CEA	Signet-ring cells; bilateral; stomach is #1 primary

Complete vs Partial Mole — Final Flash Card

	Complete Mole	Partial Mole
Karyotype	46XX (all paternal)	69XXX (triploid)
Fetus	Absent	Present (abnormal)
USS	Snowstorm	Swiss-cheese + fetus
HCG	Very high	Moderate
GTN risk	15–20%	0.5–1%
Theca-lutein cysts	Yes (25%)	Rare

GTN WHO/FIGO Score — Threshold

- **Score 0–6 → Low-risk → Single-agent MTX (± folinic acid)**
- **Score ≥ 7 → High-risk → EMA-CO**
- Cannonball lung lesions alone: may be low-risk if score < 7
- Brain/liver metastases: score 4 for site → almost always high-risk

Top 10 Exam Traps — Ovarian Cancer & GTD

1. Most common malignant germ cell tumour = dysgerminoma (NOT mature teratoma — teratoma is most common BENIGN GCT)
2. Dysgerminoma marker for MONITORING = LDH (PLAP is secreted, but LDH is monitored)
3. Clear cell carcinoma = EPITHELIAL, NOT germ cell
4. Call-Exner bodies = Granulosa cell tumour (NOT dysgerminoma — different histology)

5. Schiller-Duval bodies = Yolk sac tumour (not granulosa — pair these separately from Call-Exner)
6. Complete mole = 46XX all PATERAL (NOT 46XX maternal); empty egg fertilised by sperm
7. Partial mole = TRIPLOID 69XXX (two sperm + one egg) = NOT diploid
8. Post-molar surveillance requires CONTRACEPTION (usually OCP) — not "no contraception"
9. Meigs syndrome = FIBROMA (not granulosa, not serous) + ascites + right hydrothorax
10. GTN WHO score ≥ 7 = EMA-CO; < 7 = methotrexate; choriocarcinoma failing MTX = always EMA-CO

HPV Vaccine Recommendations for a Daughter of a Cervical Cancer Patient

Q251 — A 54-year-old woman has advanced cervical cancer and a 14-year-old daughter. Advice for the daughter?

- A. Advise HPV vaccine
- B. Screen for BRCA mutation
- C. Screen for PTEN mutation
- D. Perform cervical biopsy

► Answer & Explanation:

****A. Advise HPV vaccine**** — Cervical cancer is caused by HPV, an environmentally acquired infection — it is NOT an inherited genetic cancer. BRCA screening (option B) applies to breast and ovarian cancer (hereditary risk). PTEN screening (option C) applies to endometrial/Cowden syndrome. Cervical biopsy in a 14-year-old (option D) without abnormal cytology or clinical abnormality would be unwarranted. The appropriate, evidence-based intervention for a 14-year-old at population-level risk of HPV exposure is HPV vaccination (2 doses at 0 and 6 months in the 9–14 age group, per WHO/India MoHFW guidelines). This question tests both the understanding that cervical cancer is viral (not genetic) in origin AND the practical vaccination schedule.

About This Book

This book was generated from the NEET-PG OBG question database (2015–2025), containing 1,063 questions across all major OBG topics. Questions were sourced from PrepLadder, DigiNerve, Medicoholic PDFs, DocTutorials, firstranker.com, and coaching-site recall compilations.

Sources: Medicoholic PYQ compilations · PrepLadder PYQ banks · DigiNerve 2025 recall · DocTutorials 2024 · firstranker.com 2015–2024

For exam preparation only. Verify critical management details against current guidelines (FOGSI, WHO, ACOG).

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